

Positive Geometries in Physics

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to my friends

Disclaimers

Authorship

The results of this thesis are partially based on existing work with coauthors.

- Chapter 2 is unpublished solo-authored work, except for the introductory material in the first section. I am very grateful for discussions with Rainer Sinn and Dmitrii Pavlov.
- Chapter 3 is unpublished and solo-authored work.
- Chapter 4 is based on [KPS26a]. All authors contributed equally.
- Chapter 5 is based on [KS25]. All authors contributed equally. The exception is the last section of this chapter, which was authored solely by me.
- Chapter 6 is based on upcoming work with Mattia Arundine, Veronica Calvo Cortes, and Facundo Rost. All authors contributed equally.

Use of AI Tools

AI has been used in this thesis as a search engine, for fixing typos, and in some circumstances to generate TikZ code for figures.

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Notation

Algebraic Geometry

$\mathrm{Sing}(X)$	Singular locus of X
$\mathrm{codim}(X)$	Codimension of X
$\mathrm{Ch}(Y)$	Chow hypersurface associated to Y
$\mathcal{O}_X$	Structure sheaf of X
$\mathbb{C}[x_1, \ldots, x_n]$	Polynomial ring in n variables over $\mathbb{C}$
$\langle f_1, \ldots, f_m \rangle$	Ideal generated by $f_1, \ldots, f_m \in \mathbb{C}[x_1, \ldots, x_n]$
$\mathbb{P}^n$	Complex projective n-space
$\mathbb{V}(I)$	Vanishing locus of an ideal I
$\mathrm{Gr}(k, n)$	Grassmannian of k-planes in $\mathbb{C}^n$
$T_x X$	Intrinsic tangent space of X at x
$\mathbf{T}_x X$	Embedded tangent space of X at x

Homology and Hodge Theory

$H_\bullet(X)$	Singular homology of X with coefficients in $\mathbb{Q}$
$H_c^\bullet(X)$	Compactly supported cohomology of X with coefficients in $\mathbb{Q}$
$W_\bullet H$	Increasing weight filtration on H
$F^\bullet H$	Decreasing Hodge filtration on H
$\mathrm{gr}_q^W H$	q-th graded piece of the weight filtration
$\mathrm{gr}_F^p H$	p-th graded piece of the Hodge filtration
Ω_X^n	Sheaf of regular n-forms on X
$\Omega_{\log}^n(X \setminus D)$	Space of logarithmic n-forms with poles along D
$\mathrm{Res}_D \omega$	Poincaré residue of ω along D

Matrices and Linear Algebra

$\mathrm{GL}_n(K)$	General linear group over K
A^T	Transpose of a matrix A
$\ker$	Kernel of a map
im	Image of a map

Positive Geometry

C_d	Rational normal curve of degree d
$\mathcal{S}_{01}$	Distinguished secant line on C_d
$\mathrm{Sec}^{\ell}(C_d)$	Variety of ℓ-secants to C_d
$\partial_a S$	Algebraic boundary of a semi-algebraic set S
$\mathcal{L}(D)$	Boundary stratification of a divisor D
$\mathcal{R}(S)$	Residual arrangement of S
$\mathrm{adj}(S)$	Adjoint of S
$g(X,Y)$	Genus of the pair (X,Y)
$\mathrm{cr}(X,Y)$	Combinatorial rank of the pair (X,Y)
$\mathcal{A}_{n,k,m}(Z)$	Amplituhedron defined by Z
$\mathcal{A}_k^{\infty}$	k-th limit amplituhedron

Miscellaneous

$\langle \cdot\,, \cdot \rangle$	Standard scalar product on $\mathbb{C}^n$
span_K	Linear span over K
dlog	Logarithmic differential
$[n]$	Set of natural numbers from 1 to n
$\binom{[n]}{k}$	Set of all size k subsets of $[n]$

Sets and Fields

$\mathbb{Z}$	Integer numbers
$\mathbb{Q}$	Rational numbers
$\mathbb{R}$	Real numbers
$\mathbb{C}$	Complex numbers

Tropical and Toric Geometry

$\mathrm{trop}(U)$	Tropicalization of U
Σ	Polyhedral fan
X_Σ	Toric variety associated to Σ
$\mathrm{Lk}(\Sigma)$	Spherical link of Σ at the origin

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Introduction

Positive geometry is a novel mathematical field with close connections to particle physics that utilizes methods from complex, real, and tropical algebraic geometry. At the heart of this subject is the geometric notion of a positive geometry: essentially a real semi-algebraic set together with a differential form that satisfies a recursive condition, called the canonical form. This notion was first introduced following the most prominent object in this field, the amplituhedron, which is conjecturally a positive geometry. The amplituhedron $\mathcal{A}_{n,k,m}$ was introduced by the physicists Nima Arkani-Hamed and Jaroslav Trnka in the groundbreaking paper [AT14]. Its canonical form provides a remarkable new and geometric approach to scattering amplitudes in $\mathcal{N} = 4$ super Yang—Mills theory (SYM), whose computation is one of the central problems in modern particle physics.

What are Positive Geometries?

This most basic question about positive geometries is already a subtle one, as there are currently multiple notions of positive geometries in the literature, not all of which are compatible. This is partially the case because there is also an ongoing debate on the question: "What should a positive geometry be?"

We want to introduce the reader to the notion that first appeared in [ABL17] and is more intuitive. The starting point is a semi-algebraic set S of dimension n, which sits inside the real points of a complex ambient variety X. Then, the essential ingredient is that we associate an n-form ω_S to S, called the canonical form, which is regular on the interior of S and has simple poles along the algebraic closure of the Euclidean boundary ∂S. Now, if the canonical form is unique, and its Poincaré residue along a component of the algebraic closure of ∂S produces canonical forms of the restriction to that component, then we call the triple (X, S, ω_S) a positive geometry. The simplest positive geometries are top-dimensional polytopes $P \subset \mathbb{R}^n$ where the canonical form is the volume (function) Vol of their dual polytope. For example, the cube $C = [-1, 1]^3 \subset \mathbb{R}^3$ is shown in the figure below, and its dual $C^\vee$ with respect to the

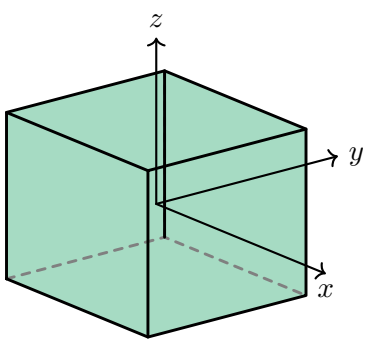
origin in $\mathbb{R}^n$ is shown to the right of it. In this case, the canonical form ω_C is

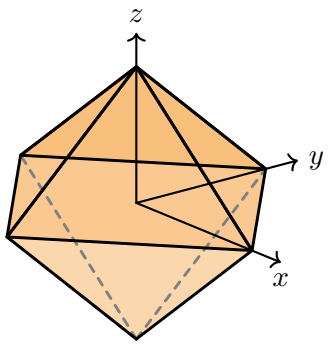
$$\begin{aligned}\omega_C &= \frac{8}{(x-1)(x+1)(y-1)(y+1)(z-1)(z+1)}\, dx \wedge dy \wedge dz \\ &= \mathrm{Vol}_{C^\vee}(x,y,z)\, dx \wedge dy \wedge dz,\end{aligned}$$

where the dual volume function $\mathrm{Vol}_{C^\vee}(x,y,z)$ is the volume of the translated dual polytope of C i.e. $\mathrm{Vol}((C-c)^\vee)$ with $c = (x,y,z)$ in the interior of C. For example, at the origin $O = (0,0,0)$, we get

$$\omega_C(O) = \mathrm{Vol}_{C^\vee}(O) = \mathrm{Vol}((C-O)^\vee) = 8,$$

which is indeed the normalized volume of the octahedron $C^\vee$ shown below in orange.





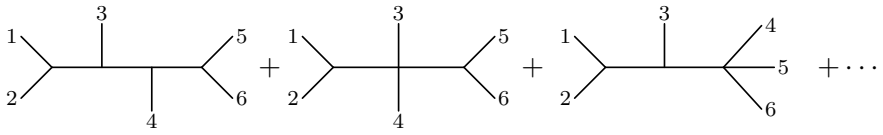
However, Brown and Dupont recently introduced a new framework for positive geometries in [BD25]. This shifts the paradigm in the study of positive geometries from semi-algebraic properties towards the mixed Hodge structures on the relative homology groups of a pair of varieties, corresponding to the Zariski closure of the boundary ∂S and the ambient variety X. The key notions are two Hodge theoretic invariants: genus and combinatorial rank. A vanishing genus relates to the uniqueness of a canonical form and a positive combinatorial rank to its existence. However, it is still an open question how best to connect these two approaches.

The Case for Positive Geometries

We will see throughout Chapter 2 that positive geometries give rise to interesting mathematical structures that showcase the interplay between tropical, toric, and real algebraic geometry, which deserve to be studied in detail. However, here we want to make a case for studying positive geometries because of their applications to theoretical physics, more precisely their role in scattering amplitudes. Here the strengths of the positive geometry machinery are twofold: computational and conceptual.

Let us deal with the computational advantages first. Loosely speaking, scattering amplitudes capture the probability of certain outcomes of n-particle interactions. The computa-

tion of these probabilities is a central problem in modern particle physics. Historically scattering amplitudes have been computed by means of mnemonic devices called a Feynman diagrams, which are planar graphs. Depending on the physical theory one wants to consider, there are rules for attaching a rational function, or more generally, an integral to each such graph. The n-point scattering amplitude A_n is then obtained by summing over the expressions associated with a family of graphs with n leaves. For example,

$A_6 =$  $+ \cdots$

This computation has a key problem, namely that the number of graphs to be summed over grows as $n!!$, hence brute force computations quickly become infeasible. However, it has been known since the groundbreaking discovery of the Parke-Taylor formula in [PT86], that this summation is very redundant, as the authors showed that this sum, spanning multiple pages, collapses into a single one-line expression. The amplituhedron, and more broadly positive geometries, promise to extend this success to further theories. More succinctly, in $\mathcal{N} = 4$ SYM, at 6 points the (tree-level maximal-helicity-violating) scattering amplitude is obtained by summing over 220 Feynman graphs as shown above. Equivalently, the canonical form of the amplituhedron $\mathcal{A}_{6,2,4}$ computes the same scattering amplitude A_6.

The second advantage of positive geometries is conceptual. The Parke-Taylor formula mentioned above reveals a hidden simplicity in scattering amplitudes, which was obscured by the summation over the Feynman graphs. Remarkably, the semi-algebraic data of a positive geometry and its canonical forms make this property apparent. Even more, they naturally encode other important properties in physics, such as factorization, which corresponds to a recursive structure of the positive geometries. More succinctly, the facets of positive geometries produce positive geometries, which capture lower-point scattering amplitudes. So studying these geometric objects offers a much more concise point of view for scattering amplitudes.

This naturally leads one to a few key challenges when dealing with positive geometries.

I. The first challenge stems from the two approaches to positive geometries. One would like to know which algebro-geometric conditions force a semi-algebraic set to be a positive geometry, and how that is related to the genus and combinatorial rank. Furthermore, the recursive definition of positive geometries is very useful, but it is often hard to verify directly. So it is important to understand when the mixed Hodge theory approach gives rise to positive geometries in this sense.
II. The second challenge is the computation of canonical forms, and more basically showing their existence and uniqueness. Since canonical forms must have simple poles along the algebraic boundary of the underlying semi-algebraic set, this involves an in-depth understanding of the algebro-geometric and semi-algebraic properties, and their interplay.

III. The third challenge is to extend the success of the amplituhedron to other branches of physics, beyond $\mathcal{N} = 4$ SYM.

Summary and Structure

The overall goal of this thesis is to tackle the challenges identified above.

After the preliminaries, in Chapter 2 we give a practitioner's introduction to the new framework for positive geometries by Brown and Dupont. First we tackle the issue arising from applying this framework to pairs with singularities. We derive several genus vanishing results for singular varieties, and establish tools to compute the combinatorial rank via tropical methods. In Section 2.4, we tackle the first challenge: We introduce (recursive) positive pairs, and with them the notion of positive geometries in the mixed Hodge theory framework. We show that a regular connected region in the complement of a recursive positive pair gives rise to a positive geometry in the original sense. This provides the first explicit bridge between the mixed Hodge theoretic framework and the definition from [ABL17], thus also making progress on the second challenge from above. Moreover, this also provides an answer to the question from the beginning of this chapter of what positive geometries should be.

In Chapter 3 we study relative homology groups and the space of logarithmic forms of an arrangement of hyperplanes $\mathcal{A}$ and a quadric Q in projective space, and the restriction of the arrangement $\mathcal{A}|_Q$ to the quadric. These pairs appear naturally in the study of the $m = k = 2$ amplituhedron $\mathcal{A}_{n,2,2} \subset \mathrm{Gr}(2,4)$, or in the study of 1-loop Feynman integrals. We focus our attention on the case where the real locus of the quadric is homeomorphic to a sphere, together with a genericity assumption; this provides combinatorial answers to several natural questions coming from the framework of positive geometries via mixed Hodge structures from Chapter 2. So in this sense this chapter serves also as a gentle introduction to the previously introduced methods. We show that both $(\mathbb{P}^n, \mathcal{A} \cup Q)$ and $(Q, \mathcal{A}|_Q)$ are recursive positive pairs. Then we show that the number of real connected regions on the quadric is the number of topes of the uniform matroid of rank k on n elements. We use this count to derive a closed binomial formula for the combinatorial rank of the pairs. Moreover, we construct an explicit basis of the space of logarithmic forms from real regions.

In Chapter 4 we study the amplituhedron in the mixed Hodge theory framework for positive geometries. For the cases where the amplituhedron is known to be a positive geometry, we show that the corresponding pair of the ambient Grassmannian and the algebraic boundary of the amplituhedron has genus zero. This includes the non-negative Grassmannian, cyclic polytopes, and the $m = k = 2$ amplituhedron. However, the situation changes for larger values of parameters. We give an explicit lower bound for the genus, coming from a positive genus curve in the residual arrangement of the amplituhedron, in the cases $m = 2$

and $m = 4$. Importantly, this does not rule out that the amplituhedron is a positive geometry in the original sense. To make this distinction concrete, we construct the Del Pezzo cube: a semi-algebraic set $P \subset \mathbb{P}^3$ which is a positive geometry, but for which the pair $(\mathbb{P}^3, \partial_a P)$ has genus one. Analogously to the amplituhedron, the genus one contribution comes from an elliptic curve in the residual arrangement. We then show how blowing up this curve produces a genus zero pair. This provides a way of tackling the central conjecture that amplituhedra are positive geometries, and it sheds some light on the first challenge from above.

In Chapter 5 we address the third challenge by introducing a limit version of the $m = 2$ amplituhedron, which gives insight into the structure of scattering amplitudes in the large-particle limit. This follows interest in the large-particle limit of scattering amplitudes, which is captured by this object. The construction is obtained by taking points on the rational normal curve over all partitions of the interval $[0, 1]$. The simplest case is the so-called pizza slice, but the construction produces a semi-algebraic subset $\mathcal{A}_k^\infty \subset \mathrm{Gr}(k, k+2)$ for all k. We determine its algebraic boundary: it consists of two Chow hypersurfaces, namely the Chow hypersurface of the rational normal curve $C_{k+1} \subset \mathbb{P}^{k+1}$ and the Chow hypersurface of the secant line through the two end points of the interval. It turns out that the boundary stratification has a very nice geometric interpretation: in terms of secant varieties and osculant planes of the rational normal curve. The main mathematical result of the chapter is that the residual arrangement is empty and that the limit amplituhedron is a positive geometry. This is remarkable, as it is a central open conjecture that finite n amplituhedra are positive geometries. Moreover, this also demonstrates the effectiveness of the notion of boundary stratification utilized in this project when dealing with the second challenge of computing canonical forms. Even further, our results also give an understanding of the limit of scattering amplitudes for large number of particles, which was an open problem for these amplitudes.

Finally, in Chapter 6 we also tackle the third challenge by providing first steps toward positive geometries in Yang-Mills cosmology. We study the cosmological Grassmannian $\mathrm{CGr}(4)$, which is the orthogonal Grassmannian in its Pfaffian embedding, together with a divisor D consisting of the Pfaffian coordinate hyperplane sections and two additional hypersurfaces. We compute its boundary stratification, show that $(\mathrm{CGr}(4), D)$ is a recursive genus zero pair, and compute its combinatorial rank. Inside the real points of $\mathrm{CGr}(4)$ we define a region R whose canonical form under the canonical map is precisely the 4-particle wavefunction coefficient in Yang—Mills theory from [Aru+26]. This marks the first establishment of a positive geometry in Yang—Mills cosmology.

Chapter 1

Preliminaries

There is no try.

Rainer

1.1 Algebraic Geometry

Let $\mathbb{C}$ denote the complex numbers, $(\mathbb{C}^\times) \subset \mathbb{C}$ the set of all non-zero points, that is the algebraic torus, and $\mathbb{P}^n \cong (\mathbb{C}^{n+1} \setminus \{0\})/(\mathbb{C}^\times)$ the projective space.

Varieties

The most essential notion from algebraic geometry needed in our work is that of a variety.

Definition 1.1.1. An *affine variety* $Y \subset \mathbb{C}^n$ is the common zero set of a list of polynomials $\{f_1, \ldots, f_m\}$ with $f_i \in \mathbb{C}[x_1, \ldots, x_n]$. We write

$$Y = \mathbb{V}(f_1, \ldots, f_m) = \{x \in \mathbb{C}^n \mid f_i(x) = 0 \text{ for all } i \in [m]\} \subset \mathbb{C}^n.$$

We will more commonly encounter another type of variety. Let $\mathbb{C}[x_0, \ldots, x_n]$ be the polynomial ring in the $n+1$ variables x_i. We say a polynomial $f \in \mathbb{C}[x_0, \ldots, x_n]$ is homogeneous of degree d if all of its monomials have the same degree d.

Definition 1.1.2. A *projective variety* $X \subset \mathbb{P}^n$ is the common zero set of a list of homogeneous polynomials $\{f_1, \ldots, f_m\}$ not necessarily all of the same degree with $f_i \in \mathbb{C}[x_0, \ldots, x_n]$. We write

$$X = \mathbb{V}(f_1, \ldots, f_m) = \{x \in \mathbb{P}^n \mid f_i(x) = 0 \text{ for all } i \in [m]\} \subset \mathbb{P}^n.$$

The subsets $U_i = \{x \in \mathbb{P}^n \mid x_i \neq 0\} \subset \mathbb{P}^n$ are called *basic open* sets, and we call $U_i \cap X$ an *affine patch* of X.

We equip $\mathbb{C}^n$ and $\mathbb{P}^n$ with the *Zariski topology*, making them topological spaces. The closed sets in the Zariski topology are precisely those which are given as zero sets of systems of homogeneous polynomials, i.e. projective varieties. This induces a topology on any variety, in which the closed sets are its subvarieties, i.e. varieties themselves. The *Zariski closure* of an open set U in this topology is defined as the smallest closed set X containing U. Lastly, varieties can be defined in much more generality; however, in this work we will only need one minor generalization.

Definition 1.1.3. A *quasi-projective variety* Y is an open subset of a projective variety X.

Remark 1.1.4. Since this is the most general notion of a variety we will need in this thesis, by variety we will henceforth mean a complex quasi-projective variety whenever we do not say otherwise.

We say a variety X has an *affine cover* $\{U_i\}$ if $X = \bigcup_i U_i$ and all the U_i are affine varieties. A variety X is *irreducible* if $X = Y_1 \cup Y_2$ implies $Y_1 = X$ or $Y_2 = X$, where Y_1, Y_2 are (closed) subvarieties of X. A variety X is *reducible* if it is not irreducible. A reducible variety X can be uniquely written as a non-trivial union of irreducible subvarieties, which are called *irreducible components*.

Example 1.1.5. The most common example of an irreducible (projective) variety in our work is the *Grassmannian* $\mathrm{Gr}(k, n)$. It is the set of all k-dimensional sub-vector spaces V of $\mathbb{C}^n$. Projectively speaking, it parameterizes the set of all $(k-1)$-dimensional linear subspaces of the projectivization $\mathbb{P}(\mathbb{C}^n) \cong \mathbb{P}^{n-1}$. In this work we will usually consider the Grassmannian with its Plücker embedding into $\mathbb{P}^N$ where $N = \binom{n}{k} - 1$. This embedding is given by taking a full-rank $k \times n$ matrix V, representing a point $[V] \in \mathrm{Gr}(k, n)$ and defining the coordinates p_I with $I \in \binom{[n]}{k}$ on $\mathbb{P}^{\binom{n}{k}-1}$ to be the $k \times k$ minors of V labeled by the columns in I. If it is clear from context, we will often write V both for the point in the Grassmannian and a matrix representative of it. We can obtain a quasi-projective variety from $\mathrm{Gr}(k, n)$ by removing the vanishing locus of the Plücker coordinate $p_{1,\ldots,k}$. That is $\mathrm{Gr}(k, n) \setminus \mathbb{V}(p_{1,\ldots,k})$ is quasi-projective, and in fact isomorphic to the affine variety $\mathbb{C}^{(n-k)k}$. ◇

Definition 1.1.6. Let $X \subset \mathbb{P}^n$ be a projective variety, and let $B \cong \mathbb{P}^\ell \subset \mathbb{P}^n$ be a linear subspace of $\mathbb{P}^n$ disjoint from X. Then we define the *join of B with X*, denoted $J(B, X)$, as

$$J(B, X) = \bigcup_{x \in X} \mathbb{P}(\mathrm{span}_{\mathbb{C}}(B, x)).$$

Geometrically speaking, $J(B, X)$ is the union of all $\mathbb{P}^{\ell+1}$ spanned by B and a point $x \in X$.

The most basic properties of a variety are its degree and dimension.

Definition 1.1.7. Let $X \subset \mathbb{P}^n$ be an irreducible variety. Then, the *dimension* of X is defined as the supremum over the length of the ascending chains of irreducible subvarieties in X. Thus, if

$$Y_0 \subsetneq Y_1 \subsetneq \ldots \subsetneq Y_d \subset X,$$

where d is the supremum, then $\dim X = d$. If $L \subset \mathbb{P}^n$ is a generic projective linear space of dimension $n-d$, i.e. $L \cong \mathbb{P}^{n-d}$, then the *degree* $\deg X$ is given as the number of points in the intersection $L \cap X$, i.e.

$$\deg X = |L \cap X|.$$

Remark 1.1.8. Similarly, the *codimension* of X is $\dim \mathbb{P}^n - \dim X$. Subvarieties $Y \subset X$ of codimension one will be called *hypersurfaces*.

Example 1.1.9. We continue with the Grassmannian $\mathrm{Gr}(k,n) \subset \mathbb{P}^N$ from Example 1.1.5. Its dimension is $k(n-k)$. Its degree in the Plücker embedding is given as

$$\deg \mathrm{Gr}(k,n) = (k(n-k))! \cdot \frac{0! \cdot 1! \cdot \ldots \cdot (k-1)!}{(n-k)! \cdot (n-k+1)! \cdot \ldots \cdot (n-1)!}. \tag{1.1.1}$$

In the case $k = 2$, that is, in the case of the Grassmannian of lines, this simplifies to

$$\deg \mathrm{Gr}(2,n) = C_{n-2}, \tag{1.1.2}$$

where C_{n-2} is the $(n-2)$-th Catalan number. See for example [Smi16, Corollary 2.33]. ◇

Let X be a variety. We say a function $f : X \to \mathbb{C}$ is *regular* if for each point $p \in X$ there exists an affine open set $U \ni p$ such that $f|_U$ can be written as a fraction g/h of two polynomials in the variables $x_1, \ldots, x_n$, such that $h(x) \neq 0$ for all $x \in U$. We say a function is *rational*, denoted by a dashed arrow $f : X \dashrightarrow \mathbb{C}$, if it is a regular function defined on some nonempty open subset $U \subset X$. Two such functions are identified if they agree on a nonempty open subset of their common domain. Morphisms between varieties $f : X \to Y$, also called *regular maps*, are those maps which are locally given by tuples of regular functions. A projective variety is *rational* if there exists a rational map $\mathbb{P}^n \dashrightarrow X$ with rational inverse map.

We turn to the notion of singularities.

Definition 1.1.10. Let $X \subset \mathbb{P}^N$ be a projective variety of dimension n with vanishing ideal $I \subset \mathbb{C}[x_0, \ldots, x_N]$. Suppose that, for a point p on an affine patch U of X, the variety X is given by the vanishing of the polynomials $f_1, \ldots, f_r$. Then, denote by J_p the *Jacobian* defined as the $r \times N$ matrix with entries $(\partial f_i / \partial x_j)$, where $\partial / \partial x_i$ denotes the partial derivatives with respect to x_i. Then, a point $p \in U$ is *singular* if $\dim \ker J_p > n$. The set of singular points $\mathrm{Sing}\, X$ is a subvariety of X, also referred to as the *singular locus* of X. We say X is *smooth* if $\mathrm{Sing}\, X = \varnothing$.

We make the following related definition for tangent spaces.

Definition 1.1.11. Let $X \subset \mathbb{P}^N$ be an irreducible variety, and $U \subset X$ an affine open. We define the *embedded tangent space* $\mathbf{T}_x X$ of X at $x \in U$ as the kernel of the Jacobian J_x.

In later chapters, certain varieties of codimension one will play an important role.

Definition 1.1.12. Let $X \subset \mathbb{P}^N$ be an irreducible variety with affine cover $\{U_i\}$. A *Cartier divisor* is a system of rational functions $f_i : U_i \to \mathbb{C}$ such that:

i. the f_i are not identically zero;

ii. f_i/f_j and f_j/f_i are both regular on $U_i \cap U_j$.

The functions $\{f_i\}$ and open sets U_i define the same divisor as the functions $\{g_j\}$ and open sets V_j if f_i/g_j and g_j/f_i are regular on $U_i \cap V_j$. We say a Cartier divisor on X is *effective* if the rational functions defining it are polynomials, i.e. $f_i \in \mathbb{C}[U_i]$ for all i. In this case we write $D \subset X$, where $D \cap U_i = \mathbb{V}(f_i)$ is the subvariety on every open U_i. We say an effective Cartier divisor $D = \bigcup_i D_i$ is *simple normal crossing* if its irreducible components D_i are smooth, and locally in analytic topology with coordinates z_i, we can write D as the vanishing locus of $\mathbb{V}(z_1 \cdots z_\ell)$ for $\ell \leq \dim X$.

Commutative Algebra

Equivalently, we can also define varieties in terms of commutative algebra. All rings R considered here are commutative, with identity, and Noetherian.

An ideal $I \subset R$ is called *radical* if for any $p^\ell \in I$ with $p \in R$ and $\ell > 1$ we find that $p \in I$. Similarly, an ideal $I \subset R$ is *prime* if for all $p, q \in R$ with $pq \in I$, we find either $p \in I$ or $q \in I$. Then, similar to the definition of dimension for a variety we say R has *Krull dimension* d if d is the supremum of the length of all strictly ascending chains of prime ideals

$$P_0 \subsetneq P_1 \subsetneq \ldots \subsetneq P_d \subset R.$$

We say an ideal $I \subset R$ is *maximal* if for any ideal $J \subset R$ with $I \subset J$, we have either $J = I$ or $J = R$. A ring R is *local* if it has a unique maximal ideal. The *localization* R_m of R at a maximal ideal $m \subset R$ is the ring defined as $R_m = \{r/r' \mid r, r' \in R \text{ and } r' \notin m\}$. The commutative algebra language and the definitions of varieties as zeros of polynomials come together through the ideal-variety correspondence, see e.g. [CLO15, Theorem 10], which we record here in its projective setting.

Theorem 1.1.13. *The projective subvarieties of $\mathbb{P}^n$ are in inclusion-reversing one-to-one correspondence with the radical ideals I in $\mathbb{C}[x_0, \ldots, x_n]$ with homogeneous generators, which are different from the ideal $(x_0, \ldots, x_n)$. For a variety $X \subset \mathbb{P}^n$, the corresponding ideal I under this bijection is called the* vanishing ideal *of X, and the quotient ring $\mathbb{C}[x_0, \ldots, x_n]/I$, denoted $\mathbb{C}[X]$, is called the (homogeneous)* coordinate ring *of X.*

We can now define tangent spaces more intrinsically.

Definition 1.1.14. Let X be an irreducible variety, and U an affine open of X. We define the *cotangent space* $T_x^\vee X$ as the $\mathbb{C}$-vector space m_x/m_x^2, where m_x is the maximal ideal of $x \in U$, seen as the maximal ideal $m_x \subset \mathcal{O}_{U,x} \cong \mathbb{C}[U]_{m_x}$. The intrinsic *tangent space* $T_x X$ is defined as the dual of the cotangent space $(T_x^\vee X)^\vee = \mathrm{Hom}(T_x^\vee X, \mathbb{C})$.

Remark 1.1.15. If $X \subset \mathbb{P}^n$ is embedded, then after a choice of basis the intrinsic tangent space is isomorphic to the embedded tangent space.

Example 1.1.16. Consider the Grassmannian $\mathrm{Gr}(k,n)$ from Example 1.1.5 with its Plücker embedding into $\mathbb{P}^N$ with $N = \binom{n}{k} - 1$. Also choose the standard basis for $\mathbb{C}^n$. Then, on the affine patch where the Plücker coordinate $p_{[k]} \neq 0$ any point in $\mathrm{Gr}(k,n)$ can be represented by a $k \times n$ matrix, with a $k \times k$ identity matrix in the first k columns, and a $k \times (n-k)$ matrix with indeterminates x_{ij} in the remaining columns. Here is the construction for $\mathrm{Gr}(2,4) \subset U_{p_{12}}$:

$$V = \begin{bmatrix} 1 & 0 & x_{11} & x_{12} \\ 0 & 1 & x_{21} & x_{22} \end{bmatrix} \in U_{p_{12}}.$$

Then, the intrinsic tangent space $T_V \mathrm{Gr}(k,n)$ is given by the set of $(n-k) \times k$ matrices, which map $V \to \mathbb{C}^n/V$. More precisely, without a choice of basis we can identify the intrinsic tangent space as

$$T_V \mathrm{Gr}(k,n) = \mathrm{Hom}(V, \mathbb{C}^n/V).$$

This point of view will be extremely helpful in Chapter 5. ◇

Definition 1.1.17. Let X be an irreducible variety, and let p be a smooth point of X contained in the affine patch $U \subset X$. Then, a sequence of elements $x_1, \ldots, x_m \in \mathbb{C}[U]_{m_p}$, where $m_p \subset \mathbb{C}[U]$ is the maximal ideal corresponding to $p \in U$, is called a *system of local parameters* if their images $dx_i \cong [x_i] \in m_p/m_p^2$ span the cotangent space $T_p^\vee X$.

Lastly, we introduce two slightly more advanced properties that will be useful later at several points. We say an element $r \in R$ is a *non-zero divisor* if $rr' = 0$ implies $r' = 0$ for $r' \in R$. A sequence $r_1, \ldots, r_m \in R$ is a *regular sequence* in R, if r_1 is a non-zero divisor in R, and r_i is a non-zero divisor in $R/(r_1, \ldots, r_{i-1})$ for all $2 \leq i \leq m$. By a theorem of Rees, see e.g. [BH98, Theorem 1.25], all maximal regular sequences inside a local ring have the same length. This allows us to make the next definition.

Definition 1.1.18. Let R be a local ring with maximal ideal m. Let ℓ be the length of a maximal regular sequence in R. Then, R is *Cohen-Macaulay* if ℓ equals the Krull dimension of R.

Remark 1.1.19. Let X be an irreducible quasi-projective variety with affine cover $\{U_i\}$. Then, X is *Cohen-Macaulay* if the localizations $\mathbb{C}[U_i]_{m_x}$, where m_x is the maximal ideal of $x \in U_i$, are Cohen-Macaulay for any point $x \in X$. Moreover, we say a subvariety $Y \subset X$ of codimension c is a local *complete intersection* if its defining equations $r_1, \ldots, r_c \in \mathbb{C}[U_i]$ are a regular sequence for all i.

Semi-algebraic Sets

The central objects in real algebraic geometry are the semi-algebraic sets; they play the role of the varieties from the preceding section. In that section it was important that all objects

were defined over $\mathbb{C}$, as this is an algebraically closed field. Here, however, we define objects over the real numbers $\mathbb{R}$, which, in contrast to $\mathbb{C}$, form an ordered field.

Definition 1.1.20. A *basic closed semi-algebraic set* in $\mathbb{R}^n$ is the set of solutions to a system of finitely many polynomial inequalities of the form

$$f(x_1, \ldots, x_n) \geq 0,$$

where $f \in \mathbb{R}[x_1, \ldots, x_n]$. A *basic open semi-algebraic set* is such a set where the inequalities are strict. A *semi-algebraic set* in $\mathbb{R}^n$ is a finite Boolean combination of basic semi-algebraic sets, either open or closed. In projective real space, a set $S \subseteq \mathbb{RP}^n \cong \mathbb{R}^{n+1}/\mathbb{R}^\times$ is called a *semi-algebraic set* if it is the image of a semi-algebraic set in $\mathbb{R}^{n+1}$ under the natural quotient map $\mathbb{R}^{n+1} \setminus \{0\} \longrightarrow \mathbb{RP}^n$. We say a semi-algebraic set is *regular* if it equals the Euclidean closure of its interior. For a complex variety X we denote its *real locus* by $X(\mathbb{R})$, which is the set of real points in X.

Example 1.1.21. We continue with the Grassmannian with its Plücker embedding from Example 1.1.5. Then, the *non-negative Grassmannian* $\mathrm{Gr}(k,n)_{\geq 0}$ is defined as the set of points in $\mathrm{Gr}(k,n)(\mathbb{R})$ whose Plücker coordinates are all non-negative up to global choice of sign. The non-negative Grassmannian is a $k(n-k)$-dimensional regular basic semi-algebraic set. For $k = 1$, for example, it is simply the non-negative orthant in $\mathbb{P}^{n-1} \cong \mathrm{Gr}(1,n)$. ◇

Definition 1.1.22. Let $S \subset \mathbb{P}^n(\mathbb{R})$ be an n-dimensional semi-algebraic set. Then, we define its *algebraic boundary* $\partial_a S$ as the complex Zariski closure in $\mathbb{P}^n$ of its Euclidean boundary ∂S.

Example 1.1.23. We continue with the non-negative Grassmannian $\mathrm{Gr}(k,n)_{\geq 0}$. Its algebraic boundary $\partial_a \mathrm{Gr}(k,n)_{\geq 0}$ is the subvariety in $\mathrm{Gr}(k,n) \subset \mathbb{P}^N$ with $N = \binom{n}{k} - 1$, given by the union of the n cyclic Plücker hyperplane sections

$$\partial_a \mathrm{Gr}(k,n)_{\geq 0} = \bigcup_{I=\{i,i+1,\ldots,i+k\}} \mathbb{V}(p_I) \subset \mathrm{Gr}(k,n),$$

where the index is read modulo n. ◇

1.2 Sheaves and Vector Bundles

We will define the main object of this section - sheaves - generally for topological spaces. However, we will be mostly interested in the application to irreducible complex projective varieties equipped with their Zariski topolgy.

Definition 1.2.1. Let X be a topological space. A *sheaf* $\mathcal{F}$ on X is an assignment of an abelian group $\mathcal{F}(U)$ to every open subset $U \subset X$, and a morphism $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$ for every inclusion $V \subset U$ of open sets, such that

i. $\mathcal{F}(\emptyset) = 0$,

ii. ρ_{UU} is the identity for every open subset U of X, and
iii. $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$ for all chains of open subsets $W \subset V \subset U$.
iv. For every open cover $\{V_i\}$ of any open subset $U \subset X$ and every $s \in \mathcal{F}(U)$ such that

$$\rho_{UV_i}(s) = 0$$

for every i, it follows that $s = 0$.
v. For every open cover $\{V_i\}$ of any open subset $U \subset X$ and all elements $s_i \in \mathcal{F}(V_i)$ with the property that

$$\rho_{V_i, V_i \cap V_j}(s_i) = \rho_{V_j, V_i \cap V_j}(s_j)$$

for all i, j, there is an element $s \in \mathcal{F}(U)$ such that

$$\rho_{UV_i}(s) = s_i.$$

The elements $s \in \mathcal{F}(U)$ are called *sections* of the sheaf $\mathcal{F}$. Elements of $\mathcal{F}(X)$ are called *global sections*, and we denote the set of global sections by $\Gamma(X, \mathcal{F})$.

Example 1.2.2. The simplest example of a sheaf is the *locally constant sheaf* $\underline{\mathbb{Q}}_X$ which assigns to every open set the abelian group of $\mathbb{Q}$-valued functions $U \mapsto \mathbb{Q}$. If X is connected, then $\underline{\mathbb{Q}}_X(U) \cong \mathbb{Q}$ for any open $U \subset X$. ◇

Example 1.2.3. The first non-trivial example of a sheaf we are going to encounter in this work is the structure sheaf. For this we let X be an irreducible complex projective variety. The *structure sheaf* $\mathcal{O}_X$ of X is the sheaf defined by assigning to every open subset $U \subset X$ the ring (and in particular abelian group)

$$\mathcal{O}_X(U) = \{\, f : U \to \mathbb{C} \mid f \text{ is regular at every point } p \in U \,\}.$$

The restriction maps

$$\rho_{UV} : \mathcal{O}_X(U) \to \mathcal{O}_X(V), \qquad f \mapsto f|_V$$

for inclusions $V \subset U$ are given by restricting regular functions. If X has affine cover $\{U_i\}$, then the structure sheaf is defined via the coordinate rings on the affine patches, i.e. $\mathcal{O}_X(U_i) \cong \mathbb{C}[U_i]$. ◇

Definition 1.2.4. Let X be an irreducible projective variety with affine cover $\{U_i\}$. The *stalk* $\mathcal{O}_{X,p}$ of the structure sheaf $\mathcal{O}_X$ at a point $p \in U_i \subset X$ is the local ring $\mathcal{O}_{X,p} = \mathbb{C}[U_i]_{m_p}$, with maximal ideal $m_p \subset \mathbb{C}[U_i]$ corresponding to $p \in U_i$.

The importance of stalks is that they already define the sheaf uniquely, see e.g. [Har77, Proposition 1.1.].

Proposition 1.2.5. *Let $\mathcal{F}$ and $\mathcal{G}$ be sheaves on X. A morphism of sheaves $\varphi : \mathcal{F} \longrightarrow \mathcal{G}$ is an isomorphism if and only if for every point $p \in X$, the induced map on stalks $\varphi_p : \mathcal{F}_p \longrightarrow \mathcal{G}_p$ is an isomorphism.*

In practice, we want more structure for sheaves on varieties.

Definition 1.2.6. A sheaf of $\mathcal{O}_X$-modules, also simply called an *$\mathcal{O}_X$-module*, is a sheaf $\mathcal{F}$ on X such that for each open subset $U \subset X$, the group $\mathcal{F}(U)$ is an $\mathcal{O}_X(U)$-module and for each inclusion $V \subset U$ of open subsets, the restriction morphism

$$\mathcal{F}(U) \to \mathcal{F}(V)$$

is compatible with the module structures via

$$\mathcal{O}_X(U) \to \mathcal{O}_X(V).$$

Let X be a connected space. Then, we say $\mathcal{F}$ is *locally free* if there is an open cover $\{U_i\}$ of X such that $\mathcal{F}(U_i)$ is a free $\mathcal{O}_X(U_i)$-module, i.e. $\mathcal{F}(U_i) \cong \mathcal{O}_X(U_i)^r$ for some r. The *rank* of a locally free $\mathcal{O}_X$-module $\mathcal{F}$ at $p \in X$ is the rank of the module $\mathcal{F}_p$ as an $\mathcal{O}_{X,p}$-module. Locally free $\mathcal{O}_X$-modules of rank 1 are also called *invertible* sheaves.

Example 1.2.7. The simplest example of a free $\mathcal{O}_X$-module of rank 1, which therefore is invertible, is the structure sheaf $\mathcal{O}_X$ itself. This can be seen from the fact that, locally at a point $p \in U \subset X$, the structure sheaf is simply the stalk $\mathbb{C}[U]_{m_p} = \mathcal{O}_{X,p}$. ◇

Example 1.2.8. The second main example of a sheaf of $\mathcal{O}_X$-modules we are going to encounter in this work is the ideal sheaf: We define this first for an irreducible affine variety X. Let $Y \subset X$ be a subvariety. Then, the *ideal sheaf* $\mathcal{I}_{Y/X}$ of Y is defined as

$$\mathcal{I}_{Y/X}(U) = \{s \in \mathcal{O}_X(U) \mid s(p) = 0 \, \forall p \in U \cap Y\},$$

where $U \subset X$ is any open subset. Note that each $\mathcal{I}_{Y/X}(U)$ is an ideal of the ring $\mathcal{O}_X(U)$, hence the name. If X is a projective variety with affine cover $\{U_i\}$, and $Y \subset X$ is a projective subvariety, then the ideal sheaf is defined by restricting to the affine open U_i first. This defines a sheaf by the gluing property v. in Definition 1.2.1. Note that most ideal sheaves are not invertible, except when the ideal is the vanishing ideal of an effective Cartier divisor Y. ◇

Lastly, we want to introduce twists of sheaves.

Definition 1.2.9. Let X be a variety with affine cover $\{U_i\}$. Let $D \subset X$ be an effective Cartier divisor with f_D the defining polynomial of $D \cap U$ for an open $U = U_i$. Then, the *twisted structure sheaf* $\mathcal{O}_X(D)$ is defined via

$$\mathcal{O}_X(D)(U) = \{s \in \mathbb{C}(X) \mid s \cdot f_D \in \mathcal{O}_X(U)\}.$$

That is to say $\mathcal{O}_X(D)(U)$ consists of all rational functions $U \dashrightarrow \mathbb{C}$ which extend to regular functions on $U \subset X$ upon multiplication with f_D. If $X = \mathbb{P}^n$ and $\deg D = d$, we also write $\mathcal{O}_{\mathbb{P}^n}(d)$ for $\mathcal{O}_{\mathbb{P}^n}(D)$. Moreover, if $X \subset \mathbb{P}^n$ is embedded, we often write $\mathcal{O}_X(d)$ for the pullback of $\mathcal{O}_{\mathbb{P}^n}(d)$ under the inclusion $\iota : X \hookrightarrow \mathbb{P}^n$.

Remark 1.2.10. Twisted structure sheaves can also be defined for any hypersurface $Y \subset X$, see e.g. [Vak25, Definition 15.4.5]. We will use this freely from now on, but as this definition involves much more language, we decided to omit it from this thesis for the sake of conciseness.

Vector Bundles

We will again restrict our attention to the case where the objects are defined over the field of complex numbers.

Definition 1.2.11. A *vector bundle* of rank r on X, the *base space*, is a variety E, called *total space*, with a morphism $\pi : E \longrightarrow X$ that satisfies the following properties.

i. The fiber $\pi^{-1}(x)$ over a point $x \in X$ is an r-dimensional $\mathbb{C}$-vector space V.

ii. There is an open cover $\{U_i\}$ of X and isomorphisms called *local trivializations*

$$\varphi_i \colon U_i \times \mathbb{C}^r \to \pi^{-1}(U_i)$$

such that:

(a) $(\pi \circ \varphi_i)(x, v) = x$ for all $v \in \mathbb{C}^r$, and

(b) $v \mapsto \varphi_i(x, v)$ is a linear isomorphism of $\mathbb{C}^r$ with $\pi^{-1}(x)$.

We will often simply refer to a vector bundle by its total space, even though the data of the morphism π are crucial. A *section* of a vector bundle is a morphism $s : U \to E$ for an open $U \subset X$ such that $\pi \circ s = \mathrm{id}_U$. A vector bundle of rank 1 is called a *line bundle*. For a vector bundle $\pi : E \to X$ we define the *dual bundle* $E^\vee$ by setting the fibers over X to be $\pi^{-1}(x)^\vee = \mathrm{Hom}(\pi^{-1}(x), \mathbb{C})$.

Example 1.2.12. Let $X \subset \mathbb{P}^N$ be a smooth irreducible variety of dimension n, and let $\{U_i\}$ be an affine covering of X. Then an important example of a vector bundle is the tangent bundle TX. Let $T_xX \subset T_x\mathbb{P}^N$ be the embedded tangent space of X at $x \in U_i$. Then, $\pi^{-1}(x) = T_xX$ defines the *tangent bundle* TX and is a vector bundle of rank n.

Similarly, we define the normal bundle on X as follows. Suppose X is defined on U_i by the polynomials $f_1, \ldots, f_m \in \mathbb{C}[U]$. Recall the definition of the Jacobian J of X (on U_i), which gives a linear morphism $\mathbb{C}^m \to \mathbb{C}^N$ via multiplication of vectors from the left. Now, take a subset $I \in \binom{[m]}{N-n}$, and let $U_I \subset U_i$ be the open set of points in U_i such that the rank of the matrix J_I, obtained by taking the rows of J labeled by I, is $N - n$. Then, the local trivializations

$$U_I \times \mathbb{C}^{N-n} \to U_i \times \mathbb{C}^N, \quad (x, v) \mapsto (x, v^T J_I)$$

define the *normal bundle* $N_{X/\mathbb{P}^N}$. ◇

For singular varieties we have an identification of the normal bundle, see e.g. [Ful98, Appendix B.7.4], which for the sake of this work, we can use as the definition. For this, define the product of an ideal sheaf $\mathcal{I}^2$ by taking products of the ideal $\mathcal{I}(U)^2$ on any open U.

Lemma 1.2.13. *Let $X \subset \mathbb{P}^N$ be a Cohen-Macaulay projective variety, and Y a (not necessarily smooth) complete intersection in X of codimension c. Denote by $\mathcal{I}$ the ideal sheaf of Y in X and suppose that its generators have degree $d_1, \ldots, d_r$. Then, the* normal bundle $N_{Y/X}$ *is*

defined as the dual to $\mathcal{I}/\mathcal{I}^2$. Moreover, its c-fold exterior product is the line bundle given by

$$\bigwedge^c N_{Y/X} \cong \bigotimes_{i=1}^c \mathcal{O}_Y(d_i).$$

Proof. The claim that $\mathcal{I}/\mathcal{I}^2$ is a locally free vector bundle of rank c is [Vak25, Proposition 21.2.16]. Hence, $\bigwedge^c N_{Y/X}$ is a locally free line bundle. Now, since $Y \subset X$ is a local complete intersection inside the Cohen-Macaulay variety X, it is also Cohen-Macaulay by [Sta26]. Therefore, the equations $f_i \in \mathcal{O}_X(U)$ cutting out Y on an affine patch $U \subset X$, more precisely their vanishing loci $Z_i = \mathbb{V}(f_i)$, are effective Cartier divisors in $X \cap U$. Thus, locally, we have $U \cap Y = U \cap \bigcap_{i=1}^c Z_i$; hence by [Vak25, 21.2.J] we have

$$N_{Z_i/X} \cong \left(\mathcal{I}_{Z_i/X}/\mathcal{I}^2_{Z_i/X}\right)^\vee \cong \mathcal{O}_{Z_i}(d_i).$$

The decomposition of $\bigwedge^c N_{Y/X}$ then follows by [Ful98, Appendix B.7.4]. □

Differential Forms

Differential forms, in the guise of canonical forms for positive geometries, play a crucial role in this thesis.

Definition 1.2.14. Let X be a smooth variety of dimension n. The *vector bundle of differential forms* $\Omega^\bullet_X$ is defined as the exterior algebra $\bigwedge T^\vee X$ of the cotangent bundle, which is formally the graded algebra

$$\Omega^\bullet_X = \bigoplus_{k=0}^n \bigwedge^k T^\vee X = \bigoplus_{k=0}^n \Omega^k_X,$$

where $\bigwedge^k T^\vee X$ denotes the space spanned by all k-fold wedge products $dx_1 \wedge \ldots \wedge dx_k$ with $dx_i \in T^\vee X$. The fibers over $x \in X$ are precisely the k-th exterior products of the cotangent space $\bigwedge^k T^\vee_x X$ at that point. The top part Ω^n_X is a line bundle called the *canonical bundle* on X, sometimes denoted by ω_X. Elements $\omega \in \Omega^k_X$ are called *k-forms*, and they are regular differential forms on open subsets of X.

Remark 1.2.15. The vector bundle $\Omega^\bullet_X$ is also a sheaf on X; moreover, Ω^k_X is a locally free $\mathcal{O}_X$-module of rank $\binom{n}{k}$. Therefore, we will sometimes refer to $\Omega^\bullet_X$ as the *sheaf of differentials* on X. Also note that Ω^0_X is the sheaf of regular 0-forms, i.e. the structure sheaf $\mathcal{O}_X$.

Example 1.2.16. Consider the Grassmannian $G = \mathrm{Gr}(2,4)$ with its Plücker embedding into $\mathbb{P}^5$. Then, its canonical line bundle $\omega_G = \Omega^4_G$ is isomorphic to the twisted structure sheaf $\mathcal{O}_G(-4)$, see e.g. [EH16, Proposition 5.25], which is the restriction of $\mathcal{O}_{\mathbb{P}^5}(-4)$. This can be seen explicitly on the affine open $U = \{p_{12} \neq 0\} \subset G$. Here any $V \in U$ can be written in the indeterminates x_i, as

$$V = \begin{bmatrix} 1 & 0 & x_1 & x_2 \\ 0 & 1 & x_3 & x_4 \end{bmatrix}.$$

Then, the 4-form $dx_1 \wedge dx_2 \wedge dx_3 \wedge dx_4 \in \Omega^4_G|_U$ locally generates the restriction of the canonical line bundle $\omega_G|_U$. Choosing a different affine chart of G, we can see that this form has a 4-fold pole at $p_{12} = 0$. Since, $p_{12} = 0$ is irreducible we must have $\Omega^4_G \cong \mathcal{O}_G(-4)$ as claimed. ◇

More generally, we can write k-forms on X in coordinates as follows. Let U be an open subset of X with $x_1, \ldots, x_n$ a local system of parameters, see Definition 1.1.17. For any $I \subset [n]$ of size k, i.e. $I \in \binom{[n]}{k}$ set $dx_I = \wedge_{i \in I} dx_i$. Then, the dx_I span $\Omega^k_X|_U$ by definition via the exterior wedge product. Hence, any k-form $\omega \in \Omega^k_X|_U$ can be written as

$$\omega = \sum_{I \in \binom{[n]}{k}} f_I \, dx_I,$$

where $f_I \in \mathcal{O}_X(U)$ are regular functions $f_I : U \to \mathbb{C}$. We also define the *exterior differentiation* $d : \Omega^\bullet_X \to \Omega^{\bullet+1}_X$ sending

$$d\omega = d\left(\sum_{I \in \binom{[n]}{k}} f_I \, dx_I\right) = \sum_{I \in \binom{[n]}{k}} \left(\sum_{i=1}^n \frac{\partial f_I}{\partial x_i} dx_i \wedge dx_I\right).$$

This is well-defined, i.e. independent of the local chart: for $\omega = \sum_J g_J dy_J$ this follows from the total derivative formula $d_y y_i = \sum_k \partial y_i / \partial x_i dx_i$ and

$$d_y f_I = \sum_{i=1}^n \frac{\partial f_I}{\partial y_i} \frac{\partial y_i}{\partial x_i} dx_i = \sum_{i=1}^n \frac{\partial f_I}{\partial x_i} dx_i = dx f_I$$

and the fact that

$$d_x(dy_i) = \sum_{r,j} \frac{\partial y_i}{\partial x_r \partial x_i} dx_i \wedge dx_r = 0,$$

as $dx_i \wedge dx_r = -dx_r \wedge dx_i$. Then,

$$d_x \omega = \sum_J (d_x g_J \wedge dy_J + g_J d_x(dy_J)) = \sum_J (d_y g_J \wedge dy_J) = d_y \omega.$$

Also note that $d^2 = 0$ essentially by the fact that partial derivatives commute.

Similarly to the twisted structure sheaf from Definition 1.2.9, we will also need a twisted sheaf of differential forms. Recall Remark 1.2.10. Then, for a hypersurface $D \subset X$, and any open U in X, we set $\Omega^\bullet_X(D) = \Omega^\bullet_X \otimes_{\mathcal{O}_X} \mathcal{O}_X(D)$. We can then identify those forms $\omega \in \Omega^\bullet_X$ which have at most simple poles along $D \subset X$ as follows.

Definition 1.2.17. Let X be a projective variety of dimension n with affine cover $\{U_i\}$, and let $D \subset X$ be a hypersurface such that $X \setminus D$ is smooth. A regular n-form $\omega \in \Omega^n_{X \setminus D}$ on $X \setminus D$ has *at most a simple pole along* D if, on every affine open U_i with $f_D \in \mathbb{C}[U_i]$ the local equation of $D \cap U_i$, the product $f_D \cdot \omega$ can be extended to a regular n-form on X. Therefore, the *space of regular n-forms on $X \setminus D$ with at most a simple pole along D* is the $\mathbb{C}$-vector space of global sections of the twisted canonical bundle $\Omega^n_X(D)$, i.e.

$$\Gamma\left(X, \Omega^n_X(D)\right).$$

Lastly, we want to define logarithmic differential forms, which play an important role in the study of positive geometries.

Definition 1.2.18. Let X be a projective variety of dimension n, and let $D \subset X$ be a hypersurface such that $X \setminus D$ is smooth. The *sheaf of logarithmic 1-forms along D* is

$$\Omega^1_X(\log D) := \left\{\omega \in \Omega^1_X(D) \;\middle|\; d\omega \in \Omega^2_X(D)\right\},$$

where $d : \Omega^\bullet_X \to \Omega^{\bullet+1}_X$ is the exterior differentiation. Then, the *sheaf of logarithmic forms along D* is the exterior algebra

$$\Omega^\bullet_X(\log D) = \bigwedge \Omega^1_X(\log D) = \bigoplus_{k\geq 0} \Omega^k_X(\log D).$$

Lastly, the *space of logarithmic forms along D*, $\Omega^n_{\log}(X \setminus D)$, is the $\mathbb{C}$-vector space given by the global sections of $\Omega^n_X(\log D)$, i.e.

$$\Omega^n_{\log}(X \setminus D) = \Gamma(X, \Omega^n_X(\log D)).$$

The elements in $\Omega^n_{\log}(X \setminus D)$ are called dlog *forms*.

We can write dlog forms in coordinates as follows. Let U be an open of X, with $x_1, \ldots, x_n$ again a local system of parameters for $x \in U$, hence the dx_i generate the cotangent space $T^\vee_x X$. Moreover, assume that $D \cap U$ is locally given by $D = \mathbb{V}(x_1 \cdot \ldots \cdot x_r)$ with $r < n$. Then, the space of regular 1-forms $\Gamma(X, \Omega^1_X(D))$ with at most simple poles along D is locally on U is given as

$$\Gamma\big(U, \Omega^1_X(D)\big) = \bigoplus_{i=1}^{n} \mathcal{O}_X(U)\, \frac{dx_i}{x_1 \cdots x_r}.$$

However, consider as a toy example

$$d\left(\frac{1}{x_n}\, dx_i\right) = -\frac{1}{x_n^2}\, dx_i \wedge dx_n \notin \Omega^2_X(D)(U),$$

in contrast

$$d\left(\frac{dx_i}{x_i}\right) = -\frac{1}{x_i}\, dx_i \wedge dx_i = 0 \in \Omega^2_X(D)(U).$$

Therefore, any dlog form $\omega \in \Omega^1_X(\log D)$ can be written as

$$\omega = \sum_{i=1}^{r} f_i \frac{dx_i}{x_i} + \sum_{i=r+1}^{n} f_i dx_i = \sum_{i=1}^{r} f_i\, \mathrm{dlog}(x_i) + \sum_{i=r+1}^{n} f_i dx_i,$$

where the $f_i \in \mathcal{O}_X$ are regular functions on X. Then, more generally any higher dlog-form $\omega \in \Omega^k_X(\log D)$ can be written as

$$\omega = \sum_{I \in \binom{[r]}{p}} \alpha_I \wedge \bigwedge_{i \in I} \frac{dx_i}{x_i},$$

where the α_I are regular q-forms on U i.e. $\alpha \in \Omega^q_X(U)$, and $k = p + q$.

We have the following relation between the two spaces of differential forms on $X \setminus D$.

Proposition 1.2.19 ([BD25, Proposition 1.13])**.** *There is a natural inclusion for the space of logarithmic forms on $X \setminus D$ into the space of forms on $X \setminus D$ with at most simple poles along D. More succinctly, we have*

$$\Omega^n_{\log}(X \setminus D) \subset \Gamma(X, \Omega^n_X(D)).$$

Equality holds if D is a simple normal crossing divisor.

Example 1.2.20. Here is a concrete example where the inclusion of the preceding result is strict. See also [BD25, Proposition 5.3]. Consider the divisor $D \subset \mathbb{P}^2$ given by the union of the plane cusp curve C and a generic line L. In coordinates, on the affine chart $U_z \subset \mathbb{P}^2$ where $z \neq 0$, we have

$$D = \mathbb{V}((y^2 - x^3)(x - 1)).$$

Thus, D is not simple normal crossing, as the cuspidal point $(0, 0)$ is a singular point of D (and not in L). Since $\deg D = 4$, we have $\Omega^2_{\mathbb{P}^2}(D) \cong \mathcal{O}_{\mathbb{P}^2}(1)$, so the space of differential forms with at most simple poles along D, i.e. $\Gamma(\mathbb{P}^2, \Omega^2_{\mathbb{P}^2}(D))$, is generated on U_z by

$$\left\{ \frac{(c_1 + c_2 x + c_3 y)}{(x-1)(y^2 - x^3)}\, dx \wedge dy \;\middle|\; c_1, c_2, c_3 \in \mathbb{C} \right\}.$$

However, for

$$\omega = \frac{c_1}{(x-1)(y^2 - x^3)}\, dx \wedge dy,$$

we can take Poincaré residue along the cusp C using its parameterization $\varphi : \mathbb{C} \to C$ sending $t \mapsto (t^2, t^3)$, and then

$$\varphi^*(\operatorname{Res}_C \omega) = \frac{c_1}{t^2(t^2 - 1)}\, dt,$$

which clearly has a double pole on the cuspidal point $t = 0 \mapsto (0, 0)$. Hence, the logarithmic condition forces $c_1 = 0$, so

$$\Omega^2_{\log}(\mathbb{P}^2 \setminus D) = \left\{ \frac{(c_2 x + c_3 y)}{(x-1)(y^2 - x^3)}\, dx \wedge dy \;\middle|\; c_2, c_3 \in \mathbb{C} \right\} \subsetneq \Gamma(\mathbb{P}^2, \Omega^2_{\mathbb{P}^2}(D)).$$

◇

1.3 Chain Complexes and (Co-)homology

In this section we introduce the homology theories that are going to play an important role in several chapters of this thesis. Our exposition here is tailored to our use case; for a more general introduction, see [Hat02]. We start by defining abstract chain complexes first, and then specialize to our situation later.

Definition 1.3.1. Let R be a ring. A *chain complex* G of R-modules is a family $G_\bullet = (G_i)_{i \in \mathbb{Z}}$ of R-modules together with homomorphisms $\partial_\bullet : G_\bullet \to G_{\bullet - 1}$ such that $\partial_{\bullet - 1} \circ \partial_\bullet = 0$:

$$\cdots \to G_{n+1} \xrightarrow{\partial_{n+1}} G_n \xrightarrow{\partial_n} G_{n-1} \xrightarrow{\partial_{n-1}} \cdots \xrightarrow{\partial_2} G_1 \xrightarrow{\partial_1} G_0 \xrightarrow{\partial_0} \cdots .$$

The maps $\partial_\bullet$ are called *differentials*. The condition $\partial_{\bullet-1} \circ \partial_\bullet = 0$ is equivalent to $\mathrm{im}(\partial_\bullet) \subseteq \ker(\partial_{\bullet-1})$. The complex is *exact at position* k if $\mathrm{im}(\partial_{k+1}) = \ker(\partial_k)$ as submodules of G_k, and is *exact* if this holds for every k.

Definition 1.3.2. We say a sequence of abelian groups $A \to B \to C$ is a *short exact sequence*, if

$$0 \to A \to B \to C \to 0,$$

and this sequence is exact. We say a sequence *splits* if $B \cong A \oplus C$. A sequence of chain complexes $A_\bullet \to B_\bullet \to C_\bullet$ is short exact if for all k

$$0 \to A_k \xrightarrow{\alpha_k} B_k \xrightarrow{\beta_k} C_k \to 0,$$

is a short exact sequence of abelian groups.

In the remainder of the thesis we are only interested in the case where the ring R is the field of rational numbers $\mathbb{Q}$.

Simplicial and Cellular Homology

We start with the arguably most intuitive homology theory: simplicial homology.

Definition 1.3.3. A *k-simplex* $\Delta^k \subset \mathbb{R}^n$ is the convex hull of an ordered set of vectors $v_0, \ldots, v_k$ in $\mathbb{R}^n$, such that the vectors

$$\{v_0 - v_1, v_0 - v_2, \ldots, v_0 - v_k\}$$

are linearly independent in $\mathbb{R}^n$. We write $\Delta^k = (v_0, v_1, \ldots, v_k)$ and sometimes identify Δ^k with its sequence of vertices. A finite *simplicial complex* X in $\mathbb{R}^n$ is a collection of simplices $\Delta_1, \ldots, \Delta_r$ in $\mathbb{R}^n$ that is downward closed and compatible in the following sense:

i. For every simplex $\Delta = (v_0, v_1, \ldots, v_k) \in C$, every ordered subsequence is again in C.
ii. If $\Delta_i, \Delta_j \in C_\bullet^\Delta$ such that they contain common vectors $v_0, v_1, \ldots, v_k$, then they appear in the same order in both sequences.

Definition 1.3.4. Let X be a simplicial complex, where X^k denotes the set of k-simplices in X. Then the *simplicial chain complex* $C_\bullet^\Delta$ is the collection of all $\mathbb{Q}$-linear combinations of k-simplices, together with the differentials $\partial_k : C_k(X) \to C_{k-1}(X)$ sending

$$\partial_k(\Delta) = \sum_{i=0}^{k} (-1)^i \Delta_{v_0,\ldots,\hat{v_i},\ldots v_k},$$

where $\Delta_{v_0,\ldots,\hat{v_i},\ldots v_k}$ denotes the $(k-1)$-simplex obtained by omitting v_i from Δ. Its *k-th simplicial homology group* $H_k^\Delta(X)$ with coefficients in $\mathbb{Q}$ is defined as

$$H_k^\Delta(X) = \frac{\ker \partial_k}{\mathrm{im}\, \partial_{k+1}}.$$

Example 1.3.5. Let X be the real circle S^1. Then we can equip X with a Δ-complex structure as follows. As a set $X = \Delta^0 \cup \Delta^1$, where we attach the edge Δ^1 to the vertex Δ^0. Then the chain complex simplifies to

$$0 \to \Delta^1 \xrightarrow{\partial_1} \Delta^0 \to 0.$$

Thus, we only need to compute $\partial_1(\Delta^1) = \Delta^0 - \Delta^0 = 0$, and hence

$$H_k^\Delta(S^1) = \begin{cases} \mathbb{Q} & , k = 0, 1 \\ 0 & , \text{else.} \end{cases}$$

More generally, for an n-dimensional sphere S^n we find similarly

$$H_k^\Delta(S^n) \cong \begin{cases} \mathbb{Q} & , k = 0, n \\ 0 & , \text{else.} \end{cases}$$

◇

We record the definition for another kind of complex for later use. Recall that a *homeomorphism* between two topological spaces $X \to Y$ is a bijection with an inverse map such that the pre-images of open sets are again open sets. A *k-cell* e^k is a topological space homeomorphic to $\mathbb{R}^k$.

Definition 1.3.6. We define a finite *CW complex* X as follows.

i. Start with a collection of points X^0, which are regarded as 0-cells.

ii. Inductively for $k \leq n$, form the k-skeleton X^k from X^{k-1} by attaching k-cells e_α^k via the *attaching maps*

$$\varphi_{e_\alpha^k} : S^{k-1} \longrightarrow X^{k-1},$$

where S^{k-1} denotes the $k-1$-dimensional sphere. This means that X^k is the quotient space of the disjoint union

$$X^{k-1} \sqcup \bigsqcup_\alpha D_\alpha^k$$

under the identifications

$$x \sim \varphi_{e_\alpha^k}(x) \qquad \text{for } x \in \partial D_\alpha^n,$$

where D_α^k denotes the k-dimensional disk, and ∂D_α^k its boundary. Thus, as a set,

$$X^k = X^{k-1} \sqcup \bigsqcup_\alpha e_\alpha^k,$$

where each e_α^k is an open k-disk.

iii. Set $X = X^n$.

We say X is a *n-dimensional* CW complex.

Example 1.3.7. We can give the projective line $\mathbb{P}^1$ a CW structure as follows. Set $\mathbb{P} = e_1^0 \sqcup e_1^2$, and we glue the single 2-cell to the single 0-cell. More precisely,

$$\varphi_{e_1^2} : S^1 \to X^1 = X^0, \quad x \mapsto e_1^0.$$

In general, for the n-dimensional projective space $\mathbb{P}^n$ we get a $2n$-dimensional CW structure by setting $\mathbb{P}^k = \mathbb{P}^{k-1} \sqcup e_1^{2k}$ with attaching maps

$$\varphi_{e_1^{2k}} : S^{2k-1} \to X^{2k-1} = \mathbb{P}^{k-1}, \quad (x_1, \dots, x_k) \mapsto [x_1 : \dots : x_k],$$

where the x_i are complex variables for the ambient space $\mathbb{R}^{2k} \cong \mathbb{C}^k$ of S^{2k-1}. ◇

Definition 1.3.8. Let X be a topological space with a CW structure, denote by X^k its k-skeleton. The *k-th cellular chain group* is the collection of $\mathbb{Q}$-linear combinations of k-cells e_α^k in $X^k \subset X$, denoted C_k^{CW}. Define the differentials $\partial_k : C_k^{CW}(X) \to C_{k-1}^{CW}(X)$ as

$$\partial_k(e_\alpha^k) = \sum_\beta \deg\left(\varphi_{e_\alpha^k}\right)_\beta e_\beta^{k-1},$$

where $(\varphi_{e_\alpha^k})_\beta$ is the β entry of the attaching maps and $\deg$ denotes the topological degree, see Remark 1.3.9. Together $C_\bullet^{CW}(X)$ and $\partial_\bullet$ make the *cellular chain complex*. Then, the *k-th cellular homology group* $H_k^{CW}(X)$ with coefficients in $\mathbb{Q}$ is given as

$$H_k^{CW}(X) = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}}.$$

Remark 1.3.9. For the formal definition of the topological degree of the attaching maps we refer the reader to [Hat02, Section 2.2], which we leave out here for the sake of conciseness of this work.

Example 1.3.10. Using the CW-complex structure from above, we can compute the cellular homology of $\mathbb{P}^n$ as follows. First, note that every boundary map ∂_k is trivial for odd k, as their domain $C_k^{CW}(\mathbb{P}^n)$ is trivial. Similarly, for even k the boundary maps ∂_k must be the trivial zero map as the codomain vanishes. Therefore, only for even k can the homology groups $H_k^{CW}(\mathbb{P}^n)$ be non-trivial, and since $\ker \partial_k = \operatorname{span}_{\mathbb{Q}}([e_1^k])$, we get

$$H_k^{CW}(\mathbb{P}^n) \cong \begin{cases} \mathbb{Q} & , k \leq 2n \text{ and } k \text{ even}, \\ 0 & , \text{ else.} \end{cases}$$

◇

Singular Homology

The most important homology theory for this work is that of singular homology.

Definition 1.3.11. Let X be a topological space, and let k be a non-negative integer. A *singular k simplex* is a continuous map $\sigma_k : \Delta^k \to X$ where $\Delta^k \subset \mathbb{R}^n$ is the standard k-simplex with vertices $v_0, \dots, v_k$. The free $\mathbb{Q}$-module $C_k(X)$ generated by all formal $\mathbb{Q}$-linear combinations of singular k simplices is referred to as the space of *(singular) k-chains*. We define the

differentials $\partial_k : C_k(X) \to C_{k-1}(X)$ by $\mathbb{Q}$-linear extension of the map

$$\partial_k(\sigma_k) = \sum_{i=0}^{k} (-1)^i \sigma_k|_{(v_0,\dots,\hat{v_i},\dots,v_k)},$$

where $\sigma_k|_{(v_0,\dots,\hat{v_i},\dots,v_k)}$ denotes the restriction of σ_k to the $(k-1)$-simplex obtained by omitting the i-th vertex of Δ^k.

Defining homology then follows from the central observation that $C_\bullet(X)$ makes a chain complex.

Definition 1.3.12. The family $C_\bullet(X)$ together with the differentials $\partial_\bullet$ is the *singular chain complex* of X. Its k-th *(singular) homology group* $H_k(X)$ with coefficients in $\mathbb{Q}$ is defined as the quotient

$$H_k(X) = \frac{\ker \partial_k}{\operatorname{im} \partial_{k+1}}.$$

Elements of $[\sigma] \in H_k(X)$ are called *singular k-chains*, and we abbreviate the notation by $\sigma = [\sigma]$ if there is no ambiguity. The *reduced homology groups* $\widetilde{H}_k(X)$ are defined via $H_0(X) = \widetilde{H}_0(X) \oplus \mathbb{Q}$ and $H_k(X) = \widetilde{H}_k(X)$ for all other k. We write $H_\bullet(X)$ and $\widetilde{H}_\bullet(X)$ for the collection of the (reduced) homology groups in all degrees k.

Unfortunately, singular homology is not readily computable; this drawback is remedied by the following results, see e.g. [Hat02, Theorem 2.27, Theorem 2.35].

Theorem 1.3.13. *Let X be an irreducible variety with a Δ-complex structure, and a CW structure. Then,*

$$H_\bullet(X) \cong H_\bullet^{\Delta}(X) \cong H_\bullet^{CW}(X).$$

Example 1.3.14. We continue with the projective space from Example 1.3.10. We immediately compute the singular homology of $\mathbb{P}^n$ as

$$H_k(\mathbb{P}^n) \cong H_k^{CW}(\mathbb{P}^n) \cong \begin{cases} \mathbb{Q} & , k \leq 2n \text{ and } k \text{ even}, \\ 0 & , \text{else}. \end{cases}$$

◇

Relative Homology

In the applications for homology in the later chapters of this thesis we need a slightly more general setting - that of relative homology.

Definition 1.3.15. Let $A \subset X$ be topological spaces. Then, the differential $\partial_k : C_k(X) \to C_{k-1}(X)$ takes any singular k-simplex $\sigma \in C_k(A)$ to a singular $(k-1)$-simplex $\partial_k(\sigma) \in C_{k-1}(A)$. Hence, we get the *relative k-chains*, denoted $C_k(X, A)$, given as the quotient

$$C_k(X, A) = C_k(X)/C_k(A),$$

with well-defined relative differentials $\overline{\partial}_k : C_k(X, A) \to C_{k-1}(X, A)$.

Then we can define relative homology analogously.

Definition 1.3.16. The relative k-chains $C_k(X, A)$ together with the relative boundary maps $\overline{\partial}_k$ make a chain complex $C_\bullet(X, A)$. The *k-th relative homology group* with coefficients in $\mathbb{Q}$ is then defined as the quotient

$$H_k(X, A) = \frac{\ker \overline{\partial}_k}{\operatorname{im} \overline{\partial}_{k+1}}.$$

We write $H_\bullet(X, A)$ for the collection of the relative homology groups in all degrees k.

We postpone an example to the end of this section in 1.3.30. Lastly, relative homology also allows us to define orientability.

Definition 1.3.17. An n-dimensional regular compact and connected semi-algebraic set $\overline{M}$ with boundary ∂M is *orientable* if $\dim H_n(\overline{M}, \partial M) = 1$. If $\overline{M}$ is orientable, we also say $M = \overline{M} \setminus \partial M$ is orientable, and M induces an orientation on its boundary ∂M.

Remark 1.3.18. This is compatible with the more intuitive definition of choosing an orientation locally, which is globally compatible. See [Hat02, Chapter 3.3] for more details.

Singular Cohomology

In applications, it is often preferable to work with a dual homology group.

Definition 1.3.19. Let $C_\bullet(X)$ be the singular chain complex of X. Then, consider the sequence obtained from taking the dual $\mathbb{Q}$-vector space to $C_k(X)$ everywhere, that is applying $\operatorname{Hom}(-, \mathbb{Q})$:

$$\cdots \to \operatorname{Hom}(C_{k-1}(X), \mathbb{Q}) \xrightarrow{\delta^k} \operatorname{Hom}(C_k(X), \mathbb{Q}) \xrightarrow{\delta^{k+1}} \operatorname{Hom}(C_{k+1}(X), \mathbb{Q}) \to \cdots,$$

where the differentials are $\delta^k(\varphi) = \varphi \circ \partial_k$ is the dual map to ∂_k for any $\varphi \in \operatorname{Hom}(C_{k-1}(X), \mathbb{Q})$. The free $\mathbb{Q}$-vector space $\operatorname{Hom}(C_k(X), \mathbb{Q})$ is called the *space of singular k-cochains* of X. The chain complex obtained by taking all such groups together with the boundary maps δ^k is called the *cochain complex* $C^\bullet(X)$. We then define the *k-th cohomology group* $H^k(X)$ with coefficients in $\mathbb{Q}$ as the quotient $H^k(X) = \ker \delta^{k+1} / \operatorname{im} \delta^k$. We write $H^\bullet(X)$ for the collection of the cohomology groups in all degrees k.

The main theorem to relate homology and cohomology is the universal coefficient theorem, which simplifies for our case of chain complexes over $\mathbb{Q}$ to the following.

Theorem 1.3.20. *Let $H_\bullet(X)$ be the homology groups of X and $H^\bullet(X)$ its cohomology groups with coefficients in $\mathbb{Q}$; then we have an isomorphism between the dual $\mathbb{Q}$-vector space $H_\bullet(X)^\vee$ and cohomology. Put in symbols,*

$$H_\bullet(X)^\vee = \operatorname{Hom}(H_\bullet(X), \mathbb{Q}) \cong H^\bullet(X).$$

The same holds for relative homology $H_\bullet(X, Y)^\vee \cong H^\bullet(X, Y)$.

Cohomology groups can also be used to detect connectedness of a space.

Definition 1.3.21. Let X be a topological space. We say X has r *connected components* if $H^0(X) \cong \mathbb{Q}^r$. If $r = 1$ we say X is *connected*.

Compactly Supported Cohomology

The next modification of the (co-)homology setup so far is going to involve compactly supported chains.

Definition 1.3.22. A k-cochain $\varphi \in C_c^k(X)$ is called *compactly supported* if there exists a compact subspace $K \subset X$ such that $\varphi(\sigma)$ vanishes for any singular chain σ with image in $X \setminus K$. We can then define the *compactly supported k cochains* $C_c^k(X)$ as the free $\mathbb{Q}$-vector space generated by all compactly supported $\varphi \in C_c^k(X)$. For any k define the differential $\delta_c^k : C_c^{k-1}(X) \to C_c^k(X)$ by linear extension of the map

$$\delta_c^k(\varphi) = \varphi \circ \partial_k,$$

where $\partial_k : C_k(X) \to C_{k-1}(X)$ is the differential for the singular chain complex.

As usual, the central result for the definition of compactly supported cohomology is that we just defined a chain complex.

Definition 1.3.23. The family $C_c^\bullet(X)$ together with the differentials $\delta_\bullet^c$ is the *singular compactly supported cochain complex*. Its *k-th singular cohomology with compact support* $H_c^k(X)$ with coefficients in $\mathbb{Q}$ is defined as the quotient

$$H_c^k(X) = \frac{\ker \delta_c^{k+1}}{\operatorname{im} \delta_c^k}.$$

We write $H_c^\bullet(X)$ for the collection of the cohomology groups with compact support in all degrees k.

We postpone an example to the end of this section in 1.3.30. Lastly, there is a very useful connection between compactly supported cohomology and relative cohomology.

Proposition 1.3.24 ([PS08, Section 5.5.2])**.** *Let X be a projective variety and Y a subvariety such that $X \setminus Y$ is smooth. Then, there is a canonical isomorphism $H_c^\bullet(X \setminus Y) \cong H^\bullet(X, Y)$.*

We will use this fact freely in the remainder of this work.

Sheaf Cohomology

Let X be an n-dimensional irreducible variety with affine cover $\{U_i\}$ in the Zariski topology. For $I \subset [m]$ denote by U_I the multiple intersections $\bigcap_{i \in I} U_i$. We will introduce sheaf cohomology only for the (twisted) structure sheaf $\mathcal{O}_X$ or the (twisted) sheaf of differential forms Ω_X^k. So let $\mathcal{F}$ be such a sheaf.

Let k be a non-negative integer, and fix $I = \{i_0, \dots, i_k\} \subset [m]$. Then, for $J \supset I$ of size $k+2$, define a linear map by

$$d_J^k : \bigoplus_I \mathcal{F}(U_I) \to \mathcal{F}(U_J), \quad d_J^k(\alpha) = \sum_{j=0}^{k+1} (-1)^j \alpha_{J \setminus i_j}|_{U_J},$$

where α_J is a section of $\mathcal{F}(U_J)$, and $\alpha_{J\setminus i_j}$ a section of $\mathcal{F}(U_{J\setminus i_j})$. Then,

$$0 \to \bigoplus_{I=\{i_0\}} \mathcal{F}(U_I) \xrightarrow{d^0} \bigoplus_{I=\{i_0,i_1\}} \mathcal{F}(U_I) \xrightarrow{d^1} \bigoplus_{I=\{i_0,i_1,i_2\}} \mathcal{F}(U_I) \xrightarrow{d^2} \cdots$$

is a chain complex where the J-th entry of d^k is given by d_J^k as above, see for example [Ara03, Section 6.3].

Definition 1.3.25. Let X be a variety and $\{U_i\}$ an open affine cover. Let $\mathcal{F}$ be the structure sheaf or the sheaf of differentials on X. The k-th *sheaf cohomology group* $H^k(X, \mathcal{F})$ is defined as

$$H^k(X, \mathcal{F}) = \frac{\ker d^k}{\operatorname{im} d^{k-1}}.$$

Remark 1.3.26. Note that strictly speaking what we defined here is Čech cohomology. However, the notion of Čech cohomology and sheaf cohomology agree on complex varieties, by [Ara03, Theorem 6.4.6], at least for our choice of (quasi-coherent) sheaves. Moreover, we also implicitly use the fact that for these sheaves, the definition does not depend on the choice of affine cover $\{U_i\}$, see *loc. cit.*

Example 1.3.27. Let X be a smooth projective variety with affine cover $\{U_i\}$ and $\mathcal{O}_X$ its structure sheaf. Then,

$$H^0(X, \mathcal{O}_X) = \ker d^0 = \left\{ \alpha = (\alpha_i) \in \bigoplus_i \mathcal{O}_X(U_i) \;\middle|\; \alpha_i|_{U_{ij}} = \alpha_j|_{U_{ij}} \right\} = \mathcal{O}_X(X).$$

◇

Tools

We end this section by collecting some standard tools for computations with (co-)homology. First, all presented (co-)homology theories here are invariant under homeomorphisms, see e.g. [Hat02, Chapter 2.1].

Theorem 1.3.28. *Let $A \subset X$ and $B \subset Y$ be topological spaces. Suppose that (X, A) and (Y, B) are homeomorphic. Then, the (compactly supported) cohomology and also homology of the two pairs are isomorphic.*

An important general result in (co-)homology theory is that short exact sequences give rise to long exact sequences.

Proposition 1.3.29. *Let $A_\bullet \to B_\bullet \to C_\bullet$ be a short exact sequence of chain complexes. Then, it induces a long exact sequence in cohomology*

$$\cdots \to H^\bullet(C) \to H^\bullet(B) \to H^\bullet(A) \to H^{\bullet+1}(C) \to \cdots .$$

We record the two most important sequences for this thesis. For a triple of subvarieties $Z \subset Y \subset X$ we get a long exact sequence in relative cohomology

$$\cdots \to H^{\bullet-1}(Y, Z) \to H^\bullet(X, Y) \to H^\bullet(X, Z) \to H^\bullet(Y, Z) \to \cdots .$$

Example 1.3.30. We can now compute an example for relative homology. Let $Z = \varnothing$ and $Y = p \subset X = \mathbb{P}^1$. Then, using the homology computations of projective spaces from Example 1.3.14 and the duality in Theorem 1.3.20, we obtain

$$H^k(\mathbb{P}^1, p) \cong \begin{cases} \mathbb{Q} & , k = 2 \\ 0 & , \text{else.} \end{cases}$$

By Proposition 1.3.24 this also gives an example of the compactly supported cohomology groups $H^k_c(\mathbb{P}^1 \setminus p) \cong H^k(\mathbb{P}^1, p)$. ◇

For two subvarieties $A, B \subset X$, whose interiors cover the interior of X, we get the *Mayer-Vietoris* long exact sequence in cohomology:

$$\to H^{\bullet-1}(A) \oplus H^{\bullet-1}(B) \to H^{\bullet-1}(A \cap B) \to H^\bullet(X) \to H^\bullet(A) \oplus H^\bullet(B) \to \tag{1.3.1}$$

We introduce the Euler characteristic, see e.g. [Hat02, Theorem 2.44].

Lemma 1.3.31. *The topological Euler characteristic $\chi(X)$ is given as*

$$\chi(X) = \sum_{k \geq 0} (-1)^k \dim H_k(X).$$

If X has a chosen CW structure where f_k records the number of k-simplices, then

$$\chi(X) = \sum_{k \geq 0} (-1)^k f_k.$$

We record the classical Andreotti-Frankel Theorem.

Theorem 1.3.32 ([AF59, Theorem 1])**.** *Let X be an n-dimensional smooth affine variety. Then,*

$$H^k(X) = 0 \quad k > n.$$

We also recall the Lefschetz hyperplane theorem.

Theorem 1.3.33 ([Voi03, Theorem 1.23])**.** *Let X be a projective variety of dimension n, and let Y be a hypersurface in X such that the complement $X \setminus Y$ is smooth. Then, we get an isomorphism*

$$H^k(X) \cong H^k(Y) \quad \text{for all } k \leq n-2,$$

and an injection $H^{n-1}(X) \hookrightarrow H^{n-1}(Y)$.

1.4 The Amplituhedron

A central object in this thesis is a semi-algebraic set called the amplituhedron. An amplituhedron depends on the choice of positive integer parameters k, m, and n, with $n \geq k+m$, and on the choice of a $n \times (k+m)$ matrix Z with positive maximal minors. In this thesis we are going to fix $m = 2$ or $m = 4$. The matrix Z induces a rational map

$$\widetilde{Z} : \mathrm{Gr}(k, n) \dashrightarrow \mathrm{Gr}(k, k+m), \quad V \mapsto VZ.$$

By the assumption on Z, this map is well-defined on $\mathrm{Gr}(k, n)_{\geq 0}$. The image $\widetilde{Z}(\mathrm{Gr}(k, n)_{\geq 0})$ of the non-negative Grassmannian under this map is called the *(tree-level) amplituhedron* $\mathcal{A}_{n,k,m}(Z)$, first defined in [AT14]. Henceforth, we will often suppress writing the dependence on Z, that is $\mathcal{A}_{n,k,m} = \mathcal{A}_{n,k,m}(Z)$. The amplituhedron is a regular semi-algebraic set in $\mathrm{Gr}(k, k+m)$, i.e. it is the Euclidean closure of its interior. In particular, its Euclidean boundary is $\partial\mathcal{A}_{n,k,m} = \mathcal{A}_{n,k,m} \setminus \operatorname{int} \mathcal{A}_{n,k,m}$.

Example 1.4.1. For $k = 1$ the amplituhedron $\mathcal{A}_{n,1,m} \subset \mathbb{P}^m$ becomes a full-dimensional (projective) polytope. Explicitly, for $m = 2$ the matrix Z has positive maximal minors since it is a Vandermonde matrix. A not-to-scale sketch of the amplituhedron $\mathcal{A}_{4,1,2}(Z)$ can be seen in Figure 1.4.1. Moreover, note that the rows of Z all define points on the rational normal curve of degree 2. In Chapter 5, this situation is going to be studied in much more detail. ◇

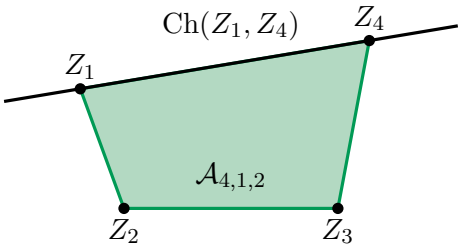


Figure 1.4.1: A 2D amplituhedron $\mathcal{A}_{4,1,2}(Z)$.

The amplituhedron, or rather its algebraic boundary, is the central object of this thesis. Therefore, we recall the descriptions of the algebraic boundary of the amplituhedron $\mathcal{A}_{n,k,m}$ for $m = 2$ and $m = 4$. These commonly use the notion of *twistor coordinates*, see e.g. [Hod13] or [Wil23, Definition 4.5]. The twistor coordinate $\langle Y i_1 \dots i_m \rangle$ (or $\langle Y Z_{i_1} \dots Z_{i_m} \rangle$) is defined as the determinant of the $(k+m) \times (k+m)$ matrix obtained by stacking a matrix representing $Y \in \mathrm{Gr}(k, k+m)$ together with the rows of Z labeled by $i_1, \dots, i_m \in [n]$. A twistor coordinate can be regarded as a linear function on $\mathrm{Gr}(k, k+m)$ and its vanishing locus is a hyperplane section of $\mathrm{Gr}(k, k+m)$. More precisely, the twistor coordinate $\langle Y i_1 \dots i_m \rangle$ on $\mathrm{Gr}(k, k+m)$ is also identified with the Chow form $\mathrm{Ch}(i_1, \dots, i_m)$, where $i_1, \dots, i_m$ is an abbreviation for the $(m-1)$-dimensional projective linear space spanned by the rows of Z with these labels. It will be convenient, for an index $i > n$, to take it modulo n, such that i

$\mathrm{mod}\ n \in [n]$, in which case we say that the indices are *read cyclically*. We now record the algebraic boundary of $\partial_a \mathcal{A}_{n,k,2}$ as in [RST24, Proposition 3.1].

Proposition 1.4.2. *For $m = 2$ the algebraic boundary $\partial_a \mathcal{A}_{n,k,2}$ of $\mathcal{A}_{n,k,2}$, with $n \geq k+2$, is the union of the following hyperplane sections of $\mathrm{Gr}(k, k+2)$:*

$$\langle Y i (i+1) \rangle = 0,$$

where $Y \in \mathrm{Gr}(k, k+2)$ and $i \in [n]$, with indices read cyclically.

Example 1.4.3. We continue with the 2-dimensional amplituhedron from Example 1.4.1, which had $m = 2$. Suppose we choose as the Z matrix defining the amplituhedron the Vandermonde matrix

$$Z = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{bmatrix}.$$

In this case, we explicitly get

$$\begin{aligned} \langle Y 14 \rangle = \mathrm{Ch}(Z_1, Z_4) &= \det \begin{bmatrix} 1 & 0 \\ 1 & 3 \end{bmatrix} p_2 - \det \begin{bmatrix} 1 & 0 \\ 1 & 9 \end{bmatrix} p_1 + \det \begin{bmatrix} 0 & 0 \\ 3 & 9 \end{bmatrix} p_0 \\ &= 3p_2 - 9p_1, \end{aligned}$$

where we identified $\mathrm{Gr}(1,3) \cong \mathbb{P}^2$ with coordinates p_0, p_1, p_2. The remaining 3 irreducible components of the boundary are precisely the lines supporting the remaining facets of $\mathcal{A}_{4,1,2}$ in Figure 1.4.1. ◇

Remark 1.4.4. One way of thinking about the algebraic boundary of $\mathcal{A}_{n,k,2}$ is that it is characterized by the cycle of lines $\mathrm{span}_{\mathbb{C}}(Z_i, Z_{i+1}) \subset \mathbb{P}^{k+1}$ for $i \in [n]$, with indices read cyclically. That is, $\partial_a \mathcal{A}_{n,k,2}$ consists of the Chow hypersurfaces of these lines. This will be important later in the proofs of Section 4.2.

Similarly, the algebraic boundary for $m = 4$ was conjectured in [ATT18, Section 5] and eventually proven in [ELT25, Corollary 8.8] to be the following.

Proposition 1.4.5. *For $m = 4$ the algebraic boundary $\partial_a \mathcal{A}_{n,k,4}$ of $\mathcal{A}_{n,k,4}$, with $n \geq k+4$, is the union of the following hyperplane sections of $\mathrm{Gr}(k, k+4)$:*

$$\langle Y i (i+1) j (j+1) \rangle = 0,$$

where $Y \in \mathrm{Gr}(k, k+4)$ and $\{i, i+1\} \cap \{j, j+1\} = \varnothing$, with $i, j \in [n]$ read cyclically.

Finally, we turn our attention back to the semi-algebraic description of the amplituhedron. There is in fact an alternative definition of the amplituhedron, which is usually referred to as the sign-flip characterization.

Definition 1.4.6. Let $m = 2r$ be even and $n \geq k + m$, and let $Z \in \mathbb{R}^{n\times k+m}$ be a totally positive matrix. The subset $\mathcal{F}_n^\circ(Z)$ is the semi-algebraic subset of $\mathrm{Gr}(k, k+m)$ consisting of points $Y \in \mathrm{Gr}(k, k+m)$ at which

i. all linear forms $\langle Y i_1(i_1+1)\dots i_r(i_r+1)\rangle$, for all $i \in [n]$ read cyclically, are nonzero and have the same sign,
ii. the sequence $(\langle Y12\rangle, \langle Y13\rangle, \dots, \langle Y1n\rangle)$ has exactly k sign flips, ignoring zeros.

It is conjectured, see e.g. [ELT25], that the closure of the set $\mathcal{F}_n^\circ(Z)$ characterizes the amplituhedron for $m = 2, 4$.

Proposition 1.4.7 ([PSW23, Theorem 5.1])**.** *The amplituhedron $\mathcal{A}_{n,k,2}(Z) \subset \mathrm{Gr}(k, k+2)$ is the closure of $\mathcal{F}_n^\circ(Z)$ in $\mathrm{Gr}(k, k+2)$. More precisely,*

$$\widetilde{Z}(\mathrm{Gr}(2,n)_{>0}) \subset \mathcal{F}_n^\circ(Z) \subset \mathcal{A}_{n,k,2}(Z),$$

and all inclusions turn into equalities when taking the closure in the Euclidean topology.

An important branch of studying the amplituhedron utilizes the notion of positroid varieties and stratifications, see e.g. [PSW23]. At several points in this work, we will make use of the results from that area, hence we fix some notation. Choose a subset $\mathcal{M} \subset \binom{[n]}{k}$, then

$$\mathring{\Pi}_{\mathcal{M}}^{\geq 0} = \{V \in \mathrm{Gr}(k,n)_{\geq 0} \mid p_I(V) > 0 \text{ for } I \in \mathcal{M},\ p_I(V) = 0 \text{ for } I \notin \mathcal{M}\}$$

is a *positroid cell* if it is not empty, and $\mathcal{M}$ is called a *positroid*. It turns out that $\mathcal{M}$ is a matroid. There are myriad combinatorial models indexing positroid cells, see e.g. [Pos06]. The complex Zariski closure in $\mathrm{Gr}(k,n)$ of this cell is called the *positroid variety*, denoted by

$$\Pi_{\mathcal{M}} = \overline{\mathring{\Pi}_{\mathcal{M}}^{\geq 0}} \subset \mathrm{Gr}(k,n).$$

We define the quasi-projective *open positroid varieties* $\mathring{\Pi}_{\mathcal{M}}$ as

$$\mathring{\Pi}_{\mathcal{M}} = \Pi_{\mathcal{M}} \setminus \left(\bigcup_{\mathcal{M}' \subset \mathcal{M}} \Pi_{\mathcal{M}'} \right).$$

One of the most remarkable results for positroids is that the open positroid varieties give rise to a very nice stratification of the Grassmannian $\mathrm{Gr}(k,n)$, see e.g. the seminal work of Postnikov [Pos06]. Explicitly, this stratification can be written as

$$\mathrm{Gr}(k,n) = \bigsqcup_{\mathcal{M}} \mathring{\Pi}_{\mathcal{M}},$$

which is a disjoint union over all open positroid varieties. Moreover, its intersection with the $\mathrm{Gr}(k,n)_{\geq 0}$ is well-behaved, that is, it gives the complete boundary structure.

1.5 Positive Geometries

In this section we elect to introduce positive geometries in the recursive sense of [ABL17], see also [Lam22], as this is most closely connected to applications in physics. For a different point of view, based on the recent work [BD25], we refer the reader to Section 2.4.

Before we can start introducing positive geometries, we need to establish the notion of Poincaré residues. Let X be an n-dimensional irreducible complex projective variety defined over $\mathbb{R}$. Suppose D is a divisor in X, with irreducible components $D_1, \dots, D_m$, such that $X \setminus D$ is smooth. Moreover, assume that $D' = D_m \cap \bigcup_{i=1}^{m-1} D_i \subset D_m$ is a divisor in D_m, such that $D_m \setminus D'$ is smooth. Next, let ω be a logarithmic form on X with poles along D, i.e. $\omega \in \Omega^n_{\log}(X \setminus D)$. Then, on $X \setminus D$ we can write

$$\omega = \frac{dD_m}{D_m} \wedge \eta + \zeta,$$

with $\eta, \zeta \in \Omega^{n-1}_{\log}(D_m \setminus D')$. Then, we define the *Poincaré residue* $\operatorname{Res}_{D_m} \omega$ as

$$\operatorname{Res}_{D_m} \omega = \eta|_{D_m} \in \Omega^{n-1}_{\log}(D_m \setminus D'). \tag{1.5.1}$$

For a simple example, see 1.5.2, and for a short discussion of the residue map on the level of cohomology, see (2.1.2).

We can now turn to positive geometries. Let S be an n-dimensional semi-algebraic subset of the real points $X(\mathbb{R})$ of X. We require that the interior $\operatorname{int} S$ of S (with respect to the analytic topology on $X(\mathbb{R})$) is an oriented manifold, see Definition 1.3.17, such that its (analytic) closure is S. Suppose the irreducible components of the algebraic boundary $\partial_a S$ are the prime divisors $D_1, \dots, D_m$ on X from above. We set $S_i = S|_{D_i} \subset D_i(\mathbb{R})$ to be the closure of the interior of $D_i \cap \partial S$.

Definition 1.5.1. A triple (X, S, ω_S) consisting of the n-dimensional ambient variety X, a top-dimensional semi-algebraic set S in the real points of X, and a differential n-form ω_S on X - the *canonical form* - is called a *positive geometry* if ω_S satisfies the following:

i. For $n > 0$ the canonical form is unique and $\omega_S \in \Omega^n_{\log}(X \setminus D)$.

ii. For each component D_i, the triple (D_i, S_i, ω_{S_i}) is a positive geometry where the Poincaré residue $\operatorname{Res}_{D_i} \omega_S$ equals ω_{S_i}, and $\operatorname{int} S_i$ has an orientation induced by that of $\operatorname{int} S$.

iii. If $n = 0$, the ambient variety $X = S$ is a point and $\omega_S = \pm 1$. The sign determines the orientation on this 0-dimensional manifold.

Example 1.5.2. Let us first consider the green-shaded quadrilateral $Q \subset \mathbb{P}^2$ in Figure 1.5.1a. Denote the coordinates of the affine patch $U_z = \mathbb{C}^2 \subset \mathbb{P}^2$ by x and y. The unique canonical form of Q on the affine patch U_z is given by

$$\omega_Q = \frac{y + x + 2}{xy(2y - x - 2)(y - 2x + 2)} dx \wedge dy.$$

Note that the Poincaré residue along $x = 0$ yields

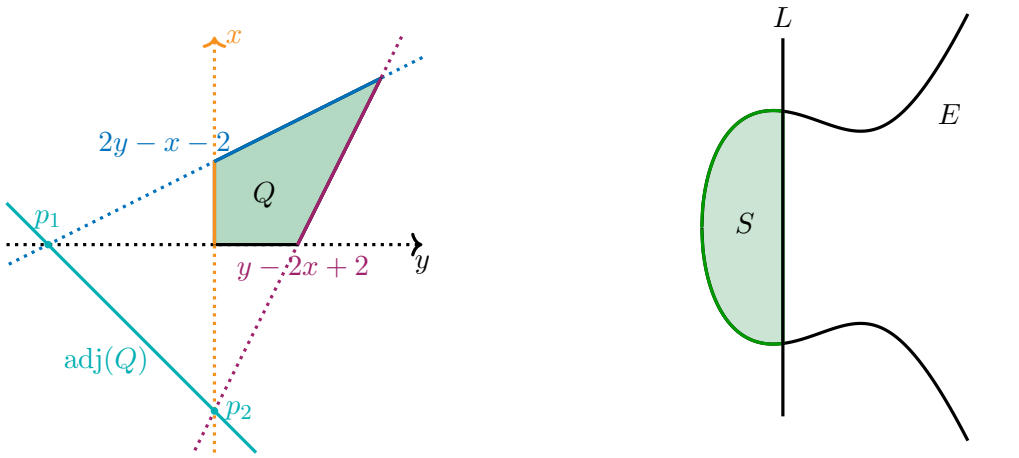
$$\operatorname{Res}_x \omega_Q = \frac{1}{y(1 - y)} dy.$$

This is precisely the canonical form of the orange line segment, as required by ii. in the above definition. We will explain in Example 1.5.10 how to obtain the canonical form ω_Q. ◇

Example 1.5.3. We also briefly want to mention a central counterexample to a positive geometry. Consider a plane elliptic curve $E \subset \mathbb{P}^2$ and a line L as seen in Figure 1.5.1b. It is well known that E supports a global regular form ω_E, implying that ω_E has no poles and no residues on E. Now, consider the green-shaded bounded region S in that figure. Suppose S had a canonical form ω_S. Then, in particular ω_S would have a simple pole along E, hence the Poincaré residue along E does not vanish. However, then

$$\mathrm{Res}_E\, \omega_S + \lambda \omega_E,$$

has the correct residues on the green curve segment in Figure 1.5.1b for all $\lambda \in \mathbb{C}$. Therefore, we would have a 1-dimensional family of "canonical" forms, contradicting the uniqueness requirement i. in Definition 1.5.1. Hence, S cannot be a positive geometry. ◇



(a) A positive geometry Q.

(b) A non-positive geometry region S.

Figure 1.5.1: (Non) Examples of positive geometries.

A central challenge in the study of positive geometries is the computation of canonical forms. One such approach utilizes the notion of adjoint hypersurfaces, residual arrangements, and boundary stratifications. We start by defining the latter.

Definition 1.5.4. The algebraic *boundary stratification* of the divisor $D \subset X$, denoted $\mathcal{L}(D)$, is the poset of irreducible subvarieties in X ordered by inclusion, obtained as follows. Take any irreducible component in D, in its multiple intersections D_I, and in their singular loci. Then, $\mathcal{L}(D)$ is the collection of these irreducible subvarieties of X, such that it is closed under taking intersections of these components. By convention $\mathcal{L}(D)$ contains X as the maximal element.

Example 1.5.5. We continue with Example 1.5.2. The algebraic boundary $\partial_a Q$ of the quadrilateral Q in Figure 1.5.1a is given by the union of the four colored lines supporting the four facets of Q, i.e.

$$\partial_a Q = \mathbb{V}(xy(2y - x - 2z)(y - 2x + 2z)) \subset \mathbb{P}^2,$$

Here we homogenized with respect to the z variable. Denote the four boundaries by L_i, and the four vertices by v_i for $i \in [4]$. Since all boundaries and their intersections are smooth, the full boundary stratification $\mathcal{L}(\partial_a Q)$ is shown in Figure 1.5.2. $\diamond$

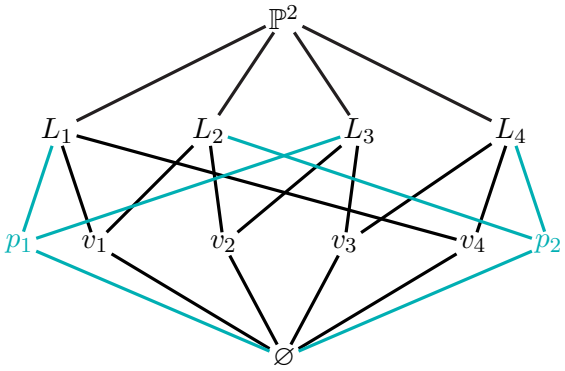


Figure 1.5.2: Boundary stratification $\mathcal{L}(\partial_a Q)$ of the quadrilateral Q.

Definition 1.5.6. Let S be a semi-algebraic set in the real points of X, and let $D = \partial_a S$ be the algebraic boundary of S. Then, an irreducible component $Z \in \mathcal{L}(D)$ is called *residual* if the dimension of the real semi-algebraic set $(Z \cap S)$ is strictly smaller than the dimension of Z as a variety. Put in symbols, Z is residual if

$$\dim_{\mathbb{R}}(Z \cap S) < \dim_{\mathbb{C}}(Z).$$

The union of all residual components is called the *residual arrangement* $\mathcal{R}(S)$.

Remark 1.5.7. When there is no ambiguity from context, we freely identify $\mathcal{R}(S)$ either as a sub-poset of $\mathcal{L}(D)$ or as a reducible subvariety of the ambient variety X. Also note that this definition of residual arrangement is slightly more general than those used elsewhere in the literature, see e.g. [RST24; KPS26a].

We can now define the adjoint hypersurface.

Definition 1.5.8. Suppose $X \subset \mathbb{P}^N$ is smooth and with canonical bundle isomorphic to $\mathcal{O}_X(a)$. Let $S \subset X$ be a semi-algebraic set, with algebraic boundary $\partial_a S \subset X$ and $\mathcal{O}_X(D) \cong \mathcal{O}_X(b)$. Write $d = a + b$. An *adjoint* $\mathrm{adj}(S)$ is a global section of the twisted structure sheaf $\mathcal{O}_X(d)$, i.e. $\mathrm{adj}(S) \in \Gamma(X, \mathcal{O}_X(d))$ which vanishes on $\mathcal{R}(S)$. We will also identify $\mathrm{adj}(S)$ as a hypersurface in X given by the zero locus of the section.

Remark 1.5.9. Pragmatically, an adjoint of a semi-algebraic set $S \subset \mathbb{P}^N$ in projective space is the lowest degree hypersurface in $\mathbb{P}^N$ that interpolates the residual arrangement of S.

Example 1.5.10. We continue Examples 1.5.2 and 1.5.5. The residual arrangement $\mathcal{R}(Q)$ is then given by the two points p_1 and p_2 that are the intersections of the lines disjoint from the quadrilateral Q. Moreover, $\mathcal{R}(Q)$ is the sub-poset of $\mathcal{L}(\partial_a Q)$ colored teal in Figure 1.5.2.

Therefore, the adjoint $\mathrm{adj}(Q)$ has degree $d-n-1=1$, and is thus the unique line given by the interpolation of the two points in $\mathcal{R}(Q)$:

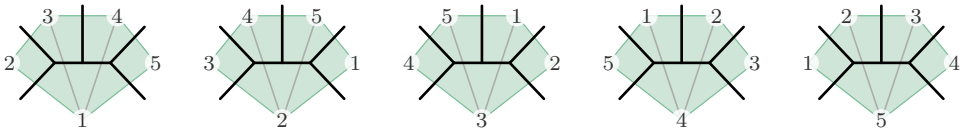
$$\mathrm{adj}(Q) = y + x + 2z.$$

Then, the canonical form ω_Q as in Example 1.5.2 is given by the adjoint in the numerator and the defining equations of the algebraic boundary in the denominator. ◇

Positive Geometries and Scattering Amplitudes

In the introduction we claimed that positive geometries provide a natural framework for scattering amplitudes. We now illustrate this for the five-particle amplitude A_5 in planar $\mathrm{Tr}\,\varphi^3$ theory. This is based on [Ark+18, Section 3].

The amplitude A_5 can be derived from the 5 graphs shown in Figure 1.5.3 as follows. The 5 graphs are dual to triangulations with chords of the pentagon. For each chord $\{i,j\}$ from vertex i to j in the pentagon, we define an indeterminate X_{ij}. Then, to each triangulation we associate the reciprocal of $X_{ij}X_{k\ell}$ if $\{i,j\}$ and $\{k,\ell\}$ were the chords used in the triangulation.



$$\frac{1}{X_{13}X_{14}} \qquad \frac{1}{X_{24}X_{25}} \qquad \frac{1}{X_{35}X_{13}} \qquad \frac{1}{X_{14}X_{24}} \qquad \frac{1}{X_{25}X_{35}}$$

Figure 1.5.3: The five planar Feynman graphs contributing to A_5.

The indeterminates X_{ij} are not algebraically independent. They depend on the parameters $c_{ij} \in \mathbb{R}$, which in turn describe the kinematics of the involved particles. We have

$$\begin{aligned} X_{13} &= x \\ X_{14} &= y \\ X_{24} &= y - x + c_{13} \\ X_{25} &= -x + c_{13} + c_{14} \\ X_{35} &= -y + c_{14} + c_{24}. \end{aligned} \tag{1.5.2}$$

Then, the 5-point scattering amplitude is obtained by summing over the expressions in Figure 1.5.3.

$$\begin{aligned} A_5 &= \frac{1}{X_{13}X_{14}} + \frac{1}{X_{14}X_{24}} + \frac{1}{X_{24}X_{25}} + \frac{1}{X_{25}X_{35}} + \frac{1}{X_{35}X_{13}} \\ &= \frac{(c_{13}+y)(c_{13}+c_{14})(c_{14}+c_{24}) - xc_{13}(c_{14}+c_{24}) + c_{14}xy}{xy(y-x+c_{13})(-x+c_{13}+c_{14})(-y+c_{14}+c_{24})}. \end{aligned} \tag{1.5.3}$$

In Figure 1.5.4 we can see the positive geometry $\mathcal{A}_5$, called the ABHY-associahedron,

which encodes the scattering amplitude as follows. First, observe that the 5 facets of $\mathcal{A}_5$ are precisely supported on the vanishing loci of the equation in (1.5.2) with $X_{ij} = 0$. Therefore, the algebraic boundary $\partial_a \mathcal{A}_5$ is given by the vanishing of

$$xy(-y + c_{14} + c_{24})(-x + c_{13} + c_{14})(y - x + c_{13}).$$

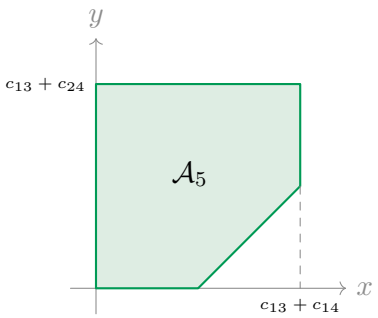


Figure 1.5.4: The five-point ABHY associahedron $\mathcal{A}_5$.

The residual arrangement consists of the $\binom{5}{2} - 5 = 5$ intersection points of the 5 lines supporting the facets of $\mathcal{A}_5$ that are not vertices. There is a unique conic interpolating these 5 points, which is the adjoint

$$\mathrm{adj}(\mathcal{A}_5) = (c_{13} + y)(c_{13} + c_{14})(c_{14} + c_{24}) - xc_{13}(c_{14} + c_{24}) + c_{14}xy.$$

This is precisely the numerator of the scattering amplitude from (1.5.3). Therefore, the canonical form can be identified with the scattering amplitude, that is

$$\omega_{\mathcal{A}_5} = \frac{\mathrm{adj}(\mathcal{A}_5)}{\partial_a \mathcal{A}_5}\, dx \wedge dy = A_5\, dx \wedge dy.$$

This shows how canonical forms can be used to compute scattering amplitudes in certain quantum field theories.

Chapter 2

Practitioner's Tools for Positive Geometries via Mixed Hodge Theory

The genus is...difficult!

Veronica

In recent work Brown and Dupont [BD25] suggest a paradigm shift for the study of positive geometries. They propose to study positive geometries via the mixed Hodge theory on the relative homology of a pair of varieties (X, Y).

The goal of this chapter is three-fold. First, in Section 2.1, we give a brief practitioner's introduction to this new framework for positive geometries. This will also provide the connection between the remaining chapters of this work. Of course, due to the depth of mixed Hodge theory we made no attempt to be self-contained. Second, in applications the pairs (X, Y) that arise are almost always singular. However, since the assumption of simple normal crossing pairs is ubiquitous in mixed Hodge theory, there are only very few tools available to assess the genus and combinatorial rank in these cases. Hence, in Sections 2.2 and 2.3, our main results are several novel tools that are effective when dealing with the challenges that arise when applying this framework to singular varieties. Lastly, in Section 2.4, our main result gives a criterion under which the new framework leads to positive geometries in the sense of Definition 1.5.1. We will also give a definition of positive geometries in this new framework and show how it is compatible with the one we gave before.

2.1 Genus of Pairs and Mixed Hodge Theory

In this section we recall the setup for positive geometries via mixed Hodge theory, that was first introduced in the similarly titled paper by Brown and Dupont [BD25].

Mixed Hodge Structures

Before introducing mixed Hodge structures, we start, consistent with the historical developments, with the simpler pure Hodge structures.

Let X be a smooth projective variety, then there is a natural decomposition of the cohomology groups on X, first identified by Hodge in [Hod52]:

$$H^k(X) = \bigoplus_{p+q=k} H^k(X)^{p,q} = \bigoplus_{p+q=k} H^q(X, \Omega_X^p),$$

where Ω_X^p is the sheaf of p-forms on X, see Definition 1.2.14. This is the *Hodge decomposition*, and it gives X, or rather its cohomology groups, a *pure Hodge structure of weight* k. The dimensions of the $\mathbb{Q}$-vector spaces $H^q(X, \Omega_X^p)$ are the Hodge numbers

$$h^{p,q}(X) = h^{p,q}(H^{p+q}(X)) = h^{q,p}(X).$$

We can organize them into a Hodge diamond.

Example 2.1.1. For a first example let X be the projective space $\mathbb{P}^n$. Then, we get

$$h^{p,q}(\mathbb{P}^n) = \begin{cases} 1 & , p = q \\ 0 & , \text{else.} \end{cases}$$

This follows from the symmetry in p and q of the decomposition above, and the fact that $\dim H^k(\mathbb{P}^n) = 1$ if k is even and zero else, see 1.3.14.

Next, let X be a smooth elliptic curve. Then, the complete Hodge diamond is given in Figure 2.1.1. Note that $h^{1,0} = \dim H^0(E, \Omega_E^1) = 1$ detects precisely the existence of a global holomorphic 1-form on E, which we used in Example 1.5.3. ◇

$$h^{1,1}(E) = 1$$

$$h^{1,0}(E) = 1 \quad h^{0,1}(E) = 1$$

$$h^{0,0}(E) = 1$$

Figure 2.1.1: Hodge diamond of an elliptic curve E.

If we want to consider singular, non-compact, or pairs of varieties we can endow their cohomology groups with a mixed Hodge structure, which roughly speaking allows for pure Hodge structures of different weights to be combined. We use the axiomatic definition as in [BD25, Definition 1.5].

Definition 2.1.2. A *mixed Hodge structure* is a finite-dimensional $\mathbb{Q}$-vector space H together with

i. an increasing filtration $W_\bullet$ on H called *weight filtration*, i.e.

$$0 = W_{-1}H \subset W_0H \subset W_1H \subset \dots$$

ii. a decreasing filtration $F^\bullet$ on the complexification $H_\mathbb{C}$ called *Hodge filtration*, i.e.

$$0 = F^{n+1}H_\mathbb{C} \subset F^n H_\mathbb{C} \subset \dots$$

such that for every integer w, the filtration induced by $F^\bullet$ on

$$\mathrm{gr}_w^W H := W_w H / W_{w-1} H$$

defines a pure Hodge structure of weight w. For $w = p+q$ we define its Hodge-filtered parts by

$$H_\mathbb{C}^{p,q} = (\mathrm{gr}_w^W H_\mathbb{C})^{p,q} = \mathrm{gr}_F^p(\mathrm{gr}_w^W H_\mathbb{C}).$$

The corresponding mixed *Hodge numbers* are the complex vector space dimensions

$$h^{p,q} = \dim H_\mathbb{C}^{p,q} = h^{q,p}.$$

Remark 2.1.3. See [Del71] for the construction of these two filtrations. In our setting, we will exclusively deal with mixed Hodge structures on the k-th relative cohomology group of a pair of complex varieties (X, Y) where X is of dimension n. Then, for the Hodge numbers we explicitly write

$$h^{p,q}(H^k(X,Y)) = \dim_\mathbb{C}\left(\mathrm{gr}_F^p\left(\mathrm{gr}_{p+q}^W H^k(X,Y)_\mathbb{C}\right)\right).$$

Moreover, in the cases where there is no ambiguity we will write $h^{p,q}(X,Y)$ for the corresponding mixed Hodge numbers on the middle cohomology group $h^{p,q}(H^n(X,Y))$. Lastly, note that gr^W and gr_F are both exact functors, which for our purposes means that applying them to exact sequences yields another exact sequence, a fact we will freely use throughout the thesis.

Example 2.1.4. Let E be an elliptic curve and $p \neq q$ two points on it. Then we will later see in Lemma 2.1.8 that we have the following long exact sequence

$$0 \to H^1(E) \to H^1(E \setminus \{p,q\}) \to H^0(p,q)(-1) \to H^2(E) \to \cdots.$$

Then we can apply the exact weight functors gr_2^W and gr_1^W to obtain

$$\mathrm{gr}_1^W H^1(E \setminus \{p,q\}) \cong H^1(E),$$

and

$$\mathrm{gr}_2^W H^1(E \setminus \{p,q\}) \cong \ker\left(H^0(p,q)(-1) \to H^2(E)\right) \cong \mathbb{Q}(-1),$$

since $H^2(E) \cong \mathbb{Q}(-1)$ and $H^0(p,q)(-1) \cong \mathbb{Q}(-1)^2$. ◇

If a mixed Hodge structure H satisfies $h^{p,q} = 0$ for all $p \neq q$, then we say H is of *Tate type*. Let i be an integer, then denote the *twisted Hodge structure* by $H(-i)$, where $H(-i)^{p,q} = H^{p-i,q-i}$. We say a map $\phi : H_1 \to H_2$ between two Hodge structures is a *morphism* of Hodge structures if $\phi(H_1^{p,q}) \subset H_2^{p,q}$ for all p and q.

We record the following general result on which weights can occur in $H^\bullet(X)$ for a variety X of dimension n, see [PS08, Section 5]:

Table 2.1.1: Weights on $H^k(X)$

	general	smooth	compact
$k \leq n$	$[0, 2k]$	$[k, 2k]$	$[0, k]$
$k \geq n$	$[2k-2n,\ 2n]$	$[k, 2n]$	$[2k-2n,\ k]$

Remark 2.1.5. Note that this implies that $H^0(X)$ is pure of weight 0 for any variety X.

In fact for pairs of varieties (X, Y) something slightly stronger follows from [Del74, Théorème 8.2.4 (i), (ii)], for the cases where non-zero Hodge numbers can occur. See also [BD25, Section 1.2.2].

Table 2.1.2: Possible values of p, q for $H^k(X, Y)^{p,q} \neq 0$

	possible values of p, q
$k \leq n$	$p, q \in \{0, \ldots, k\}$
$k \geq n$	$p, q \in \{k-n, \ldots, n\}$

Spectral Sequences

The main tool for computing Hodge numbers, next to the long exact sequences in Section 1.3, are spectral sequences. We start by introducing the essential notions for our specific setting for a spectral sequence that will converge to relative cohomology. For a broader introduction see [Hat04].

So, let X be an n-dimensional variety and $Y \subset X$ a subvariety of codimension 1, such that $X \setminus Y$ is smooth. Denote the irreducible components of Y by $Y_1, \ldots, Y_m$. Write $Y_I = \bigcap_{i \in I} Y_i$ for any $I \subset [m]$, and by convention $Y_\varnothing = X$. Now fix $I \in \binom{[m]}{p}$ and consider $J \subset I$ such that $|I \setminus J| = 1$. Then, consider the inclusions $Y_I \hookrightarrow Y_J$ and denote the maps in cohomology induced by their pullback by

$$\delta_J^q : H^q(Y_J) \longrightarrow H^q(Y_I).$$

A spectral sequence $\{E_\bullet^{p,q}\}$ is a collection of *pages* $E_r^{p,q}$ for $r \in \mathbb{Z}_{\geq 0}$, each of which is a two-dimensional array of abelian groups at position p, q, with morphisms called *differentials* d^r between them such that they form chain complexes. For us, we start by defining the first page $E_1^{p,q}$ to be $\bigoplus_{|I|=p} H^q(Y_I)$, so we set

$$E_1^{p,q} = \bigoplus_{|I|=p} H^q(Y_I).$$

Now, let $I = i_1 < \ldots < i_p \subset [m]$. Then, the first differential $d^1 : E_1^{p,q} \to E_1^{p+1,q}$ can

explicitly be written by defining its I-th component as the map

$$\begin{aligned} \delta_p^q : \bigoplus_{J\subset I} H^q(Y_J) &\longrightarrow H^q(Y_I) \\ (v_1,\dots,v_p) &\longmapsto \sum_{k=1}^{p} (-1)^k \delta^q_{I\setminus i_k}(v_k) \end{aligned} \tag{2.1.1}$$

where we ordered the direct sum on the left-hand side as $I\setminus J = i_p$ through $I\setminus J = i_1$. Then, we set $E_2^{p,q} = \ker d^1 / \operatorname{im} d^1$ understood at each position (p,q), which is well-defined, see e.g. [Hat04, Proposition 5.2]. Together, with the higher differentials $d^r : E_r^{p,q} \to E_r^{p+r,q-r+1}$, which we are not going to discuss in detail, this defines the higher pages $E_r = \ker d^r / \operatorname{im} d^r$. Now, since $H^{q-r+1}(Y_I) = 0$ for r big enough, the differentials d^r become the zero map for big enough r, hence there is a natural limit of the pages E_r as $r \mapsto \infty$, which is denoted $E_\infty^\bullet$.

Now let $H^\bullet = \bigoplus_{q\geq 0} H^q$ be a cohomology group with its grading by cohomological degree. Fix a non-negative integer $k = p+q$, then the weight filtration from Definition 2.1.2, with the convention $W_{k-p}H^k \subset W_{k-(p+1)}H^k$, defines a descending filtration on $H^\bullet$. Then, by [MP12, Chapter 24, Theorem 3.3], we say the spectral sequence *converges* to $\operatorname{gr}_q^W H^{p+q}$, that is $E_\infty^{p,q} \cong \operatorname{gr}_q^W H^{p+q}$. We write $E_r^{p,q} \Rightarrow H^{p+q}$. A converging spectral sequence *degenerates* on the r-th page if $E_r^{p,q} \cong E_\infty^{p,q}$.

We can now record the spectral sequence relevant for this work.

Proposition 2.1.6 ([BD20, Proposition 4.3])**.** *There is a spectral sequence converging to relative cohomology of (X,Y) with the weight filtration, that is*

$$E_1^{p,q} = \bigoplus_{|I|=p} H^q(Y_I) \;\Rightarrow\; H^{p+q}(X,Y).$$

Its maps are morphisms of mixed Hodge structures, so in particular $E_\infty^{p,q} \cong \operatorname{gr}_q^W H^{p+q}(X,Y)$. If Y is a simple normal crossing divisor, then the sequence degenerates on $E_2^\bullet$.

Poincaré Duality

Poincaré duality is an essential tool when dealing with this setup. We introduce it briefly in the various versions we are going to use throughout this work.

First, we rephrase the Poincaré residue from Section 1.5 in terms of cohomology. So let X be a projective n-dimensional variety with a subvariety $Y \subset X$ of codimension 1, such that $X\setminus Y$ is smooth. Let $Y_1,\dots,Y_m$ be the irreducible components of Y, and suppose that $Y_m \setminus Z$ is smooth with $Z = Y_m \cap \bigcup_{i=1}^{m-1} Y_i$. Then, there is a well-defined morphism of mixed Hodge structures, the *Poincaré residue* in cohomology, see e.g. [BD25, Section 1.5.2],

$$\operatorname{Res}_{Y_m} : H^n(X\setminus Y) \to H^{n-1}(Y_m \setminus Z)(-1). \tag{2.1.2}$$

Note that this map is locally the Poincaré residue in (1.5.1), where the cohomology groups are identified with the space of n-forms on $X\setminus Y$, and $n-1$-forms on $Y\setminus Z$ respectively;

hence the name. We will also frequently use *Poincaré duality*: for a smooth orientable quasi-projective variety U of dimension n, we have

$$H_c^k(U) \cong H^{2n-k}(U)^\vee(-n),$$

see e.g. [PS08, Corollary 6.26]. This Poincaré duality implies another useful identification, see e.g. [BD25, Equation (16)].

Proposition 2.1.7. *Let X be an n-dimensional projective variety, and $Y \subset X$ a subvariety of codimension 1, such that $X \setminus Y$ is smooth. Then, Poincaré duality gives an isomorphism between mixed Hodge structures:*

$$H_\bullet(X, Y) \cong H^{2n-\bullet}(X \setminus Y)(n). \tag{2.1.3}$$

Tools

Lastly, we record a few auxiliary results, that will be needed in later chapters.

Lemma 2.1.8 ([Mil13, Corollary 16.2])**.** *Let X be a smooth variety and Y a smooth subvariety of codimension c. Then, we get the long exact Gysin sequence of mixed Hodge structures*

$$\cdots \to H^{k-2c}(Y)(-c) \to H^k(X) \to H^k(X \setminus Y) \to H^{k+1-2c}(Y)(-c) \to \cdots.$$

Example 2.1.9. We can use this to compute the mixed Hodge structure on the algebraic torus $\mathbb{C}^\times$. Let $X = \mathbb{P}^1$, and $Y = \{p, q\} \in X$ with $p \neq q$. Then $\mathbb{P}^1 \setminus \{p, q\} \cong \mathbb{C}^\times$, and we obtain by Lemma 2.1.8

$$0 \to H^1(\mathbb{C}^\times) \to H^0(p, q)(-1) \xrightarrow{\delta^0} H^2(\mathbb{P}^1) \to H^2(\mathbb{C}^\times) \to 0,$$

where we used that $H^1(p, q)(-1) = 0$ and $H^1(\mathbb{P}^1) = 0$. Then $H^1(\mathbb{C}^\times) \cong \ker \delta^0 \cong \mathbb{Q}(-1)$, since $H^0(p, q)(-1) \cong \mathbb{Q}^2(-1)$ and $H^2(\mathbb{P}^1) \cong \mathbb{Q}(-1)$. We then also must have that $H^2(\mathbb{C}^\times) = \operatorname{coker} \delta^0 \cong 0$ as the image of δ^0 must be 1-dimensional. Hence,

$$H^k(\mathbb{C}^\times) \cong \begin{cases} \mathbb{Q}(0) & , k = 0, \\ \mathbb{Q}(-1) & , k = 1, \\ 0 & , \text{else.} \end{cases}$$

◇

A classical tool to relate the homology of a product space to its factors is the Künneth formula.

Theorem 2.1.10 ([Kün23])**.** *Let X and Y be topological spaces. Then, for all integers k we have an isomorphism*

$$H_k(X \times Y) \cong \bigoplus_{p+q=k} H_p(X) \otimes_{\mathbb{Q}} H_q(Y).$$

The same result holds for (compactly supported) cohomology.

Example 2.1.11. Combining the compactly supported long exact sequence as in [PS08, §5.5.2] with Example 2.1.9 we can compute the mixed Hodge structure on the compactly supported cohomology of the algebraic torus $\mathbb{C}^\times$:

$$H_c^k(\mathbb{C}^\times) \cong \begin{cases} \mathbb{Q}(0) & , k = 1, \\ \mathbb{Q}(-1) & , k = 2, \\ 0 & , \text{else.} \end{cases}$$

Now let X be the ℓ-fold product of the torus $\mathbb{C}^\times$. We compute $H_c^q((\mathbb{C}^\times)^\ell)$ by applying the Künneth formula:

$$H_c^q((\mathbb{C}^\times)^\ell) \cong \begin{cases} \mathbb{Q}^{\binom{\ell}{2\ell-q}}(\ell - q) & , \ell \leq q \leq 2\ell \\ 0 & , 0 \leq q < \ell \end{cases}.$$

◇

We will also need the results about the Hodge structure on complements of hyperplane arrangements, see e.g. [Sha97].

Lemma 2.1.12. *Let $\mathcal{A} \subset \mathbb{P}^n$ be an arrangement of hyperplanes. The mixed Hodge structure on $H^k(\mathbb{P}^n \setminus \mathcal{A})$ is pure of weight $2k$ and Tate type for all k. Let $\mathcal{A}_a \subset \mathbb{C}^n$ be an affine arrangement of hyperplanes. The mixed Hodge structure on $H^k(\mathbb{C}^n \setminus \mathcal{A}_a)$ is pure of weight $2k$ and Tate type for all k.*

Similarly, the Hodge structure of a smooth non-degenerate quadric hypersurface $Q \subset \mathbb{P}^n$ is as simple as possible, see e.g. [Rei72].

Lemma 2.1.13. *Let Q be a $(n-1)$-dimensional smooth irreducible quadric in $\mathbb{P}^n$. The cohomology groups $H^\bullet(Q)$ are pure of Tate type.*

The same is true for the Grassmannian.

Lemma 2.1.14. *Let $\mathrm{Gr}(k, n)$ be the Grassmannian. The cohomology groups of $\mathrm{Gr}(k, n)$ have pure Hodge structure of Tate type.*

Proof. This follows from [Mel19, Proposition 2.4.8] together with the stratification [Bri04, Proposition 1.1.3] for $\mathrm{Gr}(k, n)$. □

Genus, Combinatorial Rank, and the Canonical Map

In this paragraph let X be projective and Y be a hypersurface in X, such that the complement $X \setminus Y$ is smooth. We introduce the essential notions from [BD25] that connect mixed Hodge theory and positive geometries.

We define the *genus* of the pair (X, Y) as

$$g(X, Y) = \sum_{p>0} \dim H^n(X, Y)^{p,0} = \sum_{p>0} h^{p,0}(X, Y).$$

Let U be a smooth open quasi-projective variety, and $(\overline{U}, D_U)$ a simple normal crossing compactification with $\overline{U} \setminus D_U = U$. Then, we set $g(U) = g(\overline{U}, D_U)$, which is well-defined as the latter does not depend on the compactification, see e.g. [PS08, Theorem 4.18]. We say (X, Y) is a *genus zero pair* if $g(X, Y) = 0$.

Similarly, we denote the *combinatorial rank* of the pair (X, Y) as the dimension of the zero weighted part on the middle relative (co-)homology, i.e.

$$\mathrm{cr}(X, Y) = \dim \mathrm{gr}_0^W H^n(X, Y) = h^{0,0}(X, Y).$$

We set $\mathrm{cr}(U) = \mathrm{cr}(\overline{U}, D_U)$, which is again well-defined by *loc. cit.*

For genus zero pairs, there is a well-defined map from the middle relative homology groups of (X, Y) to the space of logarithmic forms along Y, see Definition 1.2.18, given in [BD25] as

$$\Omega : H_n(X, Y) \longrightarrow \Omega^n_{\log}(X \setminus Y). \tag{2.1.4}$$

We refer to Ω as the *canonical map*. This map is the composition of the following four maps, which we will not describe in their full details.

i. The natural surjective quotient map $H_n(X, Y) \twoheadrightarrow \mathrm{gr}_0^W H_n(X, Y)$;

ii. The isomorphism (for genus zero pairs) $\mathrm{gr}_0^W H_n(X, Y) \cong F^0 H_n(X, Y)$;

iii. The Poincaré duality on a filtered piece $F^0 H_n(X, Y) \cong F^n H^n(X, Y)$ as in (2.1.3);

iv. The isomorphism from mixed Hodge theory $F^n H^n(X, Y)_{\mathbb{C}} \cong \Omega^n_{\log}(X \setminus Y)$.

Remark 2.1.15. In the case $g(X, Y) > 0$, we can still set up the canonical map Ω as follows. Let $K = \bigoplus_{p>0} H_n(X, Y)^{-p,0}$ with $\dim K = g(X, Y)$. Then, instead of the isomorphism in ii. from above we only get an isomorphism

$$\mathrm{gr}_0^W H_n(X, Y) \cong F^0 H_n(X, Y)/K.$$

Using iii. and iv. we identify $F^0 H_n(X, Y)/K$ with a quotient space $\widetilde{\Omega}^n_{\log}(X \setminus Y)$ of the space of logarithmic forms $\Omega^n_{\log}(X \setminus Y)$. Then, we can still define the analog of the canonical map

$$\Omega : H_n(X, Y) \to \widetilde{\Omega}^n_{\log}(X \setminus Y),$$

which is well-defined modulo the kernel K. Usually we will work with genus zero pairs, but we will briefly explore the consequences of this definition for the guiding example of the Del Pezzo cube of Chapter 4.

Example 2.1.16. Let us apply this new framework to the quadrilateral Q from Example 1.5.2. Here X is the ambient projective space $\mathbb{P}^2$, so $n = 2$, and Y is the algebraic boundary $\partial_a Q$. First, note that $\mathbb{P}^2 \setminus \partial_a Q$ is a hyperplane arrangement complement, hence it follows by Lemma 2.1.12 that $g(\mathbb{P}^2, \partial_a Q) = 0$. Moreover, $\partial_a Q$ is a simple normal crossing divisor in $\mathbb{P}^2$, hence $\mathrm{cr}(\mathbb{P}^2, \partial_a Q) = \binom{\deg \partial_a Q - 1}{n} = 3$. Note that this is also the number of bounded regions as seen in Figure 1.5.1a, which is no coincidence as a classical result from Zaslavsky [Zas75, Theorem A] implies. Since $g(\mathbb{P}^2, \partial_a Q) = 0$ the canonical map $\Omega : H_2(\mathbb{P}^2, \partial_a Q) \to$

$\Omega^2_{\log}(\mathbb{P}^2 \setminus \partial_a Q)$ is well-defined. If x and y are the coordinates of the affine patch $U_z \supset Q$, we explicitly get

$$\Omega([S]) = \frac{y + x + 2}{xy(2y - x - 2)(y - 2x + 2)} dx \wedge dy.$$

This is precisely the canonical form ω_Q as we found in Example 1.5.2. ◇

Example 2.1.17. Let us also see what happens to the counterexample to a positive geometry from Example 1.5.3. In that example, $X = \mathbb{P}^2$ and Y is the union of the line L and the elliptic curve E bounding the regions S. First, note that since $E \cup L$ are simple normal crossing, we have $\mathrm{cr}(\mathbb{P}^2, E \cup L) = \binom{3}{2} = 3$. The spectral sequence in Proposition 2.1.6 then shows that

$$H^2(\mathbb{P}^2, E \cup L) \cong H^1(E) \oplus \mathbb{Q}.$$

Therefore, $g(\mathbb{P}^2, E \cup L) = g(E) = 1$, and the canonical map in (2.1.4) is only well-defined up to equivalence classes, see Remark 2.1.15. More precisely, by the same remark, we see that elements in the quotient vector space $\widetilde{\Omega}^2_{\log}(\mathbb{P}^2 \setminus (E \cup L))$ are represented by dlog forms on $\mathbb{P}^2 \setminus (E \cup L)$ modulo the canonical form of E. More precisely, if e and ℓ are the defining equations of E and L on the affine patch $U_z \subset \mathbb{P}^2$, then

$$[S] \mapsto \frac{\lambda + 1}{e\ell}\, dx \wedge dy,$$

where x, y are the coordinates on $U_z \cong \mathbb{C}^2$, and λ is any complex number.

◇

For the sake of self-containment, we end this section by recording the results for computing genus and combinatorial rank from [BD25], which we are going to use in this work.

Computing the Genus

In this paragraph we collect some tools to compute the genus of a pair of varieties. Let X be an n-dimensional variety.

The genus is insensitive to subvarieties of codimension at least two.

Lemma 2.1.18 ([BD25, Proposition 3.9])**.** *Let $Z, Y \subset X$ be subvarieties, where Z has at least codimension 2. Then,*

$$g(X, Y \cup Z) = g(X, Y).$$

The genus can be bounded recursively.

Lemma 2.1.19 ([BD25, Corollary 3.13])**.** *Let $Y \subset X$ be a hypersurface with irreducible components $Y_1, \ldots, Y_m$. Set $Z = Z_1 \cup \ldots \cup Z_m$ with $Z_i = Y_i \cap \bigcup_{j \neq i} Y_j$. Then,*

$$g(X, Y) \leq g(X) + \sum_{i=1}^{m} g(Y_i, Z_i).$$

Equality holds if $F^0 H^{n-1}(X, Z) = \sum_{p \geq 0} H^{n-1}(X, Z)^{p,0} = 0$.

For a simple normal crossing divisor we get a simpler upper bound.

Lemma 2.1.20 ([BD25, Corollary 3.14]). *Let $Y \subset X$ be a simple normal crossing divisor, with irreducible components $Y_1, \ldots, Y_m$. Then,*

$$g(X, Y) \leq \sum_{I \subset [m]} g(Y_I).$$

Computing the Combinatorial Rank

In this paragraph we collect some tools for the computation of the combinatorial rank. Let X be an n-dimensional variety.

There is a natural upper bound for the combinatorial rank of any hypersurface in projective space.

Lemma 2.1.21 ([BD25, Proposition 1.13]). *Let Y be a hypersurface in $\mathbb{P}^n$ of degree d. Then, the twisted canonical bundle $\Omega^n_{\mathbb{P}^n}(Y)$ is isomorphic to the twisted structure sheaf $\mathcal{O}_{\mathbb{P}^n}(d-n-1)$, and in particular*

$$\mathrm{cr}(\mathbb{P}^n, Y) \leq \binom{d-1}{n}.$$

Equality holds if Y is a simple normal crossing divisor.

In general the combinatorial rank can be bounded from above recursively.

Lemma 2.1.22 ([BD25, Corollary 4.11]). *Let $Y \subset X$ be a hypersurface with irreducible components $Y_1, \ldots, Y_m$. Set $Z = Z_1 \cup \ldots \cup Z_m$ with $Z_i = Y_i \cap \bigcup_{j \neq i} Y_j$. Then,*

$$\mathrm{cr}(X, Y) \leq \mathrm{cr}(X) + \sum_{i=1}^{m} \mathrm{cr}(Y_i, Z_i).$$

Equality holds if $\mathrm{gr}_0^W H^{n-1}(X, Z) = H^{n-1}(X, Z)^{0,0} = 0$.

Lastly, let X be smooth, and let $Y \subset X$ be simple normal crossing divisor, with irreducible components $Y_1, \ldots, Y_m$. For any $I \subset [m]$ write Y_I for the intersection $\bigcap_{i \in I} Y_i$. We associate a CW complex to Y, called the *boundary complex* $\Delta_{CW}(Y)$, as follows: The 0-dimensional cells are the irreducible components Y_i. Then, for any irreducible component of Y_I we get a $(|I|-1)$-dimensional cell σ_I with vertices labeled by I. We glue σ_I into $\Delta_{CW}(Y)$ by identifying the facet $I \setminus \{i\}$ with the $|I|$-dimensional cell in $Y_{I \setminus i}$ containing the original irreducible component. Note that despite the slightly different definition of the dual boundary complex, we can record the following result, since by the classical result the homology of a delta complex and a CW complex are isomorphic, see [Hat02, Theorem 2.35].

Then there is a classical identification of the top weight part of the middle cohomology of $X \setminus Y$, originally due to Deligne [Del74], see also [Hac08, Theorem 3.1].

Theorem 2.1.23. *Let $Y \subset X$ as above, and denote the boundary complex by $\Delta_{CW}(Y)$. Then,*

$$\widetilde{H}_{\bullet-1}(\Delta_{CW}(Y)) \cong \mathrm{gr}_{2n}^{W} H^{2n-\bullet}(X \setminus Y).$$

Andreotti-Frankel's theorem 1.3.32 has an important direct consequence.

Corollary 2.1.24. *In the situation from above, suppose additionally that $X \setminus Y$ is affine. Then,*

$$\widetilde{H}_k(\Delta_{CW}(Y)) = \begin{cases} \mathrm{gr}_{2n}^{W} H^{2n-k}(X \setminus Y) & , k = n-1 \\ 0 & , \textit{else}. \end{cases}$$

Proof. We apply Andreotti-Frankel's theorem 1.3.32 to $H_{2n-k}(X \setminus Y) = H^{2n-k}(X \setminus Y)^{\vee}$ after taking duals, see Theorem 1.3.20. □

This allows us to compute the combinatorial rank as follows, which is essentially a reformulation of [BD25, Proposition 4.4].

Corollary 2.1.25. *Let X and Y as above, and denote by χ the topological Euler characteristic. Suppose that Y_I is smooth for all $I \subset [m]$. Then,*

$$\mathrm{cr}(X, Y) = \dim \widetilde{H}_{n-1}(\Delta_{CW}(Y)) = 1 + (-1)^{n-1}\chi(\Delta_{CW}(Y)).$$

Proof. The first equality follows from the definition of the combinatorial rank and Theorem 2.1.23. The second equality follows from the definition of the Euler characteristic in Lemma 1.3.31, and Corollary 2.1.24. □

2.2 Combinatorial Rank in Applications

One of the main goals of this section is to introduce more tools to compute combinatorial ranks for pairs that come from applications. In particular, we will compute the combinatorial rank via tropical varieties, see Proposition 2.2.12, and for the special case where we have a (product) bundle structure on $X \setminus Y$, see Proposition 2.2.19.

These new tools that we introduce rely on the interplay between polyhedral, tropical, and toric geometry, so before we start we go on a brief detour introducing the essential notions. For a broader introduction on the connections between these subjects, see [MS15].

Polyhedral Geometry

First, we need a few basic notions from discrete geometry. A *polyhedral cone* σ in $\mathbb{R}^n$ is a subset

$$\sigma = \mathrm{Cone}(v_1, \ldots, v_r) = \{\lambda_1 v_1 + \cdots + \lambda_r v_r \mid \lambda_i \in \mathbb{R}_{\geq 0}\} \subset \mathbb{R}^n.$$

The v_i are called *vertices* of σ. The *dimension* $\dim \sigma$ of a cone is defined as

$$\dim \sigma = \dim \mathrm{span}_{\mathbb{R}}(v_1, \ldots, v_r),$$

where the latter is the dimension of the real vector space. A cone is *rational* if all its vertices $v_i \in \mathbb{R}^n$ have rational coordinates. A *face* of a polyhedral cone σ is any subset of the form

$$\tau = \sigma \cap \{x \in \mathbb{R}^n \mid u(x) = 0\},$$

where u is a linear form on $\mathbb{R}^n$ that is $u \in (\mathbb{R}^n)^\vee$ satisfying $u(x) \geq 0$ for all $x \in \sigma$. Then, a *polyhedral fan* is a finite collection Σ of polyhedral cones $\sigma \subset \mathbb{R}^n$ such that every face of a cone in Σ belongs to Σ, and the intersection of any two cones in Σ is a face of each. A polyhedral fan has *weights* if each cone comes associated with a non-negative integer $\omega \in \mathbb{Z}_{\geq 0}$. The *$f$-vector* $f \in \mathbb{Z}^n$ of Σ, or more generally an n-dimensional CW-complex, see Definition 1.3.6, is defined by setting f_k to be the number of faces of dimension k in Σ counting with weights. The *support* $|\Sigma| \subset \mathbb{R}^n$ of a fan is defined as the set of all vectors v contained in Σ. We say a fan $\widetilde{\Sigma}$ *refines* another fan Σ if $|\Sigma| = |\widetilde{\Sigma}|$, and for any cone $\sigma \in \Sigma$ there exists a cone $\widetilde{\sigma} \in \widetilde{\Sigma}$ such that $\widetilde{\sigma} \subset \sigma$. Lastly, a fan Σ has *lineality space* $L \cong \mathbb{R}^\ell$ if L is the biggest linear space contained in all cones $\sigma \subset \Sigma$. Then, for $\pi : \mathbb{R}^n \to \mathbb{R}^n/L$, the quotient Σ/L defines a fan with cones $\overline{\sigma} = \pi(\sigma)$ for every $\sigma \in \Sigma$.

Definition 2.2.1. Let $\Sigma \subset \mathbb{R}^n$ be a full-dimensional fan with weights. Then, we define the (spherical) *link at the origin* $\operatorname{Lk}\Sigma$ as the CW complex obtained by intersecting Σ with a top-dimensional Euclidean sphere $S^{n-1} \subset \mathbb{R}^n$. By convention, a weight m cone of Σ gives rise to m parallel simplices in the CW complex, whose boundaries are identified.

For fans with lineality space we have the following result, which we believe is well known, but could not find in the literature.

Lemma 2.2.2. *Let $\Sigma \subset \mathbb{R}^n$ be a fan with lineality space $L \cong \mathbb{R}^\ell$. Then, the support $|\Sigma|$ is homeomorphic to the product $L \times |\overline{\Sigma}|$, where $\overline{\Sigma} \subset \mathbb{R}^n/L \cong \mathbb{R}^{n-\ell}$ is the quotient fan Σ/L.*

Proof. We want to show that $|\Sigma| \cong L \times |\overline{\Sigma}|$ are homeomorphic. Choose a decomposition $\mathbb{R}^n \cong L \oplus W$, with the quotient maps $\pi_L : \mathbb{R}^n \to W \cong \mathbb{R}^n/L$ and $\pi_W : \mathbb{R}^n \to L \cong \mathbb{R}^n/W$. Then by construction we can identify $\pi_L(|\Sigma|)$ with the support of the quotient fan $|\overline{\Sigma}|$. Next, consider the map

$$\pi : \mathbb{R}^n \longrightarrow L \times |\overline{\Sigma}|, \quad u \mapsto (\pi_W(u), \pi_L(u)).$$

Then, it suffices to show that the map is a homeomorphism. First, this map is well-defined and continuous because π_L and π_W are linear maps between finite-dimensional vector spaces, and there is a unique way of writing $u = \ell + w$ with $\ell \in L$ and $w \in W$ for any $u \in \mathbb{R}^n \cong L \oplus W$. Next, note that π_W and π_L restrict to isomorphisms on $L \subset \mathbb{R}^n$ and $W \subset \mathbb{R}^n$, respectively. Hence, π has a continuous inverse by sending $(\ell, [w]) \mapsto \ell + w$. (This is well-defined as $W \cap L = \{0\}$, that is if $[w] = [w']$ then $w - w' \in L \cap W = \{0\}$.) This establishes the homeomorphism $|\Sigma| \cong L \times |\overline{\Sigma}|$. □

For fans with lineality space the homology of the link of the original fan and that of the quotient fan are related.

Corollary 2.2.3. *Let $\Sigma \subset \mathbb{R}^n$ be a top-dimensions fan with lineality space $L \cong \mathbb{R}^\ell$, and let $\overline{\Sigma} \subset \mathbb{R}^n/L$ be the quotient fan. Suppose that $\ell > 0$ and $\operatorname{Lk}\overline{\Sigma} \neq \varnothing$. Then,*

$$\widetilde{H}_n(\operatorname{Lk}\Sigma) \cong \widetilde{H}_{n-\ell-1}(\operatorname{Lk}\overline{\Sigma}).$$

Proof. We start by choosing a decomposition $\mathbb{R}^n \cong L \oplus W$. Then we get a homeomorphism $S^{n-1} \subset L \oplus W$ to the join $S^{\ell-1} * S^{n-\ell}$, where the former is a sphere in L and the latter in W, see e.g. [Hat02, p. 10]. Hence, by Lemma 2.2.2 we get $\operatorname{Lk}\Sigma \cong S^{\ell-1} * \operatorname{Lk}\overline{\Sigma}$. Then, by [Mil56, Lemma 2.2] we get

$$H_n(\operatorname{Lk}\Sigma) \cong H_n(S^{\ell-1} * \operatorname{Lk}\overline{\Sigma}) \cong \bigoplus_{p+q=n} H_p(S^{\ell-1}) \otimes_{\mathbb{Q}} H_q(\operatorname{Lk}\overline{\Sigma}) \cong H_{n-\ell-1}(\operatorname{Lk}\overline{\Sigma}),$$

where we used that $H_p(S^{\ell-1}) = 0$ for $p \neq \ell - 1$, see Example 1.3.5.

□

Tropical Geometry

Next, we turn to tropical geometry. For a broader introduction see [MS15, Chapter 3]. Let f be a polynomial in the Laurent polynomial ring $R^{\pm 1} = \mathbb{C}[x_1^{\pm 1}, \ldots, x_N^{\pm 1}]$. Write $f = \sum_u c_u x^u$ with $c_u \in \mathbb{C}$ in multi-index notation, that is $u \in \mathbb{Z}^N$ and $x^u = \prod_{i=1}^N x_i^{u_i}$. For a fixed *weight* $w \in \mathbb{Z}^N$, we define its *initial form* $\operatorname{in}_w f$ as

$$\operatorname{in}_w(f) = \sum_{\{u \in \mathbb{Z}^N \mid w\cdot u = \min_{c_v \neq 0}(w\cdot v)\}} c_u x^u,$$

where the sum is over the set of indices which have minimal inner product with the chosen weight w. Now, for an ideal I in the Laurent polynomial ring $R^{\pm 1}$, we define the *initial ideal* $\operatorname{in}_w I \subset R$, as

$$\operatorname{in}_w I = \langle \operatorname{in}_w f \mid f \in I \rangle \subset R^{\pm 1}.$$

Definition 2.2.4. Let $U \subset (\mathbb{C}^\times)^N$ be a very affine variety with defining ideal $I \subset R^{\pm 1}$. Then, the *tropicalization* of U, denoted $\operatorname{trop} U$, is defined as

$$\operatorname{trop} U = \{w \in \mathbb{R}^N \mid \operatorname{in}_w I \neq \langle 1 \rangle\} \subset \mathbb{R}^N.$$

We also refer to $\operatorname{trop} U$ as a *tropical variety*.

One of the central results in tropical geometry is then that tropical varieties have a polyhedral structure. In our case, with the choice of field $\mathbb{C}$, we get the following stronger statement.

Theorem 2.2.5 ([MS15, Corollary 3.5.5])**.** *Let $U \subset (\mathbb{C}^\times)^N$ be an irreducible very affine variety of dimension n. Then, $\operatorname{trop} U \subset \mathbb{R}^N$ is an n-dimensional fan, which is connected in codimension 1, and has $\mathbb{Z}_{\geq 0}$-weights on its maximal cones satisfying a balancing condition.*

Remark 2.2.6. In the preceding theorem, the fan structure of $\operatorname{trop} U$ is given by the Gröbner fan of the ideal I defining U. We will implicitly assume this when we speak of the fan $\operatorname{trop} U$ going forward. For more details see [MS15, Chapter 2].

Lastly, in general, tropical geometry depends on a choice of field and valuation. We chose to focus solely on the case where the underlying field is $\mathbb{C}$ and the valuation is trivial for the sake of simplicity.

Toric Geometry

Our main result below then computes the combinatorial rank via tropicalization. However, before stating it we introduce, very tersely, some notation from toric geometry. Let $\Sigma \subset \mathbb{R}^n$ be a full-dimensional fan with rational ray generators and trivial lineality space, and $N \cong \mathbb{Z}^n$ a lattice with $N_{\mathbb{R}} = N \otimes_{\mathbb{Z}} \mathbb{R} \cong \mathbb{R}^n$. The fan Σ determines an n-dimensional *toric variety* X_Σ via its cones $\sigma \in \Sigma$ as follows: Let $M = N^\vee \cong \mathbb{Z}^n$ be the dual lattice, and $M_{\mathbb{R}} = M \otimes_{\mathbb{Z}} \mathbb{R}$. For each $\sigma \in \Sigma$ we define the *dual cone*

$$\sigma^\vee = \{m \in M_{\mathbb{R}} \mid \langle m, u\rangle \geq 0 \text{ for all } u \in \sigma\}$$

We choose a set of rational (semi-group) generators $u_1, \dots, u_r$ of $\sigma^\vee \cap M$, which is a subset of the lattice points in M. Then, define $\varphi_\sigma : (\mathbb{C}^\times)^n \to \mathbb{C}^r$, which sends

$$t = (t_1, \dots, t_n) \mapsto (t^{u_1}, \dots, t^{u_r}), \tag{2.2.1}$$

where $t^{u_j} = \prod_{i=1}^n t_i^{(u_j)_i}$ and the t_i are torus coordinates corresponding to the fixed basis of M. The Zariski closure of the image of φ_σ is an n-dimensional affine variety denoted by U_σ. Then, X_Σ is obtained, by gluing these affine patches U_σ together according to the face structure of the fan Σ. More precisely, for cones of Σ with $\sigma' \cap \sigma = \tau$, choose (semi-group) generators $u_1, \dots, u_r$ for $\tau^\vee \cap M$. Then, the coordinates of $U_\sigma \cap U_{\sigma'}$ can be identified via the subset of the u_i corresponding to the generators of $\sigma^\vee \cap M$ and $(\sigma')^\vee \cap M$.

Any cone $\sigma \in \Sigma$ also defines a *torus orbit* $O_\sigma \cong (\mathbb{C}^\times)^c$, where c is the codimension of σ: Consider the c-dimensional subvector space of $\mathbb{R}^n = M_{\mathbb{R}}$ given as

$$\sigma^\perp = \{m \in M_{\mathbb{R}} \mid \langle u, m\rangle = 0 \text{ for all } u \in \sigma\}.$$

Take (group) generators $u_1, \dots, u_r$ of $\sigma^\perp \cap M$, then the orbit $O_\sigma \subset X_\Sigma$ is isomorphic to the image of

$$(\mathbb{C}^\times)^n \to (\mathbb{C}^\times)^r, \quad t = (t_1, \dots, t_n) \mapsto (t^{u_1}, \dots, t^{u_r}).$$

We say the *algebraic torus* T^n of X_Σ is $T^n \cong (\mathbb{C}^\times)^n$.

We record an important result under the name of cone-orbit correspondence.

Theorem 2.2.7. *[CLS11, Theorem 3.2.6] Let $\Sigma \subset \mathbb{R}^n$ be a fan without lineality space. Then any c-dimensional torus orbit $O_\sigma \subset X_\Sigma$ corresponds to a cone $\sigma \in \Sigma$ of codimension c. This bijection is called* cone-orbit correspondence*.*

Any subvariety $Y \subset (\mathbb{C}^\times)^n$, called *very affine variety*, naturally embeds into X_Σ via the embedding $Y \hookrightarrow (\mathbb{C}^\times)^n \cong U_O \hookrightarrow X_\Sigma$, where U_O is the affine subvariety of X_Σ given as in (2.2.1). Denote its Zariski closure by $\overline{Y} \subset X_\Sigma$. Explicitly, if Y is of dimension d and is

parameterized via

$$\varphi : (\mathbb{C}^\times)^d \mapsto (\mathbb{C}^\times)^n, \quad x \mapsto (x^{a_1}, \dots, x^{a_n}),$$

then the embedding $Y \hookrightarrow X_\Sigma$ is obtained by composition with the map in (2.2.1) for $\sigma = O$ the origin of Σ:

$$x \mapsto ((x^a)^{u_1}, \dots, (x^a)^{u_r}),$$

where $x^a = \varphi_O(x)$ and r is the (not unique) number of (semi-group) generators of $\sigma^\vee \cap M = M$.

We can now make the central definition that will allow us to relate the combinatorial rank to the homology of a tropical variety.

Definition 2.2.8. Let $U \subset T^n$ be a very affine variety, and $\Sigma \subset \mathbb{R}^n$ a fan with $|\Sigma| = \operatorname{trop} U$, so that $U \subset T^n \subset X_\Sigma$ is naturally embedded. We say U is *schön* if $\overline{U} \cap O_\sigma \subset X_\Sigma$ is smooth for all torus orbits $O_\sigma \subset X_\Sigma$. If additionally all positive-dimensional strata $\overline{U} \cap O_\sigma$ are also connected, we say U is *sehr schön*.

Remark 2.2.9. Note that in the literature, for example in [MS15, Definition 6.4.19], schön only requires that the orbits are smooth and not necessarily connected. Hence, sehr schön is a natural generalization of this notion. Also note that over the complex numbers smooth varieties are connected if and only if they are irreducible, hence we will use these notions interchangeably. Moreover, it should also be noted that this condition is weaker than simple normal crossing, as we shall see in Example 3.3.10.

In the case U is a hypersurface this condition can be made more explicit via the Newton polytope, see e.g. the discussion in [Tev05, pp.1-2].

Lemma 2.2.10. *Let $U \subset (\mathbb{C}^\times)^n$ be a very affine hypersurface with defining polynomial $q = \sum_u c_u x^u$ in the Laurent polynomial ring $R^\pm$. The Newton polytope $\mathcal{N}_U \subset \mathbb{R}^n$ of U is given as the convex hull of all u with $c_u \neq 0$ in q. For any face F of $\mathcal{N}_U$ write $q_F \in R^\pm$ for the Laurent polynomial given by those terms $c_u x^u$ in q, such that $u \in F$. Then, U is schön if and only if*

$$dq_F(x) = \left(\frac{\partial q_F}{\partial x_1}, \dots, \frac{\partial q_F}{\partial x_n}\right)(x) \neq 0,$$

for all $x \in \mathbb{V}(q_F)$ and for all faces $F \in \mathcal{N}_U$. In this case, we say U is non-degenerate with respect to its Newton polytope.

Corollary 2.2.11. *Let $U \subset (\mathbb{C}^\times)^n$ be a very affine hypersurface which is non-degenerate with respect to its Newton polytope. Then, U is sehr schön.*

Proof. Let Σ be the inner normal fan of the Newton polytope $\mathcal{N}$ of U, and let X_Σ be the associated toric variety. Denote by $\overline{U}$ the Zariski closure of U in X_Σ. By Lemma 2.2.10 it remains to show that all positive-dimensional strata $\overline{U} \cap O_\sigma$ with $\sigma \in \Sigma$ are connected. First, recall that the number of connected components of $\overline{U} \cap O_\sigma$ is measured by the dimension of

$H^0(\overline{U} \cap O_\sigma)$, see Definition 1.3.21. Next, note that the stratum $\overline{U} \cap O_\sigma$ is a smooth hypersurface in $O_\sigma \cong (\mathbb{C}^\times)^c$, where c is the codimension of $\sigma \in \Sigma$. Moreover, by [Mik03, Lemma 2.20] this very affine hypersurface is non-degenerate with respect to the restricted Newton polytope $\mathcal{N}|_{F_\sigma}$ where F_σ is the face of the Newton polytope of U corresponding to the cone σ. It then follows from [DK87, Remark 3.10], that

$$H^0(\overline{U} \cap O_\sigma) \cong H^0((\mathbb{C}^\times)^c) \cong \mathbb{Q},$$

for all $c \geq 2$. Since $c = \operatorname{codim}_\Sigma \sigma = 1$ corresponds to zero-dimensional strata $\overline{U} \cap O_\sigma$, which we allowed to be disconnected, we have just shown that U is sehr schön. □

Combinatorial Rank

We can now present one of our main results that computes the combinatorial rank via tropical geometry. This is essentially an application of [Hac08, Theorem 2.5] in our setting. Let X be a projective variety of dimension n, and Y a subvariety such that $X \setminus Y$ is smooth.

Proposition 2.2.12. *Let $X \setminus Y \subset (\mathbb{C}^\times)^N$ be a smooth and sehr schön very affine variety of dimension n. Then, the combinatorial rank of (X, Y) is determined by the tropicalization of $X \setminus Y$. More precisely,*

$$\operatorname{cr}(X, Y) = \dim \widetilde{H}_{n-\ell-1}\left(\operatorname{Lk} \frac{\operatorname{trop}(X \setminus Y)}{L}\right) = (-1)^{n-\ell} \sum_{k \geq 0} (-1)^k f_k,$$

where L is the ℓ-dimensional lineality space of $\operatorname{trop}(X \setminus Y)$, and f_k are the entries of the f-vector of the quotient fan $\operatorname{trop}(X \setminus Y)/L$ counted with weights.

Proof. We essentially follow the proof of [Hac08, Theorem 2.5]. So let $X \setminus Y$ be sehr schön. Assume that $\operatorname{trop}(X \setminus Y)$ has a trivial lineality space L. Let $\Sigma = \operatorname{trop}(X \setminus Y)$ have the fan structure given by the Gröbner fan as in Remark 2.2.6, and let Z_Σ be the associated toric variety, hence $X \setminus Y \subset (\mathbb{C}^\times)^N \subset Z_\Sigma$ as by our preceding discussion. Consider a toric resolution of singularities of Z_Σ given by a refinement $\widetilde{\Sigma}$ of $\operatorname{trop}(X \setminus Y)$ i.e. $\pi : \widetilde{Z} \to Z_\Sigma$, where $\widetilde{Z} = Z_{\widetilde{\Sigma}}$ is smooth. Set $\widetilde{U} = \overline{\pi^{-1}(X \setminus Y)} \subset \widetilde{Z}$. Then, $E = \widetilde{U} \setminus \pi^{-1}(X \setminus Y)$ is a simple normal crossing divisor in $\widetilde{U}$, by the proof of [Hac08, Theorem 2.5]. Note that we also get an isomorphism $\widetilde{U} \setminus E \cong X \setminus Y$.

Next, we want to establish a homeomorphism between the boundary complex $\Delta_{CW}(E)$ and the link of $\operatorname{trop}(X \setminus Y)$ where $X \setminus Y \subset Z_\Sigma$, as in our discussion above. By the sehr schön assumption on $X \setminus Y$, we know that each positive-dimensional stratum $\overline{(X \setminus Y)} \cap \mathcal{O}_\sigma$ for $\sigma \in \Sigma$ is smooth, irreducible, and thus connected. Therefore, for any cell ρ_I of codimension $k - 1 \leq n$ in the boundary complex $\Delta_{CW}(E)$ is precisely an intersection E_I with $|I| = k$, which correspond to the smooth and irreducible strata $\overline{(X \setminus Y)} \cap O_\sigma \subset Z_\Sigma$ of codimension k. On the other hand the cells ρ_i correspond bijectively to the k-dimensional cones $\sigma \in \operatorname{trop}(X \setminus Y)$. Lastly, for $k = n-1$, note that $\overline{(X \setminus Y)} \cap O_\sigma$ is zero-dimensional and smooth again by the (sehr) schön assumption, hence it is a set of $m > 0$ distinct points. By [ST07, Lemma 3.2], this

number m coincides with the tropical weight on the $(n-1)$-dimensional cone $\sigma \in \operatorname{trop}(X \setminus Y)$. Hence, $\Delta_{CW}(E)$ has m copies $(n-1)$-dimensional cells, with their boundaries identified with each other. Then, it suffices to note that we defined the link $\operatorname{Lk}(\operatorname{trop}(X \setminus Y))$, by gluing in precisely m copies of maximal $(n-1)$-dimensional cells for any cone $\sigma \in \operatorname{trop}(X \setminus Y)$, where m was the tropical weight of that maximal cone. Therefore, $\Delta_{CW}(E)$ is homeomorphic to $\operatorname{Lk}(\operatorname{trop}(X \setminus Y))$, so their homology groups are isomorphic. Hence, by Theorem 2.1.23 and the isomorphism $X \setminus Y \cong \widetilde{U} \setminus E$ we get

$$\operatorname{gr}_{2n}^{W} H^n(X \setminus Y) \cong \operatorname{gr}_{2n}^{W} H^n(\widetilde{U} \setminus E) \cong \widetilde{H}_{n-1}(\Delta_{CW}(E)) \cong \widetilde{H}_{n-1}\left(\operatorname{Lk} \operatorname{trop}(X \setminus Y)\right).$$

Now suppose we started with a fan $\operatorname{trop}(X \setminus Y)$ with non-trivial lineality space. Then, by Corollary 2.2.3

$$\widetilde{H}_{n-1}\left(\operatorname{Lk} \operatorname{trop}(X \setminus Y)\right) \cong \widetilde{H}_{n-\ell-1}\left(\operatorname{Lk} \frac{\operatorname{trop}(X \setminus Y)}{L}\right),$$

Hence, we were allowed to assume that $\operatorname{trop}(X \setminus Y)$ had only a trivial lineality space. Moreover, by combining Corollaries 2.1.24, and 2.1.25 we get

$$\operatorname{cr}(X, Y) = \dim \widetilde{H}_{n-\ell-1}\left(\operatorname{Lk} \frac{\operatorname{trop}(X \setminus Y)}{L}\right) = (-1)^{n-\ell} \sum_{k \geq 0} (-1)^k f_k,$$

and f_k are the entries of the f-vector of $\operatorname{trop}(X \setminus Y)/L$ counted again with multiplicity. This proves the claim. □

Remark 2.2.13. This result is also very closely related to [Hac08, Corollary 3.3], except that we allow for the zero-dimensional strata $\overline{X \setminus Y} \cap \mathcal{O}_\sigma$ to be a union of points. See the definition of sehr schön in 2.2.8. Also note that this result is directly related to that in Corollary 2.1.25 i.e [BD25, Proposition 4.4].

In case $X \setminus Y \subset \mathbb{C}^{n+1}$ is a very affine hypersurface, Lemmas 2.2.10 and 2.2.11 have an immediate consequence.

Corollary 2.2.14. *Let $X \setminus Y \subset (\mathbb{C}^\times)^{n+1}$ be a smooth very affine hypersurface of dimension n, which is non-degenerate with respect to its Newton polytope $\mathcal{N}$. Then, the combinatorial rank is given as*

$$\operatorname{cr}(X, Y) = \chi\left(\operatorname{Lk} \frac{\operatorname{trop}(X \setminus Y)}{L}\right) = (-1)^n \sum_{k \geq 0} (-1)^k f_k,$$

where f_k are the entries of the f-vector of $\operatorname{trop}(X \setminus Y)$ counted with weights.

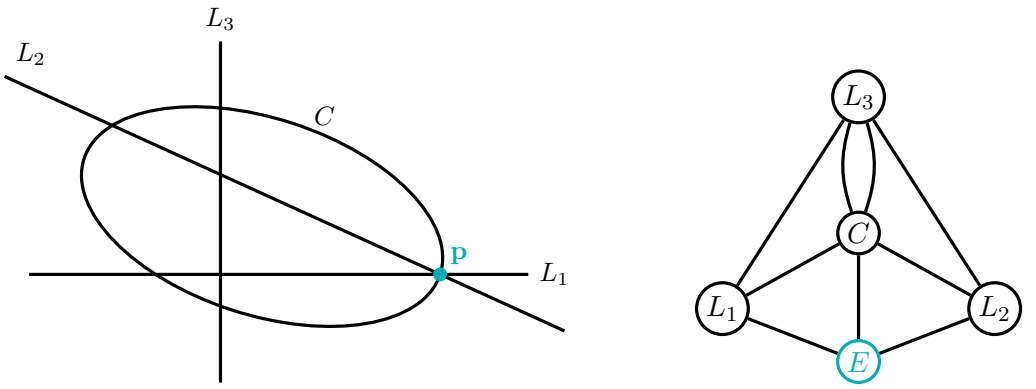
Proof. This follows from combining Corollary 2.2.11 and Proposition 2.2.12. □

Example 2.2.15. Note that Proposition 2.2.12 and Corollary 2.2.14 also apply to non-normal crossing situations, recall from Remark 2.2.9 that the schönness property was weaker than simple normal crossing. This will also demonstrate the importance of weights on the top-dimensional cones in the tropicalization.

Here is a concrete example. Let $\mathbb{P}^2$ be the projective plane with coordinates x_0, x_1 and x_2. Consider a generic smooth conic C with defining equation f, two lines L_1 and L_2 meeting C in the common point $p \in C$, and a third generic line L_3. Thus, $D = C \cup L_1 \cup L_2 \cup L_3$ as a divisor in $\mathbb{P}^2$ is not normal crossing in p, see Figure 2.2.1a. However, $U = \mathbb{P}^2 \setminus D$ is a smooth very affine hypersurface defined by $f' = \sum_u c_u x^u \in \mathbb{C}[x_0^{\pm 1}, x_1^{\pm 1}, x_2^{\pm 1}]$. Since C was generic, $dq_F(x) \neq 0$ for all x in the linear space supporting the face F of the Newton polytope $\mathcal{N}$ of f'. Hence, U is non-degenerate with respect to $\mathcal{N}$ and Corollary 2.2.14 applies. Now, $\operatorname{trop} U$ is a 2-dimensional fan, its combinatorial structure is encoded in the graph displayed in Figure 2.2.1b. This graph has 5 vertices and 7 single edges and one double edge. The double edge between L_3 and C makes a 2-dimensional cone with weight 2 in $\operatorname{trop} U$. Hence,

$$\operatorname{cr}(\mathbb{P}^2, D) = \chi(\operatorname{Lk} \operatorname{trop} U) = 1 - 5 + (7 + 2) = 5.$$

Finally, note that this graph is also the boundary complex $\Delta_{CW}(D)$, see Corollary 2.1.25, obtained from D under a blow-up in the non-simple normal crossing point p, with exceptional divisor E. This explains the choice of labels in Figure 2.2.1b. ◇



(a) The non-simple normal crossing divisor D.

(b) The combinatorics of $\operatorname{trop}(\mathbb{P}^2 \setminus D)$.

Figure 2.2.1: A non-simple normal crossing divisor and the tropicalization of its complement.

We record an important consequence for the Grassmannian of lines.

Corollary 2.2.16. *Let $D \subset \operatorname{Gr}(2,n)$ be the subvariety in $\operatorname{Gr}(2,n)$ defined by the union of all vanishing loci of the Plücker coordinates, such that $\mathring{\operatorname{Gr}}(2,n) = \operatorname{Gr}(2,n) \setminus D$ is the open Grassmannian. Let $M_{0,n}$ be the $(n-3)$-dimensional moduli space of n marked points on $\mathbb{P}^1$. Then the combinatorial rank of $(\operatorname{Gr}(2,n), D)$ and $M_{0,n}$ is given as*

$$\operatorname{cr}(\operatorname{Gr}(2,n), D) = \operatorname{cr}(M_{0,n}) = \chi(\operatorname{trop}(M_{0,n})) = (n-2)!\,.$$

Proof. For the first equality, it suffices to show that $\mathring{\operatorname{Gr}}(2,n)$ is homeomorphic (even isomorphic as varieties) to $M_{0,n} \times (\mathbb{C}^\times)^{n-1}$. First notice that $\operatorname{GL}_2(\mathbb{C})$ acts three-transitively on $\mathring{\operatorname{Gr}}(2,n)$ by multiplication from the left. Then, additionally there is a free and transitive torus action $(\mathbb{C}^\times)^n/\mathbb{C}^\times \cong (\mathbb{C}^\times)^{n-1}$ acting by multiplication on the right. Equivalently, to projective global scaling, this torus acts by column scaling of a matrix representing a point

in $\mathrm{Gr}(\mathring{2}, n)$. Hence, we can write any point in $\mathring{\mathrm{Gr}}(2, n)$ as

$$M = \begin{bmatrix} 1 & 0 & 1 & 1 & \dots & 1 \\ 0 & 1 & 1 & x_1 & \dots & x_{n-3} \end{bmatrix}.$$

Then, the isomorphism is given via the map $\phi : \mathring{\mathrm{Gr}}(2, n)/(\mathbb{C}^\times)^{n-1} \to M_{0,n}$ sending

$$M \mapsto (x_1, \dots, x_{n-3}).$$

Hence, $M_{0,n}$ is isomorphic to an affine hyperplane arrangement in $\mathbb{C}^{n-3}$ with the hyperplanes given by all 2×2 determinants of the matrix above. Denote this arrangement of hyperplanes by $\mathcal{A} \subset \mathbb{C}^{n-3}$, and the arrangement complement by $U = \mathbb{C}^{n-3} \setminus \mathcal{A}$. Then, we have shown that $\mathring{\mathrm{Gr}}(2, n) \cong (\mathbb{C}^\times)^{n-3} \times U$, hence Proposition 2.2.12 applies. This shows the first three equalities in

$$\mathrm{cr}(\mathrm{Gr}(2, n), D) = \mathrm{cr}(M_{0,n}) = \chi(\mathrm{trop}(M_{0,n})) = (n-2)!\,.$$

For the last equality, observe that the number of bounded regions of $M_{0,n}$ with a generic additional hyperplane L added equals $(n-2)!$ by [GZ83, Theorem 3.1]. This number is the combinatorial rank of $(\mathbb{P}^{n-3}, \mathcal{A} \cup L)$ by [BD25, Proposition 6.1]. □

Remark 2.2.17. The computation of the dimension of $\mathrm{gr}_0^W H_c^{n-3}(M_{0,n})$ appears in a different guise in [Eur24, Corollary 1.7]. Moreover, it is well known in Physics how to obtain an explicit basis of the space of logarithmic differential forms, see e.g. [KK89, p. 8].

Example 2.2.18. We demonstrate Corollary 2.2.16 for the case $\mathrm{Gr}(2, 5)$. Let $\mathrm{Gr}(\mathring{2}, 5)$ denote the open Grassmannian obtained by removing the divisor D, given by all Plücker coordinate hyperplane sections, from $\mathrm{Gr}(2, 5)$. Note that irreducible components of D are Schubert hyperplanes and thus singular. However, $\mathrm{Gr}(\mathring{2}, 5)$ is homeomorphic (even isomorphic) to $M_{0,5} \times (\mathbb{C}^\times)^4$, where $M_{0,5}$ is the 2-dimensional moduli space of 5 marked points on $\mathbb{P}^1$. Hence, Corollary 2.2.16 applies despite D being singular, where Corollary 2.1.25 does not apply. So it suffices to find the combinatorial rank of $M_{0,5}$. The tropical moduli space $\mathrm{trop}(M_{0,5})$ is then the 2-dimensional fan obtained by taking the cone over the Petersen graph shown in Figure 2.2.2. This graph has 10 vertices and 15 edges, hence the f-vector of $\mathrm{trop}(M_{0,5})$ is $(1, 10, 15) \in \mathbb{Z}^3_{\geq 0}$. Then by [Sch24a] it follows that the compactification of $M_{0,n}$ is schön. Therefore, we compute the combinatorial ranks as in Proposition 2.2.12 and Corollary 2.2.16 as

$$\mathrm{cr}(\mathrm{Gr}(2, 5), D) = \mathrm{cr}(M_{0,5}) = 1 - 10 + 15 = (5-2)! = 6.$$

◇

Combinatorial Rank for Bundles

Lastly, in this paragraph we take a complementary view on the interplay of combinatorial rank and tropicalizations with lineality space, through fiber bundles. In fact, we will only consider the case where a variety is homeomorphic to a product of a torus $(\mathbb{C}^\times)^\ell$ and a

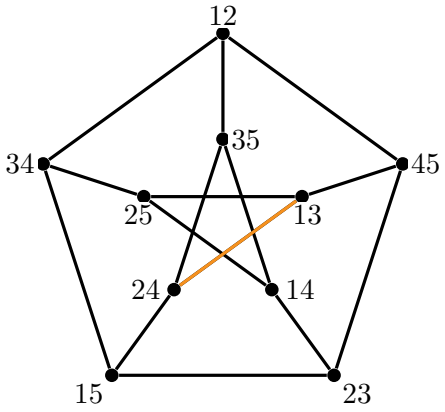


Figure 2.2.2: The tropical Grassmannian $\mathrm{trop}(\mathring{\mathrm{Gr}}(2,5))/(\mathbb{C}^\times)^4 \cong \mathrm{trop}(M_{0,5})$.

smooth base space. We have the following result, which is essentially an application of the Künneth formula, see Theorem 2.1.10. Recall that for an open quasi-projective variety B, we set $\mathrm{cr}(B) = \mathrm{cr}(\widetilde{B}, D_B)$ where $(\widetilde{B}, D_B)$ is any simple normal crossing compactification.

Proposition 2.2.19. *Suppose $X \setminus Y$ is isomorphic to $B \times (\mathbb{C}^\times)^\ell$ where B is a smooth quasi-projective variety, then*

$$\mathrm{cr}(X, Y) = \mathrm{cr}(B).$$

Proof. First, recall that $H_c^\bullet(X \setminus Y) \cong H^\bullet(X, Y)$ from Proposition 1.3.24; hence to compute the combinatorial rank of the pair (X, Y) it suffices to consider $\mathrm{gr}_0^W H_c^n(X \setminus Y)$. Since homology is invariant under homeomorphisms, we apply the Künneth formula, see Theorem 2.1.10, for compactly supported cohomology

$$H_c^n(X \setminus Y) \cong H_c^n(B \times (\mathbb{C}^\times)^\ell) \cong \bigoplus_{k=\ell}^{n} H_c^{n-k}(B) \otimes_{\mathbb{Q}} \mathbb{Q}(\ell - k)^{\binom{\ell}{2\ell-k}},$$

where we used that $H_c^k((\mathbb{C}^\times)^\ell) \cong H_c^k((S^1)^\ell) \cong \mathbb{Q}(\ell-k)^{\binom{\ell}{2\ell-k}}$ for $k \geq \ell$, which was computed in Example 2.1.11. In weight zero the only contribution comes from $k = \ell$, hence we obtain

$$\mathrm{gr}_0^W H_c^n(X \setminus Y) \cong \mathrm{gr}_0^W H_c^{n-\ell}(B) \cong \mathrm{gr}_0^W H_c^{\dim B}(B).$$

This shows $\mathrm{cr}(X, Y) = \mathrm{cr}(B)$. □

Example 2.2.20. Let $n \geq 0$, and consider $\mathbb{P}^n$ with the divisor D given by the union of all coordinate hyperplanes. Then, $\mathbb{P}^n \setminus D \cong (\mathbb{C}^\times)^n \cong \mathbb{C}^\times \times (\mathbb{C}^\times)^{n-1}$. Then, the theorem above says that $\mathrm{cr}(\mathbb{P}^n, D) = \mathrm{cr}(\mathbb{C}^\times)^n = \mathrm{cr}(\mathbb{C}^\times \times (\mathbb{C}^\times)^{n-1}) = \mathrm{cr}(\mathbb{C}^\times) = 1$, which is of course in agreement with Example 2.1.11. ◇

If the base space B is a sehr schön very affine variety, we get again a corollary of Proposition 2.2.12.

Corollary 2.2.21. *In the situation from above. Suppose additionally that B is very affine and*

sehr schön, then

$$\operatorname{cr}(X,Y) = \operatorname{cr}(B) = \dim \widetilde{H}_{\dim B-1}(\operatorname{Lk} \operatorname{trop} B) = (-1)^{\dim B-1} \sum_{k\geq 0} (-1)^k f_k,$$

where f_k are the entries of the f-vector of the fan $\operatorname{trop} B$ counted with weights.

Remark 2.2.22. We believe that one can extend the result of Proposition 2.2.19 further. For example, take fiber bundles $F \to E \to B$ with total space E and general fiber F for $E \to B$ over a simply connected base space B. In this case, the fibers should have trivial monodromy, hence the Leray spectral sequence should imply the same conclusion as in the aforementioned result. Even further we believe the result will extend to principal vector bundles over a path-connected base. We leave these generalizations for future work.

Example 2.2.23. By the proof of Corollary 2.2.16 we know that the open Grassmannian $\operatorname{Gr}(\mathring{2}, 5)$ is homeomorphic to the product (bundle) $M_{0,5} \times (\mathbb{C}^\times)^4$. Hence,

$$\operatorname{cr}(\operatorname{Gr}(2,5), D) = \operatorname{cr}(M_{0,5}) = (5-2)! = 6,$$

as in Example 2.2.18. ◇

2.3 Genus in Applications

In this section we develop tools that allow us to assess the genus even in cases where the pair (X, Y) might have (mild) singularities. We start by establishing several genus vanishing conditions for a not necessarily smooth variety X of dimension n, i.e. for the trivial pair $(X, \varnothing)$. Then, our main result is a criterion for a not necessarily simple normal crossing pair (X, D) to have vanishing genus.

Proposition 2.3.1. *The mixed Hodge numbers $h^{0,q}(X)$ are bounded by the dimension of the middle sheaf cohomology group $H^n(X, \mathcal{O}_X)$. Put in symbols, we have for all non-negative integers q*

$$h^{0,q}(X) = h^{0,q}(H^n(X)) \leq \dim H^n(X, \mathcal{O}_X).$$

In particular, if $H^n(X, \mathcal{O}_X) = 0$, then X has genus zero, that is $g(X) = 0$.

Proof. Set $H = H^n(X)$. We write $F^pW_qH = F^pH \cap W_qH$ for the filtration induced by $F^\bullet$ on W_qH. Recall that

$$\operatorname{gr}_q^W H = \frac{W_qH}{W_{q-1}H}.$$

The Hodge filtration $F^\bullet$ on H induces a filtration on $\operatorname{gr}_q^W H$, given as

$$F^p(\operatorname{gr}_q^W H) = \frac{F^pW_qH + W_{q-1}H}{W_{q-1}H} \cong \frac{F^pW_qH}{F^pW_{q-1}H},$$

where we use the second isomorphism theorem for vector spaces. Therefore,

$$\mathrm{gr}_F^p(\mathrm{gr}_q^W H) = \frac{F^p(\mathrm{gr}_q^W H)}{F^{p+1}(\mathrm{gr}_q^W H)} \cong \frac{\frac{F^p W_q H}{F^p W_{q-1} H}}{\frac{F^{p+1} W_q H + F^p W_{q-1} H}{F^p W_{q-1} H}}.$$

Hence, by the third isomorphism theorem for quotient vector spaces,

$$\mathrm{gr}_F^p(\mathrm{gr}_q^W H) \cong \frac{F^p W_q H}{F^{p+1} W_q H + F^p W_{q-1} H}.$$

Next, note there is a natural surjection

$$\mathrm{gr}_F^p(W_q H) \cong \frac{F^p W_q H}{F^{p+1} W_q H} \twoheadrightarrow \frac{F^p W_q H}{F^{p+1} W_q H + F^p W_{q-1} H} \cong \mathrm{gr}_F^p(\mathrm{gr}_q^W H).$$

Combining this with $W_q H \subset H$ we get

$$\dim \mathrm{gr}_F^p(\mathrm{gr}_q^W H) \leq \dim \mathrm{gr}_F^p(W_q H) \leq \dim \mathrm{gr}_F^p(H).$$

Lastly, setting $p = 0$, we obtain for any non-negative integer q

$$h^{0,q}(H^n(X)) = \dim \mathrm{gr}_F^0 \left(\mathrm{gr}_q^W H^n(X)\right) \leq \dim \mathrm{gr}_F^0 H^n(X) \leq \dim H^n(X, \mathcal{O}_X),$$

where the last inequality follows for example from [Kov11, Eq. (1.6)]. This claim about the genus then follows by symmetry of $h^{p,q}(X) = h^{q,p}(X)$ for all p and q. □

Remark 2.3.2. The reader familiar with the notion of rational or DuBois singularities will recognize that the numbers $h^{0,q}(X)$ in Proposition 2.3.1 sum up to $\dim H^n(X, \mathcal{O}_X)$ if and only if X has this type of singularities, see e.g. [Kov11, Eq. (1.3)]. These situations appear frequently in applications, for example Schubert varieties, which appear as boundary divisors of the amplituhedron, see Lemma 1.4.5, have rational singularities by [Ram87, Theorem 3.9].

Remark 2.3.3. Note that using Serre duality as in [Har77, Corollary 7.7] we get $H^n(X, \mathcal{O}_X)^\vee \cong H^0(X, \overset{\bullet}{\omega}_X)$, where $\overset{\bullet}{\omega}_X$ is the dualizing sheaf of X. Therefore, the result above is a natural generalization of the identification of $g(Y) = \dim H^0(Y, \omega_Y)$ as the geometric genus for a smooth variety Y, where ω_Y is the canonical line bundle, see [BD25, Section 2.1.1].

Example 2.3.4. Let us consider a quadric cone $C \subset \mathbb{P}^3$ which is singular in its vertex v. Since C is a hypersurface in projective space, its dualizing sheaf $\overset{\bullet}{\omega}_C$ is isomorphic to the twisted structure sheaf $\mathcal{O}_C(-2)$, which is the restriction of $\mathcal{O}_{\mathbb{P}^3}(-2)$ to C. (This can be computed via adjunction, see e.g. [Vak25, Proposition 29.4.8].) Since $\mathcal{O}_{\mathbb{P}^3}(-2)$ does not have global sections, neither does $\mathcal{O}_C(-2)$. Hence,

$$H^n(C, \mathcal{O}_C)^\vee \cong H^0(C, \overset{\bullet}{\omega}_C) = H^0(C, \mathcal{O}_C(-2)) = 0,$$

where we used Serre duality, see [Har77, Corollary 7.7]. Thus, $g(C) = 0$, which aligns with the more general fact that all hypersurfaces of small degree have vanishing genus, see [BD25, Theorem 3.23]. ◇

Example 2.3.5. Here is an example where the inequality in Proposition 2.3.1 is strict. Let $C \subset \mathbb{P}^2$ be a nodal plane curve of degree 3. Then, the dualizing sheaf $\overset{\bullet}{\omega}_C$ is explicitly given as

the structure sheaf $\mathcal{O}_C$. By Serre duality as in [Har77, Corollary 7.7] we have $H^n(C,\mathcal{O}_C)^\vee \cong H^0(C,\overset{\bullet}{\omega}_C) = H^0(C,\mathcal{O}_C)$. Since C is a connected curve, $H^0(C,\mathcal{O}_C)$ is isomorphic to $\mathbb{Q}$ by [Sta26, Lemma 33.9.3], and thus Lemma 2.3.1 implies the well-known fact that the (arithmetic) genus of C equals 1. However, by [BD25, Proposition 3.21], we have $g(C)=0$. ◇

In fact, both of the above examples point towards a more general criterion when genera of varieties vanish. This is captured in the main result of this section below, which is a natural generalization of [BD25, Theorem 3.25]. Recall the notions of Cohen—Macaulay varieties and complete intersections from Definition 1.1.18.

Theorem 2.3.6. *Let $X\subset\mathbb{P}^N$ be a projective Cohen-Macaulay variety of dimension n. Suppose that X has an invertible dualizing sheaf $\overset{\bullet}{\omega}_X$ which is isomorphic to the twisted structure sheaf $\mathcal{O}_X(-d)$ for some non-negative integer d. Further, let $Y\subset X$ be a (not necessarily smooth) complete intersection of dimension $n-c$, with its ideal sheaf $\mathcal{I}$ in X generated by c polynomials of degree $d_1,\dots,d_c$. Suppose $\sum_{i=1}^c d_i < d$, then $g(Y)=0$.*

Proof. By Lemma 2.3.1 it suffices to show that $\dim H^{n-c}(Y,\mathcal{O}_Y)=0$. First, note that a local complete intersection is locally cut out by a regular sequence, see Remark 1.1.19, and then [Sta26, Lemma 10.104.2] implies that Y is Cohen-Macaulay as well. Hence, we can apply the adjunction formula, as in [Vak25, Proposition 29.4.8] iteratively, yielding

$$\overset{\bullet}{\omega}_Y \cong \iota^*\overset{\bullet}{\omega}_X \otimes_{\mathcal{O}_Y} \bigwedge^c \left(\mathcal{I}/\mathcal{I}^2\right)^\vee,$$

where $\overset{\bullet}{\omega}_Y$ is the dualizing sheaf of Y, and $\iota: Y\hookrightarrow X$ is the closed embedding. Then, by Lemma 1.2.13 we get

$$\bigwedge^c \left(\mathcal{I}/\mathcal{I}^2\right)^\vee = \bigwedge^c N_{Y/X} \cong \bigotimes_{i=1}^c \mathcal{O}_Y(d_i).$$

As we assumed that $\overset{\bullet}{\omega}_X\cong\mathcal{O}_X(-d)$, we explicitly get

$$\overset{\bullet}{\omega}_Y \cong \mathcal{O}_Y(d_1+\ldots+d_c-d),$$

which is anti-ample if $d_1+\ldots+d_c-d<0$ as for an embedded variety $X\subset\mathbb{P}^N$ the line bundle $\mathcal{O}_X(1)$ is very ample. Moreover, since we assumed that $\overset{\bullet}{\omega}_X$ was invertible so must $\overset{\bullet}{\omega}_Y$ be. Hence, $(\overset{\bullet}{\omega}_Y)^{-1}$ is ample. Then, it follows from Kodaira's vanishing theorem, see e.g. [Har77, Remark 7.15], that its global sections vanish, i.e. $H^0(Y,\overset{\bullet}{\omega}_Y)=0$. As Y is Cohen-Macaulay, we can apply Serre duality as in [Har77, Corollary 7.7] with the locally free sheaf $\mathcal{O}_Y$. We get

$$H^{n-c}(Y,\mathcal{O}_Y)\cong H^0(Y,\mathcal{O}_Y^\vee\otimes_{\mathcal{O}_Y}\overset{\bullet}{\omega}_Y)^\vee \cong H^0(Y,\overset{\bullet}{\omega}_Y)^\vee = 0,$$

where we used that $\mathcal{O}_Y^\vee\cong\mathcal{O}_Y$ and its tensor product with $\overset{\bullet}{\omega}_Y$ remains $\overset{\bullet}{\omega}_Y$. Then,

$$h^{0,q}(Y)=h^{0,q}(H^{n-c}(Y))\le \dim H^{n-c}(Y,\mathcal{O}_Y)=0$$

by Lemma 2.3.1, hence $g(Y)=0$, which had to be shown. □

Remark 2.3.7. Note that the assumptions in Theorem 2.3.6: X being Cohen-Macaulay and having an invertible dualizing sheaf $\overset{\bullet}{\omega}_X$ are equivalent to requiring X to be Gorenstein, see [Sta26, Lemma 48.24.4]. Moreover, the computation of the dualizing sheaf $\overset{\bullet}{\omega}_Y$ is analogous as in [Har77, Theorem 7.11].

We record an important consequence for applications in the Grassmannian.

Corollary 2.3.8. *Let $Y \subset \mathrm{Gr}(k,n)$ be a not necessarily smooth complete intersection in the Grassmannian, whose ideal sheaf $\mathcal{I}$ is generated by forms of degree d_i such that $\sum_i d_i < n$. Then, Y has vanishing genus, that is $g(Y) = 0$.*

Proof. The canonical line bundle ω_G of $G = \mathrm{Gr}(k,n)$ is isomorphic to $\mathcal{O}_G(-n)$ by [EH16, Proposition 5.25]. Moreover, G is Cohen-Macaulay by [Hoc73, Theorem 5]. Therefore, Y and G satisfy the assumptions in Theorem 2.3.6, and the claim follows. □

Example 2.3.9. Let us consider a concrete example stemming from the $m = 2$ amplituhedron and its algebraic boundary, see Proposition 1.4.2. Consider the ambient Grassmannian $\mathrm{Gr}(k,k+2)$ where $k > 1$, and let D be the Chow form of a line $L \subset \mathbb{C}^{k+2}$. So, in particular D is a linear hyperplane section of $\mathrm{Gr}(k,k+2)$. Note that D is singular; more precisely, its singular locus is then given as the set of all k-dimensional vector spaces $V \subset \mathbb{C}^{k+2}$, i.e. $V \in \mathrm{Gr}(k,k+2)$ which contain L. Corollary 2.3.8 implies that $g(D) = 0$. ◇

Genus and Fiber Bundles

In applications fiber bundles frequently make an appearance. We restrict our attention here to two simpler cases, which give nice relations between the genera of the base space and the total space.

Proposition 2.3.10. *Let $X \setminus Y$ be smooth and isomorphic to $B \times (\mathbb{C}^\times)^\ell$, where B is a smooth quasi-projective variety. Then, the genus of (X,Y) is determined by the genus of B, that is $g(X,Y) = g(B)$.*

Proof. The proof is analogous to that of Proposition 2.2.19. Note that we got, essentially by the compactly supported Künneth formula 2.1.10,

$$H_c^n(X \setminus Y) \cong H_c^n(B \times (\mathbb{C}^\times)^\ell) \cong \bigoplus_{k=\ell}^{n} H_c^{n-k}(B) \otimes_{\mathbb{Q}} \mathbb{Q}(\ell-k)^{\binom{\ell}{2\ell-k}},$$

where we identified $H_c^\ell((\mathbb{C}^\times)^\ell)$ as in Example 2.1.11. It then suffices to note that the Hodge numbers $h^{p,0}(H_c^n(X \setminus Y))$ can only be non-zero if $k = \ell$. Hence, for all p we get

$$h^{p,0}(X,Y) = H_c^\ell((\mathbb{C}^\times)^\ell)^{p,0} = H_c^{n-\ell}(B)^{p,0},$$

which shows the claim. □

We record an important consequence for the Grassmannian of lines.

Corollary 2.3.11. *Let $\mathrm{Gr}(2,n)$ be the Grassmannian of lines, and let Y be the union of the vanishing loci of all Plücker coordinates. Then, $g(\mathrm{Gr}(2,n),Y) = g(M_{0,n}) = 0$, where $M_{0,n}$ is the moduli space of n marked points on $\mathbb{P}^1$.*

Proof. Recall the isomorphism $\mathrm{Gr}(2,n) \setminus Y \cong M_{0,n} \times (\mathbb{C}^\times)^{n-1}$ given in the proof of Corollary 2.2.16. Moreover, we also saw that $M_{0,n}$ is isomorphic to the complement of an affine hyperplane arrangement, hence its mixed Hodge structure is pure of Tate type by Lemma 2.1.12. So in particular, all Hodge numbers with $p \neq q$ vanish, hence $g(M_{0,n}) = 0$, and the claim follows from Proposition 2.3.10. □

We end this section by introducing another tool, which allows us to bound the genus of a pair (X,Y) from above.

Proposition 2.3.12. *Let X be a projective variety, and $Y \subset X$ a hypersurface, and denote by $\mathcal{L}(Y)$ the boundary stratification. Suppose that $g(Z) = 0$ for all components $Z \in \mathcal{L}(Y)$. Then, the genus of the pair (X,Y) vanishes, i.e. $g(X,Y) = 0$.*

Proof. The simple normal crossing case is covered in Lemma 2.1.20. So suppose Y is not simple normal crossing, and denote by $\mathcal{L}(Y)$ the boundary stratification of Y. We want to apply Lemma 2.1.19 recursively by dimension. The recursion starts in dimension n and is covered by Lemma 2.1.19. Hence, for the recursion step we can assume that we have an irreducible component $Z \in \mathcal{L}(Y)$ of dimension d. Without loss of generality, let $D_Z = D_1^Z \cup \ldots D_\ell^Z$ be the union of all irreducible components in $\mathcal{L}(Y)$ that have codimension one in Z. This is not restrictive, because if Z' has codimension at least 2 in Z, then $g(Z, D_Z \cup Z') = g(Z, D_Z)$ by Lemma 2.1.18. Hence, we can assume that the pairs appearing in the recursive step are of type (Z, D_Z), where D_Z is pure of dimension $d-1$. We can then apply Lemma 2.1.19 to the pair (Z, D_Z), to obtain

$$g(Z, D_Z) \leq g(Z) + \sum_{i=1}^{\ell} g(D_i^Z, U_i),$$

where $U_i = D_i^Z \cap \bigcup_{j \neq i} D_j^Z \subset D_Z$ and in particular the D_i^Z are of dimension $d-1$. Therefore, we can apply Lemma 2.1.19 again to the pairs $(D_i^Z, U_i^{(1)})$, where $U_i^{(1)}$ is the codimension 1 locus of $U_i \subset D_i^Z$, which we can assume without loss of generality by the same argument as above. Then, the recursion terminates in dimension 0. This shows the claim.

□

2.4 From Pairs to Positive Geometries

The main result in this section provides criteria under which the mixed Hodge theory framework for positive geometries developed in [BD25] gives rise to positive geometries in the sense of Definition 1.5.1. Moreover, we will also give a definition of positive geometries within

this framework, which is compatible with the original notion. For the remainder of this section let X be a projective variety defined over the real numbers, and Y a hypersurface such that $X \setminus Y$ is smooth. Also let $\iota : (X(\mathbb{R}), Y(\mathbb{R})) \hookrightarrow (X, Y)$ be the natural inclusion of the pair. Then, any semi-algebraic set S in the real points $X(\mathbb{R})$ with boundary in $Y(\mathbb{R})$ makes a cycle $\iota_*[S] \in H_\bullet(X, Y)$.

Definition 2.4.1. Let $\mathcal{L}(Y)$ be the boundary stratification of Y. For each $Z \in \mathcal{L}(Y)$, denote by $D_Z^{(1)} \subset Z$ the union of all codimension 1 components of $\mathcal{L}(Y)$ contained in Z. We say (X, Y) is a *recursive genus zero pair* if $g(Z, D_Z^{(1)}) = 0$ for all $Z \in \mathcal{L}(Y)$.

Example 2.4.2. Let $P \subset \mathbb{P}^n$ be a polytope, and $\partial_a P$ its algebraic boundary. Then, $(\mathbb{P}^n, \partial_a P)$ makes a recursive genus zero pair, as every irreducible component in $\mathcal{L}(\partial_a P)$ is a projective space $\mathbb{P}^r$ for some r. ◇

The vanishing genus result in Proposition 2.3.12 has an immediate consequence.

Corollary 2.4.3. *Let $\mathcal{L}(Y)$ be the boundary stratification of $Y \subset X$. Suppose that $g(Z) = 0$ for all irreducible components $Z \in \mathcal{L}(Y)$. Then (X, Y) is a recursive genus zero pair.*

Proof. Since $\mathcal{L}(Y)$ is closed under taking intersections, Proposition 2.3.12 applies to any pair $(Z, D_Z^{(1)})$ where D_Z is the union of all irreducible components in $\mathcal{L}(Y)$ which are of codimension one in Z. □

This definition allows us to connect the framework discussed in this section with positive geometries as in Definition 1.5.1. But before we do so, we establish a small lemma which rephrases orientability as in Definition 1.3.17, in the mixed Hodge theory framework.

Lemma 2.4.4. *Let (X, Y) be a recursive genus zero pair, and let $S \subset X(\mathbb{R})$ be a regular, connected, and compact semi-algebraic set with its Euclidean boundary ∂S contained in $Y \subset X$. Also suppose that the interior of S is disjoint from $\mathrm{Sing}(Z)$ for all $Z \in \mathcal{L}(Y)$. Denote the natural inclusion map by $\iota : (X(\mathbb{R}), Y(\mathbb{R})) \hookrightarrow (X, Y)$. Suppose $\Omega(\iota_*[S|_Z]) \neq 0$ for all $Z \in \mathcal{L}(Y)$, then S and all restrictions $S|_Z$ are orientable manifolds.*

Proof. Let Z be an irreducible component in $\mathcal{L}(Y)$ of dimension d, and denote by $D_Z^{(1)}$ the union of all irreducible components in $\mathcal{L}(Y)$ that have codimension one in Z. Then, note that $S|_Z$ defines a manifold with boundary $\partial S|_Z = S|_Z \setminus \operatorname{int} S|_Z \subset D_Z^{(1)}$, and its interior defines a smooth manifold as the interior of S avoids all singular loci. Now suppose that $S|_Z \neq \varnothing$, and $0 \neq \Omega(\iota_*[S|_Z]) \in \Omega^d_{\log}(Z \setminus D_Z^{(1)})$. Now, since (X, Y) is a recursive genus zero pair, $g(Z, D_Z^{(1)}) = 0$, and hence the canonical map $\Omega|_Z$ provides an isomorphism

$$\mathrm{gr}_0^W H_d(Z, D_Z^{(1)}) \cong \Omega^d_{\log}(Z \setminus D_Z^{(1)}).$$

Therefore, $\iota_*[S|_Z]$ is a non-zero weight 0 cycle in $H_d(Z, D_Z^{(1)})$. For the sake of contradiction, suppose that $S|_Z$ was not orientable as in Definition 1.3.17, that is $H_d(S|_Z, \partial S|_Z) = 0$ as $\dim H_d(S|_Z, \partial S|_Z) \leq 1$ by [Hat02, Theorem 3.47]. Then, $[S|_Z] = 0 \in H_d(Z(\mathbb{R}), D_Z^{(1)}(\mathbb{R}))$

and thus $\iota_*[S|_Z] = 0$ in contradiction to the above. Hence, we must have $H_d(S|_Z, \partial S|_Z) = \mathbb{Q}$. □

Our main result in this paragraph now asserts the connection between Definition 1.5.1 and the mixed Hodge theory framework laid out in the preceding sections.

Proposition 2.4.5. *Let S be a regular, connected n-dimensional semi-algebraic subset in the real points of $X \setminus Y$, where Y is the algebraic boundary of S. Also suppose that the interior of S is disjoint from $\mathrm{Sing}(Z)$ for all $Z \in \mathcal{L}(Y)$. Suppose (X, Y) is a recursive genus zero pair, and $0 \neq \Omega(\iota_*[S]) \in \Omega^n_{\log}(X \setminus Y)$. Then, the triple $(X, S, \Omega(\iota_*[S]))$ is a positive geometry in the sense of Definition 1.5.1.*

Proof. Let X and $S \subset X(\mathbb{R})$ be of dimension n. We want to show that $S \subset X$ makes a positive geometry with canonical form $\omega_S = \Omega(\iota_*[S])$. First, note that the interiors of $S|_Z$ are orientable manifolds, by Lemma 2.4.4. Second, by assumption $\iota_*[S]$ is a cycle in $H_n(X, Y)$. So it remains to verify that the three conditions in Definition 1.5.1 hold. By assumption the canonical map Ω is well-defined, as $g(X, Y) = 0$, hence $\Omega(\iota_*[S]) \neq 0$, is the unique algebraic top form on S with the correct simple poles along the algebraic boundary of S. This shows i. in Definition 1.5.1

We next show the recursive property ii. For the recursion step we need to show that the residues of $\Omega(S)$ have the correct poles. To that end, note that the restriction to an irreducible component $Z \in \mathcal{L}(Y)$, with induced codimension one locus $D_Z^{(1)} \subset Z$, corresponds to the Zariski closure of the restriction of $S|_Z$ with facets supported on $D_Z^{(1)}$ by definition of the algebraic boundary. Let

$$\Omega_Z : H_{\dim Z}(Z, D_Z^{(1)}) \to \Omega^{\dim Z}_{\log}(Z \setminus D_Z^{(1)})$$

be the canonical map for the induced pair $(Z, D_Z^{(1)})$, which is well-defined since we assumed that $g(Z, D_Z^{(1)}) = g(Z, D_Z) = 0$. Again, as (X, Y) is a recursive genus zero pair it follows from [BD25, Proposition 2.15], that $\mathrm{Res}_Z \Omega(\iota_*[S]) = \Omega_Z(\iota_*[S|_Z])$ is the unique top form on $Z \setminus D_Z^{(1)}$ with simple poles along the algebraic boundary of $S|_Z$. This shows ii. We proceed recursively to dimension 0. The values of the residues can then be fixed to ± 1 by the connectedness assumption on S. This shows iii. in Definition 1.5.1. Hence, we proved that $(X, S, \Omega(\iota_*[S]))$ is a positive geometry. □

Remark 2.4.6. It is worth emphasizing that the assumption of a recursive genus zero pair is a sufficient criterion to find a positive geometry, but it is certainly not necessary. This is of course not entirely surprising as the genus is invariant under some algebraic modifications such as blow-ups, that can be isomorphisms over the region S, and thus could resolve some of the obstructions to making (recursive) genus zero pairs. We will give an explicit example of a positive geometry which is not a (recursive) genus zero pair in Section 4.4.

We now turn to the definition of positive geometries in this framework.

Definition 2.4.7. Let X be an orientable projective variety defined over the real numbers. Let $Y \subset X$ be a subvariety of codimension one, such that $X \setminus Y$ is smooth. Then, (X, Y) is a *positive pair* if the following is satisfied:

i. the weight zero part of the subgroup $H = \iota_* H_n(X(\mathbb{R}), Y(\mathbb{R})) \subset H_n(X, Y)$ generated by the real cycles inside the real points $(X \setminus Y)(\mathbb{R})$ is not trivial, i.e. $\mathrm{gr}_0^W H \neq 0$;

ii. the genus $g(X, Y)$ vanishes.

Any $\mathbb{Q}$-linear combination of (real) n-chains $\sigma \in \mathrm{gr}_0^W H$ is called a *pre-positive geometry*. If the pair (X, Y) is additionally a recursive genus zero pair, then (X, Y) is called a *recursive positive pair*, and in this case any $\mathbb{Q}$-linear combination of (real) n-chains $\sigma \in \mathrm{gr}_0^W H$ is called a *positive geometry*.

Remark 2.4.8. We make several remarks on how this definition relates to related concepts in the literature on positive geometries.

i. In the recent work [ST26], the real cycles $\sigma \in H_n(X, Y)$ were called positive geometries, and correspond to our pre-positive geometries.

ii. Also note that any positive geometry $\sigma \in \mathrm{gr}_0^W H$ with $\Omega(\sigma) = 0$ was called a *pseudo positive geometry* in [ABL17, Section 2.2].

iii. Moreover, any positive geometry $\sigma \in \mathrm{gr}_0^W H$ was called a *weighted positive geometry* in [DHS23]. See also Example 2.4.10.

Example 2.4.9. For an example of a pre-positive geometry, we refer to [BD25, Section 5.9]. ◇

Next we want to give two examples of positive geometries in this framework.

Example 2.4.10. Consider the hyperplane arrangement in $\mathbb{P}^2$, whose restriction to the affine patch U_z is shown in Figure 2.4.1a. There are 2 bounded regions Δ_1 and Δ_2. These two simplices share two facets supported on the x and y axes, and thus are connected in codimension 2 through the origin. Individually, the two simplices are of course positive geometries, and by Definition 1.5.1, we are allowed to choose the orientations of their interiors independently as they are disjoint. Therefore, we can write their canonical forms on an affine patch as

$$\omega_{\Delta_1} = \frac{1}{xy(x+y-1)}\, dx \wedge dy, \quad \text{and } \omega_{\Delta_2} = \frac{-1}{xy(x+y+1)}\, dx \wedge dy,$$

which corresponds to counter-clockwise orientation as shown in 2.4.1a. However, note that canonical form, as in Definition 1.5.1, of the union $\Delta_1 \cup \Delta_2$ is then not simply the sum of the forms $\omega_{\Delta_1} + \omega_{\Delta_2}$. This is because the residue of the sum $\omega_{\Delta_1} + \omega_{\Delta_2}$ at the origin evaluates to ± 2, whereas the residue along the other vertices equals ± 1. This contradicts the requirements of a canonical form from Definition 1.5.1. Hence, $\Delta_1 \cup \Delta_2$ is a weighted positive geometry, see [DHS23]. The canonical form with the correct residues is simply given by the difference

$$\omega_{\Delta_1} - \omega_{\Delta_2} = \frac{2(x+y)}{xy(x+y-1)(x+y+1)}\, dx \wedge dy.$$

Note that this form will have trivial residue at the origin, which is not a contradiction as the origin is not a boundary of the restriction of $\Delta_1 \cup \Delta_2$ to the x and y axes.

Let us also briefly discuss this example in the new framework. By Lemma 2.1.12 hyperplane arrangements in projective space make recursive positive pairs. The orientation on the cycles is induced from the global choice of orientation - for us counter-clockwise. Hence, the cycle $\sigma = [\Delta_1] + [\Delta_2] \in \mathrm{gr}_0^W H$ is a positive geometry in the sense of Definition 2.4.7, with canonical form given by the sum

$$\Omega(\sigma) = \Omega([\Delta_1]) + \Omega([\Delta_2]) = \omega_{\Delta_1} + \omega_{\Delta_2}.$$

Notably, this is not the canonical form of $\Delta_1 \cup \Delta_2$, however the additional sign $(-1)^{k-1}$ can be explained by demanding that codimension k connected components have vanishing residue along the shared boundary.

◇

Example 2.4.11. Consider the two unit half-disks P_1 and P_2 in $U_z \subset \mathbb{P}^2$ shown in Figure 2.4.1b. Individually, these two regions make positive geometries with the ambient $\mathbb{P}^2$ and canonical forms

$$\omega_{P_1} = \frac{1}{x(x^2+y^2-1)}\, dx \wedge dy = \omega_{P_2}.$$

However, the disk D obtained by the union of the two regions P_1 and P_2 is not a positive geometry in the sense of Definition 1.5.1, as it does not have any 0-dimensional strata for example. Note that D is what the authors in [ABL17] call a pseudo-positive geometry with zero canonical form. Nevertheless, the cycle $\sigma = [P_1] + [P_2]$ is clearly a positive geometry, and indeed we also have $\Omega(\sigma) = \Omega([P_1]) + \Omega([P_2]) = \omega_{P_1} - \omega_{P_2} = 0$.

◇



(a) Two triangles in the plane connected in codimension 2.

(b) Two pizza slices in the plane covering a pizza.

Figure 2.4.1: Two "problematic" positive geometries.

We end this section by recording that one of the most important positive geometries, the non-negative Grassmannian, fits nicely into this new framework.

Proposition 2.4.12. *Let $\mathrm{Gr}(k,n)$ be the Grassmannian and $\partial_a \mathrm{Gr}(k,n)_{\geq 0}$ the algebraic boundary of the non-negative Grassmannian $\mathrm{Gr}(k,n)_{\geq 0}$. Let*

$$\iota : (\mathrm{Gr}(k,n)(\mathbb{R}), \partial_a \mathrm{Gr}(k,n)_{\geq 0}(\mathbb{R})) \hookrightarrow (\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0})$$

be the natural inclusion. Then, $(\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0})$ is a recursive positive pair. In particular, $\iota_*[\mathrm{Gr}(k,n)_{\geq 0}] \in H_{k(n-k)}(\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0})$ is a positive geometry.

Proof. Let m denote the dimension of $\mathrm{Gr}(k,n)$. Recall the notations for positroid varieties from the end of Section 1.4. The boundary stratification $\mathcal{L}(\partial_a \mathrm{Gr}(k,n)_{\geq 0})$ is given by the positroid stratification, see e.g. [Pos06]. Then, by [GL24, Corollary 4.13] the mixed Hodge structure of $H^d(\Pi, \partial_a \Pi_{\geq 0})$ is pure of Tate type, so all Hodge numbers with $p \neq q$ vanish. Therefore, $g(\Pi, \partial_a \Pi_{\geq 0}) = 0$ for all positroid varieties in the stratification. Hence, $(\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0})$ is a recursive genus zero pair. Let ι be the inclusion map for the real points of the pair $(\mathrm{Gr}(k,n), D)$. It remains to show that the real cycles generate a non-trivial group $\mathrm{gr}_0^W H$ where

$$H = \iota_* H_m(\mathrm{Gr}(k,n)(\mathbb{R}), \partial_a \mathrm{Gr}(k,n)_{\geq 0}(\mathbb{R})) \subset H_m(\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0}).$$

It is well known that $\mathrm{Gr}(k,n)_{\geq 0}$ has a canonical form ω, see e.g. [Lam22, Theorem 8]. Then, since the canonical map Ω gives an isomorphism

$$\mathrm{gr}_0^W H_m(\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0}) \cong \Omega^m_{\log}(\mathrm{Gr}(k,n) \setminus \partial_a \mathrm{Gr}(k,n)_{\geq 0}),$$

the group $\mathrm{gr}_0^W H$ cannot be zero, as in particular $\iota_*[\mathrm{Gr}(k,n)_{\geq 0}]$ is a non-zero cycle in weight 0. Therefore, $(\mathrm{Gr}(k,n), \partial_a \mathrm{Gr}(k,n)_{\geq 0})$ is a recursive positive pair, and the cycle $\iota_*[\mathrm{Gr}(k,n)_{\geq 0}]$ is a positive geometry. □

Chapter 3

Arrangements of Hyperplanes and a Quadric

Have you tried making it nonlinear?

Every Nonlinear Algebra group member

Quadrics with an arrangement of hyperplane sections on them appear frequently in applications. For example, they appear in the context of 1-loop Feynman integrals see [Mou26, Section 1.1] for an introduction. They also appear in case of the $m = k = 2$ amplituhedron $\mathcal{A}_{n,2,2}$ sitting inside $\mathrm{Gr}(2,4) \subset \mathbb{P}^5$ which is a quadric hypersurface, together with the algebraic boundary of the amplituhedron, see Proposition 1.4.2. In these contexts, one is naturally interested in differential forms on the complement of the arrangement on the quadric and its real regions, which give contenders for canonical forms, which in turn are important for the computation of scattering amplitudes. In this chapter we will study these arrangements and their differential forms.

Let Q be a smooth non-degenerate quadric in $\mathbb{P}^n$, with a real defining equation of signature $(n,1)$. Let $\mathcal{A}$ be an arrangement of m hyperplanes in $\mathbb{P}^n$. Denote by $\mathcal{A}_Q$ the divisor in $\mathbb{P}^n$ given by the union of $\mathcal{A}$ and Q. In this section we study the complements $\mathbb{P}^n \setminus \mathcal{A}_Q$ and $Q \setminus \mathcal{A}|_Q$ from a positive geometry viewpoint as in Chapter 2. We investigate their genus, combinatorial rank, number of real regions, and give a basis of the vector space of logarithmic differential forms. It turns out that these varieties naturally make genus zero pairs. Moreover, under some more genericity assumptions, which we call generic spherical arrangement see Definition 3.1.4, the answers to these questions are particularly nice and of combinatorial nature. We will show how all of these assumptions are necessary if we are looking for a combinatorial result. Our two main results can be stated as follows.

Theorem 3.0.1. *Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement with m hyperplanes. Then, the combinatorial rank of the restricted pair $(Q, \mathcal{A}|_Q)$ is given as*

$$\operatorname{cr}(Q, \mathcal{A}|_Q) = \binom{m}{n-1} + \binom{m-1}{n-1}.$$

Moreover, a basis $\mathcal{B}_Q$ of the weight 0 piece of middle relative homology $\operatorname{gr}_0^W H_{n-1}(Q, \mathcal{A}|_Q)$ is given by the classes of real regions in $(Q \setminus \mathcal{A}|_Q)_{\mathbb{R}}$ modulo a set of linear relations described in Section 3.3. Equivalently, the image of $\mathcal{B}_Q$ under the canonical map $\Omega : H_{n-1}(Q, \mathcal{A}|_Q) \to \Omega^{n-1}_{\log}(Q \setminus \mathcal{A}|_Q)$ is a basis of $\Omega^{n-1}_{\log}(Q \setminus \mathcal{A}|_Q)$.

We will identify real regions in $\mathbb{P}^n \setminus \mathcal{A}_Q$ by their sign-vector in $\{+,-\}^{m+1}$, where the $(m+1)$-th entry corresponds to the sign with respect to the quadric Q, which is projectively well-defined. Similarly, in Proposition 3.2.1 we show that the regions on Q induced by $\mathcal{A}$ are also naturally labeled by sign patterns in $\{+,-\}^m$. Therefore, we can extend the theorem above to the pair $(\mathbb{P}^n, \mathcal{A}_Q)$ by finding a section for the boundary map $\partial_Q : H_n(\mathbb{P}^n, \mathcal{A}_Q) \to H_{n-1}(Q, \mathcal{A}|_Q)$, denoted $\gamma : H_{n-1}(Q, \mathcal{A}|_Q) \to H_n(\mathbb{P}^n, \mathcal{A}_Q)$, see Lemma 3.3.7, which sends a region $\sigma \in \{+,-\}^m$ to $(\sigma|+) \in \{+,-\}^{m+1}$. Let $\iota : (\mathbb{P}^n, \mathcal{A}) \hookrightarrow (\mathbb{P}^n, \mathcal{A}_Q)$ be the inclusion map. For $I \subset [m]$, denote by L_I the multiple intersection of L_i for $i \in I$ and $L_i \in \mathcal{A}$ the hyperplanes. Lastly, let L be an additional hyperplane in $\mathbb{P}^n$ that meets $\mathcal{A}$ generically, that is all intersections of L with L_I are of codimension 1 in L_I. Then, we say a real connected region in $\mathbb{P}^n \setminus \mathcal{A}(\mathbb{R})$ is bounded if its intersection with L is empty. We can then state a similar result as Theorem 3.0.1 but for the pair $(\mathbb{P}^n, \mathcal{A}_Q)$.

Theorem 3.0.2. *Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement with m hyperplanes. Then, the combinatorial rank of $(\mathbb{P}^n, \mathcal{A}_Q)$ is given as*

$$\operatorname{cr}(\mathbb{P}^n, \mathcal{A}_Q) = \#\{\text{bounded regions in } (\mathbb{P}^n \setminus \mathcal{A})(\mathbb{R})\} + \binom{m}{n-1} + \binom{m-1}{n-1}.$$

Moreover, let $\mathcal{B}_{\mathbb{P}}$ be the union of the pushforwards of the classes of bounded real regions in $(\mathbb{RP}^n \setminus (\mathcal{A}_Q)_{\mathbb{R}})$ under ι_*, and the image of the basis $\mathcal{B}_Q$ from Theorem 3.0.1 under the section $\gamma : H_{n-1}(Q, \mathcal{A}|_Q) \to H_n(\mathbb{P}^n, \mathcal{A}_Q)$. Then, $\mathcal{B}_{\mathbb{P}}$ is a basis of the weight 0 piece of middle relative homology $\operatorname{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}_Q)$. Equivalently, the image of $\mathcal{B}_{\mathbb{P}}$ under the canonical map $\Omega : H_n(\mathbb{P}^n, \mathcal{A}_Q) \to \Omega^n_{\log}(\mathbb{P}^n \setminus \mathcal{A}_Q)$ is a basis of $\Omega^n_{\log}(\mathbb{P}^n \setminus \mathcal{A}_Q)$.

We illustrate these theorems via a concrete example.

Example 3.0.3. Consider the arrangement $\mathcal{A}_Q$ in $\mathbb{P}^2$, which arises as the complexification of an underlying real arrangement, with coordinates x, y, and z, given by the union of the zeros

of the following equations

$$\begin{aligned} L_1 &= y \\ L_2 &= -5x + 5y - z \\ L_3 &= x \\ L_4 &= 5x + 5y - z \\ Q &= -x^2 - y^2 + z^2. \end{aligned}$$

The arrangement is shown in Figure 3.0.1 with respect to the open patch $U_z \cong \mathbb{C}^2 \subset \mathbb{P}^2$ - where the hyperplane at infinity is given by the vanishing of z. Therein, one can see 14 projective regions in $\mathbb{P}^2 \setminus \mathcal{A}_Q(\mathbb{R})$, out of which 10 are on the positive side of Q, the interior of the green circle. We compute the canonical form ω_I of region I explicitly, using the notion of

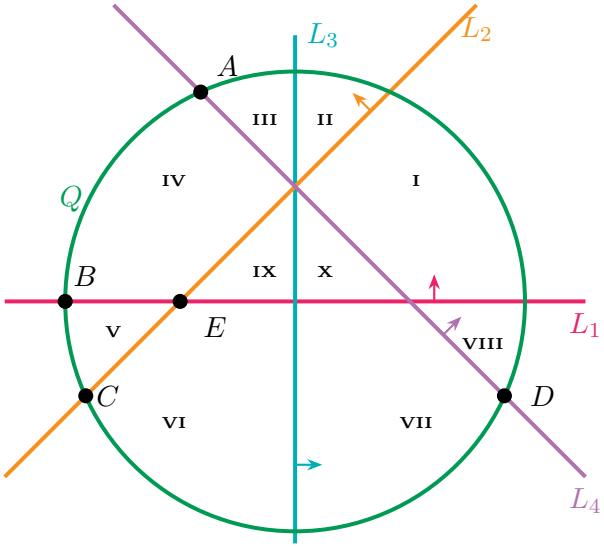


Figure 3.0.1: A (real) arrangement $\mathcal{A}_Q$ in $\mathbb{P}^2$.

adjoints, see Definition 1.5.8. Recall that on U_z we can write the canonical form of a region $\omega_I = f\, dx \wedge dy$, where f is a rational function of degree 3 - the degree of the canonical divisor on $\mathbb{P}^2$. Since the region I is bounded by L_1, L_2, and L_4 and Q, and the canonical form must have logarithmic singularities along these boundaries, the numerator of f must have degree 2. This quadric is uniquely determined by interpolating the residual points A, B, C, D, and E. We then obtain

$$\omega_I = \Omega([I]) = \frac{5x^2 + 10xy + 6x + 7y + 1}{x(5x + 5y - z)(-5x + 5y - z)(x^2 + y^2 - 1)}\, dx \wedge dy,$$

where $\Omega : H_2(\mathbb{P}^2, \mathcal{A}_Q) \to \Omega^2_{\log}(\mathbb{P}^2 \setminus \mathcal{A}_Q)$ is the canonical map from (2.1.4). Writing $\omega_\bullet = \Omega([\bullet])$ for all regions, we collect all 10 canonical forms in Table 3.0.1. Note that the signs in Table 3.0.1 are consistent with respect to a chosen orientation on $\mathbb{R}^2$. Therefore, we get, for example, that the sum $\omega_{IX} + \omega_X$ correctly computes the canonical form of the bigger triangle, given by the union of the regions IX and X. Moreover, one can see that the canonical forms

Table 3.0.1: Canonical forms $\omega_\bullet$ of the regions in Figure 3.0.1.

region	sign	canonical form
I	$+-++\,\vert\,+$	$-4\dfrac{5x^2+10xy+6x+7y+1}{y(5x+5y-1)(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
II	$++++\,\vert\,+$	$2\dfrac{x+2y+2}{x(5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
III	$++-+\,\vert\,+$	$2\dfrac{x-2y-2}{x(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
IV	$++--\,\vert\,+$	$-4\dfrac{5x^2-10xy-6x+7y+1}{y(5x+5y-1)(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
V	$-+--\,\vert\,+$	$2\dfrac{2x+y-2}{y(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
VI	$----\,\vert\,+$	$\dfrac{x^2-2xy-5y^2+6y-1}{yx(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
VII	$--+-\,\vert\,+$	$-\dfrac{x^2+2xy-5y^2+6y-1}{yx(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
VIII	$--++\,\vert\,+$	$2\dfrac{2x-y+2}{y(-5x+5y-1)(x^2+y^2-1)}\,dx\wedge dy$
IX	$+---\,\vert\,+$	$-\dfrac{1}{xy(-5x+5y-1)}\,dx\wedge dy$
X	$+-+-\,\vert\,+$	$\dfrac{1}{xy(5x+5y-1)}\,dx\wedge dy$

of the (projectively) 4 regions where Q is negative, are simple linear combinations of the 10 we wrote down above. Hence, $\dim\Omega^2_{\log}(\mathbb{P}^2\setminus\mathcal{A}_Q)\leq 10$ in agreement with the natural upper bound for the combinatorial rank of hypersurfaces in projective space by Lemma 2.1.21. In fact, we note that also the sum $\omega_I+\omega_{II}+\ldots+\omega_X=0$. We compute directly that there are no additional linear relations between the forms. Therefore,

$$\begin{aligned}\dim\Omega^2_{\log}(\mathbb{P}^2\setminus\mathcal{A}_Q) &= \#\text{bounded regions in } (\mathbb{R}^2\setminus\mathcal{A})+\binom{4}{2-1}+\binom{4-1}{2-1}\\ &= 2+4+3=9,\end{aligned}$$

as Theorem 3.0.2 predicts. To see how this basis is obtained from $(\mathbb{P}^2,\mathcal{A})$ and $(Q,\mathcal{A}|_Q)$, see Example 3.3.9. ◇

This chapter is structured as follows. First, we investigate the mixed Hodge structure of the pair $(Q,\mathcal{A}|_Q)$ in Section 3.1. We introduce a precise definition of our genericity assumptions, called generic spherical arrangement, see Definition 3.1.4. For such pairs, we then establish the genus, combinatorial rank, and a few technical results on short exact sequences, that are going to be a key ingredient for the proofs in the later sections. In Section 3.2 we prove that the number of bounded real regions in $Q\setminus\mathcal{A}|_Q$ is given by a simple combinatorial formula. Thereafter, we use this result in Section 3.3 to find an explicit basis of the weight zero homology groups. Throughout these sections we illustrate the necessity of our genericity assumptions on $\mathcal{A}$ in order to obtain nice combinatorial results. We end this chapter by indicating a few further research directions.

3.1 Mixed Hodge Structure, Genus, and Combinatorial Rank

Let $\mathcal{A} \subset \mathbb{P}^n$ be the complexification of an arrangement of m real hyperplanes. Let $Q \subset \mathbb{P}^n$ be a quadric hypersurface. In this section, we compute the genus and combinatorial rank for the pairs $(\mathbb{P}^n, \mathcal{A}_Q)$ and $(Q, \mathcal{A}|_Q)$. The first result asserts that their genera vanish, which follows essentially from [BD25].

Proposition 3.1.1. *The pairs $(\mathbb{P}^n, \mathcal{A}_Q)$ as well as $(Q, \mathcal{A}|_Q)$ have vanishing genus.*

Proof. It follows directly from [BD25, Proposition 3.26, Corollary 3.11] that $g(\mathbb{P}^n, \mathcal{A}_Q) = 0$. The claim for the second pair follows as $\mathbb{P}^n \setminus \mathcal{A}$ is smooth and affine, and $g(\mathbb{P}^n, \mathcal{A}) = 0$ this in turn implies $g(Q, \mathcal{A}|_Q) = 0$ by [BD25, Corollary 3.12]. □

Remark 3.1.2. It is worth emphasizing that we did not make any genericity assumptions for the pairs in Proposition 3.1.1.

The canonical map (2.1.4) has an immediate consequence for the combinatorial rank.

Corollary 3.1.3. *The combinatorial rank of $(\mathbb{P}^n, \mathcal{A}_Q)$ and $(Q, \mathcal{A}|_Q)$ equals the dimension of $\Omega^n_{\log}(\mathbb{P}^n, \mathcal{A}_Q)$, and $\Omega^{n-1}_{\log}(Q, \mathcal{A}|_Q)$ respectively.*

We now give a concrete definition of the genericity assumptions referred to in the introduction.

Definition 3.1.4. Let $Q_{\mathbb{R}}$ be a $(n-1)$-dimensional non-degenerate real quadric in $\mathbb{RP}^n = \mathbb{P}(\mathbb{R}^{n+1})$ of signature $(n, 1)$, and $\mathcal{A}_{\mathbb{R}}$ be an arrangement of m real hyperplanes in $\mathbb{RP}^n$, that is simple normal crossing when restricted to Q. Moreover, suppose $\mathcal{L}(\mathcal{A}|_Q)$ is the power set of $[m]$ ordered by inclusion, and that m is big enough such that there are 0-dimensional intersections. A real arrangement $\mathcal{A}_Q(\mathbb{R}) \subset \mathbb{RP}^n$ satisfying these assumptions is called *generic spherical arrangement*. A complexified arrangement $\mathcal{A}_Q \subset \mathbb{P}^n$ is a generic spherical arrangement, if its underlying real arrangement is.

We will henceforth assume that this assumption holds for the arrangements $\mathcal{A}_Q \subset \mathbb{P}^n$. In this case we get a stronger result for the pairs.

Theorem 3.1.5. *Let $\mathcal{A}_Q$ be a generic spherical arrangement. Then, $(\mathbb{P}^n, \mathcal{A}_Q)$ and $(Q, \mathcal{A}|_Q)$ are recursive positive pairs. In particular, any connected real region σ in $\mathbb{P}^n \setminus \mathcal{A}_Q$, and any connected real region $\sigma' \in Q \setminus \mathcal{A}|_Q$ gives rise to a positive geometry in sense of Definition 2.4.7.*

Proof. We start by showing the claim for the pair $(Q, \mathcal{A}|_Q)$. Let Z be an irreducible component in the boundary stratification $\mathcal{L}(\mathcal{A}|_Q)$ of the divisor and let $D^{(1)}_Z$ be the union of all irreducible components in $\mathcal{L}(\mathcal{A}|_Q)$ which have codimension one in Z. The assumptions of a generic spherical arrangement imply that any pair $(Z, D^{(1)}_Z)$ is a quadric with an arrangement of hyperplane sections. Therefore, all genera of $(Z, D^{(1)}_Z)$ vanish by Proposition 3.1.1. Hence, the pair is a recursive genus zero pair. The proof for $(\mathbb{P}^n, \mathcal{A}_Q)$ is analogous.

Next, note that any connected region in $(Q \setminus \mathcal{A}|_Q)(\mathbb{R})$ gives rise to a non-zero form in $\Omega^{n-1}_{\log}(Q \setminus \mathcal{A}|_Q)$, as by assumption any region contains a zero-dimensional stratum. This form must live in weight zero by [BD25, Proposition 2.17]. Hence, $\mathrm{gr}_0^W H(Q, \mathcal{A}|_Q)$ from Definition 2.4.7 cannot be zero. Thus, $(Q, \mathcal{A}|_Q)$ is a recursive positive pair. Analogously, we have that $(\mathbb{P}^n, \mathcal{A}_Q)$ is a recursive positive pair. □

Remark 3.1.6. One can also show that these connected regions are positive geometries in the sense of Definition 1.5.1. See also Proposition 2.4.5.

For generic spherical arrangements we can compute the combinatorial ranks explicitly as the following two main results of this section assert. Their proofs are relegated to the end of this section.

Proposition 3.1.7. *Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement with m hyperplanes. The combinatorial rank of the restriction to Q, i.e. $(Q, \mathcal{A}|_Q)$, is given by*

$$\mathrm{cr}(Q, \mathcal{A}|_Q) = \binom{m}{n-1} + \binom{m-1}{n-1}.$$

This result can also be used to compute the combinatorial rank of the full arrangement $\mathcal{A}_Q \subset \mathbb{P}^n$. Let L be a hyperplane in $\mathbb{P}^n$ that meets $\mathcal{A}$ generically, that is all intersections of L with L_I are of codimension 1 in L_I. Then, we say a real connected region in $\mathbb{P}^n \setminus \mathcal{A}$ is bounded if its intersection with L is empty. We then obtain the following result.

Corollary 3.1.8. *Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement. Then, the combinatorial rank for the pair $(\mathbb{P}^n, \mathcal{A}_Q)$ is given as*

$$\mathrm{cr}(\mathbb{P}^n, \mathcal{A}_Q) = \#(\textit{bounded regions in } (\mathbb{P}^n \setminus \mathcal{A})(\mathbb{R})) + \binom{m}{n-1} + \binom{m-1}{n-1}.$$

Before proving these results, we want to make a note on the mixed Hodge structure on $(\mathbb{P}^n, \mathcal{A}_Q)$. One might hope, similar to the result for hyperplane arrangements $(\mathbb{P}^n, \mathcal{A})$, see e.g. [Sha97], that the mixed Hodge structure on $(\mathbb{P}^n, \mathcal{A}_Q)$ is also pure of weight 0. However, this turns out not to be true as the following example shows.

Example 3.1.9. Let $\mathcal{A}_Q = Q \cup L \subset \mathbb{P}^3$, be an arrangement of a quadric surface Q and a single generic plane L. Denote $D = D_1 \cup D_2$ with $D_1 = Q$ and $L = D_2$ for easier bookkeeping. Then, we can compute $H^\bullet(\mathbb{P}^3, Q \cup L)$ via the spectral sequence converging to relative cohomology in Proposition 2.1.6, as this pair is simple normal crossing. Also recall the definition in *loc. cit.* of $\delta^q_J : H^q(D_J) \to H^q(D_I)$ induced by the signed sum of the inclusions $D_I \hookrightarrow D_J$ with $J \subset I \subset [m]$. These defined the maps δ^q_p which in turn gave the boundary maps $d^1 : E_1^{p,q} \to E_1^{p+1,q}$ by taking the alternating sum over the inclusions $J \subset I$. Explicitly, if $I = \{i_1 < \ldots < i_{p+1}\}$, then the sign of the component labeled by J with $I \setminus J = i_\ell$ is determined by $(-1)^\ell$. We want to consider weight 2, so $q = 2$ and $p = 1$. We get

$$\delta^2_1 : H^2(Q) \oplus H^2(L) \to H^2(Q \cap L), \quad v \mapsto \delta^2_{\{1\}}(v) - \delta^2_{\{2\}}(v)$$

Noting, that $H^2(Q) = \mathbb{Q}^2$ and $H^2(L) = \mathbb{Q}$, we get

$$\delta_1^2 = \begin{bmatrix} 1 & 1 & -2 \end{bmatrix}.$$

(The factor of two comes from the intersection of $Q \cap L$ being of degree 2.) Thus,

$$\ker \delta_1^2 = \begin{bmatrix} 1 & 0 \\ -1 & 2 \\ 0 & 1 \end{bmatrix}.$$

For $\delta_0^2 : H^2(\mathbb{P}^3) \to H^2(Q) \oplus H^2(L)$, we make a choice of sign so that $\delta_0^2([\mathbb{P}^2]) = ([\mathbb{P}^2], [\mathbb{P}^2])$. Hence, we get

$$\operatorname{im} \delta_0^2 = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}^T.$$

Note that $\operatorname{im} \delta_0^2 \subset \ker \delta_1^2$ as it should be for $E_1^{p,q}$. In particular, we get

$$\operatorname{gr}_2^W H^3(\mathbb{P}^3, Q \cup L) \cong \frac{\ker \delta_1^2}{\operatorname{im} \delta_0^2} \cong \mathbb{Q}(-1).$$

Therefore, in particular $H^3(\mathbb{P}^3, Q \cup L)$ is not pure of weight 0 as hoped. ◇

However, despite this counterexample in dimension 3, we have the following result for plane arrangements.

Proposition 3.1.10. *Let $\mathcal{A}_Q \subset \mathbb{P}^2$ be any arrangement (not necessarily simple normal crossing) of lines in the plane and a single smooth conic. Then, $H_2(\mathbb{P}^2, \mathcal{A}_Q)$ is pure of Tate type. In particular, the middle homology $H_2(\mathbb{P}^2, \mathcal{A}_Q)$ has weight 0.*

Proof. Let D_i denote the irreducible components of $\mathcal{A}_Q$. The spectral sequence in Proposition 2.1.6 converges to relative cohomology $H^\bullet(\mathbb{P}^2, \mathcal{A}_Q)$, explicitly

$$E_1^{p,q} = \bigoplus_{|I|=p} H^q(D_I) \Rightarrow H^{p+q}(\mathbb{P}^2, \mathcal{A}_Q).$$

Next recall from Chapter 2 that higher differentials on the pages E_r with $r > 1$ are maps $d^r : E_r^{p,q} \to E_r^{p+r,q-r+1}$. However, note that all varieties appearing on E_1 are smooth and projective (or disjoint union of points), hence all cohomology groups $H^q(D_I)$ have pure Hodge structure of weight q. Or equivalently, the rows on the first page E_1 are all pure of weight q. Then, the differentials d^r for $r > 1$ must all be the zero map, as they are morphisms of mixed Hodge structures. Hence, the spectral sequence degenerates on E_2. Note that this also implies that $H^k(\mathbb{P}^2, \mathcal{A}_Q)$ is non-trivial only in even weight $q = 0, 2$. Then, we compute via the spectral sequence from above

$$\operatorname{gr}_2^W H^2(\mathbb{P}^2, \mathcal{A}_Q) = \frac{\ker \delta_0^2}{\operatorname{im} \delta_{-1}^2} = 0$$

as $\delta_0^2 : H^2(\mathbb{P}^2) \to \bigoplus_i H^2(D_i)$ is injective, where D_i are the irreducible components of $\mathcal{A}_Q$. Then, $H^2(\mathbb{P}^2, \mathcal{A}_Q)$ must be pure of weight 0. More precisely,

$$\operatorname{gr}_0^W H^2(\mathbb{P}^2, \mathcal{A}_Q) = \frac{\ker \delta_2^0}{\operatorname{im} \delta_1^0}.$$

The claim then follows by duality $H_2(\mathbb{P}^2, \mathcal{A}_Q) \cong H^2(\mathbb{P}^2, \mathcal{A}_Q)^\vee$. □

The remainder of this section is dedicated to proving the main results Propositions 3.1.7 and 3.1.8. The proofs rely on a series of technical lemmas, which ultimately establish several short exact sequences for weighted pieces of mixed Hodge structures on the relative cohomology groups. We introduce some notation. Write $\mathcal{A}_Q = Q \cup \bigcup_{i=1}^m L_i$, denote the irreducible components of $\mathcal{A}_Q$ by $D_0, \dots, D_m$ with $D_0 = Q$ and $D_i = L_i$ for $i \in [m]$. Fix some non-negative integer i. Denote the *deletion arrangement* as $\mathcal{A}'_Q = \mathcal{A}_Q \setminus D_i$, and the *restriction arrangement* as $\mathcal{A}''_Q = \mathcal{A}_Q|_{D_i}$. Denote their respective complements by $U = \mathbb{P}^n \setminus \mathcal{A}_Q$, $U' = \mathbb{P}^n \setminus \mathcal{A}'_Q$, and $U'' = D_i \setminus \mathcal{A}''_Q$. Note that for $i = 0$, we get $\mathcal{A}'_Q = \mathcal{A}$, and $\mathcal{A}''_Q = \mathcal{A}|_Q$. Then, as $U \subset U'$ are smooth, and setting $U = U' \setminus U''$, which is also smooth, we can apply Lemma 2.1.8. We obtain the *deletion-restriction* long exact sequence of mixed Hodge structures

$$\cdots \to H^k(U') \to H^k(U) \to H^{k-1}(U'')(-1) \to H^k(U') \to \cdots . \tag{3.1.1}$$

Remark 3.1.11. In the literature on hyperplane arrangements the deletion-restriction sequence is well known. However, note that here we also allow for the restriction to and deletion of the quadric hypersurface.

We can now derive the main technical result in this section, which will allow us to recursively compute the combinatorial ranks. It is essentially a refinement of the result [BD25, Corollary 4.11] for our situation.

Proposition 3.1.12. *We get the following split short exact sequences of mixed Hodge structures on the weight 0 graded pieces in relative homology:*

i. $0 \to \mathrm{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}'_Q) \to \mathrm{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}_Q) \to \mathrm{gr}_0^W H_{n-1}(\mathbb{P}^{n-1}, \mathcal{A}''_Q) \to 0$;

ii. $0 \to \mathrm{gr}_0^W H_{n-1}(Q, \mathcal{A}'|_Q) \to \mathrm{gr}_0^W H_{n-1}(Q, \mathcal{A}|_Q) \to \mathrm{gr}_0^W H_{n-2}(Q'', \mathcal{A}''|_{Q''}) \to 0$;

iii. $0 \to \mathrm{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}) \to \mathrm{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}_Q) \to \mathrm{gr}_0^W H_{n-1}(Q, \mathcal{A}|_Q) \to 0$;

where $Q'' \subset \mathbb{P}^{n-1}$ is the smooth quadric hypersurface obtained by intersecting Q with one of the hyperplanes in $\mathcal{A}$.

Proof. Since we take homology with $\mathbb{Q}$-coefficients any short exact sequence splits. It remains to show that we get such short exact sequences. The proof for all sequences is identical, with the corresponding choice of complements U, U', and U''. Then, the deletion-restriction sequence in (3.1.1) reads

$$\begin{aligned}\cdots \to & H^{n-2}(\mathbb{P}^{n-1} \setminus \mathcal{A}''_Q)(-1) \to H^n(\mathbb{P}^n \setminus \mathcal{A}'_Q) \to H^n(\mathbb{P}^n \setminus \mathcal{A}_Q) \to \\ & H^{n-1}(\mathbb{P}^{n-1} \setminus \mathcal{A}''_Q)(-1) \to H^{n+1}(\mathbb{P}^n \setminus \mathcal{A}'_Q) \to \cdots .\end{aligned}$$

We apply the exact functor gr_{2n}^W to this sequence. Then, the left-most term,

$$\mathrm{gr}_{2n}^W H^{n-2}(\mathbb{P}^{n-1} \setminus \mathcal{A}''_Q)(-1) = \mathrm{gr}_{2n-2}^W H^{n-2}(\mathbb{P}^{n-1} \setminus \mathcal{A}''_Q)$$

vanishes, as the complement is smooth of dimension $n-1$ and thus the $(n-2)$-th cohomology group can only have weights in the range $[n-2, 2n-4]$ by Table 2.1.1. Similarly,

$\operatorname{gr}_{2n}^W H^{n+1}(\mathbb{P}^n \setminus \mathcal{A}'_Q)$ vanishes, as $\mathbb{P}^n \setminus \mathcal{A}'_Q$ is a smooth affine variety, and thus Andreotti-Frankel's theorem 1.3.32 implies that already $H^{n+1}(\mathbb{P}^n \setminus \mathcal{A}'_Q) = 0$. Then, the claim follows from Poincaré duality (2.1.3). Note the shift in weight in $\operatorname{gr}_{2n-2}^W H^{n-1}(\mathbb{P}^{n-1} \setminus \mathcal{A}''_Q)$ to $\operatorname{gr}_0^W H_{n-1}(\mathbb{P}^{n-1}, \mathcal{A}''_Q)$. The proofs for the other two sequences are identical. □

Finally, we turn to the proof of the main result about the combinatorial rank of $(Q, \mathcal{A}|_Q)$.

Proof of Proposition 3.1.7. We proceed by induction on the number of hyperplanes m, the claim then follows from the split short exact sequence *ii.* in Proposition 3.1.12 with $k = n$. For all $m < n - 1$, we have $\deg \mathcal{A}_Q \leq n$. Then, it follows from [DD90, Theorem 1] and the short exact sequence *iii.* in Proposition 3.1.12, that $\operatorname{cr}(Q, \mathcal{A}|_Q) = 0$. Moreover, since $\mathcal{A}_Q$ is a generic spherical arrangement it is in particular normal crossing on Q, hence for $m = n - 1$ we get $\operatorname{cr}(Q, \mathcal{A}|_Q) = \operatorname{cr}(\mathbb{P}^n, \mathcal{A}_Q) = 1$, this in particular establishes the induction base. So, suppose the claim is true for $m > n - 1$. Then, by Proposition 3.1.12 *ii.* we get

$$\begin{aligned}\operatorname{cr}(Q, \mathcal{A}|_Q) &= \operatorname{cr}(Q, \mathcal{A}'|_Q) + \operatorname{cr}(Q'', \mathcal{A}''|_{Q''}) \\ &= \left(\binom{m-1}{n-1} + \binom{m-2}{n-1}\right) + \left(\binom{m-1}{n-2} + \binom{m-2}{n-2}\right) \\ &= \binom{m}{n-1} + \binom{m-1}{n-1},\end{aligned}$$

where we used Pascal's triangle identity twice for the equality in the last row. This shows the claim. □

The result about the combinatorial rank of $(\mathbb{P}^n, \mathcal{A}_Q)$ follows:

Proof of Corollary 3.1.8. Consider the split short exact sequence *iii.* in Corollary 3.1.12:

$$0 \to \operatorname{gr}_0^W H_k(\mathbb{P}^n, \mathcal{A}) \to \operatorname{gr}_0^W H_k(\mathbb{P}^n, \mathcal{A}_Q) \to \operatorname{gr}_0^W H_{k-1}(Q, \mathcal{A}|_Q) \to 0$$

for $k = n$ we obtain

$$\begin{aligned}\operatorname{cr}(\mathbb{P}^n, \mathcal{A}_Q) &= \operatorname{cr}(\mathbb{P}^n, \mathcal{A}) + \operatorname{cr}(Q, \mathcal{A}|_Q) \\ &= \#(\text{bounded regions in } (\mathbb{P}^n \setminus \mathcal{A})(\mathbb{R})) + \binom{m}{n-1} + \binom{m-1}{n-1},\end{aligned}$$

where we used [BD25, Proposition 6.2] for $\operatorname{cr}(\mathbb{P}^n, \mathcal{A})$, and Proposition 3.1.7 for $\operatorname{cr}(Q, \mathcal{A}|_Q)$. This finishes the proof. □

3.2 Number of Regions

In this section we show that the number of real regions in $Q \setminus \mathcal{A}|_Q$ has a nice combinatorial count. This will be helpful, in finding an explicit basis for homology in the succeeding section. The main result can be stated as follows, and its proof is again relegated to the end of this section.

Proposition 3.2.1. *Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement with m hyperplanes. Then, the number of connected real top-dimensional regions $r(\mathcal{A}|_Q)$ in $Q \setminus \mathcal{A}|_Q$ equals*

$$r(\mathcal{A}|_Q) = 2\sum_{i=0}^{n-1} \binom{m-1}{i} = \#\mathcal{T},$$

where $\#\mathcal{T}$ is the number of topes in the uniform oriented matroid of rank n on m elements.

Remark 3.2.2. First, note that here we see why the assumption of the signature of Q is essential, as the number of regions clearly depends on the geometry of the quadric. We will also see this explicitly in the proof of this result. Second, the assumption that the boundary stratification $\mathcal{L}(\mathcal{A}|_Q)$ is the full power poset is necessary if the number of regions should have a purely combinatorial count, and similarly if the combinatorial rank should not depend more on the geometric data.

We consider a concrete problematic example. Fix $n = 3$, let the quadric be given by the vanishing of $Q = -5x_0^2 + x_1^2 + x_2^2 + x_3^2$, and $\mathcal{A}$ be the union of the vanishing loci of $L_1 = x_3 - x_0$, $L_2 = x_3 + x_0$, and $L_3 = x_2$. Then, on $U_0 = \{x_0 \neq 0\} \subset \mathbb{P}^3$ we can clearly see all 6 (bounded) regions in $Q \setminus \mathcal{A}|_Q$ labeled by sign pattern, see Figure 3.2.1. In fact, we can see

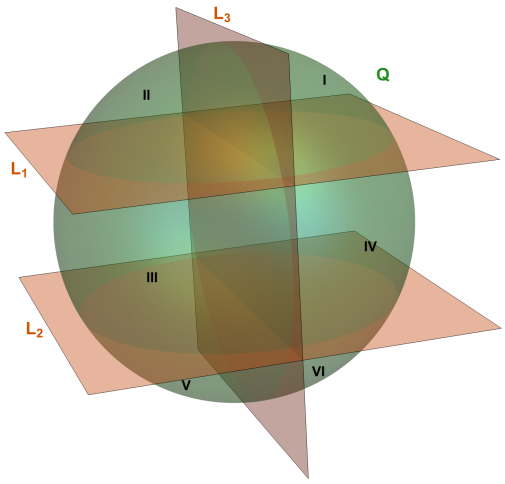


Figure 3.2.1: A non-generic spherical arrangement.

all real points of the sphere on this patch, as setting $Q = 0$ has no real roots on $x_0 = 0$. From the six bounded regions on the inside of the sphere, we get the following canonical forms in $\Omega^2_{\log}(\mathbb{P}^3 \setminus \mathcal{A}_Q)$ up to scaling

$$\omega_I = \frac{1}{L_1 L_3 \partial_1 Q} dx_2 \wedge dx_3 = -\omega_{II}$$
$$\omega_{III} = \frac{x_0}{L_1 L_2 L_3 \partial_1 Q} dx_2 \wedge dx_3 = -\omega_{IV}$$
$$\omega_V = \frac{1}{L_2 L_3 \partial_1 Q} dx_2 \wedge dx_3 = -\omega_{VI},$$

where $\partial_1 Q$ denotes the partial differential of Q with respect to the variable x_1. These 6 dif-

ferential forms span obviously a 3 dimensional space. However, as $\mathcal{A}_Q$ is a simple normal crossing divisor, $\mathrm{cr}(\mathbb{P}^3, \mathcal{A}_Q) = \binom{\deg \mathcal{A}_Q - 1}{3} = 4$ by Proposition 2.1.22. Moreover, then by Proposition 3.1.12 ii. we have $\mathrm{cr}(Q, \mathcal{A}|_Q) = \mathrm{cr}(\mathbb{P}^3, \mathcal{A}_Q) = 4$. Hence, the real regions in $(Q \setminus \mathcal{A}|_Q)(\mathbb{R})$ only generate a 3-dimensional sub vector space of the 4-dimensional space $\Omega^2_{\log}(Q \setminus \mathcal{A}|_Q)$.

We now prove the main result in this section.

Proof of Proposition 3.2.1. First note that the arrangement of hyperplane sections $\mathcal{A}|_Q$ on Q can be seen as induced by a central hyperplane arrangement, i.e. an arrangement of hyperplanes which all contain the origin, as follows. Let $\mathcal{A} \subset \mathbb{P}^n$ be the underlying hyperplane arrangement, then this makes a central arrangement $\mathcal{A}_c$ in $\mathbb{C}^{n+1}$ by taking the affine cone over $\mathcal{A}$. Similarly, denote by $CQ \subset \mathbb{C}^{n+1}$ the cone over Q. Hence, $\mathcal{A}|_Q$ is induced by $CQ \cap \mathcal{A}_c$, more precisely any affine slice is isomorphic to an affine patch of $\mathcal{A}|_Q$. Also note that all intersections of multiple hyperplane sections on Q are transversal as $\mathcal{A}|_Q$ is a simple normal crossing divisor by assumption. Then, by [Des14, p.9], the assumptions of [Des13, Theorem 3.6] are satisfied. Therefore, the number of connected top-dimensional real regions in $Q \setminus \mathcal{A}|_Q(\mathbb{R}) \subset \mathbb{R}^n$ with respect to a generic affine patch on Q can be computed as

$$r = (-1)^{n-1} \sum_{Y \in \mathcal{L}} (-1)^{n-d-1} \chi(Y), \tag{3.2.1}$$

where χ is the topological Euler characteristic, and we have used that $\mathcal{L} = \mathcal{L}(\mathcal{A}|_Q)$ is the power poset. Note that the real locus of any intersection $Y \in \mathcal{L}$ is a quadric of dimension d in Q and has signature $(d, 1)$, since Q had signature $(n, 1)$ to begin with. Then, $\chi(Y) = \chi(S^d)$, which vanishes whenever the dimension d is odd and equals 2 otherwise, see Example 1.3.5. Moreover, there are exactly $\binom{m}{n-d-1}$ many $Y \in \mathcal{L}$ of dimension d. Then (3.2.1) simplifies to

$$r = (-1)^{n-1} \sum_{d=0}^{n-1} (-1)^{n-1-d} \binom{m}{n-1-d} \chi(S^d) = \begin{cases} 2 \sum_{i=0}^{\frac{n-1}{2}} \binom{m}{2i} & , n-1 \text{ even}, \\ 2 \sum_{i=1}^{\frac{n}{2}} \binom{m}{2i-1} & , n-1 \text{ odd}. \end{cases}$$

Suppose $n-1$ is even. Rewriting the sums using Pascal's identity we get

$$\sum_{i=0}^{\frac{n-1}{2}} \binom{m}{2i} = \sum_{i \text{ even}}^{n-1} \binom{m}{i} = \sum_{i \text{ even}}^{n-1} \binom{m-1}{i-1} + \sum_{i \text{ even}}^{n-1} \binom{m-1}{i} = \sum_{i=0}^{n-1} \binom{m-1}{i}.$$

This shows the claim for $n-1$ even. For $n-1$ odd a similar computation shows the equality $\sum_{i=1}^{\frac{n}{2}} \binom{m}{2i-1} = \sum_{i=0}^{n-1} \binom{m-1}{i}$. This finishes the proof. □

3.3 Basis for the Space of Logarithmic Forms

In this section we present the main result of this chapter, that is the construction of a basis of the vector space of logarithmic top differential forms $\Omega^{n-1}_{\log}(Q \setminus \mathcal{A}|_Q)$, see Theorem 3.3.2. Equivalently, via the canonical map (2.1.4), this is a basis of the weight 0 part in relative homology $\mathrm{gr}^W_0 H_{n-1}(Q, \mathcal{A}|_Q)$. We also show how this lifts to a basis for $\Omega^n_{\log}(\mathbb{P}^n \setminus \mathcal{A}_Q)$, see Theorem 3.3.8.

To that end, we introduce some more notation. Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement with m hyperplanes. Let $\mathcal{T} = \mathcal{T}_{\mathcal{A}_Q}$ be the set of top-dimensional real regions in the complement $(Q \setminus \mathcal{A}|_Q)(\mathbb{R})$, with orientation induced from the orientation chosen on the ambient $\mathbb{R}^{n+1}$. We will also refer to elements in $\mathcal{T}$ as *topes*. So these, or rather their equivalence classes, are naturally elements in $H_{n-1}(Q, \mathcal{A}|_Q)$. By Proposition 3.2.1 elements in $\mathcal{T}$ are real regions in $Q \setminus \mathcal{A}|_Q(\mathbb{R})$ labeled by sign patterns $\tau \in \{+,-\}^m$. In an abuse of notation we write $\mathcal{T}$ also for the $\mathbb{Q}$-vector space $\mathbb{Q}\mathcal{T}$ of formal linear combinations of such regions. For $\tau \in \mathcal{T}$ write $\tau^+ = \{i \mid \mathrm{sign}(\tau_i) = +\}$. Fix $k \leq n-2$ and take a size k subset I of $[m]$, i.e. $I \in \binom{[m]}{k}$, then we write $\mathcal{T}^+_I \subset \mathcal{T}$ for the set of topes τ with $\tau^+ \subset I$. For any such I set $\ell_I = \sum_{\tau \in \mathcal{T}^+_I} \tau \in \mathcal{T}$.

Definition 3.3.1. For any non-negative $k \leq n-2$, let R_k be the subvector space of $\mathcal{T}$ spanned by all $\mathbb{Q}$-linear combinations ℓ_I, that is $R_k = \mathrm{span}_{\mathbb{Q}}\left(\ell_I \mid I \in \binom{[m]}{k}\right)$, with $R_0 = \sum_{\tau \in \mathcal{T}} \tau$. Then, denote by $\mathcal{Q}$ the $\mathbb{Q}$-vector space obtained by quotienting out $\bigoplus_{k=0}^{n-2} R_k$. Put in symbols,

$$\mathcal{Q} = \mathcal{T} / \bigoplus_{k=0}^{n-2} R_k.$$

Analogously for the deletions $\mathcal{A}'_Q$ and restrictions $\mathcal{A}''_Q$ write $\mathcal{Q}'$ and $\mathcal{Q}''$, respectively.

We can now state our first main theorem 3.0.1 more rigorously. Its proof is postponed to the end of this section.

Theorem 3.3.2. *Let $\mathcal{A}_Q \subset \mathbb{P}^n$ be a generic spherical arrangement. Then, a basis of the weight 0 middle relative homology $\mathrm{gr}^W_0 H_{n-1}(Q, \mathcal{A}|_Q)$ is given by the classes of real regions in $Q \setminus \mathcal{A}|_Q$ making a basis of $\mathcal{Q}$. Equivalently, a basis of $\Omega^{n-1}_{\log}(Q, \mathcal{A}|_Q)$ is given by the image of $\mathcal{Q}$ under the canonical map $\Omega : H_{n-1}(Q, \mathcal{A}|_Q) \to \Omega^{n-1}_{\log}(Q \setminus \mathcal{A}|_Q)$.*

Example 3.3.3. Consider an arrangement of 4 lines in the projective plane. Then, the restriction $\mathcal{A}|_Q \subset Q$ is isomorphic to an arrangement of 8 points on $\mathbb{P}^1$. This is isomorphic to the restriction of the example in Figure 3.0.1 to the conic Q. We know from Proposition 3.1.7 that $\mathrm{cr}(Q, \mathcal{A}|_Q) = \dim \Omega_{\log}(Q \setminus \mathcal{A}|_Q) = 7$. The number of topes $\tau \in \mathcal{T}$ is given by Proposition 3.2.1, which gives $\dim \mathcal{T} = 2\left(\binom{3}{0} + \binom{3}{1}\right) = 8$. Their sign pattern is induced by the choice of halfspaces for the L_i as in Figure 3.0.1. Explicitly, we get

$$\mathcal{T} = \mathrm{span}_{\mathbb{Q}}\{+-++, ++++, ++-+, ++--, -+--, ----, --+-, --++\}.$$

Note that these correspond in order to the restrictions of the regions $I, \dots, VIII$ in Figure 3.0.1 to the quadric Q. Next, since $n = 2$, we only quotient out by $R_0 = \mathbb{Q}\langle \sum_{\tau \in \mathcal{T}} \tau \rangle$. Hence, $\dim \mathcal{Q} = 7 = \dim \Omega^1_{\log}(Q \setminus \mathcal{A}|_Q)$, as predicted by Theorem 3.3.8. So in particular, $\Omega(\mathcal{Q}) \cong \Omega^1_{\log}(Q \setminus \mathcal{A}|_Q)$. Moreover, note that the generator for R_0 is precisely the only relation between the canonical forms of the 10 regions we found in Figure 3.0.1. ◇

Next, we establish the dimension of $\mathcal{Q}$.

Lemma 3.3.4. *The dimension of the $\mathbb{Q}$-vector space $\mathcal{Q}$ is given by*

$$\dim \mathcal{Q} = \binom{m}{n-1} + \binom{m-1}{n-1}.$$

Proof. Recall that $\mathcal{Q} = \mathcal{T} / \bigoplus_{k=0}^{n-2} R_k$. Note that as a set $\bigoplus_{k=0}^{n-2} R_k$ has $n_R = \sum_{k=0}^{n-2} \binom{m}{k}$ elements, and by Proposition 3.2.1 $\dim \mathcal{T} = 2 \sum_{k=0}^{n-1} \binom{m-1}{k}$. Fix a basis of $\mathcal{T}$ which orders the columns labeled by the topes $\tau \leq \tau' \in \{+, -\}^m$, with $\tau < \tau'$ if $\tau^+ \subset (\tau')^+$ breaking ties with lex order. Then, elements in $\bigoplus_{k=0}^{n-2} R_k$ can be expressed in this basis as it is a subvector space of $\mathcal{T}$. Then, we can write $\bigoplus_{k=0}^{n-2} R_k$ as the rowspan of a $n_R \times \dim \mathcal{T}$ matrix M with 0 and 1 entries. For convenience, the rows are labeled by tuples $\sigma \in \{0, +\}^m$, and ordering $\sigma < \sigma'$ for $\sigma^+ \subset (\sigma')^+$ breaking ties again by lex order. Moreover, the $n_R \times n_R$ submatrix of M corresponding to the first n_R columns is an upper triangular matrix with 1's on the diagonal, hence it is of full rank. Therefore, M has full rank n_R. Finally, using Pascal's identity and telescoping the sum we can compute

$$\begin{aligned}
\dim \mathcal{Q} &= 2 \sum_{k=0}^{n-1} \binom{m-1}{k} - \sum_{k=0}^{n-2} \binom{m}{k} \\
&= 2\binom{m-1}{n-1} + \sum_{k=0}^{n-2} \binom{m-1}{k} + \left(\sum_{k=0}^{n-2} \binom{m-1}{k} - \binom{m}{k} \right) \\
&= 2\binom{m-1}{n-1} + \sum_{k=0}^{n-2} \binom{m-1}{k} - \sum_{k=0}^{n-2} \binom{m-1}{k-1} \\
&= \binom{m-1}{n-1} + \binom{m-2}{n-1} + \binom{m-2}{n-2} + \left(\sum_{k=0}^{n-2} \binom{m-1}{k} - \binom{m-1}{k-1} \right) \\
&= \left(\binom{m-1}{n-1} + \binom{m-2}{n-2} \right) + \left(\binom{m-2}{n-1} + \binom{m-1}{n-2} \right) \\
&= \binom{m}{n-1} + \binom{m-1}{n-1},
\end{aligned}$$

finishing the proof. □

Example 3.3.5. Let $n = 3$ and $m = 3$, with $\mathcal{A} \subset \mathbb{P}^n$ given by the vanishing of the first 3 coordinates $x_0, \dots, x_2$, and let Q be given by $x_0^2 + x_1^2 + x_2^2 - x_3^2 = 0$. Then choose a basis of $\mathcal{T}$, with $\dim \mathcal{T} = 2(\binom{3}{0} + \binom{3}{1}) = 8$, ordered as described above

$$\mathcal{T} = \operatorname{span}_{\mathbb{Q}}\{---, +--, -+-, --+, ++-, +-+, -++, +++\}.$$

This then induces a basis for $\bigoplus_{k=0}^{1} R_k \subset \mathcal{T}$ which we index as

$$\bigoplus_{k=0}^{1} R_k = \operatorname{span}_{\mathbb{Q}} \{000, +00, 0+0, 00+\},$$

indicating for example that the second vector spanning this space can be written as the sum over all sign patterns with a $+$ at the first position. Then, the 4×8 matrix M constructed in the proof of Lemma 3.3.4 is given as

$$M = \begin{array}{c} \\ 000 \\ +00 \\ 0+0 \\ 00+ \end{array} \begin{array}{c} \begin{array}{cccccccc} --- & +-- & -+- & --+ & ++- & +-+ & -++ & +++ \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

We can see that the submatrix of M corresponding to the columns labeled by

$$\{---, +--, -+-, --+\}$$

is upper triangular, hence its determinant equals 1. Therefore, M is of full rank 4. ◇

To prove the main result 3.3.2, we need to introduce more notation. We define a chain map $\varphi : \mathcal{T}' \to \mathcal{T}$. Then, for $\sigma \in \mathcal{T}'$, a real region in $Q \setminus \mathcal{A}'_Q$, we set

$$\varphi(\sigma) = \begin{cases} \sigma & \text{if } \operatorname{int}\sigma \text{ does not meet } L_{\mathbb{R}}, \\ \sigma_+ + \sigma_- & \text{else,} \end{cases} \tag{3.3.1}$$

where σ_+ and σ_- are the intersections of σ with the positive and negative half-space in $Q \cap \mathbb{R}^n$ defined by $L_{\mathbb{R}}$, respectively. Note that $\varphi(\bigoplus_{k=0}^{n-2} R'_k) \subset \bigoplus_{k=0}^{n-2} R_k$ by definition of R_k. Hence, we get a well-defined chain map on the quotients $\varphi : \mathcal{Q}' \to \mathcal{Q}$. Similarly, we define $\psi : \mathcal{T} \to \mathcal{T}''$. Let $\sigma \in \mathcal{T}$, a real region in $Q \setminus \mathcal{A}_Q$, then

$$\psi(\sigma) = \begin{cases} 0 & \text{if } \sigma \text{ does not have } L_{\mathbb{R}} \text{ as a facet,} \\ \pm\sigma \cap L_{\mathbb{R}} & \text{else,} \end{cases}$$

where the sign is determined by the orientation chosen on $\mathbb{R}^{n-1} \setminus \mathcal{A}|_Q$. Again note, that $\psi(\bigoplus_{k=0}^{n-2} R_k) \subset \bigoplus_{k=0}^{n-3} R''_k$ by definition of R_k. Hence, we also get a well-defined chain map on the quotients $\psi : \mathcal{Q} \to \mathcal{Q}''$. These maps fit nicely into a short exact sequence.

Lemma 3.3.6. *There is a (split) short exact sequence of $\mathbb{Q}$-vector spaces*

$$0 \to \mathcal{Q}' \xrightarrow{\varphi} \mathcal{Q} \xrightarrow{\psi} \mathcal{Q}'' \to 0.$$

Proof. We start by showing that φ from (3.3.1) is injective. For $0 \neq \sigma \in \mathcal{T}'$ which does not have L as a boundary this is immediate. If σ does have a boundary on a hyperplane section $L \in \mathcal{A}|_Q$, then $\sigma \mapsto \sigma_+ + \sigma_- \in \mathcal{T}$. Suppose for the sake of contradiction, that $\varphi(\sigma)$ is zero modulo $\bigoplus_{k=0}^{n-2} R_k$. New linear relations in $\mathcal{Q}$ are obtained by the restriction of $\mathcal{A}_Q$ to L. However, by construction the restriction of $\varphi(\sigma) = \sigma_+ + \sigma_-$ to L vanishes trivially, as the

orientations of σ_+ and σ_- are opposing. Hence, $\varphi(\sigma) = 0 \in \mathcal{Q}$ implies $\sigma = 0 \in \mathcal{Q}'$, contrary to the assumption, thus $\ker \varphi$ is trivial. Next, we show that ψ is surjective. This follows from the fact that $\psi(\bigoplus_{k=0}^{m} R_k) = \bigoplus_{k=0}^{n-3} R''_k$ by construction, and because ψ is the identity for topes σ which have $L_{\mathbb{R}}$ as a facet.

It remains to show that $\operatorname{im} \varphi = \ker \psi$. The containment $\operatorname{im} \varphi \subset \ker \psi$ is clear, as

$$\psi(\varphi(\sigma)) = \psi(\sigma_+ + \sigma_-) = \sigma_+|_L - \sigma_-|_L = 0$$

for σ with boundary on $L|_Q$. For the other regions we have $\varphi(\sigma) = 0$ trivially. For the converse, $\operatorname{im} \varphi \supset \ker \psi$, we have $\psi(\bigoplus_{k=0}^{n-2} R_k) = \bigoplus_{k=0}^{n-3} R''_k$ implies that for $0 \neq \sigma \in \mathcal{Q}$, we have $\psi(\sigma) = 0 \in \mathcal{T}''/\bigoplus_{k=0}^{n-3} R''_k$ if and only if $\psi(\sigma) = 0 \in \mathcal{T}''$. Additionally, $\psi(\sigma) = \pm\sigma \cap L_{\mathbb{R}} = 0$ if and only if σ does not have a facet supported on L. These σ are trivially in the image of φ. Hence, $\ker \psi = \operatorname{im} \varphi$, and the sequence is exact. This finishes the proof. □

Proof of Theorem 3.3.2. The proof proceeds by an induction on the number of hyperplanes m in the arrangement $\mathcal{A} \subset \mathbb{P}^n$. The base case is $m = n-1$. Since $\mathcal{A}_Q \subset \mathbb{P}^n$ is a generic spherical arrangement, so in particular simple normal crossing on Q, we have $\operatorname{cr}(Q, \mathcal{A}|_Q) = \binom{m}{n-1} = 1$. Similarly, $\dim \mathcal{Q} = 1$ by Lemma 3.3.4. This shows the induction base case.

Now, suppose the claim is true for arbitrary but fixed $m > n-1$. Consider the following diagram of complexes, where Q'' is the smooth quadric of dimension $n-2$ obtained by intersection of Q with a hyperplane in $\mathcal{A}|_Q$.

$$\begin{array}{ccccccccc}
0 & \longrightarrow & \operatorname{gr}_0^W H_{n-1}(Q, \mathcal{A}'|_Q) & \xrightarrow{\iota_*} & \operatorname{gr}_0^W H_{n-1}(Q, \mathcal{A}|_Q) & \xrightarrow{\partial_m} & \operatorname{gr}_0^W H_{n-2}(Q'', \mathcal{A}''|_{Q''}) & \longrightarrow & 0 \\
 & & \sim\uparrow & & \sim\uparrow & & \sim\uparrow & & \\
0 & \longrightarrow & \mathcal{Q}' & \xrightarrow{\varphi} & \mathcal{Q} & \xrightarrow{\psi} & \mathcal{Q}'' & \longrightarrow & 0
\end{array}$$

Here, ι_* denotes the pushforward for the inclusion $\iota : (Q, \mathcal{A}|'_Q) \hookrightarrow (Q, \mathcal{A}|_Q)$. The first row is a short exact sequence by Corollary 3.1.12, and the second one by Lemma 3.3.6. Fix the hyperplane $L = L_m$ which is used for the restriction and deletion. Note that $\iota_*(\sigma) = \sigma_+ + \sigma_-$, for a cycle with boundary on $L|_Q$, and $\iota_*(\sigma) = \sigma$ for one without. Thus, the left square in the diagram above commutes. Similarly, $\partial_m(\sigma) = 0$ for a cycle σ without boundary on $L|_Q$, and $\partial_m(\sigma) = \pm\sigma|_L$, with the sign being determined by the choice of orientation on $\mathbb{R}^n \setminus (\mathcal{A}_Q)|_{\mathbb{R}}$ otherwise. If this choice of orientation is consistent with the one made in the definition of ψ, then also the right square in the diagram above commutes. The claim then follows by the 5-Lemma, and the results on the vector space dimensions, for $\operatorname{gr}_0^W H_{n-1}(Q, \mathcal{A}|_Q)$ and $\mathcal{Q}$, in Proposition 3.1.7 and Lemma 3.3.4, respectively. □

It remains to extend the result of Theorem 3.3.2 to obtain a basis for $\operatorname{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}_Q)$. Let $\iota : (\mathbb{P}^n, \mathcal{A}) \hookrightarrow (\mathbb{P}^n, \mathcal{A}_Q)$ be the natural inclusion of the pairs, then its pushforward induces a map $\iota_* : H_n(\mathbb{P}^n, \mathcal{A}) \to H_n(\mathbb{P}^n, \mathcal{A}_Q)$. Any real region $\sigma' \in \mathbb{P}^n \setminus \mathcal{A}_Q$ can be identified with a sign pattern in $\{+, -\}^{m+1}$, where the $(m+1)$-th entry corresponds to sign of the region with

respect to Q. This allows us to give an explicit section for the map $\partial_Q : \mathrm{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}_Q) \to \mathrm{gr}_0^W H_{n-1}(Q, \mathcal{A}|_Q)$, which exists by Proposition 3.1.12.

Lemma 3.3.7. *Let $\gamma : H_{n-1}(Q, \mathcal{A}|_Q) \to H_n(\mathbb{P}^n, \mathcal{A}_Q)$ be the map sending $[\sigma] \mapsto [\sigma']$ where $\sigma' = (\sigma \mid +) \in \{+,-\}^{m+1}$ is a region in $\mathbb{P}^n \setminus \mathcal{A}_Q$ indexed by its sign pattern. Then, $\gamma \circ \partial_Q = \mathrm{id}$, i.e. γ is a section of ∂_Q.*

Proof. First, note that γ is a well-defined map, since all real regions $\sigma \in Q \setminus \mathcal{A}|_Q$ are obtained by restricting $\mathbb{P}^n \setminus \mathcal{A}_Q$ to Q, and there is a unique region $\sigma' \in \mathbb{P}^n \setminus \mathcal{A}_Q$, whose sign pattern has $\sigma'_{m+1} = +$. Next, note that for σ' with a boundary on L_m, we have

$$\partial_Q \circ \gamma([\sigma']) = \gamma(\pm[\sigma'|_L]) = \pm[\sigma],$$

that is to say $\partial_Q \circ \gamma = \mathrm{id}$. Hence, γ is a section for $\partial_Q : H_n(\mathbb{P}^n, \mathcal{A}_Q) \to H_{n-1}(Q, \mathcal{A}|_Q)$. □

With this section γ and the pushforward ι_*, we get a basis in weight zero homology of the pair $(\mathbb{P}^n, \mathcal{A}_Q)$ as follows, which is a reformulation of the main theorem 3.0.2.

Theorem 3.3.8. *Let $B_{\mathcal{A}}$ be the homology classes of the real bounded regions in $\mathbb{R}^n \setminus \mathcal{A}$, and let B_Q be a basis of $\mathcal{Q}$ from Theorem 3.3.2. Denote by B the union of $\iota_*(B_{\mathcal{A}})$ and $\gamma(B_Q)$. Then, B is a basis of the weight 0 piece of the middle relative homology group $\mathrm{gr}_0^W H_n(\mathbb{P}^n, \mathcal{A}_Q)$. Equivalently, the image of B under the canonical map $\Omega : H_n(\mathbb{P}^n, \mathcal{A}_Q) \to \Omega^n_{\log}(\mathbb{P}^n \setminus \mathcal{A}_Q)$ is a basis for $\Omega^n_{\log}(\mathbb{P}^n \setminus \mathcal{A}_Q)$.*

Proof. First, recall that $B_{\mathcal{A}}$ gives a basis of $H_n(\mathbb{P}^n, \mathcal{A})$, see e.g. [BD25, Proposition 6.2], and $\mathcal{Q}$ is a basis of $H_{n-1}(Q, \mathcal{A}|_Q)$ by Theorem 3.3.2. Next, by Lemma 3.3.7 γ is a section of ∂_Q, hence the claim follows from the exact sequence *iii.* in Proposition 3.1.12. □

Example 3.3.9. We continue Example 3.0.3. First recall that $\dim \Omega^2_{\log}(\mathbb{P}^2 \setminus \mathcal{A}_Q) = 9$. We saw in Example 3.3.3, that we get 7 linearly independent homology classes indexed by sign patterns for the choice of orientation of the halfspaces defined by $L_1, \dots, L_4$ as in Figure 3.0.1. Explicitly, we got a basis of $\mathcal{Q} \cong H_1(Q, \mathcal{A}|_Q)$ as

$$\mathcal{Q} = \mathrm{span}_{\mathbb{Q}}(+-{+}{+}, +{+}{+}{+}, +{+}-{+}, +{+}--, -+--, ----, --+-).$$

Now, this lifts to generators of a basis for $H_2(\mathbb{P}^2, \mathcal{A}_Q)$ via the section $\gamma : H_1(Q, \mathcal{A}|_Q) \to H_2(\mathbb{P}^2, \mathcal{A}_Q)$ as

$$\gamma(\mathcal{Q}) = \mathrm{span}_{\mathbb{Q}}(+{-}{+}{+}|+, +{+}{+}{+}|+, +{+}{+}-|+, +{+}--|+, -+--|+, ----|+, --+-|+),$$

where the $|$ separates the signs of the half spaces induced by the L_i, from the sign with respect to the quadric Q. Denote these generators of $\mathcal{Q}$ by $\sigma_\bullet$ for $\bullet = I, \dots, VII$. Note that the image under the canonical map of these generators are precisely the canonical forms $\omega_\bullet$ of the regions $I, \dots, VII$ in Figure 3.0.1. More precisely, we have $\Omega(\gamma(\sigma_\bullet)) = \omega_\bullet$ for $\bullet = I, \dots, VII$ with respect to Table 3.0.1. The remaining two generators of the basis B of $H_2(\mathbb{P}^2, \mathcal{A}_Q)$ are obtained as images of the two bounded regions IX and X in Figure 3.0.1 via

$$\iota_*(+--{-}) = (+---|+) \quad \text{and } \iota_*(+-{+}-) = (+-{+}-|+).$$

Their images under the canonical map are the differential forms ω_{IX} and ω_X. Moreover, we showed in the example that these 9 differential forms are linearly independent, hence we found a basis of $\Omega^2_{\log}(\mathbb{P}^2 \setminus \mathcal{A}_Q)$ and equivalently for $H_2(\mathbb{P}^2, \mathcal{A}_Q) = \mathrm{gr}_0^W H_n(\mathbb{P}^2, \mathcal{A}_Q)$ by Proposition 3.1.10. ◇

Future Directions

In the last sections we have shown how the assumptions of a generic spherical arrangement $\mathcal{A}_Q$ lead to nice answers to the questions about genus, combinatorial rank, number of real regions, and basis in homology. We also demonstrated how most of these assumptions are necessary in order to obtain combinatorial results. However, a remaining direction is to investigate if we can relax the assumptions to allow for situations where more than codimension many hyperplanes meet on the quadric.

Consider for example the situation of Example 2.2.15. We had three generic lines in the plane, and a conic, such that 2 lines meet the conic in a common point. For such an arrangement we computed $\mathrm{cr}(\mathbb{P}^2, \mathcal{A}_Q) = 5$, which is strictly smaller than the combinatorial rank $1 + \binom{3}{1} + \binom{2}{1} = 6$ computed as in Corollary 3.1.8. However, it is easy to see that the 6 bounded regions σ_i for $i \in [6]$, seen in Figure 2.2.1a, obey the same relation $R_0 = \sigma_1 + \ldots + \sigma_6$ as before. The same observations hold true in examples in higher dimension.

Example 3.3.10. Let $n = 3$. We want to consider an arrangement of 3 planes $\mathcal{A} \subset \mathbb{P}^3$ which meets the sphere $Q = -x_0^2 + x_1^2 + x_2^2 + x_3^2$ not in a simple normal crossing divisor. Explicitly, let $L_1 = x_1$, $L_2 = x_3$ and $L_3 = x_1 + x_3$. Then, $L_1 \cap L_2 \cap L_3 \cap Q \neq \varnothing$ is not simple normal crossing in $[1 : 0 : 1 : 0]$. There are 6 bounded regions, but their canonical forms ω_i only span a 2 dimensional vector space. More explicitly, for a choice of sign for the half-spaces defined by the planes L_i we can also write down the vector space of linear relations between the canonical forms as the rowspan of the following matrix:

$$M = \begin{array}{c} \\ 000 \\ +00 \\ 0{+}0 \\ 00{+} \end{array} \begin{array}{c} \begin{array}{cccccc} {-}{-}{-} & {+}{-}{-} & {-}{+}{-} & {+}{-}{+} & {-}{+}{+} & {+}{+}{+} \end{array} \\ \begin{bmatrix} 1 & 1 & 1 & 1 & 1 & 1 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 \end{bmatrix} \end{array}$$

Note that this matrix has full rank 4, and it is a submatrix of the matrix in Example 3.3.5. In particular, the linear relations between the differential forms ω_i are the same as those given by $\bigoplus_{k=0}^{n-2} R_k$. Thus, $\mathrm{cr}(\mathbb{P}^3, \mathcal{A}_Q) = 2$. ◇

Therefore, we conjecture that the answers to the question in this Chapter are still combinatorial if we allow for some non-combinatorial normal crossing situations in our divisor $\mathcal{A}_Q$. In particular, combinatorial rank, number of real regions and a basis in homology could be obtained from the intersection poset $\mathcal{L}(\mathcal{A}_Q)$.

Another natural research direction is to understand the algebra structure on $H_n(\mathbb{P}^n, \mathcal{A}_Q)$ better. Let $H^\bullet(\mathbb{P}^n \setminus \mathcal{A}_Q)$ be the cohomology algebra over $\mathbb{C}$. Mixed Hodge theory identifies $H^1(\mathbb{P}^n \setminus \mathcal{A}_Q)$ with the space of logarithmic 1-forms $\Omega^1_{\log}(\mathbb{P}^n \setminus \mathcal{A}_Q)$. Then, for hyperplane arrangements, the ring $H^\bullet(\mathbb{P}^n \setminus \mathcal{A})$ is generated by the obvious one-forms $\mathrm{dlog}(L_j)$ for all $L_j \in \mathcal{A}$. This is not the case for our arrangements $\mathcal{A}_Q$ as the following simple example demonstrates.

Example 3.3.11. Let $\mathcal{A}$ be given by the vanishing of x and y in $\mathbb{P}^2$, and let $q = x^2 + y^2 - z^2$ be the unit circle. Then, by Proposition 3.1.10, we get

$$\mathrm{gr}_0^W H^2(\mathbb{P}^2 \setminus \mathcal{A}_Q) = H^2(\mathbb{P}^2 \setminus \mathcal{A}_Q) = \langle \omega_I, \omega_{II}, \omega_{III} \rangle,$$

with $\omega_\bullet = f_\bullet \, dx \wedge dy$ being the canonical forms of the slices in clockwise order starting in the positive orthant. Explicitly,

$$f_I = \frac{x+y-1}{xyq}, \quad f_{II} = \frac{x-y+1}{xyq}, \quad \text{and } f_{III} = \frac{x+y+1}{xyq}.$$

However, one finds

$$H^1(\mathbb{P}^2 \setminus \mathcal{A}_Q) = \left\langle \mathrm{dlog}\left(\frac{x}{y}\right), \mathrm{dlog}\left(\frac{x^2}{q}\right) \right\rangle.$$

Then, $\wedge^2 H^1(\mathbb{P}^2 \setminus \mathcal{A}_Q) \subsetneq H^2(\mathbb{P}^2 \setminus \mathcal{A}_Q)$ must be one-dimensional by the wedge construction, and it is generated by

$$\mathrm{dlog}\left(\frac{x}{y}\right) \wedge \mathrm{dlog}\left(\frac{x^2}{q}\right) = \frac{1}{xyq} dx \wedge dy = \frac{1}{2}\left(\omega_{III} - \omega_I\right).$$

Now, by Andreotti-Frankel's theorem $H^k(\mathbb{P}^2 \setminus \mathcal{A}_Q) = 0$ for $k > 2$, as $(\mathbb{P}^2 \setminus \mathcal{A}_Q)$ is a smooth affine variety. Therefore, $H^\bullet(\mathbb{P}^2, \mathcal{A}_Q)$ is generated in degree 1 and 2. ◇

It is then a natural question to ask if this holds for generic spherical arrangements of any dimension.

Chapter 4

Nonzero Genus Pairs from Amplituhedra

At the heart of any theorem is a good example.

Dmitrii

The amplituhedron $\mathcal{A}_{n,k,m}(Z)$, as introduced in Section 1.4, is the object which started the emerging branch of mathematics that is Positive Geometries. Since then, it has been a central conjecture, see [ABL17], in this field, that the amplituhedron is a positive geometry for all values of n, k, m and generic choices of Z. Despite a vast amount of mathematical work on the amplituhedron that appeared in recent years (see e.g. [PSW23; ŁPW23; Par+24; Lam24] for $m = 2$ and [ELT25; Eve+23] for $m = 4$), this conjecture is still wide open. The only settled cases are

i. $k = 1$: the amplituhedron is a cyclic polytope, see [Stu88], [Lam22, Theorem 1];
ii. $k + m = n$: the amplituhedron is isomorphic to $\mathrm{Gr}(k, n)_{\geq 0}$ [Lam22, Theorem 9];
iii. $k = m = 2$: see [RST24, Theorem 1.3].

To take a first step towards settling this conjecture, the objective in this chapter will be to study the amplituhedron from the viewpoint of positive geometries in the mixed Hodge theory setting of Chapter 2.

For the cases in which the amplituhedron is known to be a positive geometry, in Section 4.1 we give a very short proof showing that the pair $(\mathrm{Gr}(k, k + m), \partial_a \mathcal{A}_{n,k,m}(Z))$ makes a recursive positive pair as in Definition 2.4.7, see Theorem 4.1.2. Here $\partial_a \mathcal{A}_{n,k,m}(Z)$ denotes the algebraic boundary of the amplituhedron. Despite this, for $k \geq 3$ and large enough n we discover that this pair has a non-zero genus (Theorems 4.2.1 and 4.3.1), which naively suggests that the amplituhedron is not a positive geometry. However, in view of the central example constructed in Section 4.4, we note that positivity of the genus does not immediately disqualify amplituhedra from being positive geometries in the sense of Definition 1.5.1. This raises the question of how to make the framework of [BD25] match that of [ABL17]. We

comment on a possible approach to this, suggested to us by Clément Dupont, at the end of Section 4.4.

The remainder of this chapter is organized as follows. In Section 4.1 we show that for the cases $k = 1$, $k + m = n$ and $k = m = 2$, we have $g(\mathrm{Gr}(k, k + m), \partial_a \mathcal{A}_{n,k,m}(Z)) = 0$, and even more, these pairs make recursive positive pairs for a generic choice of the matrix Z. In Section 4.2 we prove that for $m = 2$ and $n \geq 2(2k - 1)$ the pair above has a non-zero genus. Section 4.3 is devoted to an analogous statement for $m = 4$. Finally, in Section 4.4 we construct an example of a semi-algebraic set $P \subset \mathbb{P}^3$ which is a positive geometry in the sense of Definition 1.5.1, but such that the pair $(\mathbb{P}^3, \partial_a P)$ has genus one. We will also comment on how these pairs give rise to a positive geometry in the mixed Hodge theory definition, see Definition 2.4.7.

4.1 Amplituhedra that make Positive Pairs

In this section we consider specific choices of the parameters n, k, and m for which the amplituhedron $\mathcal{A}_{n,k,m}$ is known to be a positive geometry in the sense of Definition 1.5.1. We show that for these choices the pair $(\mathrm{Gr}(k, k + m), \partial_a \mathcal{A}_{n,k,m})$ has genus zero, and thus fits nicely into the framework introduced in Chapter 2. Moreover, we also establish in which case these pairs are even (recursive) positive pairs, see Definition 2.4.7. We collect the results in the following theorem.

Theorem 4.1.1. *Let Z be a generic $n \times (k + m)$ matrix with positive maximal minors. Then, for any choice of integers k, m, and $n \geq k + m$, satisfying either $k = 1$, $k + m = n$, or $k = m = 2$, the boundary divisor $\partial_a \mathcal{A}_{n,k,m} \subset \mathrm{Gr}(k, k + m)$ of the amplituhedron together with the ambient Grassmannian has vanishing genus. That is,*

$$g(\mathrm{Gr}(k, k + m), \partial_a \mathcal{A}_{n,k,m}) = 0$$

for the choices of k, n and m as above. In fact, the pair $(\mathrm{Gr}(k, k+m), \partial_a \mathcal{A}_{n,k,m})$ is a recursive positive pair.

We first prove the claim for the amplituhedron $\mathcal{A}_{n,2,2}$.

Theorem 4.1.2. *The $m = k = 2$ amplituhedron $\mathcal{A}_{n,2,2}$ together with the ambient Grassmannian $\mathrm{Gr}(2, 4)$ is a recursive positive pair.*

Proof. The full boundary stratification of $\partial_a \mathcal{A}_{n,2,2}$ is given in [RST24, Table 1]. Note that only the irreducible components of $\partial_a \mathcal{A}_{n,2,2}$ are singular, and those singularities are precisely the 0-dimensional strata in *loc. cit.* Then, $(\mathrm{Gr}(2, 4), \partial_a \mathcal{A}_{n,2,2})$ makes a recursive genus zero pair by Corollary 2.4.3. The canonical form is non-zero and was computed in [RST24, Theorem 5.2]. Since $\mathrm{gr}_0^W H_d(\mathrm{Gr}(2, 4), \partial_a \mathcal{A}_{n,2,2}) \cong \Omega^d_{\log}((\mathrm{Gr}(2, 4) \setminus \partial_a \mathcal{A}_{n,2,2}))$ for genus zero pairs, and since the amplituhedron is a real cycle the claim follows. □

We collect these statements to prove the main theorem of this section.

Proof of Theorem 4.1.1. For $k = 1$, the amplituhedron $\mathcal{A}_{n,1,m} \subset \mathrm{Gr}(1, k+m) \cong \mathbb{P}^{k+m-1}$ is a cyclic polytope, see [Lam22, Theorem 1]. In this case all claims follow from [BD25, Section 6]. For the case $k+m = n$, the map $\tilde{Z} : \mathrm{Gr}(k,n)_{\geq 0} \to \mathrm{Gr}(k,n)$ is a linear isomorphism and thus the amplituhedron is isomorphic to the non-negative Grassmannian $\mathrm{Gr}(k,n)_{\geq 0}$. Therefore, the claims follow from Proposition 2.4.12. Lastly, the claim about the genus for the case of $k = m = 2$ is covered in Theorem 4.1.2. □

4.2 The $\mathrm{m} = 2$ Amplituhedron has non-zero Genus

Let $m = 2$, $k \geq 3$ and $n \geq 2(2k-1)$. The main goal of this section is to prove that the algebraic boundary $\partial_a \mathcal{A}_{n,k,2}$ and the Grassmannian $\mathrm{Gr}(k, k+2)$ make a pair with non-zero genus. To that end, let the $n \times (k+2)$ matrix Z with positive maximal minors, which defines the amplituhedron $\mathcal{A}_{n,k,2} = \mathcal{A}_{n,k,2}(Z)$, be generic. Here, generic means we pick Z in the open subset of matrices of this size, such that Kleiman's transversality theorem [EH16, Theorem 1.7] applies in the situations appearing later in this section, see Lemmas 4.2.7 and 4.2.8. The main result of this section is then the following, whose proof we postpone until the end of this section.

Theorem 4.2.1. *Let n and k be positive integers such that $n \geq 2(2k-1)$. Further let Z be generic and have positive maximal minors, and $\partial_a \mathcal{A}_{n,k,2}(Z) \subset \mathrm{Gr}(k, k+2)$ be the algebraic boundary of the amplituhedron. Then, the genus of the pair $(\mathrm{Gr}(k,k+2), \partial_a \mathcal{A}_{n,k,2}(Z))$ is at least $1 + \frac{k-3}{2} C_k$, where C_k is the k-th Catalan number. In particular, for $k \geq 3$ it is positive.*

Remark 4.2.2. The lower bound from Theorem 4.2.1 is not tight. The number $1 + \frac{k-3}{2} C_k$ is the genus of a single curve in the residual arrangement of the amplituhedron whose cohomology injects into the relative cohomology of the pair $(\mathrm{Gr}(k,k+2), \partial_a \mathcal{A}_{n,k,2}(Z))$. The residual arrangement in general contains multiple curves of positive genus. Moreover, one could in principle have non-zero contributions to the genus from higher-dimensional components of the residual arrangement. However, since our primary goal is to show that the genus of the pair is non-zero rather than to compute it exactly, we do not aim to improve this bound.

First, let us make a note of the strategy we are going to employ in this section. Since the Grassmannian is a smooth projective variety, we have $g(\mathrm{Gr}(k,k+2), \partial_a \mathcal{A}_{n,k,2}) = g(\partial_a \mathcal{A}_{n,k,2})$. Therefore, to compute the genus of the pair, we need to understand the algebraic boundary of the amplituhedron. Moreover, the boundary is reducible and hence its cohomology is essentially determined by multiple intersections of its components. Therefore, we start by establishing a series of results that will help us compute the cohomology groups of intersections of boundary divisors of the amplituhedron. First, we show that dense subsets of global complete intersections inside the Grassmannian are local complete intersections.

Lemma 4.2.3. *Let H be a (not necessarily smooth) global complete intersection in $\mathrm{Gr}(k,n)$. Then, any open subvariety $Y \subset H$ obtained as the complement of a divisor on H is a local complete intersection.*

Proof. By assumption the coordinate ring $\mathbb{C}[H]$, as a finite-dimensional $\mathbb{C}$-algebra, is a complete intersection. Any localization $\mathbb{C}[H]_g$ for $g \in \mathbb{C}[H]$ is a complete intersection by [Sta26, Lemma 10.135.2], so in particular a local complete intersection. By assumption Y is the complement of a divisor in H, therefore we can find $f \in \mathbb{C}[\mathrm{Gr}(k,n)]$ such that f is not identically zero on H and $\mathbb{V}(f) \cap H = Y$. Then, $\mathrm{Spec}(\mathbb{C}[H]_f)_0 \cong Y$ is a local complete intersection. □

The following is a direct consequence of the Lefschetz type theorem [PP24, Theorem B].

Lemma 4.2.4. *Let X be a (not necessarily smooth) global complete intersection of hyperplane sections in the Grassmannian $\mathrm{Gr}(k,n)$. Then, the ℓ-th cohomology group $H^\ell(X)$ is isomorphic to $H^\ell(\mathrm{Gr}(k,n))$ for all $\ell \leq \dim X - 1$. In particular, the Hodge structure on $H^\ell(X)$ is pure of Tate type of weight ℓ for all $\ell \leq \dim X - 1$.*

Proof. Since X is a global complete intersection of linear spaces in $\mathrm{Gr}(k,n)$, say of codimension r, we can find hyperplane sections $H_1, \ldots, H_r \subset \mathrm{Gr}(k,n)$ such that $X = H_1 \cap \ldots \cap H_r$. We also write $D_{[j]} = \bigcap_{i \in [j]} H_i$ for any $j \in [r]$. We prove the claim by induction on the codimension r of X. The case $r = 0$ is tautological. If $r = 1$, we know that $\mathrm{Gr}(k,n) \setminus X$ is smooth, and the claim follows by the Lefschetz hyperplane theorem, see Theorem 1.3.33.

Now let r be arbitrary, and $X = H_{[r]}$. So suppose $H^\ell(D_{[r-1]}) \cong H^\ell(\mathrm{Gr}(k,n))$ for $\ell < d-r+1$, where $d = k(n-k)$. Next, note that $X \subset D_{[r-1]} \subset \mathrm{Gr}(k,n)$ is a very ample effective Cartier divisor, since the Grassmannian $\mathrm{Gr}(k,n)$ has an embedding into projective space - the usual Plücker embedding. Moreover, since $D_{[r-1]}$ is a global complete intersection by definition, it follows from Lemma 4.2.3 that the complement $U = D_{[r-1]} \setminus X$ is a local complete intersection, in particular U has vanishing local cohomology defect by [PP24, Example 2.5]. Then, [PP24, Theorem B] applies, showing that $H^\ell(X) \cong H^\ell(H_{[r-1]}) \cong H^\ell(\mathrm{Gr}(k,n))$ for all $\ell < \dim X = k(n-k) - r$, finishing the proof. □

As a consequence, we obtain a result about Hodge structures on cohomology of complete intersections inside the Grassmannian.

Corollary 4.2.5. *Let X be a (not necessarily smooth) d-dimensional global complete intersection of hyperplane sections in the Grassmannian $\mathrm{Gr}(k,n)$. Further, let r be a positive integer and $D = D_1 \cup \ldots \cup D_r$ be a union of r Schubert hyperplane sections D_i in $\mathrm{Gr}(k,n)$. Then, the ℓ-th cohomology group $H^\ell(X \cap D)$ is isomorphic to $H^\ell(\mathrm{Gr}(k,n))$ for all $\ell \leq d-2$, in particular its Hodge structure is pure of weight ℓ and Tate type.*

Proof. By assumption X is a global complete intersection, hence it follows from Lemma 4.2.3, that $U = X \setminus (D \cap X)$ is a local complete intersection. In particular, its local cohomology defect vanishes by [PP24, Example 2.5]. Next, the restriction of the twisted line

bundle to X, i.e. $\mathcal{O}_{\mathrm{Gr}(k,k+2)}(r)|_X$ is ample. Then, $X \cap D$ is a section of this line bundle and as such ample itself; therefore $X \cap D$ is an ample effective Cartier divisor in X. Then, $H^\ell(X \cap D) \cong H^\ell(X)$ for all $\ell \leq d-2$ by [PP24, Theorem B]. The cohomology of $H^\ell(X)$ is isomorphic to $H^\ell(\mathrm{Gr}(k,n))$, as computed in Lemma 4.2.4, finishing the proof. □

Next, we show that, in the same setting, we get a very useful lower bound for some Hodge numbers in the Hodge-filtered parts of cohomology groups in the Mayer—Vietoris sequence.

Corollary 4.2.6. *Let X be a (not necessarily smooth) d-dimensional global complete intersection of hyperplane sections in the Grassmannian $\mathrm{Gr}(k,n)$. Further, let $D_1, \ldots, D_r$ be Schubert hyperplane sections in $\mathrm{Gr}(k,n)$, and let $D = X \cap D_1$, $R = X \cap (D_2 \cup \ldots \cup D_r)$ be the intersection of X with such a Schubert divisor, and a union of Schubert divisors, respectively. Then, we get an injection in the Mayer—Vietoris long exact sequence in cohomology*

$$H^{\ell-1}(D \cap R)^{p,0} \hookrightarrow H^\ell(D \cup R)^{p,0}$$

for all $\ell \leq d-1$ and $p > 0$. In particular, $h^{p,0}(H^{d-2}(D \cap R)) \leq h^{p,0}(H^{d-1}(D \cup R))$.

Proof. We consider the Mayer—Vietoris long exact sequence in cohomology for the union $D \cup R$, see (1.3.1):

$$\cdots \to H^{\ell-1}(D) \oplus H^{\ell-1}(R) \to H^{\ell-1}(D \cap R) \to H^\ell(D \cup R) \to H^\ell(D) \oplus H^\ell(R) \to \cdots .$$

The term $H^{\ell-1}(D) \oplus H^{\ell-1}(R)$, with $D = X \cap D_1$ and $R = X \cap (D_2 \cap \ldots \cap D_r)$, admits a pure Hodge structure of Tate type for all $\ell \leq d-1$ by Corollary 4.2.5. In particular, the Hodge numbers $h^{p,0}$ vanish for all $p > 0$. Therefore, the claim follows since the Mayer—Vietoris sequence is a long exact sequence of mixed Hodge structures, see [Dim92, p. C.30]. □

We fix some notation for the remainder of this section. For $i \in [n]$, let D_i be the irreducible components of the algebraic boundary $\partial_a \mathcal{A}_{n,k,2}$. In particular, D_i denotes the Chow hypersurface $\mathrm{CH}(L_i)$ of the line $L_i \subset \mathbb{C}^{k+2}$ spanned by the rows Z_i and Z_{i+1} of the matrix Z. Further, for $I \subset [n]$ we write D_I for the intersection $\bigcap_{i \in I} D_i$. Lastly, for any odd $j \in [n]$, we write $D_{([j])}$ for the intersection of D_i for all odd integers $i \in [j]$.

Lemma 4.2.7. *Let I be a subset of $[n]$ such that $i \in I \Rightarrow i+1 \notin I$. Then, the intersection $D_I = \bigcap_{i \in I} D_i$ is a global complete transversal intersection in $\mathrm{Gr}(k, k+2)$. Geometrically speaking, for a set of pairwise skew lines L_i the intersection of the corresponding Schubert divisors D_i is transversal and a global complete intersection. In particular, $D_{([j])}$ is a transversal and global complete intersection for any odd $j \in [n]$.*

Proof. For the Grassmannian $\mathrm{Gr}(k, k+2)$ the general linear group $\mathrm{GL}_{\mathbb{C}}(k+2)$ acts transversely on it as an algebraic group. By our choice of I, and since Z is generic, the lines L_i are in general position. Therefore, Kleiman's theorem, [EH16, Theorem 1.3], applies, showing that the intersection $\bigcap_{i \in I} D_i$ is generically transversal. This shows the first part of the statement. Lastly, notice that if we consider $D_{([j])}$, then in particular the lines L_i for $i \in ([j])$

are skew, and thus in general position, so this is a special case of the above. This finishes the proof. □

Lemma 4.2.8. *Let $k \geq 2$ and $n \geq 2(2k-1)$. Take $I \subset [n]$, with $|I| = 2k-1$, and such that the lines $L_i = \operatorname{span}(Z_i, Z_{i+1})$ for $i \in I$ are pairwise skew. Then, the intersection $E = \bigcap_{i\in I} \mathrm{CH}(L_i)$ in $\mathrm{Gr}(k, k+2)$ is a smooth, irreducible and reduced curve.*

Proof. By assumption the lines L_i are skew and since Z is generic the lines are in general position, so Lemma 4.2.7 applies, stating that the dimension of E is indeed one. Moreover, it also follows from the same lemma that for all $J \subset I$ the intersections $\bigcap_{j\in J} D_j$ are global complete intersections, and in particular are transversal. Therefore, we have

$$\operatorname{Sing}(E) = \bigcup_{i\in I} \left(\operatorname{Sing} D_i \cap \bigcap_{j\neq i} D_j \right), \tag{4.2.1}$$

since points that are smooth on each component will be smooth points of the intersection by the transversality of the intersection. We want to show that the union in (4.2.1) is empty, and to do so we consider the following incidence variety:

$$\Sigma_i = \{(p, [V]) \mid p \in V\} \subset L_i \times \mathrm{Gr}(k, k+2).$$

Since this incidence variety Σ_i is smooth, see Proposition 5.2.6, it follows from [Har92, Corollary 14.10], that a point $[V] \in D_i$ is smooth if and only if $[V] \cap L_i = \{p\}$. Conversely, the singular locus $\operatorname{Sing} D_i$ is given by the subset of $D_i \subset \mathrm{Gr}(k, k+2)$ of all k-planes $[V]$ that contain the line L_i. In particular, the codimension of the singular locus $\operatorname{Sing} D_i$ inside D_i is 3, that is $\dim \operatorname{Sing} D_i = 2k-4$. By the genericity assumption on Z, the intersection $\operatorname{Sing} D_i \cap \bigcap_{j\neq i\in I} D_j$ is also transversal. So the dimension count

$$\dim \operatorname{Sing} D_i - \operatorname{codim} \bigcap_{j\neq i} D_j = 2k - 4 - 2k + 2 = -2$$

shows that this intersection is empty. Since D_i was arbitrary, by (4.2.1) the curve E must be smooth. And since E is smooth it must be reduced as a scheme, else the Jacobian criterion for smoothness fails.

It remains to show that the curve is irreducible. For a smooth curve it suffices to show that E is connected. This follows from [MNP17, Theorem 1.1] applied to the Plücker embedding $\mathrm{Gr}(k, k+2) \hookrightarrow \mathbb{P}^{\binom{k+2}{2}-1}$. □

Example 4.2.9. Let $n = 10$, $k = 3$, $m = 2$, and set Z to be the 10×5 matrix obtained by stacking 10 points of the rational normal curve of degree 4 on top of each other. More succinctly,

$$Z = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 1^2 & 1^3 & 1^4 \\ \vdots & \vdots & \vdots & \vdots & \vdots \\ 1 & 10 & 10^2 & 10^3 & 10^4 \end{bmatrix} \in \mathbb{R}^{10\times 5}.$$

We consider the amplituhedron $\mathcal{A}_{10,3,2}(Z)$, with the curve $E = D_{([9])} = D_1 \cap D_3 \cap D_5 \cap D_7 \cap D_9$. Then, a computation in `Macaulay2` [GS] yields that the prime ideal defining the curve E in the coordinate ring $\mathbb{C}[\mathrm{Gr}(3,5)]$ is generated by the following five polynomials:

$$
\begin{aligned}
&-p_{234} - p_{235} - p_{245} + p_{345},\\
&216p_{123} - 180p_{124} + 114p_{134} - 65p_{234} + 36p_{125} + 30p_{135} - 19p_{235} + 6p_{145} - 5p_{245} + p_{345},\\
&8000p_{123} - 3600p_{124} + 1220p_{134} - 369p_{234} + 400p_{125} + 180p_{135} - 61p_{235} + 20p_{145} - 9p_{245} + p_{345},\\
&74088p_{123} - 22932p_{124} + 5334p_{134} - 1105p_{234} + 1764p_{125} + 546p_{135} - 127p_{235} + 42p_{145} - 13p_{245} + p_{345},\\
&373248p_{123} - 88128p_{124} + 15624p_{134} - 2465p_{234} + 5184p_{125} + 1224p_{135} - 217p_{235} + 72p_{145} - 17p_{245} + p_{345}.
\end{aligned}
$$

These are precisely the generators of the vanishing ideals of the divisors labeled by odd indices D_1, D_3, D_5, D_7 and D_9. We then also verify that E is smooth and has genus one, in alignment with the following lemma. The code for this example is available at [KPS26b]. ◇

We compute the (geometric) genus of the curve E from Lemma 4.2.8.

Lemma 4.2.10. *Let E be a curve obtained from the $(2k-1)$-fold transversal intersection of $D_i = \mathrm{CH}(L_i)$, where the lines $L_i \subset \mathbb{P}(\mathbb{C}^{k+2})$ spanned by Z_i and Z_{i+1} are pairwise disjoint. Then, for $k \geq 3$ the (geometric) genus of E can be computed as*

$$g(E) = 1 + \frac{k-3}{2} C_k,$$

where C_k is the k-th Catalan number. For $k = 1, 2$ the genus of the curve E is zero.

Proof. Let $k = 1$. Then, we are in $\mathrm{Gr}(1,3) \cong \mathbb{P}^2$, and the divisors D_i are lines in $\mathbb{P}^2$. In particular, E is such a line and thus has genus 0. For $k = 2$ we get a linear section of $\mathrm{Gr}(2,4)$, which is a conic and thus has genus 0. Now, let $k \geq 3$. By construction, we can apply Lemmas 4.2.7 and 4.2.8 to E, showing that this curve is a smooth irreducible complete intersection in $\mathrm{Gr}(k, k+2)$. Set $G = \mathrm{Gr}(k, k+2)$. Therefore, it follows from [Ful98, B.7.4], that the normal bundle $N_{E/G}$ of $E \subset G$ decomposes as

$$N_{E/G} = \bigoplus_{i=1}^{2k-1} N_{D_i/G}|_E, \tag{4.2.2}$$

where $N_{D_i/G}$ are the normal bundles of the divisors D_i. Since all the divisors D_i can be extended to hyperplane sections in $\mathrm{Gr}(k, k+2)$, we can identify the normal bundles $N_{D_i/G}$ with the line bundles $\mathcal{O}_G(1)$. Since E is smooth, it follows from [Har77, Proposition 8.20] that

$$\omega_E \cong \omega_{\mathrm{Gr}(k,k+2)} \otimes \wedge^{2k-1} N_{E/G},$$

where ω_E, and ω_G refer to the canonical divisors of E and G, respectively. Then, using the decomposition in (4.2.2), we identify $\wedge^{2k-1} N_{E/G}$ with the twisted line bundle $\mathcal{O}_G(2k-1)|_E$. Then, since the canonical divisor ω_G on the Grassmannian is isomorphic to $\mathcal{O}_G(-k-2)$, see [EH16, Section 5.7.4], we get

$$\omega_E \cong \mathcal{O}_G(-k-2)|_E \otimes_{\mathcal{O}_G} \mathcal{O}_G(2k-1)|_E \cong \mathcal{O}_G(k-3)|_E. \tag{4.2.3}$$

Therefore, we can now apply the Riemann—Roch theorem, see e.g. [Sha13, Theorem 3.23],

so that

$$\deg_E \omega_E = (k-3)[H][E] = 2g-2,$$

where $[E]$, and $[H]$ are the classes in the Chow ring $A^\bullet(\mathrm{Gr}(k,k+2))$ represented by E and a hyperplane section H in $\mathrm{Gr}(k,k+2)$, respectively. Therefore, we get $g = 1 + \frac{(k-3)}{2}[H]^{2k}$ since $[E] = [H]^{2k-1}$, as E is a complete intersection by Lemma 4.2.7. Finally, recall from (1.1.2) that the degree of the Grassmannian $\mathrm{Gr}(k,k+2)$ in its Plücker embedding is precisely the k-th Catalan number $C_k = [H]^{2k}$, and the claim follows. □

We are now finally in a position to prove the main result of this section.

Proof of Theorem 4.2.1. Let $\Delta = \partial_a \mathcal{A}_{n,k,2} = \bigcup_{i=1}^n D_i$ be the decomposition of the algebraic boundary of the amplituhedron into irreducible components $D_i \subset \mathrm{Gr}(k,k+2)$. For $I \subset [n]$ we write D_I for the intersection $\bigcap_{i\in I} D_i$. We start by considering the following long exact sequence in relative cohomology.

$$\cdots \to H^{2k-1}(\mathrm{Gr}(k,k+2)) \to H^{2k-1}(\Delta) \to H^{2k}(\mathrm{Gr}(k,k+2),\Delta) \to H^{2k}(\mathrm{Gr}(k,k+2)) \to \cdots.$$

By [PS08, Proposition 5.46] this is also a long exact sequence of mixed Hodge structures where the appearing maps are morphisms of such structures. Since the Hodge structure on the cohomology of the Grassmannian is pure of Tate type, it follows that the $(p,0)$ part of the terms $H^{2k-1}(\mathrm{Gr}(k,k+2))$ and $H^{2k}(\mathrm{Gr}(k,k+2))$ in the above exact sequence are zero, in turn implying that $h^{p,0}(H^{2k}(\mathrm{Gr}(k,k+2),\Delta)) = h^{p,0}(H^{2k-1}(\Delta))$ for all $p>0$. Now, we proceed by iteratively applying the Mayer—Vietoris long exact sequence to the union and intersections of the D_i. We consider $D_{([j])}$ for some odd $j \in [n]$, and let d denote its dimension. By convention we set $D_{([j])} = \mathrm{Gr}(k,k+2)$ for $j<0$. Note that the $D_{([j])}$ are global complete intersections in the Grassmannian by Lemma 4.2.7. Then, let R_j be the union of all D_i with $i \notin ([j])$ intersected with $D_{([j-2])}$, that is $R_j = D_{([j-2])} \cap \bigcup_{i\notin([j])} D_i$, and $\dim R_j = d$. The union $D_{([j])} \cup R_j$ fits into the Mayer—Vietoris long exact sequence (1.3.1) as

$$H^{d-1}(D_{([j])}) \oplus H^{d-1}(R_j) \to H^{d-1}(D_{([j])} \cap R_j) \to H^d(D_{([j])} \cup R_j) \to H^d(D_{([j])}) \oplus H^d(R_j). \tag{4.2.4}$$

Since $D_{([j])}$ and R_j as divisors in the complete intersection $D_{([j-2])}$ satisfy the assumptions in Corollary 4.2.6, we get an injection $H^{d-1}(D_{([j])} \cap R_j)^{p,0} \hookrightarrow H^d(D_{([j])} \cup R_j)^{p,0}$ and thus the inequality

$$h^{p,0}(H^{d-1}(D_{([j])} \cap R_j)) \le h^{p,0}(H^d(D_{([j])} \cup R_j)). \tag{4.2.5}$$

By noting that $D_{([j])} \cap R_j = D_{([j+2])} \cup R_{j+2}$, which are divisors in $D_{([j])}$, we can apply (4.2.4) iteratively until $D_{([j])}$ and R_j become varieties of dimension one. Then, $D_{([2k-1])}$ becomes a smooth curve E by Lemma 4.2.8. We get the following Mayer—Vietoris sequence for $E \cup R_{2k-1}$:

$$\ldots \to H^0(E \cap R_{2k-1}) \to H^1(E \cup R_{2k-1}) \to H^1(E) \oplus H^1(R_{2k-1}) \to H^1(E \cap R_{2k-1}) \to \ldots.$$

Note that zeroth-cohomology groups always have pure Hodge structure of weight zero, and

the first cohomology group on a union of points vanishes completely, i.e. $H^1(E \cap R_{2k-1}) = 0$. Therefore, we get an isomorphism of vector spaces $H^1(E \cup R_{2k-1})^{p,0} \cong H^1(E) \oplus H^1(R_{2k-1})^{p,0}$. In particular,

$$h^{1,0}(H^1(E \cup R_{2k-1})) \geq h^{1,0}(H^1(E)) = g(E) = 1 + \frac{k-3}{2} C_k,$$

where the last inequality follows from Lemma 4.2.10. Then, since $E \cup R_{2k-1} = D_{([2k-3])} \cap R_{2k-3}$, where $D_{([2k-3])}$ is of dimension two, we get a sequence of inequalities for the Hodge numbers we are interested in:

$$h^{p,0}(H^{2k-1}(\partial_a \mathcal{A}_{n,k,2})) \geq \ldots \geq h^{1,0}(H^2(D_{([2k-3])} \cup R_{2k-3})) \geq h^{1,0}(H^1(E \cup R_{2k-1})) \geq g(E)$$

by (4.2.5), and with $g(E) = 1 + (k-3)C_k/2$. Lastly,

$$h^{p,0}(H^{2k}(\mathrm{Gr}(k, k+2), \partial_a \mathcal{A}_{n,k,2})) = h^{p,0}(H^{2k-1}(\partial_a \mathcal{A}_{n,k,2}))$$

as asserted above. This finishes the proof. □

Positive Genus Curves are Residual

To conclude the section, we show that curves E stemming from generic intersections of Schubert hyperplane sections, as in Lemma 4.2.8, are entirely residual, see Definition 1.5.6. The importance of this will become clearer in the next section, where we find a positive geometry which makes a positive genus pair. Therefore, the result of this section indicates that a similar strategy can be used to show that amplituhedra might be embedded in such a way that they make genus zero pairs. This provides a clear way of showing that amplituhedra are positive geometries.

Proposition 4.2.11. *Let E be a curve as in Lemma 4.2.8. Let $k > 1$. Then, E is contained in the residual arrangement $\mathcal{R}(\mathcal{A}_{n,k,2}) \subset \mathrm{Gr}(k, k+2)$ of the amplituhedron $\mathcal{A}_{n,k,2}$. In particular, the curve E does not meet the amplituhedron.*

To prove this statement we make use of the results in [Lam24], giving a combinatorial characterization of the boundary face structure of the $m = 2$ amplituhedron. We define the *twistor embedding*

$$T_Z : \mathrm{Gr}(k, k+m) \dashrightarrow \mathrm{Gr}(m, n), \quad X \mapsto \langle X i_1 \ldots i_m \rangle,$$

where $X \in \mathrm{Gr}(k, k+2)$ is given as the rowspan of a $k \times (k+2)$ matrix, and $\langle X i_1 \ldots i_m \rangle$ are twistor coordinates, as defined in Section 1.4. By [Lam24, Lemma 1] the map T_Z is indeed an embedding. The face stratification of $\mathcal{A}_{n,k,2}$ can be studied in this embedding and [Lam24, Theorem 25] shows that it is a subposet of the positroid stratification, see the end of Section 1.4. By these results, we are left with studying rank two positroids $\mathcal{N}$ on $[n]$, which we describe as follows. Let $L \subset [n]$ denote the set of all loops of $\mathcal{N}$. Also, consider a collection of disjoint cyclic intervals $[a_1, b_1], \ldots, [a_r, b_r]$, such that $\mathrm{rank}([a_i, b_i]) = 1$. In other words, all

elements in $[a_i, b_i]$, that are not loops, are parallel. We always assume in this notation that r is taken to be minimal and each $[a_i, b_i]$ is taken as small as possible. Then, we write for a rank two matroid $\mathcal{N}$ on $[n]$

$$\mathcal{N} = \big(L, \{[a_1, b_1], \ldots, [a_r, b_r]\}\big).$$

Then, we also set

$$S = S(\mathcal{N}) = \bigcup_{i=1}^{r} [a_i, b_i], \tag{4.2.6}$$

and define

$$d(\mathcal{N}) = 2k + r - |S \setminus L| - 2|L|, \quad c(\mathcal{N}) = 2k - d(\mathcal{N}), \quad \text{and} \quad e(\mathcal{N}) = r + k - |S \cup L|. \tag{4.2.7}$$

Finally, we also denote the subset of rank two matroids $\mathcal{N}$ on $[n]$, such that $e(\mathcal{N}) \geq 0$ by $P_{n,k}$. This is in alignment with [Lam24, Definition 13] and [Lam24, Theorem 16]. We can now proceed with proving the above proposition.

Proof of Proposition 4.2.11. The stratification of the boundary of the amplituhedron $T_Z(\mathcal{A}_{n,k,2})$ in its twistor embedding for $m = 2$ was completely characterized in [Lam24]. More precisely, its residual arrangement is given in [Lam24, Proposition 38]. We proceed by translating the matroidal language used in this reference to our setting, and then show that the conditions of [Lam24, Proposition 38] are met. By assumption E is given as a complete intersection of Schubert hyperplane sections, stemming from Chow hypersurfaces $D_i = \mathrm{CH}(L_i)$ of skew lines L_i in general position for $i \in I$. Any such divisor D_i corresponds to a positroid variety Π_{ii+1} of codimension one, given by the vanishing of the Plücker coordinate p_{ii+1} on $\mathrm{Gr}(2, n)$ corresponding to D_i under the twistor embedding T_Z. More succinctly, $T_Z(D_i) = \Pi_{ii+1} \cap \mathrm{im}\, T_Z \subset \mathrm{Gr}(2, n)$. Therefore, the matroid $\mathcal{N}$ of rank two on $[n]$, characterizing Π_{ii+1}, is given by $L = \varnothing \subset [n]$ and $\{[i, i+1]\}$. In general then, the matroid $\mathcal{N}$ of rank two on $[n]$, characterizing $E = \Pi_{\mathcal{N}}$, is given by $L = \varnothing \subset [n]$, and

$$A = \{[i, i+1] \mid i \in I\},$$

where $[i, i+1] = \{i, i+1\}$ read cyclically in $[n]$. It follows from (4.2.6), that $S = \bigcup_{i \in I} [i, i+1]$ with $|S| = 2(2k-1)$, as well as $r = |A| = 2k - 1$. Similarly, by (4.2.7), we then get $e(\mathcal{N}) = r + k - |S \cup L| = 2k - 1 + k - 2(2k-1) = -k + 1$, and therefore, by [Lam24, Theorem 16], the matroid $\mathcal{N}$ is not in $P_{k,n,2}$ for $k > 1$. The other conditions in [Lam24, Proposition 38] are all satisfied for $E = \Pi_{\mathcal{N}}$, since $c(\mathcal{N}) = 2k - (2k + (2k-1) - 2(2k-1)) = 2k - 1$ is less than $2k$, and $k - |L| = k \geq 0$, and $d_i = |\{i, i+1\} \setminus L| - 1 = 1 < k$ for all $i \leq r = 2k - 1$. This shows the claim. □

4.3 The $m = 4$ Amplituhedron has non-zero Genus

In this section we record analogous results for the genus of the amplituhedron and its boundary divisor as in Section 4.2, but for the case $m = 4$. Since all of the results are derived analogously, we only point out the differences in the proofs. The main result is as follows.

Theorem 4.3.1. *Let $k \geq 3$ and n be nonnegative integers, such that $n \geq 4(4k-1)$. Further let Z be generic and have positive maximal minors, and $\partial_a \mathcal{A}_{n,k,4}(Z) \subset \mathrm{Gr}(k, k+4)$ be the algebraic boundary of the amplituhedron. Then, the genus of the pair $(\mathrm{Gr}(k, k+4), \partial_a \mathcal{A}_{n,k,4}(Z))$ is at least $1 + \frac{3k-5}{2} \deg(\mathrm{Gr}(k, k+4))$, where the degree of the Grassmannian $\mathrm{Gr}(k, k+4)$ is*

$$\deg(\mathrm{Gr}(k, k+4)) = (4k)! \cdot \frac{0! \cdot 1! \cdot \ldots \cdot (k-1)!}{4! \cdot 5! \cdot \ldots \cdot (k+3)!}.$$

For the remainder of this section, we will denote the irreducible components of the boundary divisor of $\mathcal{A}_{n,k,4}$ by $D_{ij} = D_{ii+1jj+1} = \langle Yi(i+1)j(j+1)\rangle \subset \mathrm{Gr}(k, k+4)$ with $Y \in \mathrm{Gr}(k, k+4)$, see [ELT25] and also Lemma 1.4.5. Furthermore, we define the 4-plane $V_{ij} = V_{ii+1jj+1} = \mathrm{span}_{\mathbb{C}}(Z_i, Z_{i+1}, Z_j, Z_{j+1}) \subset \mathbb{C}^{k+4}$, where the Z_i are the rows of the Z matrix. In particular, we then have $D_{ij} = \mathrm{CH}(V_{ij})$.

Remark 4.3.2. The choice of $n \geq 4(4k-1)$ in Theorem 4.3.1 is simply to ensure that we can pick $4k-1$ many 4-planes V_{ij} that are pairwise disjoint. The reason for the lower bound on the genus of the pair is a higher genus curve appearing in the intersection of boundary divisors, analogously to the main result in Section 4.2. Moreover, similarly to Proposition 4.2.11, we believe that this curve is in the residual arrangement of $\mathcal{A}_{n,k,4}$, but since the tools used in that proposition are unavailable for the case $m = 4$, it is beyond the scope of this article to prove this statement.

To prove Theorem 4.3.1, we establish the lemmas analogous to those in Section 4.2 that helped us prove its main result.

Lemma 4.3.3. *Let $\mathcal{I}$ be a collection of sets $I = \{i, i+1, j, j+1\}$ such that $I \cap I' = \varnothing$ for all $I, I' \in \mathcal{I}$. Then, the intersection $D_{\mathcal{I}} = \bigcap_{I \in \mathcal{I}} D_I$ is a global complete transversal intersection in $\mathrm{Gr}(k, k+4)$. Geometrically speaking, for a set of pairwise disjoint 4-planes V_{ij} the intersection of D_I is transversal and a global complete intersection. In particular, if $|\mathcal{I}| = 4k-1$, then $D_{\mathcal{I}}$ is a smooth irreducible and reduced curve.*

Proof. For $|\mathcal{I}| < 4k-1$ the proof is identical to that of Lemma 4.2.7 with $\mathrm{Gr}(k, k+4)$ replacing the Grassmannian $\mathrm{Gr}(k, k+2)$, and Z being generic enough with respect to the planes V_{ij}, instead of the lines L_i. For $|\mathcal{I}| = 4k-1$ the proof of Lemma 4.2.8 applies, upon noting that the singular locus of D_{ij} is the subset of $D_{ij} \subset \mathrm{Gr}(k, k+4)$ given by k-planes meeting V_{ij} in dimension at least two. □

Lemma 4.3.4. *Let $\mathcal{I}$ be a collection of $4k-1$ sets $I = \{i, i+1, j, j+1\}$ such that $I \cap I' = \varnothing$ for all $I, I' \in \mathcal{I}$. Let E be a curve obtained from the $(4k-1)$-fold intersection $D_{\mathcal{I}} = \bigcap_{I \in \mathcal{I}} D_I$*

of boundary divisors D_{ij}. Then, for $k \geq 2$ the (geometric) genus of E can be computed as

$$g(E) = 1 + \frac{3k-5}{2} \deg(\mathrm{Gr}(k, k+4)).$$

For $k = 1$ the genus of the curve E is zero.

Proof. The proof is identical to that of Lemma 4.2.10 upon noting the following. First, the canonical divisor of the curve E obtained from $4k-1$ intersections of Schubert hyperplane sections is given as $\omega_E \cong \mathcal{O}_{\mathrm{Gr}(k,k+4)}(-k-4) \otimes \mathcal{O}_{\mathrm{Gr}(k,k+4)}(4k-1) \cong \mathcal{O}_{\mathrm{Gr}(k,k+4)}(3k-5)$, as in (4.2.3). Second, the degree of $\mathrm{Gr}(k, k+4)$ is now given by (1.1.1) for $n = k+4$. Then, applying the Riemann—Roch theorem we obtain the result. □

We are now in a position to prove the main result of this section.

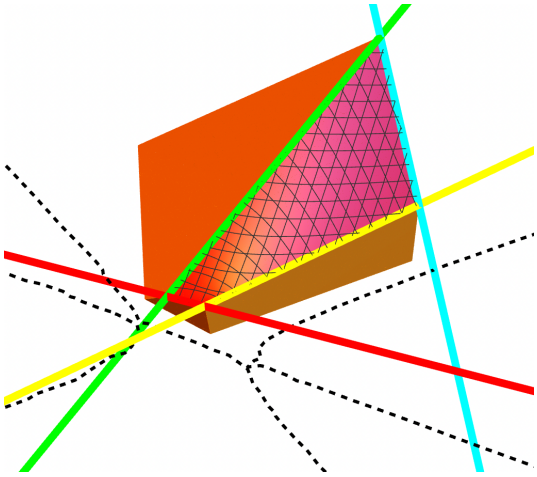
Proof of Theorem 4.3.1. Note that Lemmas 4.2.3, 4.2.4, and Corollary 4.2.5 apply to any Grassmannian and union of Schubert hyperplane sections. Therefore, in particular, they apply to $D = \partial_a \mathcal{A}_{n,k,4} \subset \mathrm{Gr}(k, k+4)$. Hence, the proof of Theorem 4.2.1 applies here, where we replace $D_{([j])}$ for any odd $j \in [n]$ with an appropriate subset of $\mathcal{I}$, defined as in Lemma 4.3.4. □

4.4 The Del Pezzo Cube: A Genus One Positive Geometry

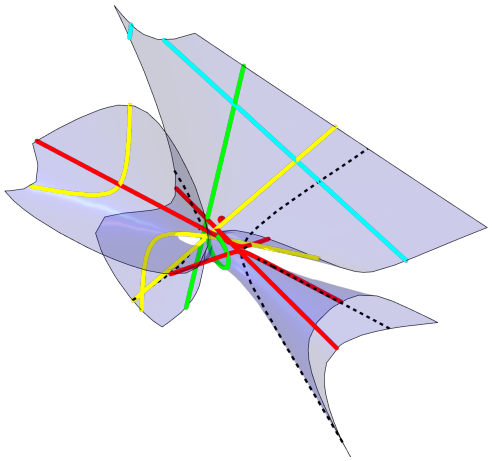
In this section we present an explicit example of a semi-algebraic set $P \subset \mathbb{P}^3$ such that $(\mathbb{P}^3, P)$ is a positive geometry in the sense of Definition 1.5.1, but the pair $(\mathbb{P}^3, \partial_a P)$ has genus one. This demonstrates that being a genus zero pair is not a necessary condition to be a positive geometry, and that the amplituhedron might still be one. In fact, together with Proposition 4.2.11, this example provides a novel approach to tackling the central conjecture that amplituhedra make positive geometries. The code to reproduce the computations of this section is available at [KPS26b].

Our example P, called *Del Pezzo cube*, seen in Figure 4.4.1a, is combinatorially a full-dimensional cube in $\mathbb{P}^3$, with one of its linear facets replaced by a cubic hypersurface in such a way that the edges are preserved. In analogy to the amplituhedron, as discussed in Section 4.2, P has a genus one curve in its residual arrangement.

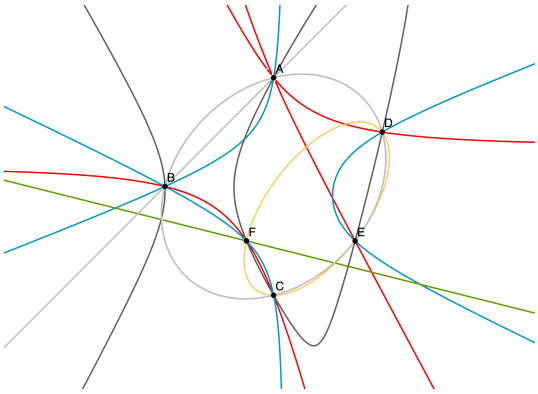
We start by describing the six facets of the Del Pezzo cube P. There are five linear facets contained in the hyperplanes denoted by $L_0, \ldots, L_4$. The remaining facet of P is contained in a cubic Del Pezzo surface S obtained from a blow-up of six real points in the plane. The four linear boundary divisors $L_1, \ldots, L_4$ intersect the surface S in four lines, which bound a quadrilateral face of P on S, and in four residual conics. This can be seen in Figure 4.4.1b, where S is the blue-shaded surface, and the quadrilateral is the region with a red point inside. The semi-algebraic set P has eight more edges: four come from intersecting the facet contained in L_0 with the other four linear facets, and the remaining four are the intersections $L_2 \cap L_4$, $L_1 \cap L_3$, $L_1 \cap L_4$ and $L_2 \cap L_3$.



(a) Semi-algebraic set P, with a curvy quadrilateral facet on S in red and its boundary lines.



(b) Del Pezzo surface S in blue, the quadrilateral face of P, bounded by four lines, and the residual elliptic curve in black (dashed).



(c) Six points A through F in $\mathbb{P}^2$ and five cubic curves interpolating these points.

Figure 4.4.1: Example of a positive geometry with genus one.

Construction of the Del Pezzo Cube

Next, we are going to explicitly construct the Del Pezzo cube in coordinates, such that we will be able to take residues later on. As the Del Pezzo surface S is the blow-up of $\mathbb{P}^2$ in six points, the combinatorial structure of the semi-algebraic set P can be read off from a plane picture (see Figure 4.4.1c), and we now discuss this in detail. We pick six points A, B, C, D, E, and F in $\mathbb{P}^2$ in generic position with coordinates

$$\begin{aligned} A &= [0:1:1], & B &= [-1:0:1], & C &= [0:-1:1], \\ D &= [1:1/2:1], & E &= [3/4:-1/2:1], & F &= [-1/4:-1/2:1]. \end{aligned}$$

The blow-up of $\mathbb{P}^2$ in these six points is a smooth irreducible cubic Del Pezzo surface $S \subset \mathbb{P}^3$. This $\mathbb{P}^3$ is dual to the space of homogeneous cubics vanishing at these six real points.

Explicitly, we choose a basis $c_0, \ldots, c_3 \in \mathbb{C}[x, y, z]_3$ with

$$\begin{aligned}
c_0 &= -16x^3 - 48x^2y + 118xy^2 - 16x^2z + 53xyz, \\
c_1 &= -32x^3 + 10x^2y - 188xy^2 + 21x^2z + 53xz^2, \\
c_2 &= 84x^3 - 172x^2y - 487xy^2 - 106y^3 + 84x^2z + 106yz^2, \\
c_3 &= -108x^3 - 6x^2y + 717xy^2 - 161x^2z + 106xyz - 53y^2z + 53z^3.
\end{aligned}$$

We write $u_0, \ldots, u_3$ for homogeneous coordinates of this $\mathbb{P}^3$. Then, the Del Pezzo surface S is the vanishing set of

$$\begin{aligned}
&540u_0^3 - 381u_0^2u_1 + 19u_0u_1^2 + 58u_1^3 - 402u_0^2u_2 + 94u_0u_1u_2 + 27u_1^2u_2 - 188u_0u_2^2 \\
&- 118u_1u_2^2 - 879u_0^2u_3 + 84u_0u_1u_3 + 32u_1^2u_3 - 106u_1u_2u_3 + 212u_0u_3^2.
\end{aligned}$$

In the following we will freely identify a hypersurface with its defining equation. For the five linear boundary divisors we choose

$$\begin{aligned}
L_0 &= 300u_0 - 900u_1 - 100u_2 - 691u_3, & L_1 &= 267u_0 + 38u_1 + 38u_2 - 76u_3, \\
L_2 &= 6u_0 + u_1 + 11u_3, & L_3 &= 27u_0 - 8u_1 - 32u_2 - 36u_3, \\
L_4 &= -257u_0 + 359u_1 - 248u_2 + 16u_3.
\end{aligned}$$

If we let $U_3 \subset \mathbb{P}^3$ be the affine open corresponding to points where $u_3 \neq 0$, then P is the set of points $q \in U_3$ for which $S(q) \geq 0$ and $L_i(q) \geq 0$ for $i = 0, \ldots, 4$, see Figure 4.4.1a. We now relate Figure 4.4.1b to Figure 4.4.1c by blowing down to $\mathbb{P}^2$. The hyperplanes L_1 and L_2 correspond to the red and blue cubics with nodes in A and B in Figure 4.4.1c, respectively. The boundary divisor L_3 blows down to the green cubic containing the conic through the points $A, \ldots, E$ and the line through F. The hyperplane L_4 corresponds to the reducible cubic in yellow, consisting of the line through A and B and the conic through the remaining four points in the plane. Therefore, the four lines bounding the facet of P contained in S are obtained as follows: two, red and blue, are the exceptional divisors over the points A and B. One, in yellow, is the strict transform of the line through A and B, and the fourth one, in green, is the strict transform of the conic through A, B, C, D, E. Finally, the hyperplane L_0 corresponds to the elliptic curve through the six points, shown in black (dashed). Note that all the curves need to be chosen in a way that they do not intersect the region P', bounded by the line through A and B and the part of the conic through $A, \ldots, E$ between A and B. All the relevant plane curves are depicted in Figure 4.4.1c.

Remark 4.4.1. Our construction depended on a particular choice of a quadrilateral on the Del Pezzo surface, of which there are generically 90. Of course, the results of the further analysis will not depend on this choice. This point has been made explicit in the recent paper [ST26].

The Del Pezzo Cube is a Positive Geometry

We now describe the residual arrangement of P. It is combinatorially related to that of the cube: the residual arrangement of a three-dimensional cube consists of three lines, namely

the intersection of every pair of boundary divisors containing opposing facets. Similarly, for our set P, the residual arrangement contains two lines, namely $L_1 \cap L_2$ and $L_3 \cap L_4$ as well as the curve $S \cap L_0$, a smooth curve of genus one, seen in black in Figure 4.4.1b. Moreover, it also contains the four conics, which arise as irreducible components of the intersections of the four boundary divisors $L_1, \dots L_4$ with S. (Recall, every such intersection decomposes into a conic and a line containing an edge of P.) The residual arrangement is interpolated by a unique homogeneous polynomial of degree 4 in $u_0, \dots, u_3$, which is the adjoint $\mathrm{adj}(P)$ of P.

We compute that P is a positive geometry in the sense of Definition 1.5.1, with the unique canonical form ω_P of P given by

$$\omega_P = \frac{\mathrm{adj}(P)}{L_0 L_1 L_2 L_3 L_4 S}\, du_0 \wedge du_1 \wedge du_2,$$

where the scaling of $\mathrm{adj}(P)$ is chosen so that all its residues at the vertices of P evaluate to ± 1. In particular, taking the residues of ω_P along the facets of P recovers their canonical forms. Our code verifying this is available at [KPS26b]. We will address the uniqueness of ω_P below.

The Genus of the Del Pezzo Cube is One

Next, let us show that the genus of the pair $(\mathbb{P}^3, \partial_a P)$ equals one. By the proof of Proposition 2.3.12 the genus of the pair $(\mathbb{P}^3, \partial_a P)$ is bounded from above by the sum of the genera of the intersections of the irreducible components of $\partial_a P$. Since all boundary divisors are rational smooth projective varieties which have vanishing genus, the only non-zero genus contribution stems from two-fold intersections. As described above, we have a single elliptic curve $S \cap L_0$ of genus one, and a collection of reducible cubic curves and lines, all of which have vanishing genus. Therefore, $g(\mathbb{P}^3, \partial_a P) \leq 1$.

It remains to show that this bound is tight. Let $D = \bigcup_{i=0}^{5} D_i = \partial_a P$, with $D_5 = S$ and $D_i = L_i$ for $i = 0, 1, 2, 3, 4$. Consider the spectral sequence converging to relative cohomology from Proposition 2.1.6.

$$E_1^{p,q} = \bigoplus_{|I|=p} H^q(D_I) \quad \Longrightarrow \quad H^{p+q}(\mathbb{P}^3, D),$$

where $I \subset \{0,1,2,3,4,5\}$ and $D_I = \bigcap_{i \in I} D_i$. Note that all two-fold intersections D_I with $I \neq \{0,5\}$ are pure of Tate type, which is seen as follows. First, note that $H^0(D_I) \cong \mathbb{Q}^2$ since we always have two irreducible smooth and thus connected components, see Definition 1.3.21. Secondly, $H^2(D_I) \cong \mathbb{Q}(-1)^2$, which follows from the Mayer—Vietoris sequence in (1.3.1) which simplifies in this case to

$$0 \to H^2(Z_1 \cup Z_2) \to H^2(Z_1) \oplus H^2(Z_2) \to 0,$$

where Z_i are the irreducible components of D_I, i.e. lines and conics both of which have top homology group isomorphic to $\mathbb{Q}(-1)$, see Lemma 2.1.13 and Example 1.3.14. Lastly,

for $H^1(D_I)$ consider D_I as a union of curves inside the projective plane defined by one of the linear boundary divisors. Then, $H^1(D_I) \cong \mathbb{Q}^\beta$, where $\beta = \sum_p (r_p - 1) - m + 1$ and r_p is the number of curves going through the point $p \in D_I$, which follows from the dual graph, see the discussion before Corollary 2.1.25, and also [BMS06, Section 1]. Therefore, all higher differentials $d^r : E_r^{p,q} \to E_r^{p+r,q-r+1}$ for $r > 3$ must be zero as they preserve weights. Moreover, the only differential contributing to weight 1 is d^1.

We now show that the weight one part of $H^3(\mathbb{P}^3, D)$ is at least one-dimensional. Recall the definition of the maps δ_p^q in (2.1.1). Let $q = 1$, and thus $p = 2$. Then, note that $\ker \delta_2^1 = H^1(S \cap L_0) \oplus \bigoplus_{1 \le i < j \le 4} H^1(L_i \cap L_j)$, as the codomain of δ_2^1 is the first cohomology group of a collection of points, which vanishes trivially. Similarly, the domain of δ_1^1 is the direct sum of the first cohomology groups of the Del Pezzo surface S and the five planes L_i. Since all of these are smooth rational varieties, their first cohomology vanishes. We have $\operatorname{im} \delta_1^1 = 0$. Using the upper bound for the genus of $(\mathbb{P}^3, D)$ from above, we get that

$$\operatorname{gr}_1^W (H^3(\mathbb{P}^3, D)) \cong \frac{\ker \delta_2^1}{\operatorname{im} \delta_1^1} \cong H^1(E), \tag{4.4.1}$$

where E is the elliptic curve obtained as $S \cap L_0$. This shows that $g(\mathbb{P}^3, D) = g(E) = 1$.

Uniqueness of Canonical Form

Next, we show that despite the fact that $g(\mathbb{P}^3, \partial_a P) = 1$, we have a unique canonical form on P. Start by noticing that the argument above implies that the map $R : \Omega^3_{\log}(\mathbb{P}^3 \setminus \partial_a P) \to \operatorname{gr}_0^W H_3(\mathbb{P}^3, \partial_a P)$, see [BD25, Definition 2.4], has a one-dimensional kernel. We can see that the kernel comes from the genus one curve $E = S \cap L_0$ in the residual arrangement of P as follows. Consider the 3-form $\kappa \in \Omega^3_{\log}(\mathbb{P}^3 \setminus \partial_a P)$ given as

$$\kappa = \frac{1}{SL_0} du_0 \wedge du_1 \wedge du_2,$$

with a local system of parameters on $\mathbb{P}^3 \setminus \partial_a P$ given by u_0, u_1 and u_2. By [BD25, Equation (21)] we identify $\Omega^3_{\log}(\mathbb{P}^3 \setminus \partial_a P) \cong F^3 H^3(\mathbb{P}^3 \setminus \partial_a P)$, whose weight one part, see (4.4.1), is then given by κ under this identification. Then, $R(\kappa) = 0$, as the codomain of R is $\operatorname{gr}_0^W H_3(\mathbb{P}^3, \partial_a P)$, and since the kernel of R is one-dimensional, κ is a generator of it.

In Remark 2.1.15, we suggested considering the one-dimensional space $\omega_P + \lambda\kappa$ with $\lambda \in \mathbb{C}$ as the space of candidate canonical forms for P. However, for any $\lambda \neq 0$ the residue of the form $\omega_P + \lambda\kappa$ along the linear divisor L_0 has a pole along the residual elliptic curve $E = S \cap L_0$ and thus does not recover the canonical form of the facet of P contained in L_0. Therefore, we must have $\lambda = 0$, and thus recover the unique canonical form ω_P.

Reembedding the Del Pezzo Cube into a Genus Zero Pair

Although the pair $(\mathbb{P}^3, \partial_a P)$ has positive genus, one can pick a different ambient variety for the semi-algebraic set P to obtain a genus zero pair. Namely, let X be the blow-up of $\mathbb{P}^3$

along the elliptic curve E, and $\widetilde{Y}$ be the strict transform of $\partial_a P$. By Lemma 2.1.19, we can bound the genus of the pair $(X, \widetilde{Y})$ as the sum of the genus of X and the genera of all intersections of the irreducible components of $\widetilde{Y}$. To that end, write $\widetilde{Y}_i$ for the strict transforms of L_i for $i = 0, \ldots, 4$, and $\widetilde{Y}_S$ for that of S. First, note that $\widetilde{Y}_S \cong S$ and $\widetilde{Y}_0 \cong L_0$, as these two boundary divisors contain the blow-up center E as an irreducible subvariety of codimension one. Since the divisors L_i for $i = 1, \ldots, 4$ intersect E in three points each, their strict transforms are isomorphic to the blow-up of $\mathbb{P}^2$ in three points, which is a smooth compact rational variety. Then, note that $\widetilde{Y}_S \cap \widetilde{Y}_0 = \varnothing$, as S and L_0 intersect transversally in the blow-up center E. Moreover, $\widetilde{Y}_S \cap \widetilde{Y}_i \cong S \cap L_i$ for $i = 1, \ldots, 4$, as the curves $S \cap L_i$ meet E in a finite collection of smooth points. Similarly, we get $\widetilde{Y}_i \cap \widetilde{Y}_j \cong L_i \cap L_j \cong \mathbb{P}^1$ for $i, j \neq 0$, as these lines do not intersect E. We have just shown that X as well as all intersections of the components of $\widetilde{Y}$ are smooth rational varieties, hence their genera vanish and $g(X, \widetilde{Y}) = 0$.

The corresponding canonical map would produce a canonical form for the preimage of P under the blow-up map. Since the blow-up is a morphism from X to $\mathbb{P}^3$, by functoriality of the canonical map the pushforward of this form would then be the canonical form of P in $\mathbb{P}^3$. This suggests that one can sometimes show that a pair $(X, X_{\geq 0})$ is a positive geometry by considering the pair of varieties (X', Y'), where X' and Y' are related to X and $\partial_a X_{\geq 0}$ in a natural way. We thank Clément Dupont for pointing this out. We hope that a similar strategy can also be deployed to the amplituhedron with its residual arrangement, which provides a concrete way of tackling the conjecture that all amplituhedra make positive geometries.

Chapter 5

Limit Amplituhedron

We don't want to take you to the limit.

Elia

In this chapter we take a spin on the amplituhedron for $m = 2$, see Section 1.4, which will allow us to define a limiting object for large n. We define the limit amplituhedron, which is a semi-algebraic subset of the Grassmannian $\mathrm{Gr}(k, k+2)$, and show that it is a positive geometry. A simple case of this construction is the mysterious "pizza slice", see Example 5.0.2. It is also a special version of the master amplituhedron that already appeared in [AT14]. This work also follows recent interest in the study of large n limits of scattering amplitudes, see e.g. [Ark+24].

The definition of a positive geometry is recursive in dimension, see 1.5.1. In the case of polytopes, this means that we successively take residues of the canonical form along the flats of the facet hyperplane arrangement. This gives ± 1 at the vertices of the polytope. In the nonlinear setting of this chapter, we need to replace the hyperplane arrangement by the algebraic boundary of the limit amplituhedron. The flats in the nonlinear setting correspond to irreducible components in the boundary stratification induced by the algebraic boundary of the limit amplituhedron. The algebraic boundary of the limit amplituhedron is the union of two Chow hypersurfaces in the Grassmannian, namely the Chow hypersurface of the rational normal curve (the projective closure of the moment curve) and a secant line. We describe the boundary stratification explicitly via iterated singular loci, that is in terms of varieties of secant planes and osculant planes to the rational normal curve.

This chapter is organized accordingly. In Section 5.1 we introduce the limit amplituhedron formally, establish some basic properties, and determine its algebraic boundary in terms of the Chow hypersurfaces of the rational normal curve and a line in $\mathbb{P}^d$. In Section 5.2 we describe all strata of the iterated singular loci of the algebraic boundary stratification, generalizing the matroid of the facet hyperplane arrangement of a polytope to our setting. Finally, in Section 5.3, we show that the residual arrangement of the limit amplituhedron is empty.

In particular, we prove that all 0-dimensional strata are vertices of the limit amplituhedron, and thus find a canonical form satisfying the requirements for a positive geometry.

In summary, our main result states the following.

Theorem 5.0.1. *The algebraic boundary of the limit amplituhedron $\mathcal{A}_k^\infty$ has two irreducible components, the Chow hypersurface $\mathrm{Ch}(C_{k+1})$ of the rational normal curve $C_{k+1} \subset \mathbb{P}^{k+1}$ and the Chow hypersurface $\mathrm{Ch}(\mathcal{S}_{01})$ of the secant line $\mathcal{S}_{01} \subset \mathbb{P}^{k+1}$ spanned by two points on C_{k+1}. The limit amplituhedron makes a positive geometry $(\mathrm{Gr}(k, k+2), \mathcal{A}_k^\infty, \omega_\infty)$ in the Grassmannian $\mathrm{Gr}(k, k+2)$, with canonical form ω_∞ given as*

$$\omega_\infty = \frac{1}{\mathrm{Ch}(C_{k+1})\,\mathrm{Ch}(\mathcal{S}_{01})}\, dx_1 \wedge \dots dx_{2k},$$

where the dx_i are a local system of parameters on $\mathrm{Gr}(k, k+2)$.

The special case $k = 1$ gives a positive geometry in $\mathbb{P}^2$ that also appears in [ABL17, Example 6.1] under the name of pizza slice.

Example 5.0.2 (Pizza slice). For $k = 1$, we pick a totally positive matrix $Z \in \mathbb{R}^{n\times 3}$ (that is a real $n \times 3$ matrix such that all its 3×3 minors are positive). In fact, we pick the rows of Z to be on the rational normal curve in $\mathbb{P}^2$ by choosing a partition $I = (t_1, \dots, t_n)$ of real numbers $0 \leq t_1 < t_2 < \dots < t_n \leq 1$ and defining the i-th column of Z to be $\gamma_2(t_i) = (1, t_i, t_i^2)$. The amplituhedron $\mathcal{A}_1^I$ is, by definition, the image $\widetilde{Z}(\mathrm{Gr}(1,n)_{\geq 0})$ of the non-negative Grassmannian $\mathrm{Gr}(1,n)_{\geq 0}$ under the linear map $\widetilde{Z}\colon \mathrm{Gr}(1,n)_{\geq 0} \to \mathbb{R}^3$, $V \mapsto VZ$. Since $\mathrm{Gr}(1,n) \cong \mathbb{P}^{n-1}$, the set $\mathrm{Gr}(1,n)_{\geq 0}$ is the non-negative orthant, hence the image under $\widetilde{Z}$ is a n-gon in $\mathbb{P}^2$ with vertices given by the rows of the matrix Z. So the vertices are all on the rational normal curve parameterized by $t \mapsto (1, t, t^2)$. To define the limit amplituhedron $\mathcal{A}_1^\infty$, we take the union over all amplituhedra $\mathcal{A}_1^{t_1,\dots,t_n}$ that we obtain by choosing any n real numbers $t_i \in [0,1]$ and any $n \in \mathbb{N}$. This union is the convex hull of the interval $\{(1, t, t^2) \mid t \in [0,1]\}$ on the rational normal curve. It is projectively equivalent to the set shown in Figure 5.0.1, essentially equal to the "pizza slice" in [ABL17, Example 6.1]. The algebraic boundary of $\mathcal{A}_1^\infty$ has two irreducible components, namely the rational normal curve itself and the line spanned by $\gamma_2(0)$ and $\gamma_2(1)$. For higher values of k, the two components are Chow hypersurfaces; for $k = 1$, in $\mathbb{P}^2$ they coincide with the varieties themselves up to duality. Let p_{12}, p_{13}, p_{23} be the Plücker coordinates of $\mathrm{Gr}(1,n)$. Then, the canonical form of $\mathcal{A}_1^\infty$ is

$$\omega_\infty = \frac{1}{(p_{12}p_{23} - p_{13}^2)(p_{12} - p_{23})} dp_{12} \wedge dp_{23}.$$

◇

This example already illustrates that we need to choose an interval $[0,1]$ on the rational normal curve. If we dropped our restriction that t_i lies in $[0,1]$, the limit amplituhedron would be the convex hull of the rational normal curve, so up to a projective change of coordinates the unit disk. This is, however, not a positive geometry in $\mathbb{P}^2$ because it cannot have a rational

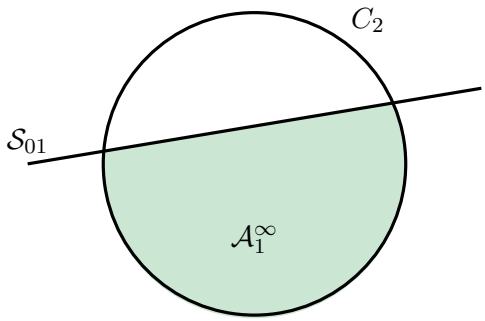


Figure 5.0.1: The limit amplituhedron $\mathcal{A}_1^\infty$ with its algebraic boundary.

canonical form (for instance for degree reasons: there is no rational section of the canonical divisor of $\mathbb{P}^2$ with a denominator of degree 2), see [ABL17, Section 10]. However, adding the secant line $\mathcal{S}_{01}$ increases the degree by 1 and we get a rational canonical form as in the definition of a positive geometry. The same happens in Grassmannians for higher values of k: we need to pick an interval on the rational normal curve so that we get a special secant line between the end points that gives the correct degree of the denominator to have a chance of finding a rational canonical form. The first nonlinear example is for $k = 2$ which gives a limit amplituhedron in $\mathrm{Gr}(2, 4)$ defined in terms of the twisted cubic $C_3 \subset \mathbb{P}^3$, see Example 5.1.7.

Remark 5.0.3. It is worth noting that studying the limit amplituhedron and its ambient Grassmannian from the point of view of mixed Hodge theory, as in Chapter 2, is very subtle. This is because the boundary stratification makes extensive use of the singularities, which appear even in codimension one, which poses formidable difficulties from the point of view of the mixed Hodge theory approach, even despite our advances in Chapter 2 establishing some results for singular varieties. Hence, this investigation warrants its own dedicated follow-up projects, which we will leave for the future.

5.1 Algebraic Boundary

Let us make the definition of the limit amplituhedron from the introduction more rigorous. Let $\gamma_{k+1} : \mathbb{C} \to \mathbb{P}^{k+1} \cap U_0$ be the map sending $t \mapsto (1, t, t^2, \ldots, t^{k+1})$, be the parametrisation of the rational normal curve C_{k+1} of degree $k + 1$ on the affine patch U_0 corresponding to a non-vanishing first coordinate in $\mathbb{P}^{k+1}$. A *partition* I_n of the interval $[0, 1] \subset \mathbb{R}$ of length n is an increasingly ordered sequence of real numbers $0 = t_1 < \ldots < t_n = 1$. We say a partition J_m *refines* I_n, and write $I_n \leq J_m$ if J_m contains I_n as a subsequence. For fixed positive integers k and n, and any partition $I_n = (t_1, t_2, \ldots, t_n)$ of $[0, 1]$, we define the real

totally positive $n \times (k+2)$ matrix

$$Z = Z(I_n) = \begin{pmatrix} \gamma_{k+1}(t_1) \\ \vdots \\ \gamma_{k+1}(t_n) \end{pmatrix},$$

whose i-th row is given by $\gamma_{k+1}(t_i)$. The maximal minors of this matrix are Vandermonde determinants and therefore $Z(I_n)$ is a totally positive matrix. We denote the amplituhedron which is the image of the induced map $\widetilde{Z}$ by $\mathcal{A}_k^I$. This notation coincides with the previous one as $\mathcal{A}_k^I = \mathcal{A}_{|I_n|,k,2}(Z(I_n))$. We can now define the limit amplituhedron.

Definition 5.1.1. The *limit amplituhedron*, $\mathcal{A}_k^\infty$ is the union of all amplituhedra $\mathcal{A}_k^I$, over all partitions I of the interval $[0,1]$.

Remark 5.1.2. The totally positive matrices $Z(I_n)$ that we consider here are special. They correspond to realizations of cyclic polytopes in $\mathbb{R}^{k+1}$ with n vertices on the rational normal curve. The convex hull of n points on the rational normal curve is always a cyclic polytope but not every cyclic polytope is affinely equivalent to such a realization (as soon as the number of vertices is big enough), see [Stu88] for more details. We restrict to totally positive matrices $Z(I_n)$ constructed from the rational normal curve to get a better handle on the limit amplituhedron.

Note that we have the following characterization of the limit amplituhedron.

Proposition 5.1.3. *Let $\mathcal{I}$ be the set of all partitions of $[0,1]$ and $\leq$ the refining order on $\mathcal{I}$, that is the order induced by inclusions. Then, there is a well-defined direct limit with*

$$\varinjlim \mathcal{A}_k^I = \mathcal{A}_k^\infty,$$

where $I \in \mathcal{I}$.

Proof. Let $\mathcal{I}$ and $\leq$ be as above. First, notice that $\leq$, as the order induced by containment of subsequences, is reflexive and transitive, thus a preorder, implying that $(\mathcal{I}, \leq)$ is a directed set. Next, for partitions $I \leq J \in \mathcal{I}$ let $D = J \setminus I$ be their refining set. Then, for each $V_I = (v_1, \ldots, v_{|I|}) \in \mathrm{Gr}(k, |I|)_{\geq 0}$ we can define $V_J = (v'_1, \ldots, v'_{|J|}) \in \mathrm{Gr}(k, |J|)_{\geq 0}$ where $v'_i = v_i$ if $i \notin D$ and $v'_i = 0$ otherwise. Thus, $V_I Z(I) = V_J Z(J)$, and we get well-defined inclusion maps

$$\iota_{IJ} : \mathcal{A}_k^I \hookrightarrow \mathcal{A}_k^J.$$

Therefore, $\langle \mathcal{A}_k^I, \iota_{IJ} \rangle$ is a directed system, as it also satisfies the additional axioms $\iota_{II} = \mathrm{id}$ and $\iota_{IK} = \iota_{JK} \circ \iota_{IJ}$ for all $I \leq J \leq K \in \mathcal{I}$; hence the direct limit is well-defined and the equality in $\mathcal{A}_k^\infty$ is by construction. □

Remark 5.1.4. Note that the limit amplituhedron is connected by construction. This follows from the fact that amplituhedra are connected, and the direct limit is a product of quotients given by a continuous map, hence connectedness is preserved.

The main result of this section, which is one part of Theorem 5.0.1, is as follows. We postpone its proof to the end of this section.

Theorem 5.1.5. *The algebraic boundary $\partial_a \mathcal{A}_k^\infty \subset \mathrm{Gr}(k, k+2)$ of $\mathcal{A}_k^\infty$ decomposes into the Chow hypersurface $\mathrm{CH}(C_{k+1})$ of the rational normal curve C_{k+1} of degree $k+1$ and the Chow hypersurface $\mathrm{Ch}(\mathcal{S}_{01})$ of the secant line $\mathcal{S}_{01}$. In symbols, we have*

$$\partial_a \mathcal{A}_k^\infty = \mathrm{CH}(C_{k+1}) \cup \mathrm{CH}(\mathcal{S}_{01}) \subset \mathrm{Gr}(k, k+2).$$

Remark 5.1.6. First, notice that this implies that the limit amplituhedron $\mathcal{A}_k^\infty$ is indeed a semi-algebraic set. Second, to match the notation more commonly found in physics literature, notice that we can write

$$\mathrm{CH}(\mathcal{S}_{01}) = \{(A_1 \dots A_k) \in \mathrm{Gr}(k, k+2) \mid \langle A_1 \dots A_k \mathcal{S}_{01} \rangle = 0\},$$

where $\langle A_1 \dots A_k \mathcal{S}_{01} \rangle$ denotes the determinant of the $(k+2) \times (k+2)$ matrix whose rows are given by the A_i's and $\mathcal{S}_{01}$. In an abuse of notation we will also write $\mathrm{Ch}(\mathcal{S}_{01}) = \langle A_1 \dots A_k \mathcal{S}_{01} \rangle$.

Let us start by considering an explicit example.

Example 5.1.7. For $k = 2$ the limit amplituhedron $\mathcal{A}_2^\infty$ lives in $\mathrm{Gr}(2, 4)$, and its algebraic boundary is given by the Chow hypersurfaces of the twisted cubic C_3 in $\mathbb{P}^3$ and its secant line $\mathcal{S}_{01}$ spanned by $\gamma_3(0)$ and $\gamma_3(1)$. If we denote the six Plücker coordinates on $\mathrm{Gr}(2, 4)$ by $p_{01}, \dots, p_{23}$, then the Chow hypersurface of $\mathcal{S}_{01}$ is given by the vanishing of

$$p_{23} - p_{13} + p_{12}.$$

The Chow hypersurface of C_3 in turn is given by

$$\begin{aligned} \mathrm{Ch}(C_3) &= \det \begin{bmatrix} p_{01} & p_{02} & p_{03} \\ p_{02} & p_{03} + p_{12} & p_{13} \\ p_{03} & p_{13} & p_{23} \end{bmatrix} \\ &= p_{01}p_{23}p_{03} + p_{01}p_{23}p_{12} - p_{01}p_{13}^2 - p_{02}^2 p_{23} + 2p_{02}p_{03}p_{13} - p_{03}^3 - p_{03}^2 p_{12}. \end{aligned}$$

See the following remark on details for obtaining this representation. ◇

Remark 5.1.8. By Theorem 5.1.5 there are two Chow hypersurfaces of interest to us: $\mathrm{Ch}(C_{k+1})$ and $\mathrm{Ch}(\mathcal{S}_{01})$. The latter has a rather simple structure: the intersection condition, $L \cap \mathcal{S}_{01}$, gives us a linear condition on the Plücker coordinates of $\mathrm{Gr}(k, k+2)$, and is thus a hyperplane section of the Grassmannian. The Chow hypersurface of the rational normal curve, on the other hand, is the vanishing locus of the determinant of the Bézout matrix, as we shall see now. Consider C_{k+1} as the closure of the image of

$$\phi : \mathbb{C} \to \mathbb{P}^{k+1}, \quad t \mapsto [1 : t : \dots : t^{k+1}].$$

Also, consider a codimension 2 plane $H \in \mathbb{P}^{k+1}$ given by the intersection of two codimension

1 hyperplanes, i.e.

$$H = \ker \begin{bmatrix} a_0 & \dots & a_{k+1} \\ b_0 & \dots & b_{k+1} \end{bmatrix}, \tag{5.1.1}$$

where $[a_0 : \ldots : a_{k+1}], [b_0 : \ldots : b_{k+1}]$ are points in the dual space $(\mathbb{P}^{k+1})^\vee$. We then define two univariate polynomials in t as

$$f = a_0 + a_1 \cdot t + \ldots + a_{k+1} \cdot t^{k+1} \quad \text{and} \quad g = b_0 + b_1 t + \ldots + b_{k+1} \cdot t^{k+1}, \tag{5.1.2}$$

where the coefficients live in $\mathbb{C}$. Then, H intersects C_{k+1} in $\gamma_{k+1}(t)$, that is $H \in \mathrm{Ch}(C_{k+1})$, if and only if $f(t) = g(t) = 0$, that is to say when the two polynomials have a common root in $\mathbb{C}$. This, in turn, is precisely the case when the resultant of f and g vanishes. The resultant is the determinant of the Sylvester matrix of f and g, which equals the determinant of the *Bézout matrix* $B(f, g)$. For $0 \leq i, j \leq k$, let $m_{ij} = \min(i, k - j)$ and $M_{ij} = \max(0, i - j)$, then we can define the entries of the complex $(k + 1) \times (k + 1)$ Bézout matrix as

$$B_{ij} = \sum_{\ell=M_{ij}}^{m_{ij}} p_{(i-\ell),(j+\ell+1)},$$

where the $p_{(i-\ell),(j+\ell+1)}$ are the Plücker coordinates of H. These are given by the 2×2 minors of the matrix representing H in Equation (5.1.1), obtained by deleting all columns except those corresponding to the subscript. This shows that $\mathrm{Ch}(C_{k+1})$ is given by the vanishing locus of $\det B(f, g)$.

The proof of Theorem 5.1.5 proceeds via a series of Lemmas, beginning with a direct generalization of [RST24, Lemma 2.2].

Lemma 5.1.9. *Fix a partition $I_n = (0 = t_1 < \ldots < t_n = 1)$ of the interval $[0, 1]$, and let Z be the totally positive $n \times (k+2)$ matrix with rows $\gamma_{k+1}(t_1)$ through $\gamma_{k+1}(t_n)$. Let $A \in \mathrm{Gr}(k, k+2)$ be a real k-space in $\mathbb{R}^{k+2}$ given by the matrix with rows A_1 through A_k. The projection $\mathbb{R}^{k+2} \to \mathbb{R}^2$ away from A is represented by a matrix $A^\perp \in \mathbb{R}^{2\times(k+2)}$, whose rowspan is the orthogonal complement of the rowspan of A. Let $z_i = A^\perp Z_i \in \mathbb{R}^2$, where $Z_i \in \mathbb{R}^{k+2}$ is the i-th row of Z, and let $M = (z_1 z_2 \ldots z_n) \in \mathbb{R}^{2\times n}$. There is a nonzero constant $c \in \mathbb{R}$ such that for all $1 \leq i < j \leq n$ the (i, j) minor of M is*

$$\det(M_{ij}) = \det(z_i z_j) = c \cdot \langle A_1 \ldots A_k i j \rangle.$$

Proof. The matrix $A \in \mathbb{R}^{k\times(k+2)}$ has rank k. Therefore, its transpose $A^T \in \mathbb{R}^{(k+2)\times k}$ has a left inverse $A^\dagger \in \mathbb{R}^{k\times(k+2)}$ so that $A^\dagger A^T = \mathrm{id}_{k\times k}$. Block multiplication of matrices shows

$$\begin{pmatrix} A^\dagger \\ A^\perp \end{pmatrix} \cdot \begin{pmatrix} A^T & Z_i & Z_j \end{pmatrix} = \begin{pmatrix} \mathrm{id}_{k\times k} & A^\dagger(Z_i \; Z_j) \\ 0 & M_{ij} \end{pmatrix}.$$

Taking determinants gives $\det(M_{ij})$ on the right hand side and

$$\det \begin{pmatrix} A^\dagger \\ A^\perp \end{pmatrix} \cdot \langle A_1 \ldots A_k i j \rangle$$

on the left hand side. Since the image of $A^\dagger$ is orthogonal to $A^\perp$, the constant $c = \det \begin{pmatrix} A^\dagger \\ A^\perp \end{pmatrix}$ is nonzero. □

Similarly there is an immediate generalization of [RST24, Theorem 3.1].

Lemma 5.1.10. *Let $n > k+2$ and Z totally positive. The algebraic boundary $\partial_a \mathcal{A}_{n,k}$ of the amplituhedron $\mathcal{A}_{n,k} = \mathcal{A}_{n,k}(Z)$ is the union of the hyperplane sections of $\mathrm{Gr}(k, k+2)$ given by $\langle A_1 \dots A_k i i{+}1\rangle = 0$ for $i \in [n{-}1]$, and $\langle A_1 \dots A_k 1n\rangle = 0$. There is a Zariski dense set of k-spaces A given as a rank k matrix with rows A_1 through A_k in the boundary stratum $\mathcal{A}_{n,k} \cap \langle A_1 \dots A_k i i{+}1\rangle$ which intersect the convex hulls $\mathrm{conv}(Z_i, Z_{i+1})$ and $\mathrm{conv}(Z_{i+2}, Z_{i+3}, Z_{i+4})$ for all $i \in [n]$, read cyclically.*

Proof. We follow [RST24]. Set $i = 1$ and write e_i for the i-th standard basis row-vector of $\mathbb{R}^n$. Consider the $2k{-}1$-dimensional semi-algebraic set of rank k matrices $\mathcal{Y} \subset \mathrm{Gr}(k,n)_{\geq 0}$ with non-negative maximal minors, comprising $k \times n$ matrices Y whose rows Y_j are given as

$$Y_j = \begin{cases} e_1 + y_{1,1} e_2 & j = 1, \\ y_{j,1} e_{j+2} + y_{j,2} e_{j+3} + e_{j+4} & 1 < j \leq k, \end{cases}$$

where $y_{i,j} \in \mathbb{R}_{\geq 0}$. Since the intersection $\partial_a \mathcal{A}_{n,k} \cap \langle A_1 \dots A_k 12\rangle$ is irreducible and of dimension $2k{-}1$ we get that $\widetilde{Z}(\mathcal{Y})$ is Zariski dense in $\partial_a \mathcal{A}_{n,k} \cap \langle A_1 \dots A_k 12\rangle$. For a generic $Y \in \mathcal{Y}$, denote its image under $\widetilde{Z}$ in $\mathrm{Gr}(k, k{+}2)$ by $A = (A_1 \dots A_k)$. The space $A \in \mathrm{Gr}(k, k{+}2)$ given by a rank k matrix with rows A_1 through A_k, intersects $\mathrm{conv}(Z_1, Z_2)$ and $\mathrm{conv}(Z_{i+1}, Z_{i+2}, Z_{i+3})$ for all $i \in [k] \setminus \{1\}$. It follows that $\langle A_1 \dots A_k 12\rangle = 0$ as well as $\langle A_1 \dots A_k i j\rangle \neq 0$ for other $i < j \in [n]$. To get the other cases, $i \neq 1$, define the operator $\sigma : \mathbb{R}^{k\times n} \to \mathbb{R}^{k\times n}$ which applies a circular shift to the columns and then flipping the signs in the first column. After that, simply consider $\sigma^{i-1}(\mathcal{Y})$. It remains to show that there are no other boundary strata, that is $\langle A_1 \dots A_k 1j\rangle \neq 0$ for $j \in [n]$ for a space $A \in \mathcal{A}^{n,k}$ whose rows are given by A_1 through A_k. This is immediate from the proof of [RST24, Theorem 3.1] and Lemma 5.1.9. □

Lemma 5.1.11. *For any $t \in [0,1]$ consider the subset $\mathcal{V}_t$ of $\mathrm{Ch}(C_{k+1}) \subset \mathrm{Gr}(k, k{+}2)$ consisting of k-spaces containing the point $\gamma_{k+1}(t)$ of the rational normal curve C_{k+1}. Then, the Euclidean boundary of $\mathcal{A}_k^\infty$ intersects $\mathcal{V}_t$ in a Zariski dense subset of $\mathcal{V}_t$.*

Proof. Fix $n_0 > k+2 \in \mathbb{N}$, a partition I_{n_0} of $[0,1]$ and $t \in [0,1]$ such that $t_i \leq t \leq t_{i+1}$ for some $i < n_0$. First, consider the case where we set $i = 1$. Let $\mathcal{Y}$ be the $(2k{-}1)$-dimensional family of $k \times n$ matrices in $\mathrm{Gr}(k,n)_{\geq 0}$ from the proof of Lemma 5.1.10 with the shifting operator σ. $\mathcal{Y}$ comprised $k \times n$ matrices Y whose rows Y_j are given as

$$Y_j = \begin{cases} e_1 + y_{1,1} e_2 & j = 1, \\ y_{j,1} e_{j+2} + y_{j,2} e_{j+3} + e_{j+3} & 1 < j \leq k, \end{cases}$$

where $y_{i,j} \in \mathbb{R}_{\geq 0}$, and we consider e_i as the standard basis row-vector of $\mathbb{R}^n$. Now take a sequence of partitions $\mathcal{I} = \{I_n\}_{n \geq n_0}$ of $[0,1]$ with $I_{n_0} < I_n < I_{n+1}$ in refinement order for all n and where we insert s_{i_n} between the original t_1 and t_2 of the partition I_{n_0}, while keeping the other t_j's unchanged. For any $I_n \in \mathcal{I}$ we can define a family $\mathcal{Y}_n$ as above. Since we can always find $s_{i_j} \in [0,1]$ with $s_{i_{j-1}} < s_{i_j} < t$ for $j > n_0$, the pointwise limit, $\lim_{n\to\infty} Y_n Z$, converges to a k-space $A \in \mathrm{Gr}(k, k{+}2)$ intersecting C_{k+1} in $\gamma_{k+1}(t)$. This space will also intersect $\mathrm{conv}(\gamma_{k+1}(t_j), \gamma_{k+1}(t_{j+1}), \gamma_{k+1}(t_{j+2}))$ for all $j \in [k{-}1]{+}1$ by construction. For the cases where $t \in [0,1]$ such that $t_i \leq t \leq t_{i+1}$ for $i > 1$, we simply use the shift σ and consider $\sigma^{i-1}Y$ for $Y \in \mathcal{Y}$. Then, using the same family of partitions $\mathcal{I}$ and family of $k \times n$ matrices $\mathcal{Y}_n$, where $(\sigma^{i-1}Y)_n = \sigma^{i-1}Y_n$, we get $A = \lim_{n\to\infty}(\sigma^{i-1}Y)_n Z \in \mathrm{Gr}(k, k{+}2)$ which intersects C_{k+1} in $\gamma_{k+1}(t)$. Denote the set of all of these pointwise limits by $\mathcal{Y}_t \subset \mathrm{Gr}(k, k{+}2)$; then we have just shown that $\mathcal{Y}_t \subset \mathcal{V}_t$. As $\mathrm{codim}(\widetilde{Z}(\mathcal{Y}_t)) = 2$ which equals $\mathrm{codim}(\mathcal{V}_t) = 2$, the set $\mathcal{Y}_t$ is Zariski dense in $\mathcal{V}_t$, finishing the proof. □

Equipped with the results from Lemmas 5.1.10 and 5.1.11 we can now prove the main Theorem of this section.

Proof of Theorem 5.1.5. We use the notation from Lemma 5.1.11. Let Σ be the incidence defined as

$$\Sigma = \{(t, V) \mid \gamma_{k+1}(t) \in V\} \subset \mathbb{C}^1 \times \mathrm{Ch}(C_{k+1}),$$

with the restrictions of the canonical projections π_1 and π_2 onto the first and second factor, respectively. We get $\pi_1^{-1}(t) = \mathcal{Y}_t \subset \mathrm{Ch}(C_{k+1})$, and $\mathcal{Y}_t \subset \mathcal{A}_k^\infty$ is irreducible by the proof of Lemma 5.1.11. Hence, for any fixed $t \in [0,1]$ the preimage $\mathcal{Y}_t$ is of codimension 1 in $\mathrm{Ch}(C_{k+1})$. Since the codomain of π_1 is 1-dimensional, we get a top-dimensional irreducible semi-algebraic subset in $\mathrm{Ch}(C_{k+1})$. Thus, the set $\pi_2(\pi_1^{-1}([0,1]))$ is Zariski dense in $\mathrm{Ch}(C_{k+1})$, more succinctly $\partial_a \mathcal{A}_k^\infty \supset \mathrm{Ch}(C_{k+1})$. Finally, for any partition I_n of $[0,1]$ it remains to consider the remaining boundary stratum $\langle A_1 \dots A_k \gamma_{k+1}(t_1)\gamma_{k+1}(t_n)\rangle = 0$ of the algebraic boundary $\partial_a \mathcal{A}_{I_n}$, as in Lemma 5.1.10. However, as $t_1 = 0$ and $t_n = 1$ for all partitions, we get $\mathcal{S}_{01}$ spanned by $\gamma_{k+1}(0)$ and $\gamma_{k+1}(1)$. Therefore, $\partial_a \mathcal{A}_k^\infty = \mathrm{Ch}(C_{k+1}) \cup \mathrm{Ch}(\mathcal{S}_{01})$, where we identified $\langle A_1 \dots A_k \mathcal{S}_{01}\rangle = 0$ with $\mathrm{Ch}(\mathcal{S}_{01})$ as in Remark 5.1.6. □

Euclidean Boundary

We end this section with a description of the Euclidean boundary $\partial \mathcal{A}_k^\infty$, where we write $\mathrm{int}(-)$ for the Euclidean interior operator on $\mathrm{Gr}_{\mathbb{R}}(k, k{+}2)$.

Proposition 5.1.12. *Let $\mathcal{I}$ be the set of all partitions of the interval $[0,1] \subset \mathbb{R}$. Then, the Euclidean interior $\mathrm{int}(\mathcal{A}_k^\infty)$ of the limit amplituhedron is given by the union of the Euclidean interiors $\mathrm{int}(\mathcal{A}_k^I)$ over all $I \in \mathcal{I}$ i.e.*

$$\mathrm{int}\,(\mathcal{A}_k^\infty) = \bigcup_{I \in \mathcal{I}} \mathrm{int}(\mathcal{A}_k^I).$$

Proof. We show both inclusions. First, notice that any interior point of $\mathcal{A}_k^I$ for a fixed partition I is clearly an interior point of $\mathcal{A}_k^\infty$. For the other inclusion, let $p \in \operatorname{int} \mathcal{A}_k^\infty$ be an interior point. Thus, there exists an open set $U \subset \operatorname{Gr}(k, k+2)$, such that $p \in U \subsetneq \mathcal{A}_k^\infty$, moreover we can choose U even smaller, such that there is another open set $V \subsetneq \mathcal{A}_k^\infty$ with $\operatorname{cl}(U) \subset V$, where $\operatorname{cl}(\cdot)$ denotes the Euclidean closure operator on $\operatorname{Gr}(k, k+2)$. Then, $\bigcup_{I \in \mathcal{I}} \operatorname{int}(V \cap \mathcal{A}^I) \supset U$ is an open cover of U, since $V \supset U$ follows from the monotonicity of the Euclidean interior operator. Moreover, since $\mathcal{A}_k^\infty$ is compact by Tychonoff's theorem, there exists a finite subcover labelled by $\mathcal{F} \subset \mathcal{I}$ with $U \subset \bigcup_{I \in \mathcal{F}} \operatorname{int}(V \cap \mathcal{A}_k^I)$. If there exists an $I_0 \in \mathcal{F}$ such that $U \subset \operatorname{int}(V \cap \mathcal{A}_k^{I_0})$, then we are done as U is an open set of $\operatorname{Gr}(k, k+2)$ with $p \in \operatorname{int}(V \cap \mathcal{A}_k^{I_0}) \subset \mathcal{A}_k^{I_0}$, that is to say p is an interior point of $\mathcal{A}_k^{I_0}$. If there does not exist such a single I_0 we can collect all of the $I \in \mathcal{F}$ with $U \cap \operatorname{int}(V \cap \mathcal{A}_k^I) \neq \varnothing$ in a finite subset $\mathcal{E}$ of $\mathcal{F}$. Then, we can denote by $\overline{I}$ the common refinement of all partitions in $\mathcal{E}$, and since taking interiors is monotone we then have $U \subset \operatorname{int}(V \cap \mathcal{A}^{\overline{I}})$, which is what we ought to show. □

We write $\partial -$ for the Euclidean boundary operator on the real locus $\operatorname{Gr}(k, k+2)(\mathbb{R})$.

Proposition 5.1.13. *The limit amplituhedron $\mathcal{A}_k^\infty$ is a closed set in the Euclidean topology. Every k-space in the Euclidean boundary $\partial \mathcal{A}_k^\infty \subset \operatorname{Gr}(k, k+2)$ intersects the rational normal curve C_{k+1} of degree $k+1$ in the interval $\gamma_{k+1}([0,1])$ or it intersects the convex hull of $\{\gamma_{k+1}(0), \gamma_{k+1}(1)\}$ taken in the affine chart U_0 of $\mathbb{P}^{k+1}$, which is an interval on the secant line $\mathcal{S}_{01}$. Conversely, every such k-space is in the Euclidean boundary $\partial \mathcal{A}_k^\infty$.*

Proof. By Proposition 5.1.12 it follows that a point is in the boundary of $\mathcal{A}_k^\infty$ if and only if it is a boundary point of $\mathcal{A}_k^I$ for all partitions I of $[0,1]$ such that $\mathcal{A}_k^I$ contains that point. Let I be a partition of $[0,1]$ of length n. For $i \in [n]$ denote by Γ_i the 2-space spanned by $\gamma_{k+1}(t_i), \gamma_{k+1}(t_{i+1})$, where the indexing t_{i+1} is read cyclically in $[n]$. Then, it follows from Lemma 5.1.10 that the (Euclidean) boundary strata $\partial \mathcal{A}_k^I$ of $\mathcal{A}_k^I$ are given by the intersection $\operatorname{Ch}(\Gamma_i) \cap \mathcal{A}_k^I \subset \operatorname{Gr}(k, k+2)$ for any $i \in [n]$ read cyclically. Therefore, a k-space is a boundary point of $\mathcal{A}_k^\infty$ if and only if it intersects $\gamma_{k+1}([0,1])$ or the line segment $[\gamma_{k+1}(0), \gamma_{k+1}(1)] \subset \mathcal{S}_{01}$ non-trivially. Since these k-spaces are, by Proposition 5.1.11, contained in $\mathcal{A}_k^\infty$, the limit amplituhedron is a closed set with boundary points as described above. □

5.2 Boundary Stratification

The purpose of this section is to systematically study the boundary stratification $\mathcal{L}(\partial_a \mathcal{A}_k^\infty)$, see Definition 1.5.4, of the limit amplituhedron. The singularity structure of the algebraic boundary $\partial_a \mathcal{A}_k^\infty$ will be the most important input. To that end, we use tangent spaces of incident varieties to study iterated singularities of Chow hypersurfaces. In the paragraph 5.2, we focus on the relevant incidence varieties. In paragraph 5.2, we apply these tools to compute the iterated singular loci and give the relevant boundary stratification from the point of view of Positive Geometry. Note that in this section we have $d = k+1$ in an effort to

Table 5.2.1: Summary of $\mathcal{L}(\partial_a \mathcal{A}_k^\infty)$ by codimension in $\mathrm{Gr}(k, k+2)$.

name	codimension	geometric interpretation
$\mathrm{Ch}(\mathcal{S}_{01})$	1	k-spaces meeting $\mathcal{S}_{01}$
$\mathrm{Sec}^\ell(C_{k+1})$	ℓ	k-spaces meeting C_{k+1} to order ℓ
$O^\ell_{0,j}$	$j+\ell+1$	k-spaces containing the j-th osculant plane at $\gamma_{k+1}(0)$, meeting $L_{0,j}$ and intersecting C_{k+1} to order ℓ
$O^\ell_{1,j}$	$j+\ell+1$	k-spaces containing the j-th osculant plane at $\gamma_{k+1}(1)$, meeting $L_{1,j}$ and intersecting C_{k+1} to order ℓ
$O^\ell(i,j)$	$i+j+\ell$	k-spaces containing the i-th osculant plane at $\gamma_{k+1}(0)$, the j-th at $\gamma_{k+1}(1)$ and intersecting C_{k+1} to order ℓ

simplify notation. A summary of our results, with the notation from 5.2.11, is given in Table 5.2.1, which collects all the strata that appear in $\partial_a \mathcal{A}_k^\infty$.

Secants and Tangent Spaces of Grassmannians

Let X be a smooth irreducible projective variety in $\mathbb{P}^d$ with vanishing ideal $I \subset \mathbb{C}[z_0, \ldots, z_d]$. Recall the definition of the tangent space from Section 1.1. It will be convenient to denote the projective tangent space $\mathbf{T}_x X$, where we made a choice of basis for the ambient $\mathbb{P}^d$, in boldface. So in particular, we have $\mathbf{T}_x X \subset \mathbf{T}_x \mathbb{P}^d$. For the Grassmannian we identified its tangent space in Example 1.1.16 as $T_V \mathrm{Gr}(k,n) = \mathrm{Hom}(V, \mathbb{C}^n/V)$, we will henceforth freely use this identification. In particular, for any $p \in \mathbb{P}^n$, we get $T_p \mathbb{P}^n = \mathrm{Hom}(p, \mathbb{C}^{n+1}/p)$, where we think of p as a 1-dimensional subspace of $\mathbb{C}^{n+1}$. For the remainder of this section we are interested in the case where X is the rational normal curve $C_d \subset \mathbb{P}^d$ of degree d.

We start by introducing a certain divisor associated to Chow forms.

Definition 5.2.1. Let $C \subset \mathbb{P}^d$ be a curve and $\mathrm{Ch}(C) \subset \mathrm{Gr}(d-1, d+1)$ be its Chow hypersurface. To a space $V \in \mathrm{Ch}(C)$, we associate a divisor D_V defined as the intersection divisor $H \cap C$ for a general hyperplane $H \subset \mathbb{P}^d$ containing V. Concretely, if H is defined by the linear form ℓ, then the restriction of ℓ to C is a section of the line bundle $\mathcal{O}_{\mathbb{P}^d}(1)|_C$ and D_V is the (zero) divisor of this section. We write S^ℓ for the symmetric ℓ-fold product C^ℓ/Sym_ℓ, the quotient of the Cartesian product modulo the action of the symmetric group Sym_ℓ acting by permutation on the factors.

Remark 5.2.2. The divisor D_V that we just defined is equal to $D_V = \sum_{p \in C} m_p p$, where m_p is the smallest order of vanishing of any linear form ℓ vanishing on V. Therefore, the *degree* $\deg D_V$ of such a divisor is the sum $\sum_{p \in C} m_p \in \mathbb{Z}$.

For any $\ell \in \mathbb{N}$, we denote by $\mathrm{Sec}^\ell(C_d)$ the *variety of ℓ secants* in $\mathrm{Gr}(d-1, d+1)$, which consists of all subspaces intersecting C_d in D_V such that $\deg(D_V) = \ell$. Note $\mathrm{Sec}^\ell(C_d) =$

$\varnothing$ for any $\ell > d-1$. Also recall that points in the ℓ-fold symmetric product $S^\ell(C_d)$ are in correspondence with effective divisors on C_d of degree ℓ, therefore we can then define the incidence

$$\Sigma_\ell = \left\{ (p_1 + \ldots + p_\ell, V) \;\middle|\; \sum_{i=1}^{\ell} p_i \leq D_V \right\} \subset S^\ell(C_d) \times \mathrm{Gr}(d-1, d+1), \quad (5.2.1)$$

with the canonical projections π_1 and π_2 into the first and second factor, respectively. Then we have $\pi_2\left(\Sigma_\ell \cap \pi_1^{-1}(S^\ell(C_d))\right) = \mathrm{Sec}^\ell(C_d)$. Following [Har92, Chapter 14], we describe the tangent space to Σ_ℓ in terms of complex analytic geometry. Any tangent vector to Σ_ℓ at the point $(p_1 + \ldots + p_\ell, V)$ is equal to $\sigma'(0)$, as an element in the tangent space of $S^\ell(C_d) \times \mathrm{Gr}(d-1, d+1)$, i.e.

$$\sigma'(0) \in \prod_{i=1}^{\ell} \mathrm{Hom}(p_i, \mathbb{C}^{d+1}/p_i) \times \mathrm{Hom}(V, \mathbb{C}^{d+1}/V),$$

for a suitable holomorphic arc $\sigma(t) = (p_1(t) + \ldots + p_\ell(t), V(t))$. Here, $p_i(t)$ is a holomorphic arc in C_d for each $i \leq \ell$ and $V(t)$ is a holomorphic arc in $\mathrm{Gr}(d-1, d+1)$ such that $p_i(t) \in V(t)$ for all t and $i \leq \ell$, and with $p_i(0) = p_i$ and $V(0) = V$. Write $\sigma'(0) = (\theta_1, \ldots, \theta_\ell, \phi)$. First consider a subspace $V \in \mathrm{Sec}^\ell(C_d)$ that meets C_d in ℓ distinct points. Then the containment $p_i(0) = p_i \in V$ implies that the restrictions of ϕ to the 1-dimensional linear space p_i must coincide with θ_i modulo V i.e. $\phi|_{p_i} = \theta_i + V$ for all $i \leq \ell$. Therefore, the tangent space of Σ_ℓ is

$$T_{(p_1+\ldots+p_\ell, V)}\Sigma_\ell = \{(\theta_1, \ldots, \theta_\ell, \varphi) \mid \varphi|_{p_i} = \theta_i + V \text{ for all } i \leq \ell\},$$

as a subspace of $\prod_{i=1}^{\ell} \mathrm{Hom}(p_i, \mathbb{C}^{d+1}/p_i) \times \mathrm{Hom}(V, \mathbb{C}^{d+1}/V)$. Moreover, we also get an intrinsic characterization of the tangent space of $\mathrm{Sec}^\ell(C_d)$ at points with $p_i \neq p_j$ $(i \neq j)$ as

$$T_V \,\mathrm{Sec}^\ell(C_d) = \{\varphi \mid \varphi(p_i) \subset \mathbf{T}_{p_i}(C_d) + V \text{ for all } i \leq \ell\} \subset \mathrm{Hom}(V, \mathbb{C}^{d+1}/V). \quad (5.2.2)$$

Next, we deal with multiplicity of intersection points.

Definition 5.2.3. Let $p \in C_d$, and denote by $\mathrm{rowspan}_{\mathbb{C}}(v, v^{(1)}, \ldots, v^{(m-1)})$ the row span of the matrix of higher derivatives, where $v(t)$ is a holomorphic arc through $p \in C_d$, and $v^{(i)}$ denotes the i-th derivative of that arc. We define the *m-th Gauss map* as follows

$$\mathcal{G}^{(m)} : C_d \longrightarrow \mathrm{Gr}(m, d+1), \quad p \mapsto (v, v^{(1)}, \ldots, v^{(m-1)}),$$

which is well-defined since C_d is smooth. The image of $p \in C_d$ under $\mathcal{G}^{(m)}$ is then called the *m-th osculant plane* $\mathcal{O}^{(m)}(p)$.

Now suppose we have $V \in \mathrm{Gr}(d-1, d+1)$ which has intersection points with C_d of higher multiplicity. Then, we can describe the tangent space as follows.

Lemma 5.2.4. *Let $V \in \mathrm{Gr}(d-1, d+1)$ with $D_V = \sum_{i=1}^{r} m_i p_i$ such that $\deg(D_V) = \ell$ and its preimage under π_2 is $(m_1 p_1 + m_2 p_2 + \ldots + m_r p_r, V) \in \Sigma_\ell$, where each p_i appears exactly m_i times. Then, the tangent space to Σ_ℓ at the unique point in $\pi_2^{-1}(V) = (D, V)$ with $D =$*

$\sum_{i=1}^{r} m_i p_i$ *is*

$$T_{(D,V)}\Sigma_\ell = \{(\theta_1, \ldots, \theta_r, \varphi) \mid \varphi|_{\mathcal{O}^{(m_i)}(p_i)} = \theta_i + V \text{ for all } i \leq r\}, \tag{5.2.3}$$

where the vectors θ_i *are in* $\mathrm{Hom}(\mathcal{O}^{(m_i)}(p_i), \mathbb{C}^{d+1}/\mathcal{O}^{(m_i)}(p_i))$.

Proof. Geometrically speaking $m_i > 1$ means that V will contain the m_i-th osculant plane $\mathcal{O}^{(m_i)}(p_i)$. Therefore, the inequality $\sum_{i=1}^{\ell} p_i \leq D_V$ implies that $\mathcal{O}^{(m_i)}(p_i) \subset V$, which in turn implies that $\varphi|_{\mathcal{O}^{(m_i)}(p_i)} = \theta_i + V$. Here $\theta_i \in \mathrm{Hom}(\mathcal{O}^{(m_i)}(p_i), \mathbb{C}^{d+1}/\mathcal{O}^{(m_i)}(p_i))$ is the tangent vector of the holomorphic arc $v_i(t)$ in $\mathrm{Gr}(m_i, d+1)$ with $v_i(t) \subset V(t)$ for all t and $v_i(0) = \mathcal{O}^{(m_i)}(p_i)$. Then, the claim follows from (5.2.1). □

The linear spans of multiple osculating planes to the rational normal curve always have the expected dimension by a generalization of Vandermonde matrices.

Remark 5.2.5. Fix a partition I_n of $[0, 1]$ and $r = (r_1, \ldots, r_n) \in \mathbb{N}^n$ with $r_1 + \ldots + r_n = d+1$. We define the *generalized Vandermonde matrix* $V_r \in \mathbb{C}^{(d+1)\times(d+1)}$ as

$$V_r = (\gamma_d^{(0)}(t_1), \ldots, \gamma_d^{(r_1)}(t_1), \ldots, \gamma_d^{(0)}(t_n), \ldots, \gamma_d^{(r_n)}(t_n)),$$

where $\gamma_d^{(k)}(t)$ denote the k-th derivative of $\gamma_d(t)$. Geometrically, speaking this Vandermonde matrix is constructed from the r_i-osculant planes $\mathcal{O}^{(r_i)}(\gamma_d(t_i))$ for all $i \leq n$. The determinant of this matrix is

$$\det(V_r) = \prod_{i=1}^{n}(1! \ldots r_i!) \prod_{j>i}(t_j - t_i)^{(r_j+1)(r_i+1)},$$

see e.g. [Cha18]. In particular, $\det V_r = 0$ if and only if $t_j = t_i$ for some $i \neq j$. This result implies that osculant planes to distinct points of the rational normal curve are transversal (and in particular do not intersect as long as $\sum m_i < d+1$).

Proposition 5.2.6. *For* $\ell \leq d-1$*, the incidence* Σ_ℓ *from* (5.2.1) *is smooth of codimension* 2ℓ.

Proof. Fix $\ell \leq d-1$. Let π_1 be the projection from the incidence Σ_ℓ into the first factor $S^\ell(C_d)$ and π_2 the projection into the Grassmannian. First, we determine the dimension of Σ_ℓ. The variety $S^\ell(C_d) \times \mathrm{Gr}(d-1, d+1)$ has dimension $\ell + 2(d-1)$. Fix $D = p_1 + \ldots + p_\ell \in S^\ell(C_d)$ and consider

$$\mathcal{V}_D = \left\{ V \in \mathrm{Gr}(d-1, d+1) \;\middle|\; \sum_{i=1}^{\ell} p_i \leq D_V \right\}.$$

Denote by O the complex vector space spanned by all the osculant planes $\mathcal{O}_i = \mathcal{O}^{(m_i)}(p_i)$; it then follows from the generalized Vandermonde determinant (5.2.5), that $\dim(O) = \ell$. Therefore, we have that $\mathcal{V}_D \cong \mathrm{Gr}(d-\ell-1, d-\ell+1)$, which is $2(d-1) - 2\ell$ dimensional. Since D varies in an ℓ-dimensional variety, we conclude $\dim(\Sigma_\ell) = \dim(\mathcal{V}_D) + \ell$; thus $\mathrm{codim}(\Sigma_\ell) = 2\ell$.

To show that Σ_ℓ is smooth recall the notation for the tangent space (5.2.3) in Lemma 5.2.4. For $V \in \mathrm{Gr}(d-1, d+1)$ with $D_V = \sum_{i=1}^{r} m_i p_i$, all p_i distinct and $\deg(D_V) = \ell$:

$$T_{(D,V)}\Sigma_\ell = \{(\theta_1, \ldots, \theta_r, \varphi) \mid \varphi|_{\mathcal{O}_i} = \theta_i + V \quad \forall i \leq r\}.$$

Then, the condition $\mathcal{O}_i \subset V$ for the incidence Σ_ℓ implies $\varphi|_{\mathcal{O}_i} = \theta_i + V$. Since the osculant planes are transversal (see Remark 5.2.5) and $\deg(D_V) = \sum_{i=1}^r m_i \leq d$, the conditions $\varphi|_{\mathcal{O}_i} = \theta_i + V$ on the linear map φ are independent. To count the conditions for each i, we choose any basis of $\mathcal{O}_i$. Then, $\varphi|_{\mathcal{O}_i} = \theta_i + V$ fixes the images of $\varphi(\mathcal{O}_i)$, that is to say, it fixes m_i columns of the $2 \times (d{+}1)$ matrix ($\dim(\mathbb{C}^{d+1}/V) = 2$) representing φ with respect to the chosen basis. Thus placing $2m_i$ linear conditions on elements in $T_{(D,V)}\Sigma_\ell$. In total, we therefore have 2ℓ linear constraints for a linear map $\varphi \in \mathrm{Hom}(V, \mathbb{C}^{d+1}/V)$ to be in $T_{(D,V)}(S^\ell(C_d) \times \mathrm{Gr}(d{-}1, d{+}1))$. Since we have shown above that $\mathrm{codim}(\Sigma_\ell) = 2\ell$, and since D_V was arbitrary, we proved that Σ_ℓ is smooth everywhere. □

Strata

We now turn to the study of the strata of the algebraic boundary $\partial_a \mathcal{A}^\infty_{d-1}$. Recall that the variety of ℓ-secants $\mathrm{Sec}^\ell(C_d)$ was the subset of $\mathrm{Gr}(d{-}1, d{+}1)$ consisting of planes V with $\deg(D_V) \geq \ell$. We start with an example.

Example 5.2.7. We continue Example 5.1.7. So $\mathcal{A}^\infty_2 \subset \mathrm{Gr}(2,4)$. In this case, we can compute the stratification of the algebraic boundary into singular loci symbolically. The results of this computation, using `Macaulay2` [GS], are displayed in Table 5.2.2. We can interpret

Table 5.2.2: Boundary stratification $\mathcal{L}(\mathcal{A}^\infty_2)$ in $\mathrm{Gr}(2,4)$ ordered by codimension.

codim 1	codim 2	codim 3	codim 4
$\mathrm{Ch}(C_3)$	$\mathrm{Sing}(\mathrm{Ch}(C_3))$	$O^1_{0,1}$	$\mathcal{S}_{01}$
$\mathrm{Ch}(\mathcal{S}_{01})$	$O^1_{0,0}$	$O^1_{1,1}$	$\mathbf{T}_{\gamma_3(0)}(C_3)$
	$O^1_{1,0}$	$O^2_{0,0}$	$\mathbf{T}_{\gamma_3(1)}(C_3)$
	$S_{0,1}$	$O^2_{1,0}$	

each stratum geometrically, see Section 5.2. The top-dimensional irreducible components are two Chow hypersurfaces: first, the variety of lines in $\mathbb{P}^3$ that intersect C_3, that is $\mathrm{Ch}(C_3)$; second, the variety of lines in $\mathbb{P}^3$ intersecting the secant line $\mathcal{S}_{01}$, that is $\mathrm{Ch}(\mathcal{S}_{01})$. It is then well known that the singular locus of the variety of lines meeting the twisted cubic, is precisely the variety of secant lines (i.e. those lines that meet the twisted cubic in two points), which is of codimension 2. We have 3 more strata in this codimension which come from the intersection of the codimension 1 strata: $O^1_{0,0}$ and $O^1_{1,0}$ are the surfaces of lines going through the points $\gamma_3(0)$ and $\gamma_3(1)$, respectively. $S_{0,1}$ is the surface of lines intersecting both $\mathcal{S}_{01}$ and C_3, generically in one point each. In codimension 3, which is dimension 1, we have four irreducible components: $O^1_{0,1}$ and $O^1_{1,1}$ are the 1-dimensional linear spaces in $\mathrm{Gr}(2,4)$ consisting of lines containing the point $\gamma_3(0)$, $\gamma_3(1)$ and intersecting

$$\mathrm{rowspan}\begin{pmatrix}\gamma_3(1)\\ \partial_t\gamma_3(0)\end{pmatrix} \quad \text{and} \quad \mathrm{rowspan}\begin{pmatrix}\gamma_3(0)\\ \partial_t\gamma_3(1)\end{pmatrix},$$

respectively, where ∂_t means taking the derivative with respect to t. On the other hand, $O^2_{0,0}$ and $O^2_{1,0}$ are the conics in $\mathrm{Gr}(2,4)$ consisting of secants going through the points $\gamma_3(0)$ and $\gamma_3(1)$, respectively. Finally, the three vertices are precisely the tangent lines $\mathbf{T}_{\gamma_3(0)}(C_3)$ and $\mathbf{T}_{\gamma_3(1)}(C_3)$, as well as the secant line $\mathcal{S}_{01}$ itself, all being part of the singular locus of the 1 dimensional strata. ◇

We proceed by considering the irreducible component of $\partial_a \mathcal{A}^{\infty}_{d-1}$ which comes from the Chow hypersurface of the rational normal curve C_d.

Proposition 5.2.8. *For any ℓ in $\mathbb{N}$ the singular locus of $\mathrm{Sec}^\ell(C_d)$ is given by*

$$\mathrm{Sing}(\mathrm{Sec}^\ell(C_d)) = \mathrm{Sec}^{\ell+1}(C_d).$$

Proof. Fix $\ell < d-1$. For brevity we write $\mathrm{Sing}^\ell(C_d) = \mathrm{Sing}(\mathrm{Sec}^\ell(C_d))$. We recall the setup for the incidence Σ_ℓ from (5.2.1) and its tangent space for ease of readability:

$$\Sigma_\ell = \left\{ (p_1 + \ldots + p_\ell, V) \;\middle|\; \sum_{i=1}^{\ell} p_i \leq D_V \right\} \subset S^\ell(C_d) \times \mathrm{Gr}(d{-}1, d{+}1)$$

We have, by construction

$$\mathrm{Sec}^\ell(C_d) = \pi_2\left(\Sigma_\ell \cap \pi_1^{-1}\left(S^\ell(C_d)\right)\right).$$

Also recall that by (5.2.3) the tangent space of Σ_ℓ at (D,V) for $V \in \mathrm{Gr}(d{-}1,d{+}1)$ such that $D_V = \sum_{i=1}^r m_i p_i = D$ and $\deg(D_V) = \ell$ and $\pi_2^{-1}(V) = (D,V)$ is given by

$$T_{(D,V)}\Sigma_\ell = \{(\theta_1,\ldots,\theta_r,\varphi) \mid \varphi|_{\mathcal{O}^{(m_i)}(p_i)} = \theta_i + V \quad \forall i \leq r\},$$

where the θ_i are elements in $\mathrm{Hom}(\mathcal{O}^{(m_i)}(p_i), \mathbb{C}^{d+1}/\mathcal{O}^{(m_i)}(p_i))$. Then, as the symmetric product $S^\ell(C_d) \cong \mathbb{P}^\ell$ of the rational normal curve C_d is smooth, see [EH16, Proposition 10.6], and as we further assume that all p_i are distinct, we get a unique smooth point in $\Sigma_\ell \cap \pi_1^{-1}\left(S^\ell(C_d)\right)$ as the preimage over V. It therefore follows that

$$\pi_2 : \Sigma_\ell \cap \pi_1^{-1}(S^\ell(C_d)) \longrightarrow \mathrm{Sec}^\ell(C_d) \subset \mathrm{Gr}(d{-}1,d{+}1)$$

is one-to-one over (D,V). To characterise the smooth points, we want to make use of [Har92, Corollary 14.10], so we need to understand when the differential map induced by π_2 is injective. Observe that we just showed that if $(\theta_1,\ldots,\theta_\ell,0) \in T_{(D,V)}\Sigma_\ell$, then by $\varphi|_{\mathcal{O}^{(m_i)}(p_i)} = \theta_i + V$, we must have $V \subset \mathcal{O}^{(m_i)}(p_i)$, this gives us a contradiction by dimensions. Therefore, π_2 has injective differential, and it follows from [Har92, Corollary 14.10] that $\mathrm{Sec}^\ell(C_d)$ is smooth at V, as chosen above. It remains to show that every smooth point of $\mathrm{Sing}^\ell(C_d)$ is of this form. To that end, we employ Zariski's weak main theorem, [Har92, Proposition 16.8]. Then, it suffices to note that any point W in $\mathrm{Sing}^{\ell-1}(C_d)$ such that $\deg(D_W) > \ell$ will have at least two distinct (disconnected) preimages $\pi_2^{-1}(W)$, and thus be a singular point. Since ℓ and D_V were arbitrary this finishes the proof. If $\ell \geq d-1$, then $\mathrm{Sec}^\ell(C_d)$ is smooth and as $\mathrm{Sec}^{\ell+1}(C_d) = \varnothing$, the claim follows immediately for all ℓ. □

In the proof of the preceding Proposition 5.2.8 we actually determine the tangent space

to the variety of ℓ-secants.

Corollary 5.2.9. *Let $V \in \mathrm{Gr}(d{-}1, d{+}1)$ with $D_V = \sum_{i=1}^{r} m_i p_i$ where all p_i distinct and $\deg(D_V) = \ell$. Then, $\mathrm{Sec}^\ell(C_d)$ is smooth at V with*

$$T_V \,\mathrm{Sec}^\ell(C_d) = \{\varphi \in \mathrm{Hom}(V, \mathbb{C}^{d+1}/V) \mid \varphi(\mathcal{O}^{(m_i)}(p_i)) \subset \mathcal{O}^{(m_i+1)}(p_i) + V \quad \forall i \leq r\}.$$

In light of Remark 5.1.8 we have another immediate consequence.

Corollary 5.2.10. *Let f and g be the univariate polynomials as in Equation (5.1.2). Then, the ℓ-th iterated singular locus $\mathrm{Sing}^\ell(\mathrm{Ch}(C_d))$ is the set of codimension 2 subspaces, where f and g have precisely $\ell + 1$ common roots, counting with multiplicity.*

Notation 5.2.11. In the analysis of the singular loci of the variety of ℓ-secants, special lines will play a role. Let $i \leq d-1$. We fix the symmetric notations

$$L_{0,i} = \mathrm{rowspan}\begin{pmatrix} \gamma_d(1) \\ \gamma_d^{(i)}(0) \end{pmatrix} \quad \text{and} \quad L_{1,i} = \mathrm{rowspan}\begin{pmatrix} \gamma_d(0) \\ \gamma_d^{(i)}(1) \end{pmatrix}, \tag{5.2.4}$$

where $\gamma_d^{(i)}(t)$ denotes the i-th derivative of γ_d at t. Set $O_{0,0}$ to be the codimension 2 variety of subspaces containing $\gamma_d(0)$, and similarly for $O_{1,0}$. For $i > 0$ define $O_{0,i} \subset \mathrm{Gr}(d{-}1, d{+}1)$ to be the codimension $2i{+}1$ variety of $(d{-}1)$-spaces containing the osculant plane $\mathcal{O}^{(i)}(\gamma_d(0))$ and intersecting the line $L_{0,i}$. In terms of the variety $\mathcal{V}_{\mathcal{O}^{(i)}}$, the set of subspaces $V \in \mathrm{Gr}(d{-}1, d{+}1)$ containing $\mathcal{O}^{(i)}(\gamma_d(0))$, we have

$$O_{0,i} = \mathrm{Ch}(L_{0,i}) \cap \mathcal{V}_{\mathcal{O}^{(i)}}.$$

For any $i \leq \ell \leq d-1$, we write

$$O_{0,i}^\ell = O_{0,i} \cap \mathrm{Sec}^\ell(C_d),$$

which has codimension $i + \ell + 1$. Symmetrically, we write $O_{1,i}^\ell$. For $0 < i, j < \ell$ and $i + j < \ell$ the intersection $O_{0,i}^\ell \cap O_{1,j}^\ell$ is the codimension $i{+}j{+}\ell$ variety in $\mathrm{Gr}(d{-}1, d{+}1)$ consisting of $(d{-}1)$-spaces containing both $\mathcal{O}^{(i)}(\gamma_d(0))$ and $\mathcal{O}^{(j)}(\gamma_d(1))$, and meeting C_d to order at least ℓ. Therefore, we denote this intersection by $O^\ell(i, j)$.

We can now study the remaining strata of $\partial_a \mathcal{A}_k^\infty$.

Proposition 5.2.12. *For any $\ell \leq d-1$, the singular locus of the intersection $\mathrm{Sec}^\ell(C_d) \cap \mathrm{Ch}(\mathcal{S}_{01})$ decomposes as follows*

$$\mathrm{Sing}(\mathrm{Sec}^\ell(C_d) \cap \mathrm{Ch}(\mathcal{S}_{01})) = \left(\mathrm{Sec}^{\ell+1}(C_d) \cap \mathrm{Ch}(\mathcal{S}_{01})\right) \cup \bigcup_{i=1}^{\ell} \left(O_{0,i}^\ell \cup O_{1,i}^\ell\right).$$

Proof. For a projective variety $X \subset \mathbb{P}^n$ and a hyperplane $H \subset \mathbb{P}^n$ we have $p \in \mathrm{Sing}(X \cap H)$ if and only if $p \in \mathrm{Sing}(X) \cap H$ or $T_p X \subset T_p H$, see [CLO15, Chapter 9]. We apply this fact to $X = \mathrm{Sec}^\ell(C_d)$ and $H = \mathrm{Ch}(\mathcal{S}_{01})$. So Proposition 5.2.8 implies that the intersection

$\mathrm{Sec}^{\ell+1}(C_d) \cap \mathrm{Ch}(\mathcal{S}_{01})$ is contained in the singular locus. Moreover, the incidence

$$\Sigma = \left\{ (s, p_1 + \ldots + p_\ell, V) \;\middle|\; s \in V \cap \mathcal{S}_{01} \wedge \sum_{i=1}^{\ell} p_i \leq D_V \right\} \subset \mathcal{S}_{01} \times S^\ell(C_d) \times \mathrm{Gr}(d{-}1, d{+}1)$$

implies that this intersection is irreducible. Therefore, it remains to study the inclusion $T_V \,\mathrm{Sec}^\ell(C_d) \subset T_V(\mathrm{Ch}(\mathcal{S}_{01}))$ for $V \in \mathrm{Sec}^\ell(C_d) \cap \mathrm{Ch}(\mathcal{S}_{01})$. Fix $V \in \mathrm{Gr}(d{-}1, d{+}1)$ with $D_V = \sum_{i=1}^r m_i p_i$ and $\deg(D_V) = \ell$, and abbreviate $\mathcal{O}_i = \mathcal{O}^{(m_i)}(p_i)$. The relevant tangent spaces are

$$T_V \,\mathrm{Sec}^\ell(C_d) = \{\varphi \in \mathrm{Hom}(V, \mathbb{C}^{d+1}/V) \mid \varphi(\mathcal{O}_i) \subset \mathcal{O}_i + V \text{ for all } 1 \leq i \leq r\}$$

by Corollary 5.2.9 and from (5.2.2)

$$T_V \,\mathrm{CH}(\mathcal{S}_{01}) = \{\varphi \in \mathrm{Hom}(V, \mathbb{C}^{d+1}/V) \mid \varphi(q) \subset \mathbf{T}_q(\mathcal{S}_{01}) + V\},$$

where q is an intersection point of V with $\mathcal{S}_{01}$. So we have containment $T_V \,\mathrm{Sec}^\ell(C_d) \subset T_V \,\mathrm{Ch}(\mathcal{S}_{01})$ if and only if we have for the subspace

$$\Lambda_V = \mathrm{span}_{\mathbb{C}}\{\varphi \in \mathrm{Hom}(V, \mathbb{C}^{d+1}/V) \mid \varphi(q) \subset \mathbf{T}_q(\mathcal{S}_{01}) + V \wedge \varphi(\mathcal{O}_i) \subset \mathcal{O}_i + V \quad \forall i \leq r\},$$

that $\dim(\Lambda_V) = \dim(T_V \,\mathrm{Sec}^\ell(C_d))$. First, notice that D_V dictates that the dimension of $\mathcal{O}_i \cap V$ equals m_i for all i. Now suppose q and the p_i's are all distinct, then we can extend the q and the corresponding osculant planes $\mathcal{O}_i$ to a basis of V. The conditions $\varphi(O_i) \subset \mathcal{O}_i + V$ and $\varphi(q) \subset \mathcal{S}_{01} + V$ then fix, up to scaling, $\ell+1$ columns of the $2 \times (d{-}1)$ matrix representing φ with respect to the chosen basis. Therefore,

$$\dim(\Lambda_V) = \ell + 1 + 2(d - \ell - 2) = 2(d-1) - \ell - 1,$$

which is strictly smaller than the dimension of $T_V \,\mathrm{Sec}^\ell(C_d)$, which equals $2(d-1) - \ell$ by Proposition 5.2.6. So it remains to consider V such that $q = p_i$ for some i. In this case, we can again extend q and the p_i's to a basis of V. However, as $p_i = q$, the inclusions will only fix, up to scaling, 2ℓ columns of the matrix representing φ with respect to this basis. Thus, $\dim(\Lambda_V) = 2(d-1) - \ell$, and thus in this case V is a singular point. Since there are only two points which lie in $\mathcal{S}_{01}$ and C_d, i.e. $\gamma_d(0)$ and $\gamma_d(1)$, we only need to study these two cases explicitly. We start with $p = q = \gamma_d(0)$, where wlog. $p = p_1$. This implies $\gamma_d(0) \in V$ and $\mathcal{O}^{(m_1)}(\gamma_d(0)) \subset V$ additionally we must have $\mathbf{T}_{\gamma_d(0)}(\mathcal{S}_{01}) + V \subset \mathcal{O}^{(m_1+1)}(\gamma_d(0)) + V$. This is the case if and only if the other basis element of $\mathbf{T}_{\gamma_d(0)}(\mathcal{S}_{01})$, that is $\gamma_d(1)$ lies in the span of $\mathcal{O}^{(m_1+1)}(\gamma_d(0))$ mod V. This, in turn, is true if and only if V contains a linear combination of $\gamma_d(1)$ and the basis vectors of $\mathcal{O}^{(m_1+1)}(\gamma_d(0))$. For $m_1 = 1$, we have actually an equality condition, $\mathbf{T}_{\gamma_d(0)}(\mathcal{S}_{01}) + V = \mathbf{T}_{\gamma_d(0)} C_d + V$, such that $A = [0 : s : t : \ldots : t] \in \mathbb{P}^d$ must be contained in V for some $[s : t] \in \mathbb{P}^1$. Then it suffices to notice that this point A lies on $L_{0,1}$ from (5.2.4). If $m_1 > 1$ and as $\mathcal{O}^{(m_1)}(\gamma_d(0)) \subset V$, we can project away from $\mathcal{O}^{(m_1)}(\gamma_d(0))$, under this projection $\gamma_d^{(m_1)}(0)$ becomes $\gamma_{d-m_1}^{(1)}(0)$ and $\gamma_d(1) = \gamma_{d-m_1}(1)$. Therefore, again we find that $[0 : s : t : \ldots : t] \in \mathbb{P}^{d-m_1}$ must be contained in the projection of V for some $[s : t] \in \mathbb{P}^1$, which lifts to $[s : \ldots : s : t : \ldots : t] \in \mathbb{P}^d$. Then, analogously notice that this

point lies on L_{0,m_1} from (5.2.4). Finally, we can use the incidence

$$\Sigma = \left\{ (s, p_1 + \ldots + p_\ell, V) \middle| s \in V \cap L_{0,m_1} \wedge \sum_{i=1}^{\ell} p_i \leq D_V \right\} \subset L_{0,m_1} \times S^\ell(C_d) \times \mathrm{Gr}(d-1, d+1)$$

to show that $O^\ell_{0,1}$ is irreducible. The reasoning for $p = q = \gamma_d(1)$ is identical. This finishes the proof. □

Proposition 5.2.13. *For any $j \leq \ell \leq d-1$ we have*

$$\mathrm{Sing}(O^\ell_{0,j}) = O^{\ell+1}_{0,j} \cup O^\ell_{0,j+1} \cup O^\ell(j,1),$$

and symmetrically

$$\mathrm{Sing}(O^\ell_{1,j}) = O^{\ell+1}_{1,j} \cup O^\ell_{1,j+1} \cup O^\ell(1,j)$$

Proof. Let $\pi_0 : \mathbb{C}^{d+1} \to \mathbb{C}^{d-j+1}$ be the projection away from the osculant plane $O = \mathcal{O}^{(j)}(\gamma_d(0))$, and let $\mathcal{V}_O = \{V \mid O \subset V\} \subset \mathrm{Gr}(d-1, d+1)$. Then, π_0 induces the regular map

$$\overline{\pi}_0 : \mathcal{V}_O \subset \mathrm{Gr}(d-1, d+1) \to \mathrm{Gr}(d-j-1, d-j+1)$$

which is, in fact, an isomorphism. Set $\overline{d} = d - j - 1$. First, notice that $\pi_0(C_d) = C_{\overline{d}}$, as we can recursively project away from $\gamma_d(0)$, under which $\gamma^{(1)}_d(0)$ becomes $\gamma_{d-1}(0)$, and then we repeat $(j-1)$-times. Secondly, we also get $\overline{\pi}_0(L_{0,j}) = \overline{\mathcal{S}_{01}} \subset \mathbb{P}^{\overline{d}}$, where $\overline{\mathcal{S}_{01}}$ is spanned by $\gamma_{\overline{d}}(0)$ and $\gamma_{\overline{d}}(1)$. We therefore get the following identification

$$O^\ell_{0,j} \cong \overline{\pi}_0(\mathcal{V}_O \cap \mathrm{Sec}^\ell(C_d) \cap \mathrm{Ch}(L_{0,j})) = \mathrm{Sec}^{\ell-j}(C_{\overline{d}}) \cap \mathrm{Ch}(\overline{\mathcal{S}_{01}}).$$

We can now apply Proposition 5.2.12 to find the singular locus:

$$\mathrm{Sing}(\mathrm{Sec}^{\ell-j}(C_{\overline{d}}) \cap \mathrm{Ch}(\mathcal{S}_{01})) = (\mathrm{Sec}^{\ell-j+1}(C_{\overline{d}}) \cap \mathrm{Ch}(\overline{\mathcal{S}_{01}})) \cup \overline{O}^{\ell-j}_{0,1} \cup \overline{O}^{\ell-j}_{1,1},$$

where the $\overline{O}^{\ell-j}_{0,1}$ is analogously defined as in $O^{\ell-j}_{0,1}$, but in the space $\mathbb{P}^{\overline{d}}$. Similarly for $\overline{O}^{\ell-j}_{1,1}$. Then first notice that

$$\overline{\pi}_0^{-1}\left(\mathrm{Sec}^{\ell-j+1}(C_{\overline{d}}) \cap \mathrm{Ch}(\overline{\mathcal{S}_{01}})\right) = O^{\ell+1}_{0,j}.$$

Next, observe that any $\overline{V} \in \overline{O}^{\ell-j}_{1,1}$ contains a line L spanned by $\gamma_{\overline{d}}(0)$ and $\gamma_{\overline{d}}(1) - s\gamma^{(1)}_{\overline{d}}(0)$ for some $s \neq 0 \in \mathbb{C}$. Then, $\gamma_{\overline{d}}(0)$ clearly lifts to $\gamma^{(j)}_d(0)$ under $\overline{\pi}_0$, and $\gamma^{(1)}_{\overline{d}}(0)$ to $\gamma^{(j+1)}_d(0)$. Therefore, the preimage $\overline{\pi}_0^{-1}(\overline{V})$ corresponds precisely to a $(d-1)$-space containing $\mathcal{O}^{(j+1)}(0)$ and a point on the line $L_{0,j+1}$, and thus $\overline{\pi}_0^{-1}(\overline{O}^{\ell-j}_{0,1}) = O^\ell_{0,j+1}$. For $\overline{O}^{\ell-j}_{1,1}$ similarly observe that $\gamma^{(1)}_{\overline{d}}(1)$ lifts to $\gamma^{(1)}_d(1)$ and $\gamma_{\overline{d}}(1)$ of course to $\gamma_d(1)$. Therefore, we can identify the preimage $\overline{\pi}_0^{-1}(\overline{O}^{\ell-j}_{1,1})$ with the subset of $\mathrm{Sec}^\ell(C_d)$ containing $L_{1,1}$ and $\mathcal{O}^{(j)}(\gamma_d(0))$, which by definition is $O^\ell(j,1)$. The proof for $\mathrm{Sing}(O^\ell_{1,j})$ is analogous after a change of basis, interchanging the roles of $\gamma_d(0)$ and $\gamma_d(1)$. □

Proposition 5.2.14. *Let $0 < i, j < d-1$ and $\ell \geq i + j$. Then, for $\ell \leq d-1$ we have*

$$\mathrm{Sing}(O^\ell(i,j)) = O^{\ell+1}(i,j)$$

and for $\ell \geq d$ the singular locus is empty.

Proof. Let P be the plane spanned by the two osculant planes $\mathcal{O}^{(i)}(\gamma_d(0))$ and $\mathcal{O}^{(j)}(\gamma_d(1))$, and let $\mathcal{V}_P = \{V \mid P \subset V\} \subset \mathrm{Gr}(d-1, d+1)$. Notice that the generalized Vandermonde determinant, (5.2.5), implies that $\dim(P) = i + j$. Therefore, let $\pi : \mathbb{C}^{d+1} \to \mathbb{C}^{d-i-j+1}$ be the projection away from P. Then, π induces the regular map

$$\overline{\pi} : \mathcal{V}_P \subset \mathrm{Gr}(d-1, d+1) \to \mathrm{Gr}(d-i-j-1, d-i-j+1)$$

which in fact is an isomorphism. We then get the following identification

$$O^\ell(i,j) \cong \overline{\pi}(\mathcal{V}_P \cap \mathrm{Sec}^\ell(C_d)) = \mathrm{Sec}^{\ell-i-j}(C_{d-i-j}).$$

Therefore, the singular locus is given by Proposition 5.2.8 as $\mathrm{Sec}^{\ell-i-j+1}(C_{d-i-j})$. This lifts under $\overline{\pi}$ to $O^{\ell+1}(i,j)$. The claim for $\ell \geq d$ is trivial as then both $\mathrm{Sec}^\ell(C_d)$ and $O^\ell(i,j)$ are empty. □

5.3 Residual Arrangement

In this section we characterize the residual arrangement $\mathcal{R}(\mathcal{A}_k^\infty)$, see Definition 1.5.6, of the limit amplituhedron. This will be a key ingredient in showing that the limit amplituhedron makes a positive geometry. In fact our main result shows that the residual arrangement is trivial.

Theorem 5.3.1. *The residual arrangement $\mathcal{R}(\mathcal{A}_k^\infty)$ is empty.*

To prove this claim, we show that $\mathcal{A}_k^\infty$ intersects each stratum discovered in Section 5.2 in a dense subset (in the Zariski topology), type by type. Therefore, the proof of our main result is postponed until the end of this section.

Proposition 5.3.2. *For $\ell \leq k$ let Y be a linear ℓ-space in $\mathbb{C}^{k+2}$ spanned by ℓ points on C_{k+1}, and let $\mathcal{V}_Y$ be the subvariety of $\mathrm{Gr}(k, k+2)$ comprising all k-spaces containing Y. Then, $\mathcal{A}_k^\infty$ intersects $\mathcal{V}_Y$ in a Zariski dense subset.*

Proof. Fix $i_1 < \ldots < i_n \in \mathbb{N}$, and let Y be the vector space spanned by $\gamma_{k+1}(t_{i_1})$ up to $\gamma_{k+1}(t_{i_\ell})$ for $t_{i_j} \in [0,1]$. Denote by Z the totally positive $n \times (k+2)$ matrix obtained by stacking all $\gamma_{k+1}(t_{i_j})$ on top of each other. Further let, π_Y be the projection of $\mathbb{C}^{k+2}$ away from Y, and π'_Y be the projection of $\mathbb{C}^n$ away from the span of $e_{i_1}, \ldots, e_{i_\ell}$, where e_j is the j-th standard basis (row) vector of $\mathbb{C}^n$. With the notation $\mathcal{V}_Y$ as above, we have $\dim(\mathcal{V}_Y) = 2(k-\ell)$. Since the $\gamma_{k+1}(t_j)$ lie on the rational normal curve C_{k+1}, the projection induces the isomorphism $\pi_Y(C_{k+1}) \cong C_{k+1-\ell}$ in $\mathbb{P}^{k+1-\ell}$. By deleting the rows Z_{i_j} of Z we get a totally positive $(n-\ell) \times (k-\ell+2)$ matrix $Z|_Y$, inducing an amplituhedron map

$$\widetilde{Z}|_Y : \mathrm{Gr}(k-\ell, n-\ell)_{\geq 0} \to \mathcal{A}^I_{k-\ell} \subset \mathrm{Gr}(k-\ell, k-\ell+2),$$

sending $V' \mapsto V'Z|_Y$. Its image is a semi-algebraic set with non-empty interior i.e. of dimension $2(k-\ell)$. Now, let $V \in \mathrm{Gr}(k,n)_{\geq 0}$ such that there is the $\ell \times \ell$ identity matrix in

the columns $i_1, \ldots, i_\ell$. Let V' be a matrix corresponding to the subvector space $\pi'_Y(V) \in \mathrm{Gr}(k-\ell, n-\ell)_{\geq 0}$. Then $\widetilde{Z}(V) = VZ$ contains Y and thus $VZ \in \mathcal{V}_Y$. More precisely, the image under $\widetilde{Z}$ can be written as the rowspan of the matrix obtained by concatenation of Y and $\widetilde{Z}|_Y(V')$. Therefore, we get a top-dimensional semi-algebraic set contained in $\mathcal{A}_k^\infty \cap \mathcal{V}_Y$ (more precisely, in $\mathcal{A}_k^I \cap \mathcal{V}_Y$) with non-empty interior, proving the claim. □

Lemma 5.3.3. *For all $\ell \leq k$ the limit amplituhedron $\mathcal{A}_k^\infty$ intersects the variety of ℓ-secants $\mathrm{Sec}^\ell(C_{k+1})$ in a Zariski dense subset.*

Proof. Pick real numbers $0 < s_1 < s_2 < \ldots < s_\ell < 1$, write $s = (s_1, \ldots, s_\ell)$, and define the ℓ-space

$$Y(s) = \mathrm{rowspan} \begin{pmatrix} \gamma_{k+1}(s_1) \\ \vdots \\ \gamma_{k+1}(s_\ell) \end{pmatrix}.$$

It follows from Proposition 5.3.2 that $\mathcal{A}_k^\infty$ intersects $\mathcal{V}_{Y(s)}$ in a dense subset. Any element of $\mathcal{V}_{Y(s)}$ is an element of $\mathrm{Sing}^\ell(\mathrm{Ch}(C_{k+1})) = \mathrm{Sec}^\ell(C_{k+1})$ by construction. The fiber of $\mathcal{V}_{Y(s)}$ over s has dimension $2(k-\ell)$. Since the base is ℓ-dimensional, the union of $\mathcal{V}_{Y(s)}$ for all $s \in [0,1]^\ell$ is irreducible of dimension $2k - \ell = \dim(\mathrm{Sec}^\ell(C_{k+1}))$, this shows the claim. □

Lemma 5.3.4. *For all $\ell \leq k$ the limit amplituhedron $\mathcal{A}_k^\infty$ contains a Zariski dense subset of the intersection $\mathrm{Sec}^\ell(C_{k+1}) \cap \mathrm{Ch}(\mathcal{S}_{01})$.*

Proof. Again pick real numbers $0 < s_1 < s_2 < \ldots < s_{\ell-1} < 1$, write $s = (s_1, \ldots, s_{\ell-1})$, and define the ℓ-space

$$Y(0,s) = \mathrm{rowspan} \begin{pmatrix} \gamma_{k+1}(0) \\ \gamma_{k+1}(s_1) \\ \vdots \\ \gamma_{k+1}(s_{\ell-1}) \end{pmatrix}.$$

It follows from Proposition 5.3.2 that $\mathcal{A}_k^\infty$ intersects $\mathcal{V}_{Y(0,s)}$ in a dense subset and any element of $\mathcal{V}_{Y(0,s)}$ is an element of $\mathrm{Sec}^\ell(C_{k+1}) \cap \mathrm{Ch}(\mathcal{S}_{01})$ by construction. The fiber of $\mathcal{V}_{Y(0,s)}$ over s has dimension $2(k-\ell)$. Since the base is $(\ell-1)$-dimensional, the union of $\mathcal{V}_{Y(0,s)}$ for all $s \in [0,1]^{\ell-1}$ is irreducible of dimension $2k - \ell - 1 = \dim\left(\mathrm{Sec}^\ell(C_{k+1}) \cap \mathrm{Ch}(\mathcal{S}_{01})\right)$, this shows the claim. □

Lemma 5.3.5. *For all $j < \ell \leq k$ the limit amplituhedron $\mathcal{A}_k^\infty$ intersects both $O_{0,j}^\ell$ and $O_{1,j}^\ell$ in Zariski dense subsets.*

Proof. Recall that $\dim(O_{0,j}^\ell) = 2k-j-\ell-1$ and set $c = \ell-j-1$. Fix $s = (s_1, \ldots, s_c) \in (0,1)^c$, take a partition I of $[0,1]$ refining s, and let $j < i_1 < \ldots < i_c$ be the labels of the s_i's in I. Consider a family of partitions $\mathcal{I} = \{I_n\}_{n \in \mathbb{N}}$ with $I_n \leq I_{n+1}$ for all n refining I, in such a way

that $\gamma_{k+1}(0), \ldots, \gamma_{k+1}(t_j)$ span the osculant plane $\mathcal{O}^{(j)}(\gamma_{k+1}(0))$ for $n \to \infty$. Next consider the ℓ-space

$$Y(I_n) = \text{rowspan} \begin{pmatrix} \gamma_{k+1}(0) \\ \vdots \\ \gamma_{k+1}(t_j) \\ \gamma_{k+1}(t_{i_1}) \\ \vdots \\ \gamma_{k+1}(t_{i_c}) \\ \gamma_{k+1}(1) \end{pmatrix}. \tag{5.3.1}$$

Notice that $Y(I_n)$ intersects $L_{0,j}$ in the limit for $n \to \infty$ by construction. As usual, let $\mathcal{V}_{Y(I_n)}$ denote the set of k-spaces containing $Y(I_n)$ for any fixed $n \in \mathbb{N}$. Then, notice that $\dim(\mathcal{V}_{Y(I_n)}) = 2k-2\ell$ and therefore the union of fibers over all s is of dimension $2k-l-j-1$ which equals $\dim(O^{\ell}_{0,j})$. Consequently, $\mathcal{V}_{Y(I_n)}$ is dense in $O^{\ell}_{0,j}$ for $n \to \infty$, and thus $\mathcal{A}^{\infty}_k$ must in turn be dense in $\mathcal{V}_{Y(I_n)}$ for all $n \in \mathbb{N}$ by Proposition 5.3.2. It follows that $\mathcal{A}^{\infty}_k$ is dense in $O^{\ell}_{0,j}$, as ought to be shown. The proof for $O^{\ell}_{1,j}$ is analogous upon defining

$$Y(I_n) = \text{rowspan} \begin{pmatrix} \gamma_{k+1}(0) \\ \gamma_{k+1}(t_{i_1}) \\ \vdots \\ \gamma_{k+1}(t_{i_c}) \\ \gamma_{k+1}(t_{n-j}) \\ \vdots \\ \gamma_{k+1}(1) \end{pmatrix}. \tag{5.3.2}$$

□

Lemma 5.3.6. *The intersection of the limit amplituhedron $\mathcal{A}^{\infty}_k$ with any stratum from Section 5.2 is Zariski dense in that stratum.*

Proof. For $\text{Ch}(\mathcal{S}_{01})$, $\text{Sec}^{\ell}(C_{k+1})$ and their intersection $\text{Ch}(\mathcal{S}_{01}) \cap \text{Sec}^{\ell}(C_{k+1})$ this follows from Lemmas 5.3.3 and 5.3.4. For $O^{\ell}(i,j)$ the subsets defined in (5.3.1) and (5.3.2) can be combined to show that the intersection with $\mathcal{A}^{\infty}_k$ is Zariski dense as well. Similarly, for the intersections $O^{\ell}(i,j) \cap O^{\ell}_{0,m}$ and $O^{\ell}(i,j) \cap O^{\ell}_{1,m}$. Finally, noting that $O^{\ell}_{0,i} \cap O^{\ell'}_{1,j} = O^{\max(\ell,\ell')}(i,j)$ by definition, finishes the proof. □

Finally the main result of this section, Theorem 5.3.1 follows easily.

Proof of Theorem 5.3.1. Combining Theorem 5.1.5 with Lemmas 5.3.3-5.3.6 proves the claim. □

The remainder of this section is devoted to proving Theorem 5.0.1. With the intention of proving the existence of a canonical form for $\mathcal{A}^{\infty}_k$ we make the following observations.

Proposition 5.3.7. *Let $k \geq 3$. There are precisely k boundary strata of $\mathcal{A}_k^\infty$ of dimension 1. They are $O_{0,k-1}^{k-1}$, $O_{1,k-1}^{k-1}$, and $O^k(i,j)$ for all $1 \leq i,j \leq k-2$ with $i+j=k-1$.*

Proof. Notice that $\mathrm{Sec}^\ell(C_{k+1})$ is of codimension k at most, and intersecting with $\mathrm{Ch}(\mathcal{S}_{01})$ only increases the codimension by 1. Therefore, all higher codimension parts come from intersections with $O_{0,i}^j$ and $O_{1,i}^j$ for some suitable i and j. Now it is easy to see that both $O_{0,k-1}^{k-1}$ and $O_{1,k-1}^{k-1}$ are two distinct 1-dimensional strata of $\partial_a\mathcal{A}_k^\infty$, as they are contained in the $(k-1)$-dimensional osculant plane at $\gamma_{k+1}(0)$ and $\gamma_{k+1}(1)$, respectively while also need to intersect C_{k+1} in an additional point. Fix $1 \leq \ell \leq k-2$. Then, the k-spaces in the intersection $O^\ell(i,j) = O_{0,i}^k \cap O_{1,j}^k$ are precisely those which contain the two osculant planes $\mathcal{O}^{(i)}(\gamma_{k+1}(0))$ and $\mathcal{O}^{(j)}(\gamma_{k+1}(1))$ and additionally meet C_{k+1} in another point. Notice that the additional condition of meeting the lines $L_{0,i}$ and $L_{1,j}$ is implied as $\gamma_{k+1}(1)$ is contained in the former and $\gamma_{k+1}(0)$ in the latter. □

Lemma 5.3.8. *The union of the 1-dimensional strata of the algebraic boundary $\partial_a\mathcal{A}_k^\infty$ is connected. In particular, $\mathcal{A}_k^\infty$ intersects each 1-dimensional stratum in a single connected component.*

Proof. For $k=1$, the claim is discussed in Example 5.0.2. For $k=2$, the claim follows from Table 5.2.2 in Example 5.1.7, as all 1-dimensional strata contain the point $\mathcal{S}_{01}$. For $k>2$ we argue as follows. Recall the strata from Proposition 5.3.7

$$O^k(i,j) = O_{0,i}^k \cap O_{1,j}^k,$$

for all $1 \leq i,j \leq k-2$ with $i+j=k-1$. Then, we claim that for each i and j, there is a unique k-space $V \in \mathrm{Gr}(k,k+2)$ containing $\mathcal{O}^{(i)}(\gamma_{k+1}(0))$ and $\mathcal{O}^{(j)}(\gamma_{k+1}(1))$, which is contained in the two 1-dimensional strata $O^k(i-1,j) = O_{0,i-1}^k \cap O_{1,j}^k$ and $O^k(i,j-1) = O_{0,i}^k \cap O_{1,j-1}^k$. First, note that this means that we can choose a matrix representative of V for which $k-1$ of its rows are given by $\mathcal{O}^{(i)}(\gamma_{k+1}(0))$ and $\mathcal{O}^{(j)}(\gamma_{k+1}(1))$. Then the claim follows because the strata are Zariski closed, so we can assume that the additional point on C_{k+1} to become the i-th or j-th derivative of γ_{k+1} at 0 or 1, respectively. Finally, for $O_{0,k-1}^{k-1}$ notice that by the same reasoning as above, it contains the k-space spanned by $\mathcal{O}^{(k-1)}(\gamma_{k+1}(0))$ and $\mathcal{O}^{(0)}(\gamma_{k+1}(1)) = \gamma_{k+1}(1)$, which is also contained in the 1-dimensional stratum $O_{0,k-2}^k \cap O_{1,1}^k$. An analogous reasoning for $O_{1,k-1}^{k-1}$ proves the first claim. For the latter, note that we have just shown that each $\mathcal{A}_k^\infty$ has precisely 2 vertices on each 1-dimensional stratum. Therefore, $\mathcal{A}_k^\infty$ must have a single connected component on each of those strata. Lastly, $\mathcal{A}_k^\infty$ does not contain the whole curve, as any k-space meeting the rational normal curve outside the interval $[0,1]$ is not contained in $\mathcal{A}_k^\infty$, hence the connected component must be a curve segment. □

We can now prove the following version of our main result Theorem 5.0.1.

Theorem 5.3.9. *The limit amplituhedron gives rise to a positive geometry* $(\mathrm{Gr}(k,k+2),\mathcal{A}_k^\infty,\omega_\infty)$. *In particular, its canonical form* $\omega_\infty \in \Omega^{2k}_{\log}(\mathrm{Gr}(k,k+2)\setminus\partial_a\mathcal{A}_k^\infty)$ *can be expressed via the Chow forms of the rational normal curve* C_{k+1} *and its secant line* $\mathcal{S}_{01}$ *as*

$$\omega_\infty = \frac{1}{\mathrm{Ch}(C_{k+1})\,\mathrm{Ch}(\mathcal{S}_{01})} dx_1\wedge\cdots\wedge dx_{2k}, \tag{5.3.3}$$

where the x_i *are local coordinates of the Grassmannian* $\mathrm{Gr}(k,k+2)$ *away from the hyperplane section* $\mathrm{Ch}(\mathcal{S}_{01})$.

Proof. By Theorem 5.3.1 the numerator of a canonical form, if it exists, must be a constant. Similarly, by Theorem 5.1.5 the denominator of the canonical form is fixed, by demanding that it has simple poles along the boundary. In combination these observations imply that any contender for a canonical form must be non-zero multiple of ω_∞ as in (5.3.3).

We start by showing that this form has the correct residues. All boundary strata of $\mathcal{A}_k^\infty$ are given in Table 5.2.1, see e.g. the explicit parameterizations in Lemmas 5.3.3 to 5.3.6. A subset of $\mathcal{A}_k^\infty$ supported on one of these strata is called a face. Hence, all faces of $\mathcal{A}_k^\infty$ are supported on rational varieties by the results in *loc. cit.* Let F' be a face of $\mathcal{A}_k^\infty$ and let $F\subset F'$ be a facet i.e. of codimension 1 in F'. Start by considering the residue of ω_∞ along $\mathrm{Ch}(C_{k+1})$ which is an effective Cartier divisor in $\mathrm{Gr}(k,k+2)$ as it is a hypersurface in a smooth variety. This form has the correct poles along the restriction of $\mathrm{Ch}(\mathcal{S}_{01})$ as well as the singular locus of $\mathrm{Ch}(C_{k+1})$, see e.g. [Web05, Section 2]. The same holds true for the residue along $\mathrm{Ch}(\mathcal{S}_{01})$. Now, the singular locus in $\mathrm{Ch}(C_{k+1})$ has codimension 1 and contributes to facets of $\mathcal{A}_k^\infty\cap\mathrm{Ch}(C_{k+1})$, hence for the residue computations along the singular locus we must replace $\mathrm{Ch}(C_{k+1})$ by its normalization, see [Lam22, Remark 3]. More explicitly, let $\pi:\Sigma_2\to\mathrm{Ch}(C_{k+1})$ be the blow-up of $\mathrm{Ch}(C_{k+1})$ in its singular locus $\mathrm{Sec}^2\subset\mathrm{Ch}(C_{k+1})$ as in the proof of Proposition 5.2.8. Let $E_2=\pi^{-1}(\mathrm{Sec}^2)\subset\Sigma_2$ be the exceptional divisor, which is effective and Cartier since Σ_2 is smooth. Then, the residue of the pullback $\pi^*(\mathrm{Res}(\omega_\infty))$ has again poles along the pullbacks of $\mathrm{Ch}(\mathcal{S}_{01})\cap\mathrm{Ch}(C_{k+1})$ and the singular locus $\mathrm{Sec}^3\subset\mathrm{Sec}^2$, which are the facets of $\mathcal{A}_k^\infty\cap\mathrm{Sec}^2$. The higher iterated residues of ω_∞ follow analogously.

It remains to show that there is a unique way of orienting the interior of $\mathcal{A}_k^\infty$ such that taking successive residues of ω_∞ along faces of $\mathcal{A}_k^\infty$ results in ±1 at the vertices. An orientation of the interior of $\mathcal{A}_k^\infty$, induces an orientation on its boundary. Thus, we get an induced orientation on the flag of the strata containing a vertex of the residual arrangement. The iterated residue of the canonical form of $\mathcal{A}_k^\infty$ on a 1 dimensional stratum must be a non-zero 1-form with simple poles along the two vertices of the curve segment, see Lemma 5.3.8. Therefore, we can choose the scalar in the numerator of ω_∞ in a way which makes the residue equal to $+1$ at one vertex and -1 at the other vertex of that 1-dimensional stratum. Hence, the canonical form for $\mathcal{A}_k^\infty$ exists and is unique, and thus $(\mathrm{Gr}(k,k+2),\mathcal{A}_k^\infty,\omega_\infty)$ is a positive geometry. □

Chapter 6

Towards Positive Geometries for Cosmology

We have to go to the stars now.

Bernd

Despite the vast success of positive geometries, and more broadly positive geometry methods, in the field of scattering amplitudes for particle physics, one major hurdle has been the adoption of these notions in other areas of physics. Theories for high-energy cosmology have not seen any applications of positive geometry beyond toy models and the Cosmohedron, see e.g. [AFV25]. In this chapter, we present experimental mathematical results which provide first steps toward establishing these methods in this field. We introduce a positive geometry which naturally captures the wavefunction coefficient at 4 particles - the analogue of the scattering amplitude in particle physics - see Definition 6.1.2. We also show how this fits into the framework laid out in Chapter 2, by finding a recursive positive pair in which this positive geometry lives.

In recent work [Aru+26], a big breakthrough led to the discovery of the wavefunction coefficient, Definition 6.1.2, for 4 particles inside an orthogonal Grassmannian. The main result in this section will be the introduction of a positive geometry whose canonical form, analogously to the amplituhedron for particle physics, captures this wavefunction coefficient.

6.1 The Wavefunction Coefficient as a Canonical Form

First we introduce the ambient space. For the details see [Man09, §2].

Definition 6.1.1. Let $\mathbb{C}^{2n}$ be the $2n$-dimensional complex vector space with a symmetric bilinear form given in the standard basis via

$$Q = \begin{bmatrix} 0 & \mathrm{id}_n \\ \mathrm{id}_n & 0 \end{bmatrix}.$$

The *orthogonal Grassmannian* $\mathrm{OGr}(n, 2n)$ is defined as the set of n-dimensional subvector spaces $V \subset \mathbb{C}^{2n}$ that are isotropic with respect to Q, that is

$$\mathrm{OGr}(n, 2n) = \{V \in \mathrm{Gr}(n, 2n) \mid V^T Q V = 0\}.$$

It has two irreducible components:

$$\mathrm{OGr}^+(n, 2n) = \{V \in \mathrm{OGr}(n, 2n) \mid \dim(E \cap V) = n \bmod 2\},$$
$$\mathrm{OGr}^-(n, 2n) = \{V \in \mathrm{OGr}(n, 2n) \mid \dim(E \cap V) = n - 1 \bmod 2\},$$

where $E \cong \mathbb{C}^n$ is the vector space spanned by the first n standard basis vectors of $\mathbb{C}^{2n}$. We write $\mathrm{CGr}(n) = \mathrm{OGr}^+(n, 2n)$, and refer to this variety as the *cosmological Grassmannian*.

From now on, we will only consider the case $n = 4$. We choose to embed the cosmological Grassmannian $\mathrm{CGr}(4)$ via its *Pfaffian embedding* (also called the spinor embedding) into $\mathbb{P}^7$: Denote the coordinates on $\mathbb{P}^7$ by $\mathrm{pf}_\varnothing, \mathrm{pf}_{1234}, \mathrm{pf}_{12}, \ldots, \mathrm{pf}_{34}$. Explicitly, on the affine patch with $\mathrm{pf}_\varnothing \neq 0$, any point of the cosmological Grassmannian can be represented by a 4×8 matrix, as in (6.1.1), and then take the *Pfaffians*, i.e. the (positive) square roots of all even-sized principal minors of the skew-symmetric matrix to the right in

$$A = \begin{array}{c} \begin{matrix} \overline{1} & \overline{2} & \overline{3} & \overline{4} & 1 & 2 & 3 & 4 \end{matrix} \\ \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & \mathrm{pf}_{12} & \mathrm{pf}_{13} & \mathrm{pf}_{14} \\ 0 & 1 & 0 & 0 & -\mathrm{pf}_{12} & 0 & \mathrm{pf}_{23} & \mathrm{pf}_{24} \\ 0 & 0 & 1 & 0 & -\mathrm{pf}_{13} & -\mathrm{pf}_{23} & 0 & \mathrm{pf}_{34} \\ 0 & 0 & 0 & 1 & -\mathrm{pf}_{14} & -\mathrm{pf}_{24} & -\mathrm{pf}_{34} & 0 \end{bmatrix} \end{array}. \tag{6.1.1}$$

For example, the Pfaffian corresponding to the columns labeled 12 is precisely pf_{12}. In this case $\mathrm{CGr}(4)$ is a smooth quadric hypersurface in $\mathbb{P}^7$, given by

$$\mathrm{CGr}(4) = \mathbb{V}(\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234} - \mathrm{pf}_{12}\, \mathrm{pf}_{34} + \mathrm{pf}_{13}\, \mathrm{pf}_{24} - \mathrm{pf}_{14}\, \mathrm{pf}_{23}). \tag{6.1.2}$$

We call the defining quadric polynomial the *Pfaffian constraint*. Denote by $\mathrm{CGr}(4)_{\geq 0}$ the *Pfaffian non-negative* cosmological Grassmannian, which is the set of all points in $\mathrm{CGr}(4)$ whose Pfaffians have all the same sign. We consider the divisor D on $\mathrm{CGr}(4)$ which is given by the union of all the Pfaffian coordinate hyperplane sections and the vanishing loci of

$$\begin{aligned} S &= \det A_{\overline{12}12} = \mathrm{pf}_{13}\, \mathrm{pf}_{24} - \mathrm{pf}_{14}\, \mathrm{pf}_{23} \\ T &= \det A_{\overline{23}23} = \mathrm{pf}_{13}\, \mathrm{pf}_{24} - \mathrm{pf}_{12}\, \mathrm{pf}_{34}\,, \end{aligned} \tag{6.1.3}$$

More succinctly,

$$D = \bigcup_I \mathbb{V}(\mathrm{pf}_I) \cup \mathbb{V}(S) \cup \mathbb{V}(T) \subset \mathrm{CGr}(4). \tag{6.1.4}$$

Then, the wavefunction coefficient on the cosmological Grassmannian was first introduced in [Aru+26].

Definition 6.1.2. The $n = 4$ Yang-Mills *wavefunction coefficient* A_4 is the logarithmic 6-form on the cosmological Grassmannian $\mathrm{CGr}(4)$ given as

$$A_4 = \frac{S+T}{\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234}\, \mathrm{pf}_{13}\, \mathrm{pf}_{24}\, ST}\, \mathrm{dpf} \in \Omega^6_{\log}(\mathrm{CGr}(4) \setminus D),$$

where $\mathrm{dpf} = \mathrm{dpf}_\varnothing \wedge \mathrm{dpf}_{1234} \wedge \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14}$.

Remark 6.1.3. To see that A_4 is a dlog form, note that, using the total derivative of the Pfaffian constraint (6.1.2), we can rewrite A_4 as

$$A_4 = \frac{\mathrm{dpf}_\varnothing}{\mathrm{pf}_\varnothing} \wedge \frac{\mathrm{dpf}_{1234}}{\mathrm{pf}_{1234}} \wedge \frac{\mathrm{dpf}_{12}}{\mathrm{pf}_{12}} \wedge \frac{\mathrm{dpf}_{23}}{\mathrm{pf}_{23}} \wedge \left(\frac{\mathrm{dpf}_{13}}{\mathrm{pf}_{13}} + \frac{\mathrm{dpf}_{24}}{\mathrm{pf}_{24}}\right) \wedge \left(\frac{\mathrm{dS}}{S} + \frac{\mathrm{dT}}{T}\right),$$

which is clearly in $\Omega^6_{\log}(\mathrm{CGr}(4) \setminus D)$. However, the form as in Definition 6.1.2 will be more convenient later on.

We can give a geometric interpretation to this form as follows. Denote by $\mathrm{CGr}(4)_{\geq 0}$ the points in $\mathrm{CGr}(4)$ whose Pfaffians all have the same sign, up to global scaling. Then, consider the semi-algebraic set

$$R_{\pm\pm} = \mathrm{CGr}(4)_{\geq 0} \cap \{x \in \mathrm{CGr}(4) \mid \pm S(x) \geq 0 \wedge \pm T(X) \geq 0\} \subset \mathrm{CGr}(4)(\mathbb{R}), \tag{6.1.5}$$

whose boundaries $\partial R_{\pm\pm}$ is in $D(\mathbb{R})$, let $R = R_{++} \cup R_{--}$ and set $[R] = [R^{++}] - [R^{--}]$. Then the main result in this section is the following.

Theorem 6.1.4. *The pair $(\mathrm{CGr}(4), D)$ is a recursive positive pair. The set R gives rise to a well-defined cycle $[R] = [R_{++}] - [R_{--}] \in H_6(\mathrm{CGr}(4), D)$, which is a positive geometry with the following canonical form*

$$\Omega([R]) = \frac{S+T}{\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234}\, \mathrm{pf}_{13}\, \mathrm{pf}_{24}\, ST}\, \mathrm{dpf} = \frac{\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234} - \mathrm{pf}_{13}\, \mathrm{pf}_{24}}{\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234}\, \mathrm{pf}_{13}\, \mathrm{pf}_{24}\, ST}\, \mathrm{dpf} \in \Omega^6_{\log}(\mathrm{CGr}(4) \setminus D),$$

where $\mathrm{dpf} = \mathrm{dpf}_\varnothing \wedge \mathrm{dpf}_{1234} \wedge \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14}$. *Moreover,* $(\mathrm{CGr}(4), R, \Omega([R]))$ *makes a positive geometry in the sense of Definition 1.5.1.*

The remainder of this section is dedicated to proving this theorem.

Algebraic Boundary Stratification

The first goal is to study the pair $(\mathrm{CGr}(4), D)$, and in particular to assert that this makes a recursive genus zero pair. More precisely, we study the boundary stratification $\mathcal{L}(D)$ of the

divisor $D \subset \mathrm{CGr}(4)$ introduced in the preceding section. We start by studying the varieties $\mathbb{V}(S), \mathbb{V}(T) \subset \mathrm{CGr}(4)$.

We introduce some notation. Let $\mathcal{S}$ be the Segre embedding of $\mathbb{P}^3 \times \mathbb{P}^3 \hookrightarrow \mathbb{P}^{15}$, and denote the coordinates of the codomain by z_{ij} for $1 \leq i, j \leq 4$. Denote by $\pi : \mathbb{P}^{15} \dashrightarrow \mathbb{P}^7$ the projection onto the subspace with coordinates labeled by $11, 12, 21, 22, 33, 34, 43, 44$. Let $\varphi = \pi \circ \mathcal{S}$ be the composition. The 5-dimensional closure $Y \subset \mathbb{P}^7$ of the image of $\varphi : \mathbb{P}^3 \times \mathbb{P}^3 \dashrightarrow \mathbb{P}^7$ is precisely given by the restriction of the usual Segre relations:

$$Y = \mathbb{V}(z_{11}z_{22} - z_{12}z_{21}, z_{33}z_{44} - z_{34}z_{43}). \tag{6.1.6}$$

Note that φ has generically 1-dimensional fibers, which come from a torus action on $\mathbb{P}^3 \times \mathbb{P}^3$. Let $\mathbb{C}^\times$ act on $\mathbb{P}^3 \times \mathbb{P}^3$ via

$$t \bullet [x] \times [y] \mapsto [x_1 : x_2 : tx_3 : tx_4] \times [y_1 : y_2 : t^{-1}y_3 : t^{-1}y_4]. \tag{6.1.7}$$

This gives rise to a GIT quotient $(\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$, which is constructed as follows. Let $\mathbb{P}^3 \times \mathbb{P}^3$ be equipped with its natural line bundle $\mathcal{O}(1,1)$, corresponding to the bi-grading of the variables $\deg x_i = (1,0)$ and $\deg y_i = (0,1)$. Let T be the ring of invariants

$$T = \bigoplus_{n \geq 0} \Gamma(\mathbb{P}^3 \times \mathbb{P}^3, \mathcal{O}(n,n))^{\mathbb{C}^\times} \cong \bigoplus_{n \geq 0} \mathbb{C}[x_i, y_i \mid i \in [4]]^{\mathbb{C}^\times}_{(n,n)},$$

where the subscript (n,n) denotes the homogeneous part of $\mathbb{C}[x_i, y_i \mid i \in [4]]$ in bi-degree (n,n), and the superscript $\mathbb{C}^\times$ means taking only the invariants under the torus action as defined in (6.1.7). In our case the only invariant sections of $\mathcal{O}(1,1)$ are spanned by the products

$$x_1y_1, x_1y_2, x_2y_1, x_2y_2, x_3y_3, x_3y_4, x_4y_3, x_4y_4.$$

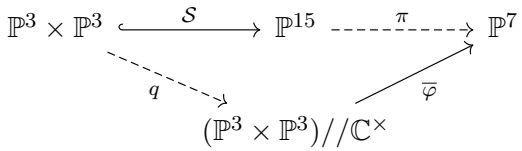
These are precisely the coordinates of the $\mathbb{P}^7$, and their relations are precisely the Segre relations as in (6.1.6). Moreover, these sections generate T as an algebra, since for any invariant monomial $x^\alpha y^\beta$ with $\alpha, \beta \in \mathbb{Z}^4_{\geq 0}$ we must have $\alpha_1 + \alpha_2 = \beta_1 + \beta_2$ and $\alpha_3 + \alpha_4 = \beta_3 + \beta_4$. So x_i and y_j always appear as pairs in $\{1,2\}$ and $\{3,4\}$. Then, by definition the GIT quotient $(\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$ is given by the projectivization of its coordinate ring as

$$T \cong \frac{\mathbb{C}[z_{ij} \mid i \leq j \leq 4]}{\langle z_{11}z_{22} - z_{12}z_{21}, z_{33}z_{44} - z_{34}z_{43} \rangle}.$$

We have just proven the following isomorphism.

Corollary 6.1.5. *The GIT quotient $(\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$ is isomorphic to $Y \subset \mathbb{P}^7$.*

Therefore, we get the following diagram, where $\overline{\varphi} : (\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times \to \mathbb{P}^7$ is the isomorphism from above, and $q : \mathbb{P}^3 \times \mathbb{P}^3 \dashrightarrow (\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$ denotes the quotient map.



Lemma 6.1.6. *The subvarieties S and T in $\mathrm{CGr}(4)$ are isomorphic to $(\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$.*

Proof. We prove the case for $S \subset \mathrm{CGr}(4)$ explicitly. Recall that

$$S = \mathbb{V}(\mathrm{pf}_{\varnothing}\,\mathrm{pf}_{1234} - \mathrm{pf}_{12}\,\mathrm{pf}_{34}, \mathrm{pf}_{13}\,\mathrm{pf}_{24} - \mathrm{pf}_{14}\,\mathrm{pf}_{23}) \subset \mathbb{P}^7.$$

and $Y = \overline{\mathrm{im}\,\varphi} \subset \mathbb{P}^7$ from (6.1.6). The isomorphism $Y \cong S$ then can be seen by comparing the defining ideals after the following change of coordinates on $\mathbb{P}^7$:

$$\begin{aligned} z_{11} &\mapsto \mathrm{pf}_{\varnothing}, \quad z_{22} \mapsto \mathrm{pf}_{1234}, \quad z_{33} \mapsto \mathrm{pf}_{13}, \quad z_{44} \mapsto \mathrm{pf}_{24}, \\ z_{12} &\mapsto \mathrm{pf}_{12}, \quad z_{21} \mapsto \mathrm{pf}_{34}, \quad z_{34} \mapsto \mathrm{pf}_{23}, \quad z_{43} \mapsto \mathrm{pf}_{14}\,. \end{aligned}$$

Then the claim for S follows from Corollary 6.1.5. The proof for $T \subset \mathrm{CGr}(4)$ is analogous. Here the change of coordinates is given as

$$\begin{aligned} z_{11} &\mapsto \mathrm{pf}_{\varnothing}, \quad z_{22} \mapsto \mathrm{pf}_{1234}, \quad z_{33} \mapsto \mathrm{pf}_{13}, \quad z_{44} \mapsto \mathrm{pf}_{24}, \\ z_{12} &\mapsto \mathrm{pf}_{12}, \quad z_{21} \mapsto \mathrm{pf}_{34}, \quad z_{34} \mapsto \mathrm{pf}_{23}, \quad z_{43} \mapsto \mathrm{pf}_{14}\,. \end{aligned}$$

This finishes the proof. □

Similarly, we want to understand the intersection $\mathbb{V}(S) \cap \mathbb{V}(T) \subset \mathrm{CGr}(4)$. To that end, let $\mathcal{S} : (\mathbb{P}^1)^4 \hookrightarrow \mathbb{P}^{15}$ be the fourfold Segre embedding, sending

$$\big([a_1 : a_2], [b_1 : b_2], [c_1 : c_2], [d_1 : d_2]\big) \longmapsto [z_{1111} : z_{1112} : \cdots : z_{2222}],$$

where $z_{ijkl} = a_i b_j c_k d_l$, for $i, j, k, l \in \{1, 2\}$. Let $\pi : \mathbb{P}^{15} \dashrightarrow \mathbb{P}^7$ be the projection to the coordinates z_{ijkl} with indices $\{1111, 1122, 1212, 1221, 2112, 2121, 2211, 2222\}$. Note that the Zariski closure of the projection of $\mathrm{im}\,\mathcal{S}$ under π is a 4-dimensional irreducible variety which is a complete intersection of the three quadric determinantal Segre relations:

$$Y = \mathbb{V}(z_{1111}z_{2222} - z_{1122}z_{2211}, z_{1111}z_{2222} - z_{1212}z_{2121}, z_{1111}z_{2222} - z_{1221}z_{2112}) \tag{6.1.8}$$

as a subvariety in $\mathbb{P}^7$. Then, we can identify $\mathbb{V}(S) \cap \mathbb{V}(T) \subset \mathrm{CGr}(4)$ with $\pi \circ \mathcal{S}\left((\mathbb{P}^1)^4\right)$, as the following Lemma asserts.

Lemma 6.1.7. *The variety $\mathbb{V}(S) \cap \mathbb{V}(T) \subset \mathrm{CGr}(4)$ is isomorphic to the Zariski closure of $\pi \circ \mathcal{S}\left((\mathbb{P}^1)^4\right)$. In particular, $\mathbb{V}(S) \cap \mathbb{V}(T)$ is an irreducible complete intersection of three quadric hypersurfaces in $\mathbb{P}^7$.*

Proof. Consider the coordinate change on $\mathbb{P}^7$ given by

$$\begin{aligned} z_{1111} &\mapsto \mathrm{pf}_{13}, \quad z_{2222} \mapsto \mathrm{pf}_{24}, \quad z_{1122} \mapsto \mathrm{pf}_{14}, \quad z_{2211} \mapsto \mathrm{pf}_{23}, \\ z_{1212} &\mapsto \mathrm{pf}_{12}, \quad z_{2121} \mapsto \mathrm{pf}_{34}, \quad z_{1221} \mapsto \mathrm{pf}_{\varnothing}, \quad z_{2112} \mapsto \mathrm{pf}_{1234}\,. \end{aligned}$$

Then, after this transformation the equations of S and T in (6.1.3) are precisely the first two generators of the variety cutting out $Y = \overline{\pi(\mathrm{im}\,\mathcal{S})}$ from (6.1.8). Similarly, the last generator is precisely the transform of the Pfaffian constraint (6.1.2) modulo the other generators. More succinctly, on the vanishing locus of the Pfaffian constraint P from (6.1.2) we have

$$\mathrm{pf}_{\varnothing}\,\mathrm{pf}_{1234} - \mathrm{pf}_{13}\,\mathrm{pf}_{24} = S + T.$$

Therefore, $Y \cong \mathbb{V}(S, T, P) \subset \mathbb{P}^7$. This shows the claim. □

We can now compute the whole boundary stratification of D. Recall the notion of joins of varieties from Definition 1.1.6.

Proposition 6.1.8. *The boundary stratification $\mathcal{L}(D)$ has 150 components, and its poset structure is sketched in Figure 6.1.1, where the small green numbers indicate the number of strata of that type.*

Proof. This can be directly computed in any computer algebra system with algorithms for irreducible decomposition and the determination of singular loci. For our implementation in *Macaulay2*, see [Koe26]. Note that the singular loci of the subvarieties appear as irreducible components of the intersections. □

We can also prove this result algebraically. Here is a sketch.

Sketch of Proposition 6.1.8. Recall that $\mathrm{CGr}(4) \subset \mathbb{P}^7$ was cut out by the quadric Pfaffian constraint in (6.1.2). Also, recall that $D \subset \mathrm{CGr}(4)$ was the union of coordinate hyperplane sections and the vanishing loci of S and T. Then, everything follows from the structure and symmetries of the Pfaffian constraint (6.1.2). First, note that we obtain Grassmannians $\mathrm{Gr}(2,4)$ by restricting to any codimension 2 coordinate space of $\mathbb{P}^7$, e.g. on $\mathrm{pf}_{1234} = \mathrm{pf}_{\varnothing} = 0$ we get

$$-\mathrm{pf}_{12}\,\mathrm{pf}_{34} + \mathrm{pf}_{13}\,\mathrm{pf}_{24} - \mathrm{pf}_{14}\,\mathrm{pf}_{23}, \tag{6.1.9}$$

Hence, there are $\binom{4}{1} = 4$ such strata. Similarly, we obtain $\mathbb{P}^1 \times \mathbb{P}^1$ by setting all Pfaffian coordinates of two monomials in (6.1.2) simultaneously to zero, e.g.

$$\mathrm{pf}_{13}\,\mathrm{pf}_{24} - \mathrm{pf}_{14}\,\mathrm{pf}_{23}\,.$$

Hence, there are $\binom{4}{2} = 6$ of these strata. The projective spaces $\mathbb{P}^3$ are obtained by setting a single coordinate of each monomial in (6.1.2) to zero. Hence, there are $2^4 = 16$ of these strata. The counts for the other projective spaces are similar.

Now we turn to the slightly more subtle strata in Figure 6.1.1. First, $J(\mathbb{P}^0, \mathrm{Gr}(2,4)) \subset \mathbb{P}^5$ denotes the join over a Grassmannian with a base point corresponding to any of the coordinate vertices in $\mathbb{P}^7$. So $J(\mathbb{P}^0, \mathrm{Gr}(2,4))$ is the set of lines going through one of the coordinate points of $\mathbb{P}^7$ and a point in the Grassmannian, which is obtained by deleting one of the monomials in the Pfaffian constraint (6.1.2). Hence, there are 8 of these strata. For example, we get the projective cone over p of the Grassmannian in (6.1.9). Next, $J(\mathbb{P}^1, \mathbb{P}^1 \times \mathbb{P}^1)$ denotes the join of $\mathbb{P}^1 \times \mathbb{P}^1$ with $\mathbb{P}^1$ as the base, i.e. it is the set traced out by all 2-planes containing the base and a point on $\mathbb{P}^1 \times \mathbb{P}^1$. The base $\mathbb{P}^1$ is spanned by any pair of the vertices of $\mathbb{P}^7$, except for the four lines corresponding to the four monomials in the Pfaffian constraint. More succinctly, $\mathrm{pf}_{\varnothing}\,\mathrm{pf}_{1234}$, $\mathrm{pf}_{12}\,\mathrm{pf}_{34}$, $\mathrm{pf}_{13}\,\mathrm{pf}_{24}$, and $\mathrm{pf}_{14}\,\mathrm{pf}_{23}$, do not give rise to such a stratum. Hence, there are $\binom{8}{2} - 4 = 24$ of these strata. Similarly, we get a stratum of type $J(\mathbb{P}^0, \mathbb{P}^1 \times \mathbb{P}^1)$, the join of $\mathbb{P}^1 \times \mathbb{P}^1$ over a vertex $\mathbb{P}^0$, for each binomial in the Pfaffian constraint and choice of three vanishing coordinates that are not in the support of the binomial, that is $\binom{4}{2} \cdot \binom{4}{3} = 24$. The remaining counts for the strata are simpler to derive. Lastly, note

that the singular loci of the strata so far are lower-dimensional strata in Figure 6.1.1. Hence, we have 150 distinct strata, i.e. $|\mathcal{L}(D)| = 150$. □

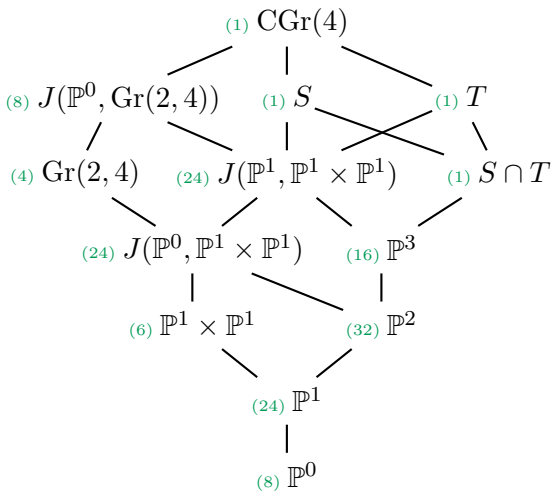


Figure 6.1.1: Algebraic boundary stratification of D.

Genus and Combinatorial Rank

In this section we connect the setup in Cosmology with the positive geometry framework laid out in Chapter 2. The main result is the following.

Theorem 6.1.9. *The pair $(\mathrm{CGr}(4), D)$ is a recursive genus zero pair.*

Proof. First, note that all projective spaces, $\mathbb{P}^1 \times \mathbb{P}^1$ and Grassmannians that appear in the boundary stratification $\mathcal{L}(D)$ shown in Figure 6.1.1, are trivially genus zero varieties. See Lemma 2.1.13 for $\mathbb{P}^1 \times \mathbb{P}^1$ and Lemma 2.1.14 for $\mathrm{Gr}(2,4)$. The same is true for the cones and joins, $J(\mathbb{P}^0, \mathrm{Gr}(2,4))$, $J(\mathbb{P}^0, \mathbb{P}^1 \times \mathbb{P}^1)$ and $J(\mathbb{P}^1 \times \mathbb{P}^1)$, appearing in $\mathcal{L}(D)$, which can be seen as follows. These varieties are complete intersections of at most two coordinate hyperplane sections in $\mathrm{CGr}(4)$. Since $\mathrm{CGr}(4)$ is a smooth hypersurface of degree 2 inside $\mathbb{P}^7$, its canonical bundle is isomorphic to $\mathcal{O}_{\mathrm{CGr}(4)}(-6)$, which follows by adjunction, see e.g. [Vak25, Proposition 29.4.8]. Hence, $J(\mathbb{P}^0, \mathrm{Gr}(2,4))$, $J(\mathbb{P}^0, \mathbb{P}^1 \times \mathbb{P}^1)$ and $J(\mathbb{P}^1 \times \mathbb{P}^1)$ as subvarieties of $\mathrm{CGr}(4)$ all satisfy the assumptions in Theorem 2.3.6, and therefore have vanishing genus. Lastly, we need to consider $\mathbb{V}(S)$, $\mathbb{V}(T)$ and $\mathbb{V}(S,T)$. However, $\mathbb{V}(S), \mathbb{V}(T) \subset \mathrm{CGr}(4)$ are quadric hypersurfaces in $\mathrm{CGr}(4)$, hence their genera vanish by Theorem 2.3.6. And $\mathbb{V}(S) \cap \mathbb{V}(T) \subset \mathrm{CGr}(4)$ is a complete intersection of two quadrics in $\mathrm{CGr}(4)$, hence it too has vanishing genus by Theorem 2.3.6. This shows that all irreducible components in the boundary stratification $\mathcal{L}(D)$ have vanishing genus. Therefore, $(\mathrm{CGr}(4), D)$ makes a recursive genus zero pair by Corollary 2.4.3. □

To compute the combinatorial rank, we establish that the complement $\mathrm{CGr}(4) \setminus D$ is a complement of a line arrangement times an algebraic torus $(\mathbb{C}^\times)^4$. The line arrangement is shown in Figure 6.1.2. We will study the canonical forms of the regions in there in the next subsection.

Lemma 6.1.10. *The complement $\mathrm{CGr}(4) \setminus D$ is isomorphic to the product $(\mathbb{C}^\times)^4 \times U$, where*

$$U = \mathbb{C}^2 \setminus \mathbb{V}(xy(1-x+y)(x-1)(x-y)),$$

is the complement of an affine line arrangement.

Proof. Let $\mathring{\mathrm{CGr}}(4) = \mathrm{CGr}(4) \setminus D_{\mathrm{pf}}$, where D_{pf} is the divisor in $\mathrm{CGr}(4)$ given by the union of the Pfaffian hyperplane sections. Then, $\mathring{\mathrm{CGr}}(4)$ is a smooth very affine variety, and has $(\mathbb{C}^\times)^4$ acting on it via

$$(t_1,t_2,t_3,t_4) \bullet A = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & t_1^{-1}t_2\,\mathrm{pf}_{12} & t_1^{-1}t_3\,\mathrm{pf}_{13} & t_1^{-1}t_4\,\mathrm{pf}_{14} \\ 0 & 1 & 0 & 0 & -t_1t_2^{-1}\,\mathrm{pf}_{12} & 0 & t_2^{-1}t_3\,\mathrm{pf}_{23} & t_2^{-1}t_4\,\mathrm{pf}_{24} \\ 0 & 0 & 1 & 0 & -t_1t_3^{-1}\,\mathrm{pf}_{13} & -t_2t_3^{-1}\,\mathrm{pf}_{23} & 0 & t_3^{-1}t_4\,\mathrm{pf}_{34} \\ 0 & 0 & 0 & 1 & -t_1t_4^{-1}\,\mathrm{pf}_{14} & -t_2t_4^{-1}\,\mathrm{pf}_{24} & -t_3t_4^{-1}\,\mathrm{pf}_{34} & 0 \end{bmatrix}.$$

Note that the vanishing loci of S and T are invariant under this group action, by their definition as minors of the matrix A in (6.1.1). Then we get an isomorphism $\mathring{\mathrm{CGr}}(4) \cong (\mathbb{C}^\times)^4 \times W$ via $\varphi : \mathring{\mathrm{CGr}}(4) \to (\mathbb{C}^\times)^4 \times W$ sending

$$(1, \mathrm{pf}_{1234}, \mathrm{pf}_{12}, \ldots, \mathrm{pf}_{34}) \mapsto (\mathrm{pf}_{12}, \mathrm{pf}_{13}, \mathrm{pf}_{14}, \mathrm{pf}_{23}) \times \left(\frac{\mathrm{pf}_{13}\,\mathrm{pf}_{24}}{\mathrm{pf}_{14}\,\mathrm{pf}_{23}}, \frac{\mathrm{pf}_{12}\,\mathrm{pf}_{34}}{\mathrm{pf}_{14}\,\mathrm{pf}_{23}} \right),$$

where $W = \mathbb{C}^2 \setminus \mathbb{V}(xy)$ and x, y are the coordinates on $\mathbb{C}^2$. The inverse is given by $\psi : (\mathbb{C}^\times)^4 \times W \to \mathring{\mathrm{CGr}}(4)$ sending

$$(a,b,c,d) \times (x,y) \longmapsto \left(a, b, c, d, \frac{cd}{b}x, \frac{cd}{a}y\right), \tag{6.1.10}$$

Indeed, since

$$\frac{\mathrm{pf}_{14}\,\mathrm{pf}_{23}}{\mathrm{pf}_{13}} \frac{\mathrm{pf}_{13}\,\mathrm{pf}_{24}}{\mathrm{pf}_{14}\,\mathrm{pf}_{23}} = \mathrm{pf}_{24}, \quad \text{and} \quad \frac{\mathrm{pf}_{14}\,\mathrm{pf}_{23}}{\mathrm{pf}_{12}} \frac{\mathrm{pf}_{12}\,\mathrm{pf}_{34}}{\mathrm{pf}_{14}\,\mathrm{pf}_{23}} = \mathrm{pf}_{34}\,,$$

the maps φ and ψ are inverse to each other, where we recover the missing coordinate pf_{1234} via the Pfaffian constraint (6.1.2). Hence, $\mathrm{im}\,\psi = \mathring{\mathrm{CGr}}(4)$. Then, we can set $\mathrm{pf}_{12} = \mathrm{pf}_{13} = \mathrm{pf}_{14} = \mathrm{pf}_{23} = 1$, to obtain

$$\varphi(\mathrm{pf}_{1234}) = 1 - x + y, \qquad \varphi(S) = x - 1, \qquad \varphi(T) = x - y.$$

Hence, we get an isomorphism of $\mathrm{CGr}(4) \setminus D \cong (\mathbb{C}^\times)^4 \times U$, where

$$U = \mathbb{C}^2 \setminus \mathbb{V}(xy(1-x+y)(x-1)(x-y)) = \mathbb{C}^2 \setminus \mathcal{A},$$

and x, y are the coordinates on $\mathbb{C}^2$. So U is a complement of an affine line arrangement $\mathcal{A} \subset \mathbb{C}^2$. □

Remark 6.1.11. Note that the proof of Lemma 6.1.10 shows that we can fix $t_1, \ldots, t_4$ such

that any generic point in $\mathring{\mathrm{CGr}}(4)$, can be written by taking the Pfaffians of

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & -1 & 0 & 1 & x \\ 0 & 0 & 1 & 0 & -1 & -1 & 0 & y \\ 0 & 0 & 0 & 1 & -1 & -x & -y & 0 \end{bmatrix}.$$

Proposition 6.1.12. *The combinatorial rank of the pair* $(\mathrm{CGr}(4), D)$ *equals* 6.

We give two separate, albeit related, proofs.

Proof. First, recall the isomorphism $\mathrm{CGr}(4) \setminus D \cong (\mathbb{C}^\times)^4 \times U$ from Lemma 6.1.10. Denote the affine line arrangement by $\mathcal{A} \subset \mathbb{C}^2$. Then, consider the projectivization of $\mathcal{A}$, denoted by $\widetilde{\mathcal{A}} \subset \mathbb{P}^2$. Note that $(\mathbb{P}^2 \setminus \widetilde{\mathcal{A}}) \cup \mathbb{V}(z) \cong U$, where we identify $\mathbb{C}^2$ with $\mathbb{P}^2 \setminus \mathbb{V}(z)$ by Proposition 1.3.24. Then, $H^2_c(U) \cong H^2_c(\mathbb{P}^2 \setminus \widetilde{\mathcal{A}} \cup \mathbb{V}(z))$, and the latter is isomorphic to $H^2(\mathbb{P}^2, \widetilde{\mathcal{A}} \cup \mathbb{V}(z))$. Therefore, by [BD25, Proposition 6.2] the combinatorial rank of $(\mathbb{P}^2, \widetilde{\mathcal{A}} \cup \mathbb{V}(z))$, and hence that of $U = \mathbb{C}^2 \setminus \mathcal{A}$ is given by the number of bounded regions in U with a generic line added, see Figure 6.1.2. This number is easily seen to be 6. Hence, the combinatorial rank of $(\mathrm{CGr}(4), D) = 6$ by Proposition 2.2.19. □

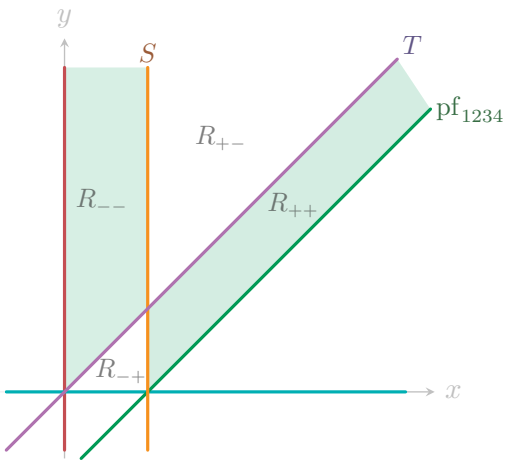


Figure 6.1.2: The line arrangement U underlying $\mathring{\mathrm{CGr}}(4)$, with regions corresponding to (6.1.5).

Proof of Proposition 6.1.12. Recall the isomorphism of $\mathrm{CGr}(4) \setminus D$ with a product of a line arrangement complement and a torus from Lemma 6.1.10. Since the normal vectors of the lines in this arrangement span $\mathbb{C}^2$, i.e. the arrangement is essential, it follows that the compactification of $\mathring{\mathrm{CGr}}(4)$ is sehr schön for example by [Sch24b, Theorem 1.1]. Therefore, the Euler characteristic of the tropicalization of $\mathrm{CGr}(4) \setminus D$ is the combinatorial rank by Proposition 2.2.12. We compute the tropicalization in MACAULAY2 and obtain combinatorially the cone over the 1-dimensional fan F obtained from contracting the orange edge of the Petersen graph in Figure 2.2.2. Hence, $\mathrm{trop}(\mathrm{CGr}(4) \setminus D)$ has (modulo lineality space) 9 vertices

and 14 edges all of weight 1, so its f-vector is $f = (1, 9, 14)$. Then, by Proposition 2.2.12 the combinatorial rank of $(\mathrm{CGr}(4), D)$ is given as

$$\mathrm{cr}(\mathrm{CGr}(4), D) = \chi(\mathrm{Lk}\,\mathrm{trop}(\mathrm{CGr}(4) \setminus D)) = 1 - 9 + 14 = 6.$$

□

Semi-algebraic Boundary

From now on we will focus on the particular region R inside the real points of $\mathrm{CGr}(4)$ which will give a geometric interpretation of the wavefunction coefficient A_4 for $n = 4$ in Definition 6.1.2. Recall the definitions of $R_{\pm\pm}$ in (6.1.5), where we also set

$$R = R_{++} \cup R_{--} \subset \mathrm{CGr}(4)(\mathbb{R}),$$

where S, T were defined in (6.1.3). This is a 6-dimensional semi-algebraic set. It is connected in codimension two through $\mathbb{V}(S, T) \cap R \subset \mathrm{CGr}(4)$. This can be seen from direct computation using quantifier elimination in *Mathematica* [Koe26]. A low-dimensional sketch for R is seen as the green-shaded area in Figure 6.1.3.

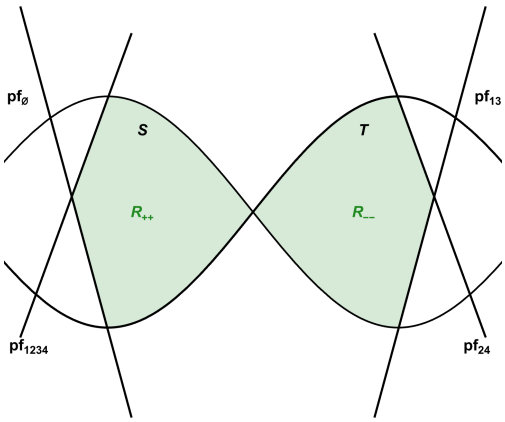


Figure 6.1.3: Sketch of $R \subset \mathrm{CGr}(4)(\mathbb{R})$.

We establish its algebraic boundary first.

Lemma 6.1.13. *The algebraic boundary of R is*

$$\partial_a R = \mathbb{V}(\mathrm{pf}_\varnothing) \cup \mathbb{V}(\mathrm{pf}_{1234}) \cup \mathbb{V}(\mathrm{pf}_{13}) \cup \mathbb{V}(\mathrm{pf}_{24}) \cup \mathbb{V}(S) \cup \mathbb{V}(T) \subset \mathrm{CGr}(4).$$

Therefore, we have $\partial_a R \subset D$ from (6.1.4). The other Pfaffian hyperplane sections have lower-dimensional intersection with R over the reals.

Proof. This is computed in *Mathematica*, see [Koe26]. □

Remark 6.1.14. Note that this can also be seen from the isomorphism in Lemma 6.1.10. Then R is isomorphic to the union of the two regions R_{++} and R_{--} shown in Figure 6.1.2 up to the torus orbit $(\mathbb{C}^\times)^4$.

Therefore, we have the following direct consequence.

Corollary 6.1.15. *Let $\iota : (\mathrm{CGr}(4)(\mathbb{R}), \partial_a(R)(\mathbb{R})) \hookrightarrow (\mathrm{CGr}(4), D)$ be the inclusion map. Then, $\iota_*[R]$ is a well-defined cycle in $H_6(\mathrm{CGr}(4), D)$.*

We can also compute the residual arrangement of R.

Lemma 6.1.16. *The residual arrangement $\mathcal{R}(R) \subset \mathrm{CGr}(4)$ of R is pure of dimension 4 and is given by*

$$\mathcal{R}(R) = \mathbb{V}(\mathrm{pf}_{\varnothing}, \mathrm{pf}_{13}) \cup \mathbb{V}(\mathrm{pf}_{\varnothing}, \mathrm{pf}_{24}) \cup \mathbb{V}(\mathrm{pf}_{1234}, \mathrm{pf}_{13}) \cup \mathbb{V}(\mathrm{pf}_{1234}, \mathrm{pf}_{24}).$$

Proof. This is computed directly in *Mathematica* using quantifier elimination, see [Koe26]. □

There is a unique adjoint-like polynomial.

Lemma 6.1.17. *There is a unique global section $\mathrm{adj}(R) \in \Gamma(\mathrm{CGr}(4), \mathcal{O}_{\mathrm{CGr}(4)}(2))$ interpolating the residual arrangement $\mathcal{R}(R)$ together with $\mathbb{V}(S, T)$. Explicitly, we have*

$$\mathrm{adj}(R) = \mathrm{pf}_{12}\,\mathrm{pf}_{34} - \mathrm{pf}_{14}\,\mathrm{pf}_{23}\,.$$

Proof. Recall that $\mathrm{OGr}(4, 8)$ is a smooth hypersurface in $\mathbb{P}^7$ given by (6.1.2), its canonical divisor is $\omega_{\mathrm{OGr}(4,8)} \cong \mathcal{O}_{\mathrm{OGr}(4,8)}(-6)$, which can be computed directly by the adjunction formula, see e.g. [Vak25, Proposition 29.4.8]. Thus, by Definition 1.5.8 the degree of the adjoint hypersurface $\deg \mathrm{adj}(R)$ is given as $\deg \partial_a R + \deg \omega_{\mathrm{OGr}(4,8)} = 2$, where we used $\deg \partial_a R$ from Lemma 6.1.13. The uniqueness then can be directly verified in *Macaulay2*, see [Koe26]. □

Remark 6.1.18. Note that $\mathrm{adj}(R)$ is precisely the numerator of the wavefunction coefficient A_4 in Definition 6.1.2. Moreover, as the Lemma above asserts it cancels all spurious poles expected by Physics. However, despite the suggestive name $\mathrm{adj}(R)$, this does not satisfy the definition of an adjoint polynomial 1.5.8, since we need to demand the vanishing on $\mathbb{V}(S, T)$, which is not part of the residual arrangement.

Canonical Form

In this section, we can finally prove the main result 6.1.4, which establishes a geometric description of the wavefunction coefficient A_4 for $n = 4$ through the lens of positive geometry as in Chapter 2. Recall the definitions of $R_{\pm\pm}$ in (6.1.5), where we also set

$$R = R_{++} \cup R_{--} \subset \mathrm{CGr}(4)(\mathbb{R}),$$

where S, T were defined in (6.1.3). Also recall that $D \subset \mathrm{CGr}(4)$ from (6.1.4), was the union of all Pfaffian hyperplane sections and the vanishing loci of S and T, from (6.1.3). We restate the main result here for convenience.

Theorem 6.1.19. *The pair $(\mathrm{CGr}(4), D)$ is a recursive positive pair. The set R gives rise to a well-defined cycle $[R] = [R_{++}] - [R_{--}] \in H_6(\mathrm{CGr}(4), D)$, which is a positive geometry with the following canonical form*

$$\Omega([R]) = \frac{S+T}{\mathrm{pf}_\varnothing\,\mathrm{pf}_{1234}\,\mathrm{pf}_{13}\,\mathrm{pf}_{24}\,ST}\,\mathrm{dpf} = \frac{\mathrm{pf}_\varnothing\,\mathrm{pf}_{1234} - \mathrm{pf}_{13}\,\mathrm{pf}_{24}}{\mathrm{pf}_\varnothing\,\mathrm{pf}_{1234}\,\mathrm{pf}_{13}\,\mathrm{pf}_{24}\,ST}\,\mathrm{dpf} \in \Omega^6_{\log}(\mathrm{CGr}(4)\setminus D),$$

where $\mathrm{dpf} = \mathrm{dpf}_\varnothing \wedge \mathrm{dpf}_{1234} \wedge \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14}$. *Moreover,* $(\mathrm{CGr}(4), R, \Omega([R]))$ *makes a positive geometry in the sense of Definition 1.5.1.*

Remark 6.1.20. The rational function of the form above is precisely the $n = 4$ wavefunction coefficient A_4 from Definition 6.1.2, as first introduced in [Aru+26]. Also note that $\Omega(\iota_*[R])$ equals the pullback of the canonical form for $R_{++} \cup R_{--}$ under ψ from (6.1.10).

Before we can prove this result we need to establish another intermediate result. Recall the notation from Lemma 6.1.6.

Lemma 6.1.21. *Denote by $\iota : (Q(\mathbb{R}), D_Q(\mathbb{R})) \hookrightarrow (Q, D_Q)$ the inclusion map, where Let Q be the GIT quotient $(\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$ and D_Q the subvariety of Q given in the invariant coordinates $z_{ij} = x_i y_j$ from Lemma 6.1.6 as $D_Q = \mathbb{V}(z_{12}z_{21}z_{33}z_{43}z_{44})$. Denote by $Q_{\geq 0} \subset Q$ the region given by $(\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{R}_{>0}$. Then, $\iota_*[Q_{\geq 0}]$ is a positive geometry in the sense of Definition 2.4.7 with canonical form*

$$\begin{aligned}\omega_Q &= \frac{1}{z_{12}z_{21}z_{33}z_{43}z_{44}}\, dz_{12} \wedge dz_{21} \wedge dz_{33} \wedge dz_{43} \wedge dz_{44} \\ &= \mathrm{dlog}(z_{12}) \wedge \mathrm{dlog}(z_{21}) \wedge \mathrm{dlog}(z_{33}) \wedge \mathrm{dlog}(z_{43}) \wedge \mathrm{dlog}(z_{44}) \in \Omega^5_{\log}(Q \setminus D_Q).\end{aligned}$$

Proof. We set $Q = (\mathbb{P}^3 \times \mathbb{P}^3)//\mathbb{C}^\times$. First, note that on the affine patch $U_1 \times U_1 \subset \mathbb{P}^3 \times \mathbb{P}^3$, we can write the canonical form of the orthants as

$$\omega = \frac{1}{x_2x_3x_4}\frac{1}{y_2y_3y_4}\, d\mathbf{x} \wedge d\mathbf{y},$$

where $d\mathbf{x} = dx_2 \wedge dx_3 \wedge dx_4$ and $d\mathbf{y} = dy_2 \wedge dy_3 \wedge dy_4$. This can be rewritten in terms of the $\mathbb{C}^\times$-invariant coordinates $z_{ij} = x_i y_j$ with $\{i, j\} \in \{11, 12, 21, 22, 33, 34, 43, 44\}$:

$$\omega = dy_4 \wedge \frac{1}{z_{12}z_{21}z_{33}z_{43}z_{44}}\, dz_{12} \wedge dz_{21} \wedge dz_{33} \wedge dz_{43} \wedge dz_{44}$$

where we used $dz_{ij} = y_j dx_i + x_i dy_j$. Now, notice that on $U \subset \mathbb{P}^3 \times \mathbb{P}^3$, the complement of all coordinate hyperplanes, the torus $\mathbb{C}^\times$ acts freely. Then, by [Sta26, Lemma 37.11.11] we get a split short exact sequence of the differential sheaves

$$0 \to q^*\Omega^1_Q \to \Omega^1_U \to \Omega^1_{\mathbb{C}^\times} \to 0,$$

where we used that $U \cong Q \times \mathbb{C}^\times$. By [Sta26, Lemma 17.26.1] the top wedge power splits, i.e.

$$\Omega^6_U \cong q^*\Omega^5_Q \otimes_{\mathcal{O}_U} \Omega^1_{\mathbb{C}^\times}.$$

Choosing as a generator of $\Omega^1_{\mathbb{C}^\times}$ the 1-form dy_4, we identify via the isomorphism above

$\omega = dy_4 \wedge \omega_Q$ with

$$\begin{aligned}\omega_Q &= \frac{1}{z_{12}z_{21}z_{33}z_{43}z_{44}}\, dz_{12} \wedge dz_{21} \wedge dz_{33} \wedge dz_{43} \wedge dz_{44} \\ &= \operatorname{dlog}(z_{12}) \wedge \operatorname{dlog}(z_{21}) \wedge \operatorname{dlog}(z_{33}) \wedge \operatorname{dlog}(z_{43}) \wedge \operatorname{dlog}(z_{44}).\end{aligned}$$

In particular, $\omega_Q \in \Omega^5_{\log}(Q \setminus D_Q)$. Finally, since restricting q to the non-negative points yields a bijection with the region $(\mathbb{P}^3_{\geq 0} \times \mathbb{P}^3_{\geq 0})//\mathbb{R}_{>0}$, the form ω_Q is the canonical form of that region. In particular, $\Omega(\iota_*[(\mathbb{P}^3_{\geq 0} \times \mathbb{P}^3_{\geq 0})/\mathbb{R}_{>0}]) = \omega_Q$. Since $(\mathrm{CGr}(4), D)$ is a recursive genus zero pair by Theorem 6.1.9, $g(Q, D_Q) = 0$ by the isomorphism in Lemma 6.1.6. This shows then that $\iota_*[(\mathbb{P}^3_{\geq 0} \times \mathbb{P}^3_{\geq 0})//\mathbb{R}_{>0}]$ makes a non-zero weight 0 cycle, hence a positive geometry in the sense of Definition 2.4.7. □

We can now turn to the proof of our main result.

Proof of Theorem 6.1.4. Recall the definition of the divisor $D \subset \mathrm{CGr}(4)$ from (6.1.4). The pair $(\mathrm{CGr}(4), D)$ makes a recursive genus zero pair by Theorem 6.1.9. The region R makes a well-defined cycle $\iota_*[R] \in H_6(\mathrm{CGr}(4), D)$ by Corollary 6.1.15.

To show that $\iota_*[R]$ makes a positive geometry, it suffices to prove that

$$\omega_R = \frac{S+T}{\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234}\, \mathrm{pf}_{13}\, \mathrm{pf}_{24}\, ST}\ \mathrm{dpf},$$

with $\mathrm{dpf} = \mathrm{dpf}_\varnothing \wedge \mathrm{dpf}_{1234} \wedge \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14}$ is a logarithmic form for each component of the boundary D, then $\Omega(\iota_*[R]) = \omega_R$.

First, consider the case $\mathrm{pf}_{1234} = 0$. Geometrically, we obtain the cone $J(\mathbb{P}^0, \mathrm{Gr}(2,4))$ over the Grassmannian

$$-\mathrm{pf}_{12}\, \mathrm{pf}_{34} + \mathrm{pf}_{13}\, \mathrm{pf}_{24} - \mathrm{pf}_{14}\, \mathrm{pf}_{23} \subset \mathbb{P}^5,$$

with coordinates $\mathrm{pf}_{12}, \mathrm{pf}_{13}, \mathrm{pf}_{14}, \mathrm{pf}_{23}, \mathrm{pf}_{24}, \mathrm{pf}_{34}$ and base $e_\varnothing \in \mathbb{P}^7$. See also Figure 6.1.1. Its intersection with R yields the positive sub-cone $J(\mathbb{P}^0, \mathrm{Gr}(2,4)_{\geq 0})$ over the non-negative Grassmannian $\mathrm{Gr}(2,4)_{\geq 0}$. Moreover, taking the Poincaré residue of ω_R along $\mathrm{pf}_{1234} = 0$ yields

$$\begin{aligned}\mathrm{Res}_{\mathrm{pf}_{1234}}\, \omega_R &= \frac{1}{\mathrm{pf}_\varnothing}\, \mathrm{dpf}_\varnothing \wedge \frac{\mathrm{pf}_\varnothing\, \mathrm{pf}_{1234} - \mathrm{pf}_{13}\, \mathrm{pf}_{24}}{\mathrm{pf}_{13}\, \mathrm{pf}_{24}\, ST}\bigg|_{\mathrm{pf}_{1234}=0} \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14} \\ &= \frac{1}{\mathrm{pf}_\varnothing}\, \mathrm{dpf}_\varnothing \wedge \frac{1}{ST}\bigg|_{\mathrm{pf}_{1234}=0} \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14} \\ &= \frac{1}{\mathrm{pf}_\varnothing}\, \mathrm{dpf}_\varnothing \wedge \frac{1}{\mathrm{pf}_{12}\, \mathrm{pf}_{23}\, \mathrm{pf}_{34}\, \mathrm{pf}_{14}}\ \mathrm{dpf}_{12} \wedge \mathrm{dpf}_{23} \wedge \mathrm{dpf}_{34} \wedge \mathrm{dpf}_{14}\ . \qquad (6.1.11)\end{aligned}$$

This is clearly the canonical form of $J(\mathbb{P}^0, \mathrm{Gr}(2,4)_{\geq 0})$ by identifying each pf_{ij} with the Plücker coordinate p_{ij} in $\mathrm{Gr}(2,4)$. The cases $\mathrm{pf}_\varnothing = 0$, $\mathrm{pf}_{13} = 0$, and $\mathrm{pf}_{24} = 0$ follow analogously.

By symmetry in S and T, it remains to show the case of $S = 0$. First note that using the

total differential formula, and the Pfaffian constraint (6.1.2), we can write

$$dS = -\operatorname{pf}_{1234}\operatorname{dpf}_{\varnothing} - p_{\varnothing}\operatorname{dpf}_{1234} + \operatorname{pf}_{34}\operatorname{dpf}_{12} + \operatorname{pf}_{12}\operatorname{dpf}_{34}\,.$$

Hence, we can write the differential form ω_R as

$$\omega_R = \frac{dS}{S} \wedge \frac{S+T}{\operatorname{pf}_{\varnothing}\operatorname{pf}_{1234}\operatorname{pf}_{13}\operatorname{pf}_{24}\,T}\frac{1}{\operatorname{pf}_{34}}\ \operatorname{dpf}_{\varnothing}\wedge\operatorname{dpf}_{1234}\wedge\operatorname{dpf}_{23}\wedge\operatorname{dpf}_{34}\wedge\operatorname{dpf}_{14}\,.$$

Then, we can take the Poincaré residue along $S=0$ to obtain

$$\operatorname{Res}_S\omega_R = -\frac{1}{\operatorname{pf}_{\varnothing}\operatorname{pf}_{1234}\operatorname{pf}_{13}\operatorname{pf}_{24}\operatorname{pf}_{34}}\ \operatorname{dpf}_{\varnothing}\wedge\operatorname{dpf}_{1234}\wedge dp_{23}\wedge dp_{34}\wedge dp_{14}.$$

This is the pushforward of the canonical form of the products of orthants in $\mathbb{P}^3\times\mathbb{P}^3$ under the quotient map $q:\mathbb{P}^3\times\mathbb{P}^3\to(\mathbb{P}^3\times\mathbb{P}^3)//\mathbb{C}^\times$. See Lemmas 6.1.6 and 6.1.21. Therefore, we have shown that ω_R has the correct residues along each facet, and thus $\omega_R=\Omega(\iota_*[R])$. In particular, this implies that R is a real weight 0 cycle in $H_6(\mathrm{CGr}(4),D)$, hence $(\mathrm{CGr}(4),D)$ is a recursive positive pair by Theorem 6.1.9. Hence, $\iota_*[R]$ is a positive geometry in the sense of Definition 2.4.7. These computations also show explicitly the recursive property of the canonical form as in Defintion 1.5.1, showing the last claim. □

Remark 6.1.22. Note that taking residue of the canonical form of $J(\mathbb{P}^0,\mathrm{Gr}(2,4)_{\geq 0})$ as in (6.1.11), along $\operatorname{pf}_{\varnothing}=0$ yields the canonical form of the non-negative Grassmannian $\mathrm{Gr}(2,4)_{\geq 0}$:

$$\operatorname{Res}_{\operatorname{pf}_{\varnothing}}\left(\operatorname{Res}_{\operatorname{pf}_{1234}}\omega_R\right) = \frac{1}{\operatorname{pf}_{12}\operatorname{pf}_{23}\operatorname{pf}_{34}\operatorname{pf}_{14}}\ \operatorname{dpf}_{12}\wedge\operatorname{dpf}_{23}\wedge\operatorname{dpf}_{34}\wedge\operatorname{dpf}_{14}\ .$$

This is the flat-space limit of the wavefunction coefficient A_4 reproducing the MHV scattering amplitude in planar $\mathcal{N}=4$ SYM for 4 particles, as first discovered in [Aru+26]. Moreover, it is worth pointing out that $R\cap\{\operatorname{pf}_{\varnothing}=\operatorname{pf}_{1234}=0\}=\mathrm{Gr}(2,4)_{\geq 0}$ due to our choice of R, hence we obtain the non-negative Grassmannian with its canonical form as a sub-positive geometry.

We end this section with the following remark on the two regions seen in Figure 6.1.3. Recall that we defined them at the beginning of this subsection.

Corollary 6.1.23. *The regions R_{++} and R_{--} are positive geometries in the sense of both Definitions 1.5.1 and 2.4.7. Their canonical forms are given by*

$$\Omega(\iota_*[R_{++}]) = \frac{-1}{ST\operatorname{pf}_{\varnothing}\operatorname{pf}_{1234}}\operatorname{dpf},$$

and

$$\Omega(\iota_*[R_{--}]) = \frac{1}{ST\operatorname{pf}_{13}\operatorname{pf}_{24}}\operatorname{dpf}\,.$$

The sign comes from a choice of a globally compatible orientation.

Proof. Write $R_{\pm}$ for both regions R_{++} and R_{--}. First, we verify in *Mathematica*, see [Koe26], that these two semi-algebraic sets have the algebraic boundary as indicated by the differen-

tial forms from above. In particular, they are regular connected 6-dimensional semi-algebraic sets, and all faces contain 0-dimensional strata. They also correspond to the regions in Figure 6.1.2. Then, analogous computations as in the proof of Theorem 6.1.4, show that the residues along some boundary component of the two forms have simple poles along the algebraic boundary of the restriction of the regions $R_\pm$ along that component. This shows that $R_\pm$ or rather their cycles $\iota_*[R_\pm]$ make positive geometries with the canonical forms as above in the sense of Definition 2.4.7. Lastly, note that $R_\pm$ are orientable manifolds by Lemma 2.4.4, since their canonical forms $\omega_\pm = \Omega(\iota_*[R_\pm])$ are non-zero. Then, it follows from Proposition 2.4.5 that the triples $(\mathrm{CGr}(4), R_\pm, \omega_\pm)$ are positive geometries in the stronger sense of Definition 1.5.1. □

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